US Army Corps of Engineers®
KANSAS CITY DISTRICT

Affinis corp
8900 Indian Creek Parkway, Suite 450
Overland Park, Kansas 66210
Phone: 913-239-1100
Toll Free: 877-527-5468
Fax: 913-239-1111
www.affinis.us
ENGINEERS • SURVEYORS
Building long-lasting relationships

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

LOCATION MAP — PROJECT LOCATION

SOLICITATION NO.: Task Order # W912DQ24F1019
CONTRACT NO.: W912DQ19D1012
ISSUE DATE: 23 February 2026

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly licensed Professional Engineer under the laws of the State of Kansas.

SAMUEL MICHAEL BRISCOE LICENSED PE31009 03-16-2026 KANSAS PROFESSIONAL ENGINEER

SAMUEL M. BRISCOE, P.E. (Date)
My license renewal date is April 30, 2026.
Pages or sheets covered by this seal:
G-001 THRU G-107, CD101 THRU CD103, CP101 THRU CP303, CG101 THRU CG102, C-101 THRU C-519, C-521 THRU C-526, C-601 THRU C-602

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly licensed Professional Engineer under the laws of the State of Kansas.

MADISON E. CROWL LICENSED 03-16-2026 30488 KANSAS PROFESSIONAL ENGINEER

MADISON E.CROWL, P.E. (Date)
My license renewal date is April 30, 2026.
Pages or sheets covered by this seal:
GC111 THRU GC113, CU101 THRU CU307

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly licensed Professional Engineer under the laws of the State of Kansas.

ADDISON RAY MILLER LICENSED 3-16-2026 29189 KANSAS PROFESSIONAL ENGINEER

ADDISON R. MILLER, PTOE (Date)
My license renewal date is April 30, 2027.
Pages or sheets covered by this seal:
GC101 THRU GC110, C-541 THRU C-542

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly licensed Professional Land Surveyor under the laws of the State of Kansas.

JONATHAN AUBREY MEYER LICENSED 3-16-2026 1579 KANSAS PROFESSIONAL SURVEYOR

JONATHAN AUBREY MEYER, PLS (Date)
My license renewal date is March 31, 2027.
Pages or sheets covered by this seal:
V-101 THRU V-105

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912DQ24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

COVER

SHEET ID
G-001

READY TO ADVERTISE

SHEET INDEX

No.	Sheet Number	Sheet Title	AEC_FILENAME
1.	G-001	COVER	53708_ATCH_CSO_G001.DWG
2.	G-002	SHEET INDEX	53708_ATCH_CSO_G002.DWG
3.	G-003	ABBREVIATIONS AND LEGEND	53708_ATCH_CSO_G003_LEGND.DWG
4.	G-004	GENERAL NOTES	53708_ATCH_CSO_G004_GNOTES.DWG
5.	G-105	EASEMENT PLAN (15TH STREET)	53708_ATCH_CSO_G105-G106_15TH-ESMT.DWG
6.	G-106	EASEMENT PLAN (15TH STREET)	53708_ATCH_CSO_G105-G106_15TH-ESMT.DWG
7.	G-107	EASEMENT PLAN (16TH STREET)	53708_ATCH_CSO_G107_16TH-ESMT.DWG
8.	GC101	TRAFFIC CONTROL PLAN (PHASE 1 & 2)	53708_ATCH_CSO_GC101_MAIN-S.DWG
9.	GC102	TRAFFIC CONTROL PLAN (PHASE 3)	53708_ATCH_CSO_GC102_MAIN_N.DWG
10.	GC103	TRAFFIC CONTROL PLAN (PHASE 4 & 5)	53708_ATCH_CSO_GC103_16TH.DWG
11.	GC104	TRAFFIC CONTROL (PHASE 6)	53708_ATCH_CSO_GC104-GC105_15TH.DWG
12.	GC105	TRAFFIC CONTROL (PHASE 6)	53708_ATCH_CSO_GC104-GC105_15TH.DWG
13.	GC106	TRAFFIC CONTROL DETOUR PLAN (SHEET 1 OF 3)	53708_ATCH_CSO_GC106-GC108_DTR.dwg
14.	GC107	TRAFFIC CONTROL DETOUR PLAN (SHEET 2 OF 3)	53708_ATCH_CSO_GC106-GC108_DTR.dwg
15.	GC108	TRAFFIC CONTROL DETOUR PLAN (SHEET 3 OF 3)	53708_ATCH_CSO_GC106-GC108_DTR.dwg
16.	GC109	PAVEMENT MARKING AND SIGNING PLAN (MAIN ST)	53708_ATCH_CSO_GC109-GC110_PM-SGN.dwg
17.	GC110	PAVEMENT MARKING AND SIGNING PLAN (15TH STREET)	53708_ATCH_CSO_GC109-GC110_PM-SGN.dwg
18.	GC111	EROSION CONTROL PLAN (15TH STREET)	53708_ATCH_CSO_GC111-GC112_15TH-EC.dwg
19.	GC112	EROSION CONTROL PLAN (15TH STREET)	53708_ATCH_CSO_GC111-GC112_15TH-EC.dwg
20.	GC113	EROSION CONTROL PLAN (16TH STREET)	53708_ATCH_CSO_GC113_16TH-EC.dwg
21.	V-101	PROJECT CONTROL (15TH STREET)	53708_ATCH_CSO_V101_15TH-CNTRL.DWG
22.	V-102	PROJECT CONTROL (16TH STREET)	53708_ATCH_CSO_V102_16TH-CNTRL.DWG
23.	V-103	EXISTING CONDITIONS (15TH STREET)	53708_ATCH_CSO_V103-V104_15TH-EXIST.DWG
24.	V-104	EXISTING CONDITIONS (15TH STREET)	53708_ATCH_CSO_V103-V104_15TH-EXIST.DWG
25.	V-105	EXISTING CONDITIONS (16TH STREET)	53708_ATCH_CSO_V105_16TH-EXIST.DWG
26.	CD101	DEMOLITION PLAN (15TH STREET)	53708_ATCH_CSO_CD101-CD102_15TH-DEMO.DWG
27.	CD102	DEMOLITION PLAN (15TH STREET)	53708_ATCH_CSO_CD101-CD102_15TH-DEMO.DWG
28.	CD103	DEMOLITION PLAN (16TH STREET)	53708_ATCH_CSO_CD103_16TH-DEMO.DWG
29.	CP101	CIVIL PLAN (15TH STREET)	53708_ATCH_CSO_CP101-CP103_15TH.DWG
30.	CP102	CIVIL PLAN (15TH STREET)	53708_ATCH_CSO_CP101-CP103_15TH.DWG
31.	CP103	CIVIL PLAN (15TH STREET)	53708_ATCH_CSO_CP101-CP103_15TH.DWG
32.	CP104	CIVIL PLAN AND PROFILE (16TH STREET)	53708_ATCH_CSO_CP104_16TH.DWG
33.	CP105	CIVIL PLAN AND PROFILE (ALLEY)	53708_ATCH_CSO_CP105_ALLEY.DWG
34.	CP106	INTERSECTION DETAILS (15TH STREET)	53708_ATCH_CSO_CP106-CP108_15TH.DWG
35.	CP107	INTERSECTION DETAILS (15TH STREET)	53708_ATCH_CSO_CP106-CP108_15TH.DWG
36.	CP108	INTERSECTION DETAILS (15TH STREET)	53708_ATCH_CSO_CP106-CP108_15TH.DWG
37.	CP109	INTERSECTION DETAILS (16TH STREET)	53708_ATCH_CSO_CP109_16TH.DWG
38.	CP110	RAMP DETAILS (15TH STREET)	53708_ATCH_CSO_CP110-CP112_15TH.DWG
39.	CP111	RAMP DETAILS (15TH STREET)	53708_ATCH_CSO_CP110-CP112_15TH.DWG
40.	CP112	RAMP DETAILS (15TH STREET)	53708_ATCH_CSO_CP110-CP112_15TH.DWG
41.	CP113	RAMP DETAILS (16TH STREET)	53708_ATCH_CSO_CP113_16TH.DWG
42.	CP301	CIVIL CROSS SECTIONS (16TH STREET)	53708_ATCH_CSO_CP301-CP302_16TH.DWG
43.	CP302	CIVIL CROSS SECTIONS (16TH STREET)	53708_ATCH_CSO_CP301-CP302_16TH.DWG
44.	CP303	CIVIL CROSS SECTIONS (ALLEY)	53708_ATCH_CSO_CP303_ALLEY.DWG
45.	CU101	UTILITY PLAN AND PROFILE (15TH STREET)	53708_ATCH_CSO_CU101-CU103_15TH.DWG
46.	CU102	UTILITY PLAN AND PROFILE (15TH STREET)	53708_ATCH_CSO_CU101-CU103_15TH.DWG
47.	CU103	UTILITY PLAN AND PROFILE (15TH STREET)	53708_ATCH_CSO_CU101-CU103_15TH.DWG
48.	CU104	UTILITY PLAN AND PROFILE (15TH STREET)	53708_ATCH_CSO_CU104_15TH.DWG
49.	CU105	UTILITY PROFILE (15TH STREET)	53708_ATCH_CSO_CU105_15TH.DWG
50.	CU106	UTILITY PLAN AND PROFILE (16TH STREET)	53708_ATCH_CSO_CU106-CU108_16TH.DWG
51.	CU107	UTILITY PLAN AND PROFILE (16TH STREET)	53708_ATCH_CSO_CU106-CU108_16TH.DWG
52.	CU108	UTILITY PROFILE (16TH STREET)	53708_ATCH_CSO_CU106-CU108_16TH.DWG
53.	CU109	UTILITY PLAN AND PROFILE (ALLEY)	53708_ATCH_CSO_CU109_ALLEY SAN.dwg
54.	CU301	UTILITY CROSS SECTIONS (15TH STREET)	53708_ATCH_CSO_CU301-CU305_15TH.DWG
55.	CU302	UTILITY CROSS SECTIONS (15TH STREET)	53708_ATCH_CSO_CU301-CU305_15TH.DWG
56.	CU303	UTILITY CROSS SECTIONS (15TH STREET)	53708_ATCH_CSO_CU301-CU305_15TH.DWG
57.	CU304	UTILITY CROSS SECTIONS (15TH STREET)	53708_ATCH_CSO_CU301-CU305_15TH.DWG
58.	CU305	UTILITY CROSS SECTIONS (15TH STREET)	53708_ATCH_CSO_CU301-CU305_15TH.DWG
59.	CU306	UTILITY CROSS SECTIONS (16TH STREET)	53708_ATCH_CSO_CU306-CU307_16TH.DWG
60.	CU307	UTILITY CROSS SECTIONS (16TH STREET)	53708_ATCH_CSO_CU306-CU307_16TH.DWG
61.	CG101	CIVIL GRADING PLAN (15TH STREET)	53708_ATCH_CSO_CG101_15TH.DWG
62.	CG102	CIVIL GRADING PLAN (SOUTH OF MAIN STREET)	53708_ATCH_CSO_CG102_16TH.DWG
63.	C-501	CIVIL DETAILS	53708_ATCH_CSO_C501-TYP.DWG
64.	C-502	CIVIL DETAILS	53708_ATCH_CSO_C502-CURB.DWG
65.	C-503	CIVIL DETAILS	53708_ATCH_CSO_C503-RAMP.DWG
66.	C-504	CIVIL DETAILS	53708_ATCH_CSO_C504-505_DRIVE.DWG
67.	C-505	CIVIL DETAILS	53708_ATCH_CSO_C504-505_DRIVE.DWG
68.	C-506	CIVIL DETAILS	53708_ATCH_CSO_C506-ISRW.DWG
69.	C-507	CIVIL DETAILS	53708_ATCH_CSO_C507-STEPS.DWG
70.	C-508	CIVIL DETAILS	53708_ATCH_CSO_C508-STR REPAIR.dwg
71.	C-509	CIVIL DETAILS	53708_ATCH_CSO_C509-RCP INSTL.dwg
72.	C-510	CIVIL DETAILS	53708_ATCH_CSO_C510-CINLET.dwg
73.	C-511	CIVIL DETAILS	53708_ATCH_CSO_C511-CI-SW.dwg
74.	C-512	CIVIL DETAILS	53708_ATCH_CSO_C512-NON-SETBK.dwg
75.	C-513	CIVIL DETAILS	53708_ATCH_CSO_C513-SPCL INLET.dwg
76.	C-514	CIVIL DETAILS	53708_ATCH_CSO_C514-SPCL CI.dwg
77.	C-515	CIVIL DETAILS	53708_ATCH_CSO_C515-SPCL CI.dwg
78.	C-516	CIVIL DETAILS	53708_ATCH_CSO_C516-JBOX.dwg
79.	C-517	CIVIL DETAILS	53708_ATCH_CSO_C518-CIGRATE.DWG
80.	C-518	CIVIL DETAILS	53708_ATCH_CSO_C519-GINET.DWG
81.	C-519	CIVIL DETAILS	53708_ATCH_CSO_C519-MHC.dwg
82.	C-520	CIVIL DETAILS	53708_ATCH_CSO_C520-HDWL.DWG
83.	C-521	CIVIL DETAILS	53708_ATCH_CSO_C521-SAN-DET.dwg
84.	C-522	CIVIL DETAILS	53708_ATCH_CSO_C522-SAN-DET-2.dwg
85.	C-523	CIVIL DETAILS	53708_ATCH_CSO_C523-EC-CI.dwg
86.	C-524	CIVIL DETAILS	53708_ATCH_CSO_C524-EC-FNC.dwg
87.	C-525	CIVIL DETAILS	53708_ATCH_CSO_C525-EC-WASH.dwg
88.	C-526	CIVIL DETAILS	53708_ATCH_CSO_C526-GRADE-TYP.dwg
89.	C-527	CIVIL DETAILS	53708_ATCH_CSO_C527-TC-NOTES.dwg
90.	C-528	CIVIL DETAILS	53708_ATCH_CSO_C528-TC-DEVICE.dwg
91.	C-529	CIVIL DETAILS	53708_ATCH_CSO_C529-TC-CLS.dwg
92.	C-530	CIVIL DETAILS	53708_ATCH_CSO_C530-TC-WRKZN.dwg
93.	C-531	CIVIL DETAILS	53708_ATCH_CSO_C531-TC-SWDTR.dwg
94.	C-532	CIVIL DETAILS	53708_ATCH_CSO_C532-TC-SGNINF.dwg
95.	C-533	CIVIL DETAILS	53708_ATCH_CSO_C533-TC-SGNPST.dwg
96.	C-534	CIVIL DETAILS	53708_ATCH_CSO_C534-TC-LNSHFT.dwg
97.	C-535	CIVIL DETAILS	53708_ATCH_CSO_C535-TC-MLHWY.dwg
98.	C-536	CIVIL DETAILS	53708_ATCH_CSO_C536-TC-TCSB.dwg
99.	C-537	CIVIL DETAILS	53708_ATCH_CSO_C537-TC-TCSB.dwg
100.	C-538	CIVIL DETAILS	53708_ATCH_CSO_C538-TC-ANCHOR.dwg
101.	C-539	CIVIL DETAILS	53708_ATCH_CSO_C539-PM-UNDVD.dwg
102.	C-540	CIVIL DETAILS	53708_ATCH_CSO_C540-PM-MISC.dwg
103.	C-541	CIVIL DETAILS	53708_ATCH_CSO_C541-SGN-NOTE.dwg
104.	C-542	CIVIL DETAILS	53708_ATCH_CSO_C542-SGN-GNDMT.dwg
105.	C-601	HORIZONTAL ALIGNMENT (15TH STREET)	53708_ATCH_CSO_C601-C602_ALIGN.DWG
106.	C-602	HORIZONTAL ALIGNMENT (16TH STREET)	53708_ATCH_CSO_C601-C602_ALIGN.DWG

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

SHEET INDEX

SHEET ID
G-002

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_G002.dwg
9/9/2026 3:37:18 PM

READY TO ADVERTISE

DRAWING SET TABLE OF CONTENTS

DISCIPLINE DESIGNATION

G-	GENERAL
GC	TRAFFIC CONTROL / EROSION CONTROL
V-	SURVEY PROJECT CONTROL / EXISTING CONDITIONS
CD	CIVIL DEMOLITION
CP	CIVIL SITE PLAN / INTERSECTION DETAILS / RAMP DETAILS
CU	CIVIL SITE UTILITY
CG	CIVIL SITE GRADING
C-	CIVIL DETAILS / HORIZONTAL ALIGNMENT

SHEET NAMING LEGEND

SAMPLE SHEET NAME:

CS501
- 2-DIGIT DISIPLINE DESIGNATOR (SEE ABOVE)
- 1-DIGIT SHEET TYPE DESIGNATOR (0-9)
- 2-DIGIT SHEET SEQUENCE (00-99)

SAMPLE CADD FILE NAME:

K1001_CS501.DWG
- 5-DIGIT PROJECT CODE (TRI-SERVICE STANDARD)
- 2-DIGIT DISIPLINE DESIGNATOR (SEE ABOVE)
- 1-DIGIT SHEET TYPE DESIGNATOR (0-9)
- 2-DIGIT SHEET SEQUENCE (00-99)

INTERDISCIPLINARY LEGENDS

TERMS AND ABBREVIATIONS

AI	AREA INLET
APPROX	APPROXIMATE
ASPH	ASPHALT
AVE	AVENUE
AVG	AVERAGE
BM	BENCH MARK
BDRY	BOUNDARY
BL	BASELINE
BLDG	BUILDING
CL	CENTER LINE
C.I.	CAST IRON
C.O.E.	CORPS OF ENGINEERS
C.O.R.	CONTRACTING OFFICER REPRESENTATIVE
CAP	CAPACITY
CFS	CUBIC FEET PER SECOND
CI	CURB INLET
CL	CENTERLINE
CLF	CHAIN LINK FENCE
CMP	CORRUGATED METAL PIPE
CO	CLEAN OUT
COMM	COMMUNICATIONS
CONC	CONCRETE
CP	CONTROL POINT
CRCE	CAGED REINFORCED CONCRETE ENCASEMENT
CU YD	CUBIC YARDS
C/E	CONSTRUCTION EASEMENT
DB	DRAIN BASIN
DEG	DEGREE
DI	DROP INLET
DIA OR Ø	DIAMETER
DIM	DIMENSION
DIP	DUCTILE IRON PIPE
DTL	DETAIL
DWG	DRAWING
DWLS	DOWELS
E	ELECTRIC / EAST / EASTING
EL/ELEV	ELEVATION (GRADE OR BLDG)
EOP	EDGE OF PAVEMENT
EOR	ENGINEER OF RECORD
EQ	EQUAL
ES	END SECTION
EW	EACH WAY
E/EX./EXIST	EXISTING
FH	FIRE HYDRANT
FI	FIELD INLET
FL	FLOW LINE
FM	FORCE MAIN
FT	FEET
G	GAS
G.V.	GATE VALVE
GA	GAUGE
GRND	GROUND
GRVL	GRAVEL
GV	GAS VALVE
H OR HT	HEIGHT
HDPE	HIGH-DENSITY POLYETHYLENE
HP	HIGH POINT
HYD	HYDRAULIC
I.D.	INSIDE DIAMETER
IN	INCH
INV	INVERT
L OR LGTH	LENGTH
LBS	POUND
L.F.	LINEAL FEET
(L.I.P.)	LEAVE IN PLACE
LL	LOT LINE
LP	LOW POINT
LT	LEFT
ME	MATCH EXISTING
MECH	MECHANICAL
MH	MANHOLE
MIN	MINIMUM
MON	MONUMENT
MUTCD	MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES
MAX	MAXIMUM
N	NORTH / NORTHING
NIC	NOT IN CONTRACT
NO.	NUMBER
OC	ON CENTERS
OD	OUTSIDE DIAMETER
OFF	OFFSET
OH	OVERHEAD
OHE	OVERHEAD ELECTRIC
P	POLE
PD	PAVEMENT DRAIN
PERF	PERFORATED
PL	PROPERTY LINE
POB	POINT OF BEGINNING
POE	POINT OF ENDING
PP	POWER POLE
PROP	PROPOSED
PSI	POUNDS/SQUARE INCH
PT	POINT
PVC	POLYVINYL CHLORIDE
R OR RAD	RADIUS
RCB	REINFORCED CONCRETE BOX
RCE	REINFORCED CONCRETE ENCASEMENT
RCP	REINFORCED CONCRETE PIPE
RCPE	REINFORCED CONCRETE PIPE ELLIPTICAL
RD	ROAD
REINF	REINFORCMENT
REQ'D	REQUIRED
REV	REVISIONS
RT	RIGHT
R/W	RIGHT OF WAY
S	SOUTH / SOUTHING
SW	SIDEWALK
SD	STORM DRAIN
SDMH	STORM DRAIN MANHOLE
SECT	SECTION
SP	SPACING
SPEC	SPECIFICATIONS
SAN	SANITARY SEWER
SSMH	SANITARY SEWER MANHOLE
STA	STATION
STD	STANDARD
TAB	TOP AND BOTTOM
TEL	TELEPHONE
T/C	TOP OF CONCRETE
TOS	TOP OF STEEL
TP	TEST PIT
TYP	TYPICAL
U.S.	UNITED STATES
UG	UNDERGROUND
UGE	UNDERGROUND ELECTRIC
(U.I.P.)	USE IN PLACE
UKN	UNKNOWN
UTY	UTILITY
VCP	VITRIFIED CLAY PIPE
VERT	VERTICAL
VOL	VOLUME
W	WATER LINE / WIDTH / WEST / WESTING
W/	WITH
W/O	WITHOUT
WMH	WATER MANHOLE
WV	WATER VALVE

SITE PLAN LINE STYLE LEGEND

- (SS)8"VCP SAN — EXISTING SANITARY SEWER PIPE
- (ST)15"HDPE SD — EXISTING STORM SEWER PIPE
- COMM — EXISTING BURIED COMMUNICATION LINE
- E E E — EXISTING OVERHEAD ELECTRICAL LINE
- W W W — EXISTING WATERLINE
- G G — EXISTING GAS
- MAJOR CONTOUR
- MINOR CONTOUR
- EXISTING EDGE OF PAVEMENT
- EXISTING CHAIN LINK FENCE
- R/W — EXISTING RIGHT-OF-WAY
- EXISTING PROPERTY LINE
- EXISTING LOT LINE
- P L — EXISTING PROPERTY/LOT LINE
- CENTER LINE

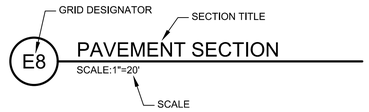
SITE PLAN SYMBOL LEGEND

- (S) EXISTING SANITARY SEWER STRUCTURE
- (CO) EXISTING SANITARY SEWER CLEANOUT
- (D) EXISTING STORM SEWER STRUCTURE
- EXISTING COMMUNICATION POLE
- EXISTING LIGHT POLE
- (M) EXISTING POWER METER
- (W) EXISTING WATER METER
- (FH) EXISTING FIRE HYDRANT
- EXISTING WATER VALVE
- (G) EXISTING GAS METER
- EXISTING STREET SIGN
- EXISTING MAILBOX
- EXISTING BORE HOLE
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- EXISTING SHRUBLINE
- PROPERTY PIN
- CONTROL POINT
- •BM BENCH MARK

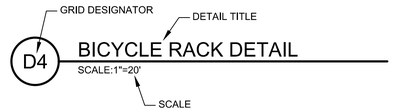
MATERIAL SYMBOL LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- FULL DEPTH ASPHALT REPLACEMENT
- GRAVEL DRIVE
- RIPRAP
- PAVEMENT & CURB REMOVAL
- SIDEWALK & DRIVEWAY REMOVAL
- 6" REINFORCED SIDEWALK

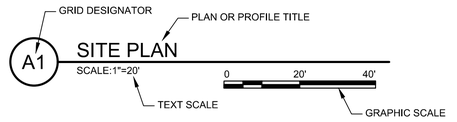
SHEET REFERENCE SYMBOL LEGEND



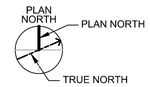
SECTION TITLE



DETAIL TITLE



DRAWING TITLE (TYP)



NORTH ARROW

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912D219D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

ABBREVIATIONS AND LEGEND

SHEET ID
G-003

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_G003_LEGND.dwg
9/9/2026 3:37:37 PM

READY TO ADVERTISE

GENERAL NOTES:

1. THE CONSTRUCTION DOCUMENTS SHALL GOVERN THE CONSTRUCTION OF ALL PUBLIC IMPROVEMENTS FOR THIS PROJECT.
2. THE CONTRACTOR SHALL HAVE ONE (1) SIGNED COPY OF THE IMPROVEMENT PLANS AND SPECIFICATIONS, APPROVED BY USACE, AT THE PROJECT SITE AT ALL TIMES.
3. ALL EXISTING UTILITIES INDICATED ON THESE IMPROVEMENT PLANS ARE TAKEN FROM UTILITY COMPANY RECORDS AND ARE APPROXIMATE ONLY. ALL UTILITIES ACTUALLY EXISTING MAY NOT BE SHOWN; SOME SERVICE LINES FROM BUILDING TO MAIN HAVE UNKNOWN LOCATIONS AND MAY CONFLICT WITH THE PLANS. THE CONTRACTOR, PRIOR TO THE COMMENCEMENT OF WORK, SHALL NOTIFY THOSE COMPANIES WHICH HAVE FACILITIES IN THE NEAR VICINITY OF THE CONSTRUCTION TO BE PERFORMED, AND SHALL VERIFY THE LOCATION AND DEPTH OF ALL UNDERGROUND UTILITIES. ANY UTILITIES FOUND TO BE IN CONFLICT WITH THE PROPOSED CONSTRUCTION THAT ARE NOT ADDRESSED ON THE PLANS SHALL BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE ENGINEER. THE CONTRACTOR SHALL MAKE ALL PROVISIONS NECESSARY TO ADEQUATELY SUPPORT AND MAINTAIN EXPOSED UTILITIES, AND TO PROTECT EXPOSED UTILITIES FROM FREEZING TEMPERATURES. THE CONTRACTOR IS RESPONSIBLE FOR UTILITY DAMAGES, RELOCATION NEEDS, AND ADEQUATE COMPACTION AROUND DISTURBED UTILITIES. ENCROACHMENTS SUCH AS TELEPHONE, POWER, AND CABLE T.V. POLES SHALL BE MOVED BY THE UTILITY OWNER AND COORDINATED BY THE CONTRACTOR AFTER WRITTEN APPROVAL FROM THE OWNER, UNLESS OTHERWISE NOTED.
4. ALL STORM SEWER PIPE LENGTHS SHOWN ON THE PLANS ARE MEASURED FROM CENTER OF DOWNSTREAM STRUCTURE TO CENTER OF UPSTREAM STRUCTURE. ALIGNMENT STATIONING, BEARINGS, PIPE SLOPES AND STRUCTURE FLOW LINES SHOWN ON THE PLANS ARE MEASURED FROM CENTER OF STRUCTURE TO CENTER OF STRUCTURE (FOR DESIGN PURPOSES).
5. ALL EXISTING SIGNS, MAILBOXES, SPRINKLERS, FENCING, AND OTHER ITEMS IN CONFLICT WITH THE PROPOSED CONSTRUCTION SHALL BE PROTECTED, REMOVED AND RESET, OR RESTORED TO PRIOR CONDITIONS UPON COMPLETION OF THE WORK IN ACCORDANCE WITH THE CONSTRUCTION SPECIFICATIONS, WHETHER SO NOTED ON THE PLANS OR NOT. TEMPORARY FENCING SHALL BE INSTALLED AT LOCATIONS WHERE EXISTING PERMANENT FENCING IS TO BE REMOVED. THE CONTRACTOR SHALL COORDINATE TEMPORARY FENCING LOCATIONS WITH THE RESIDENT, CITY, USACE AND ENGINEER.
6. ALL EXISTING DRIVEWAYS, SIDEWALKS, PLANTER WALLS, AND OTHER AREAS OUTSIDE THE TEMPORARY EASEMENT, PERMANENT EASEMENT, OR RIGHT-OF-WAY LIMITS OR NOT SHOWN TO BE DISTURBED IN THE PLANS THAT ARE DAMAGED OR DISTURBED BY THE CONTRACTOR'S OPERATIONS SHALL BE RESTORED TO A CONDITION EQUAL TO OR BETTER THAN THE ORIGINAL CONDITION BEFORE DAMAGE OCCURRED, AT THE CONTRACTOR'S EXPENSE IN ACCORDANCE WITH THE CONSTRUCTION SPECIFICATIONS.
7. THE EXISTING ITEMS SHOWN ON THE PLANS REPRESENT THE BEST INFORMATION AVAILABLE TO THE ENGINEER. ALL ITEMS NOT SHOWN ON THE PLANS, BUT WHICH MAY INTERFERE WITH COMPLETION OF THE WORK INDICATED, SHALL BE SALVAGED OR REMOVED AND DISPOSED OF AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
8. THE CONTRACTOR IS REQUIRED TO PHASE CONSTRUCTION AND TRAFFIC CONTROL ON ALL AFFECTED STREETS TO PROVIDE ACCESS TO ALL RESIDENTS AFFECTED BY CONSTRUCTION. ACCESS TO PROPERTIES SHALL BE MAINTAINED AT ALL TIMES. IF CONSTRUCTION ACROSS A DRIVEWAY IS TO TAKE MORE THAN 24 HOURS, ONE HALF THE DRIVE SHALL REMAIN IN SERVICE UNTIL THE OTHER HALF IS RECONSTRUCTED AND IN SERVICE.
9. ALL SILTATION AND EROSION CONTROL MEASURES SPECIFIED ON THE PLANS, OR OTHERWISE REQUIRED, SHALL BE IN PLACE PRIOR TO COMMENCING ANY LAND DISTURBANCE ACTIVITIES ON THE SITE. THE CONTRACTOR SHALL BE REQUIRED TO SIGN AND ABIDE BY THE STORMWATER POLLUTION PREVENTION PLAN (SWPPP). THE CONTRACTOR SHALL BE REQUIRED TO ABIDE BY LAND DISTURBANCE PERMIT CONDITIONS, AS ISSUED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) IN ACCORDANCE WITH THE CONSTRUCTION SPECIFICATIONS.
10. ALL EXISTING DRAINAGE TILES, FOUNDATION DRAINS, AND/OR ROOF DRAINS, WHICH ARE PRESENTLY CONNECTED TO THE EXISTING STORMWATER CONVEYANCE SYSTEM, THAT ARE ENCOUNTERED OR DAMAGED DURING CONSTRUCTION SHALL BE RESTORED TO THEIR ORIGINAL CONDITION OR PROPERLY REROUTED AND/OR CONNECTED TO THE STORMWATER CONVEYANCE SYSTEM AS DIRECTED BY THE CITY OF ATCHISON OR ITS REPRESENTATIVE. THE LOCATION OF DRAIN LINES MAY OR MAY NOT BE NOTED IN THE PLANS.
11. THE CONTRACTOR SHALL MAINTAIN DRAINAGE DURING CONSTRUCTION AND IS RESPONSIBLE FOR ANY DEWATERING NECESSARY FOR CONSTRUCTION.
12. THE CONTRACTOR SHALL CONTACT CITY OF ATCHISON FOR INSPECTION PRIOR TO ANY SANITARY WORK. ALL ADJUSTMENTS OF FACILITIES ARE SUBJECT TO THE APPROVAL OF CITY OF ATCHISON "CONSTRUCTION AND MATERIAL SPECIFICATION FOR SANITARY SEWERS, KDHE NO. 20969" AND THE CITY OF ATCHISON STANDARD DETAIL SHEETS. FOR NEW CONSTRUCTION, BACKFILL FOR LATERAL LINES TO SANITARY SEWER MAINS MAY BE COMPACTED SOIL BACKFILL, IN ACCORDANCE WITH SPECIFICATIONS AND REQUIREMENTS. INSPECTION OF ALL REPAIRS, MODIFICATIONS OR ADJUSTMENTS BY CITY OF ATCHISON AND THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) IS REQUIRED.
13. THE CONSTRUCTION OF THE IMPROVEMENTS SHOWN OR IMPLIED BY THESE PLANS SHALL NOT BE INITIATED, OR ANY PART THEREOF UNDERTAKEN, UNTIL THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) IS NOTIFIED OF SUCH INTENT, AND ALL REQUIRED PERMITS AND FEES, PROPERLY EXECUTED BONDS, AND CONTRACT AGREEMENTS ARE RECEIVED AND APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
14. THE CONTRACTOR SHALL CONTACT THE ENGINEERING INSPECTOR AT LEAST FORTY-EIGHT (48) HOURS IN ADVANCE TO SCHEDULE A REQUIRED INSPECTION.
15. THE CONTRACTOR SHALL NOT WORK ON SUNDAYS OR HOLIDAYS. WORK ON SATURDAYS IS ALLOWED WHEN REQUESTED AT LEAST FORTY-EIGHT (48) HOURS IN ADVANCE AND WHEN APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
16. CONTRACTOR SHALL CLEAR AND GRUB ALL TREES MARKED FOR REMOVAL. REMOVAL OF THE TREE INCLUDES REMOVAL OF ALL BRANCHES, ROOTS, AND BRUSH WITHIN THE CANOPY LIMITS OF THE TREE. ALL TREES NOT MARKED FOR REMOVAL SHALL BE PROTECTED AND SAVED. THE CONTRACTOR IS REQUIRED TO EXERCISE PARTICULAR CARE DURING EXCAVATION TO PREVENT UNNECESSARY DAMAGE TO TREES, RETAINING WALLS, STRUCTURES, UTILITIES, ETC. IN THE VICINITY OF CONSTRUCTION. ONLY THOSE TREES IN DIRECT CONFLICT WITH CONSTRUCTION OPERATIONS SHALL BE REMOVED, AS AGREED UPON BETWEEN THE CONTRACTOR AND CITY REPRESENTATIVE. ALL TREES THAT ARE DEFINITELY TO BE SAVED SHALL BE MARKED "SAVE" AND SHALL BE FENCED AT THE DRIP LINE WITH T-POSTS AND ORANGE CONSTRUCTION FENCING. TRIMMING OF TREES TO COMPLETE THE WORK SHALL BE APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
17. THE CONTRACTOR IS RESPONSIBLE FOR THE PROTECTION OF ALL PROPERTY CORNERS AND SECTION CORNERS. ANY PROPERTY CORNERS OR SECTION CORNERS DISTURBED OR DAMAGED BY CONSTRUCTION ACTIVITIES SHALL BE RESET BY A REGISTERED LAND SURVEYOR LICENSED IN THE STATE OF KANSAS, AT THE CONTRACTOR'S EXPENSE.
18. ALL STORM SEWER STRUCTURES USED FOR THIS PROJECT SHALL BE PRECAST IN ACCORDANCE WITH THE CONSTRUCTION SPECIFICATIONS UNLESS INDICATED OTHERWISE AND APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR). CONTRACTOR SHALL PROVIDE SHOP DRAWINGS TO THE ENGINEER FOR APPROVAL PRIOR TO CASTING OF STRUCTURES. ALL STORM SEWER STRUCTURE LIDS SHALL BE CAST-IN-PLACE.
19. ALL TRUCKS UTILIZED BY THE CONTRACTOR OR ANY SUBCONTRACTORS SHALL BE ROUTED ALONG TRUCK ROUTES AS APPROVED BY THE CITY OF ATCHISON AND THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
20. ALL EQUIPMENT USED BY THE CONTRACTOR HAVING METAL TRACKS SHALL NOT BE DRIVEN OVER CITY STREETS OTHER THAN THOSE SHOWN ON THE PLANS AS BEING DISRUPTED.
21. THE CONTRACTOR SHALL NOTIFY EACH IMPACTED PROPERTY OWNER OF THE ANTICIPATED CONSTRUCTION ACTIVITY 48 HOURS PRIOR TO BEGINNING WORK. THE NOTIFICATION METHOD SHALL BE OF SUITABLE MEANS AND APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
22. CONCRETE DRIVEWAYS SHALL BE REPLACED TO THE NEAREST JOINT UP TO 6 FEET BEYOND THE AREA THAT NEEDS TO BE REPLACED. IF NO JOINT IS WITHIN 6 FEET THEN A SAW-CUT WILL BE MADE AT THE PLANNED REPLACEMENT IN ACCORDANCE WITH THE CONSTRUCTION SPECIFICATIONS.
23. ALL ITEMS MARKED TO SALVAGE SHALL BE PROVIDED TO OWNER OR CITY FOR THEIR ACCEPTANCE AND USE. ANY ITEMS MARKED FOR SALVAGE, BUT WHICH ARE REFUSED BY THE RESIDENT AND CITY, SHALL BECOME THE PROPERTY OF THE CONTRACTOR FOR REMOVAL AND/OR DISPOSAL IN ACCORDANCE WITH THE PLANS AND CONSTRUCTION SPECIFICATIONS.
24. UNLESS SPECIFIED, USE APWA SPECIFICATIONS. WHERE APWA SPECIFICATIONS DO NOT APPLY REFERENCE KDOT SPECIFICATIONS, UNLESS OTHERWISE DIRECTED IN THE PLANS AND CONSTRUCTION SPECIFICATIONS.
25. SEE SPECIFICATION 01 57 19 FOR THREATENED AND ENDANGERED SPECIES. POTENTIAL ROOSTING TREES FOR THE FEDERALLY ENDANGERED INDIANA BAT ARE PRESENT ON THE PROJECT SITE AND HAVE NOT BEEN REMOVED. IF CONSTRUCTION REQUIRES REMOVAL OR DISTURBANCE OF THESE TREES, THE CONTRACTOR MAY ONLY PERFORM THIS WORK BETWEEN NOVEMBER 1 AND MARCH 31.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

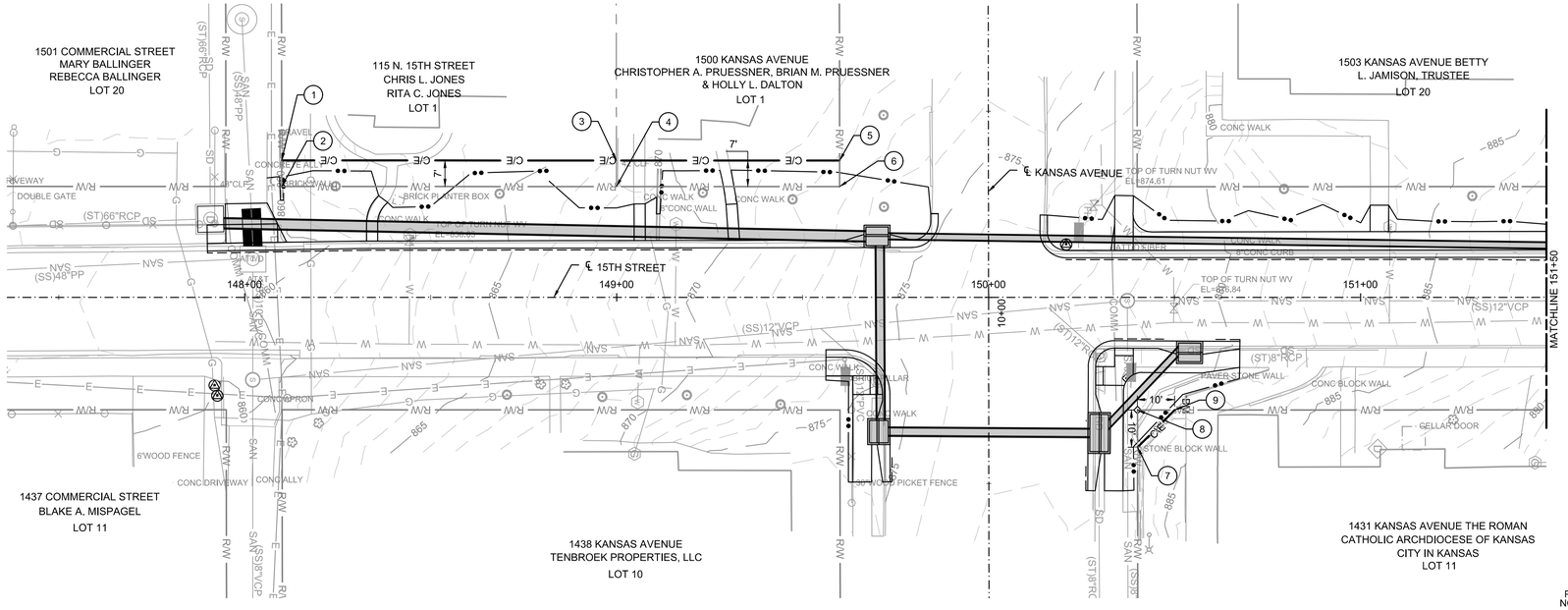
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

GENERAL NOTES

SHEET ID
G-004

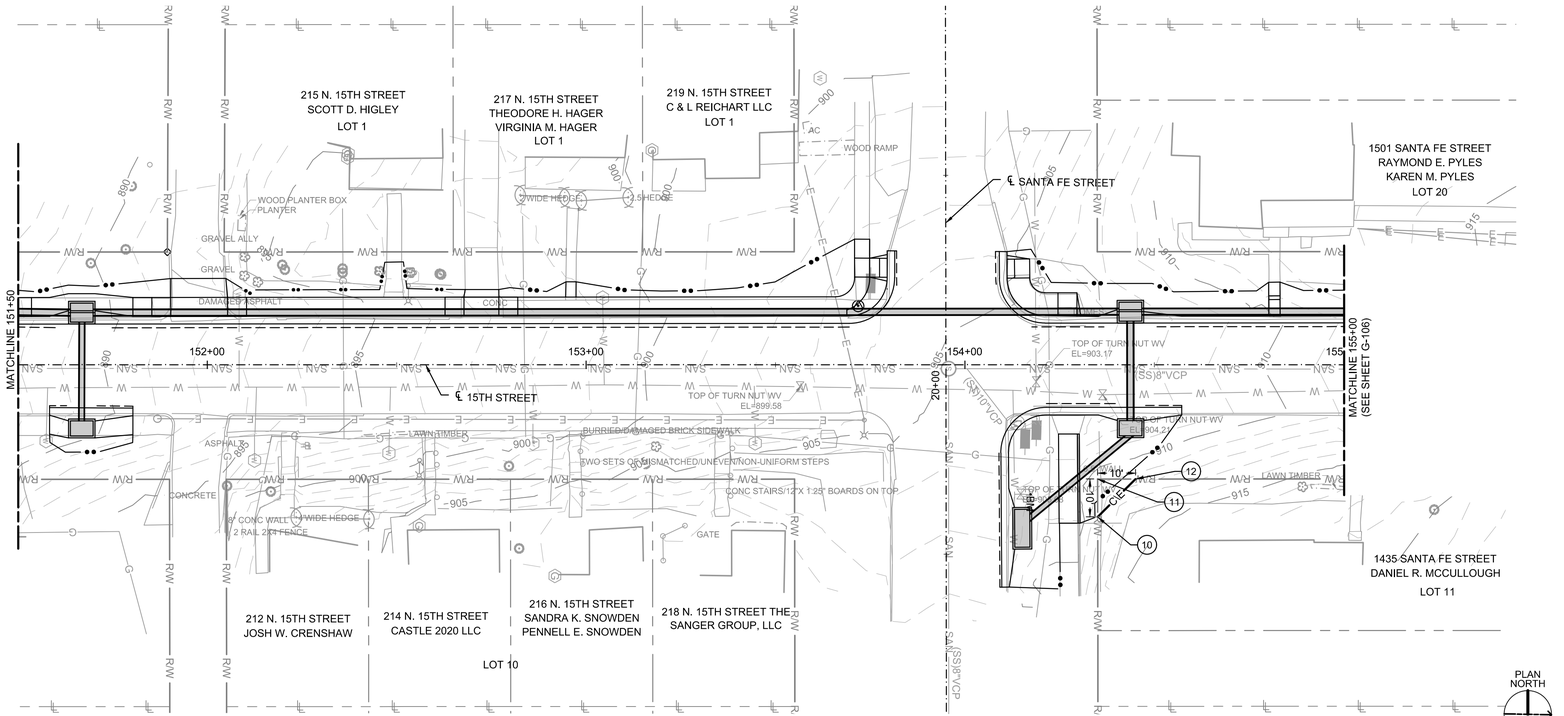
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_G004_GNOTES.dwg
9/9/2026 3:37:57 PM

READY TO ADVERTISE



H1 EASEMENT PLAN, 15TH STREET STA. 147+90 TO 151+50

SCALE:1"=20'



A1 EASEMENT PLAN, 15TH STREET STA. 151+50 TO 155+00

SCALE:1"=20'

SHEET NOTES

1. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

CALLOUTS

(##) SPOT COORDINATE REFERS TO COORDINATE-PIPE TABLE ON SHEET

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- CONSTRUCTION EASEMENT (CE)
- PROPERTY PIN

SPOT COORDINATES - CONSTRUCTION EASEMENT

POINT NO.	NORTHING	EASTING
1.	N: 460462.04	E: 2119110.82
2.	N: 460462.19	E: 2119117.81
3.	N: 460552.02	E: 2119109.02
4.	N: 460552.17	E: 2119116.02
5.	N: 460612.01	E: 2119107.82
6.	N: 460612.16	E: 2119114.82
7.	N: 460693.65	E: 2119183.21
8.	N: 460693.44	E: 2119173.21
9.	N: 460703.43	E: 2119173.00
10	N: 461088.57	E: 2119175.09
11.	N: 461088.35	E: 2119165.09
12.	N: 461098.35	E: 2119164.89

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

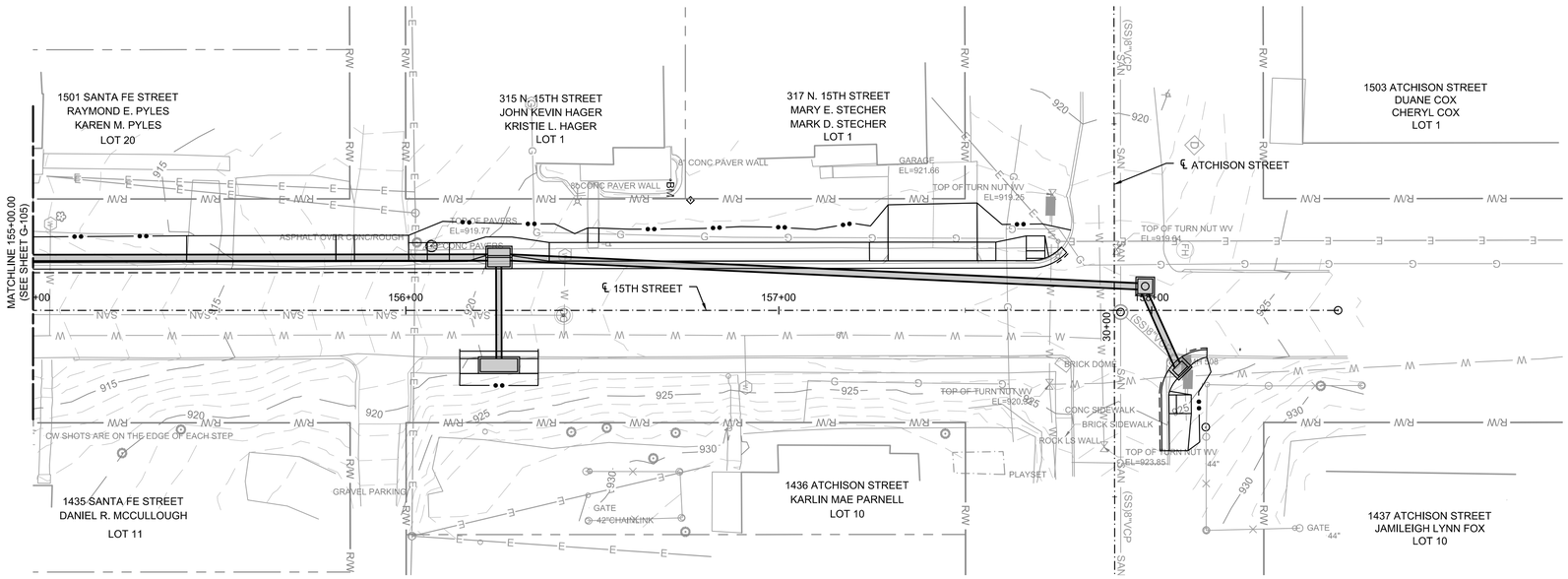
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
EASEMENT PLAN

SHEET ID
G-105

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_G105-G106_15TH-ESMT.dwg
9/9/2026 3:39:17 PM

READY TO ADVERTISE



H1 EASEMENT PLAN, 15TH STREET STA 155+00 TO 158+37

SCALE:1"=20'

SHEET NOTES

1. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E CONSTRUCTION EASEMENT
- PROPERTY PIN

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

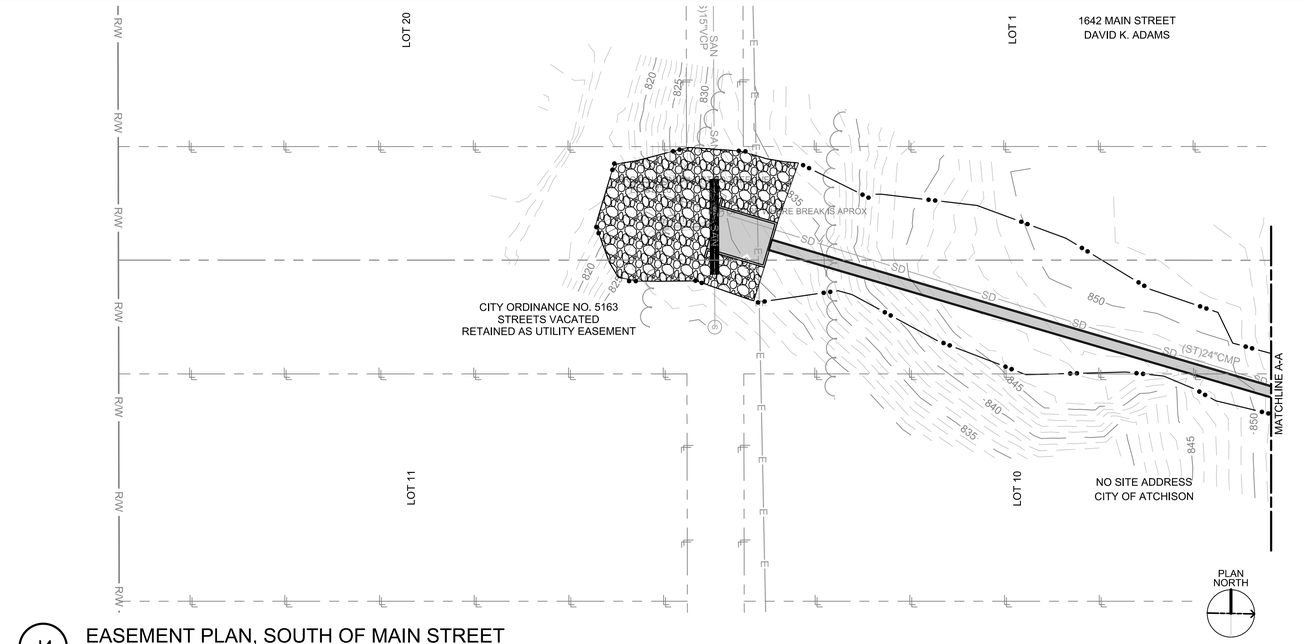
15TH STREET
EASEMENT PLAN

SHEET ID

G-106

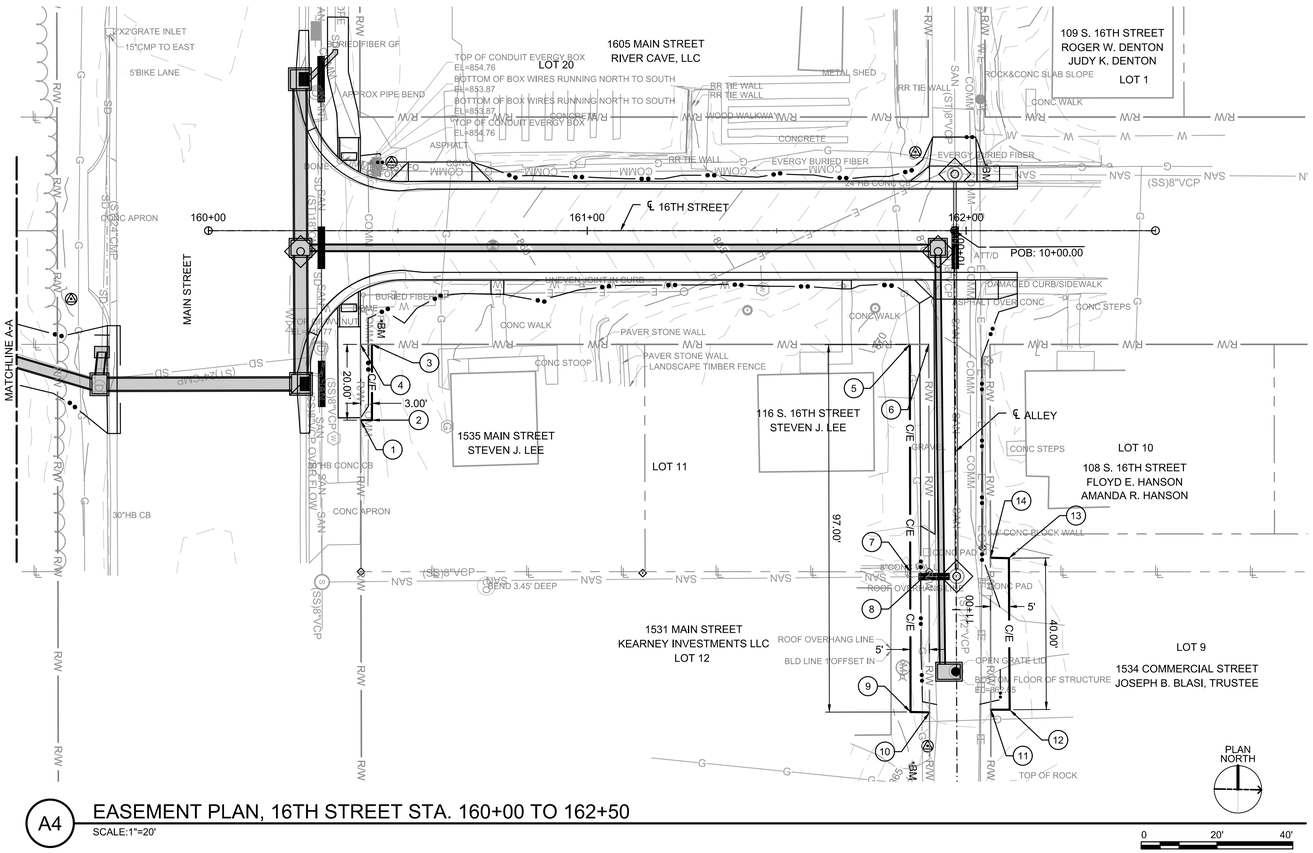
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_G105-G106_15TH-ESMT.dwg
9/9/2026 3:39:31 PM

READY TO ADVERTISE



J4 EASEMENT PLAN, SOUTH OF MAIN STREET

SCALE:1"=20'



A4 EASEMENT PLAN, 16TH STREET STA. 160+00 TO 162+50

SCALE:1"=20'

SHEET NOTES

1. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

CALLOUTS

(##) SPOT COORDINATE REFERS TO COORDINATE-PIPE TABLE ON SHEET

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- CONSTRUCTION EASEMENT
- FULL DEPTH ASPHALT REPLACEMENT
- PROPERTY PIN

SPOT COORDINATES - CONSTRUCTION EASEMENT

POINT NO.	NORTHING	EASTING
1.	N: 459888.60	E: 2118549.00
2.	N: 459891.60	E: 2118548.93
3.	N: 459891.17	E: 2118528.94
4.	N: 459888.17	E: 2118529.00
5.	N: 460033.14	E: 2118526.18
6.	N: 460038.14	E: 2118526.08
7.	N: 460034.44	E: 2118586.16
8.	N: 460039.44	E: 2118586.06
9.	N: 460035.24	E: 2118623.16
10	N: 460040.24	E: 2118623.16
11.	N: 460056.38	E: 2118622.05
12.	N: 460061.38	E: 2118622.05
13.	N: 460060.38	E: 2118582.06
14.	N: 460055.52	E: 2118582.06

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

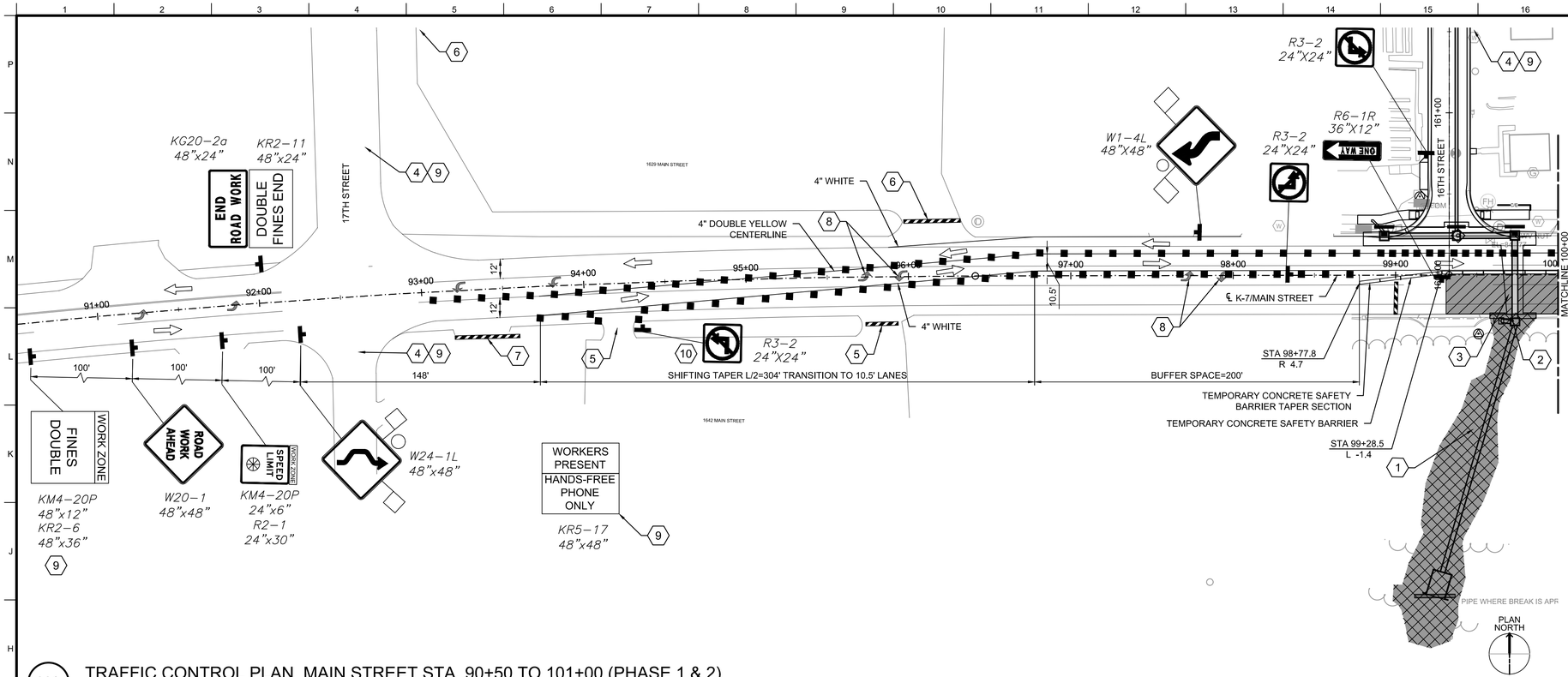
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
EASEMENT PLAN

SHEET ID
G-107

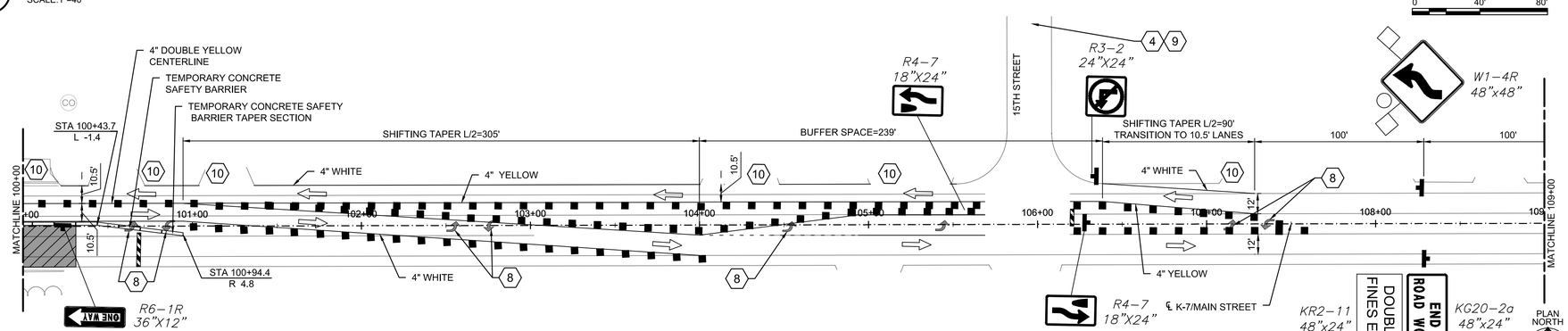
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_G107_16TH-ESMT.dwg
9/9/2026 3:40:37 PM

READY TO ADVERTISE



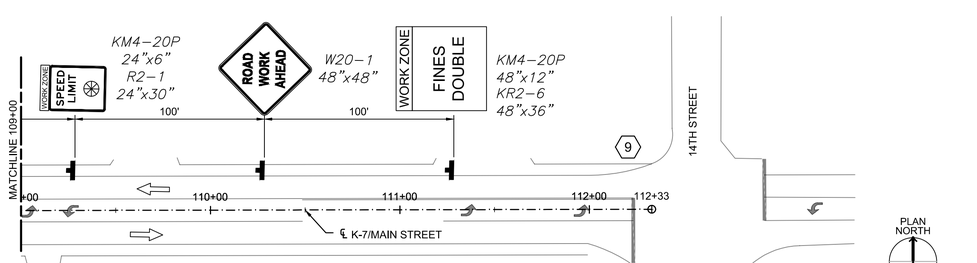
H1 TRAFFIC CONTROL PLAN, MAIN STREET STA. 90+50 TO 101+00 (PHASE 1 & 2)

SCALE:1"=40'



D1 TRAFFIC CONTROL PLAN, MAIN STREET STA. 101+00 TO 109+50 (PHASE 1 & 2)

SCALE:1"=40'



A1 TRAFFIC CONTROL PLAN, MAIN STREET STA. 109+50 TO 112+00 (PHASE 1 & 2)

SCALE:1"=40'

SHEET NOTES

1. STORM SEWER PIPE, STRUCTURES, AND GRADING SOUTH OF MAIN ST (PHASE 1) SHALL BE COMPLETED BEFORE THE START OF PHASE 2.
2. WHEN WORK ON PHASE 1 ENCROACHES TRAFFIC ON MAIN ST TRAFFIC CONTROL FOR PHASE 2 SHALL BE IMPLEMENTED.
3. STORM SEWER ACROSS MAIN STREET TO BE COMPLETED HALF AT A TIME BEGINNING WITH THE SOUTH SIDE OF THE STREET.
4. 16TH STREET INTERSECTION SHALL REMAIN OPEN DURING PHASE 1 & 2. NO LEFT TURNS SHALL BE PERMITTED DURING PHASE 2.
5. CONTRACTOR SHALL PROVIDE ACCESS TO ALL RESIDENTIAL PROPERTIES THROUGH THE DURATION OF CONSTRUCTION.
6. TEMPORARY CONCRETE SAFETY BARRIER SHALL BE PINNED.
7. SEE SHEETS C-527-C-538 FOR TRAFFIC CONTROL DETAILS.
8. ALIGNMENT FOR REFERENCE ONLY TO AID IN CONSTRUCTION.
9. FULL CLOSURE MAY BE NECESSARY WHEN TRANSITIONING BETWEEN PHASES 2 AND 3. SEE DETOUR SHEETS WHEN FULL CLOSURE IS IMPLEMENTED.

SHEET KEYNOTES

1. CONSTRUCT PHASE 1 STORM PIPE, STRUCTURES AND HILLSIDE GRADING.
2. SET-UP TRAFFIC CONTROL FOR PHASE 2 WHEN PHASE 1 CONSTRUCTION ENCROACHES MAIN STREET TRAFFIC.
3. CONSTRUCT PHASE 2 STORM PIPE, SOUTH SIDE OF MAIN STREET.
4. PLACE W20-1 (ROAD WORK AHEAD) & KG20-2a (END ROAD WORK) 100' FROM INTERSECTION.
5. WEST ENTRANCE TO REMAIN OPEN AT 1642 MAIN ST. EAST ENTRANCE TO BE CLOSED DURING PHASE 2 AND 3.
6. ACCESS TO MAIN STREET TO BE CLOSED DURING PHASE 2 AND 3 AT 1629 MAIN STREET. BUSINESS STILL HAS ACCESS FROM 17TH STREET.
7. ACCESS TO MAIN STREET TO BE CLOSED DURING PHASE 2 AND 3 FOR STORAGE UNITS. ACCESS FROM 17TH STREET.
8. REMOVE CONFLICTING PAVEMENT MARKINGS.
9. INSTALL KR5-17 100' AHEAD OF KR2-6 OR W20-1. SEE KDOT SPECIAL PROVISION 15-08029
10. INSTALL R3-2 AT ALL RIGHT-OUT ONLY ACCESS POINTS.

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- CHANNELIZING DEVICE
- TYPE 3 BARRICADES
- TRAFFIC CONTROL SIGN
- TRAFFIC CONTROL PHASE 1
- TRAFFIC CONTROL PHASE 2

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

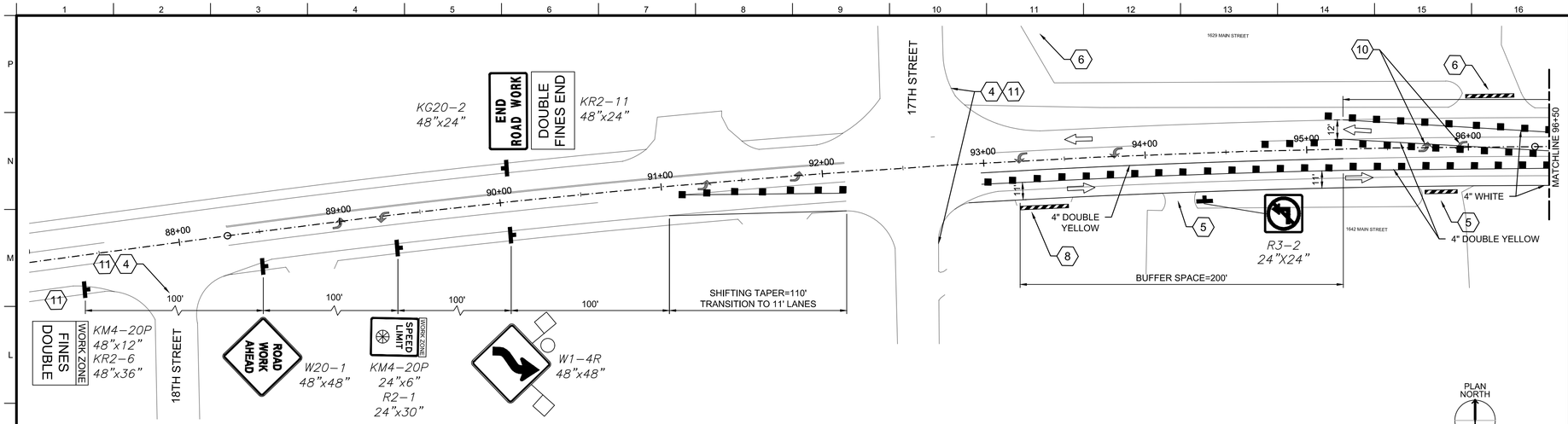
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

K-7 / MAIN STREET SOUTH
TRAFFIC CONTROL PLAN (PHASE 1 & 2)

SHEET ID
GC101

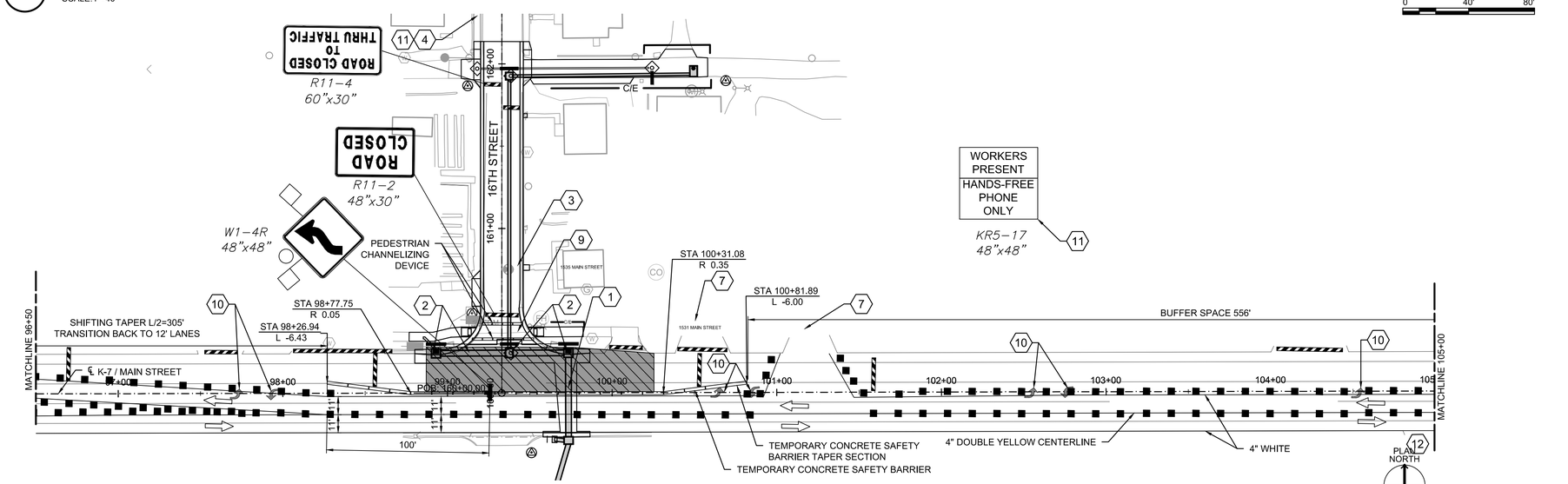
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC101_MAIN-S.dwg
9/9/2026 3:42:18 PM

READY TO ADVERTISE



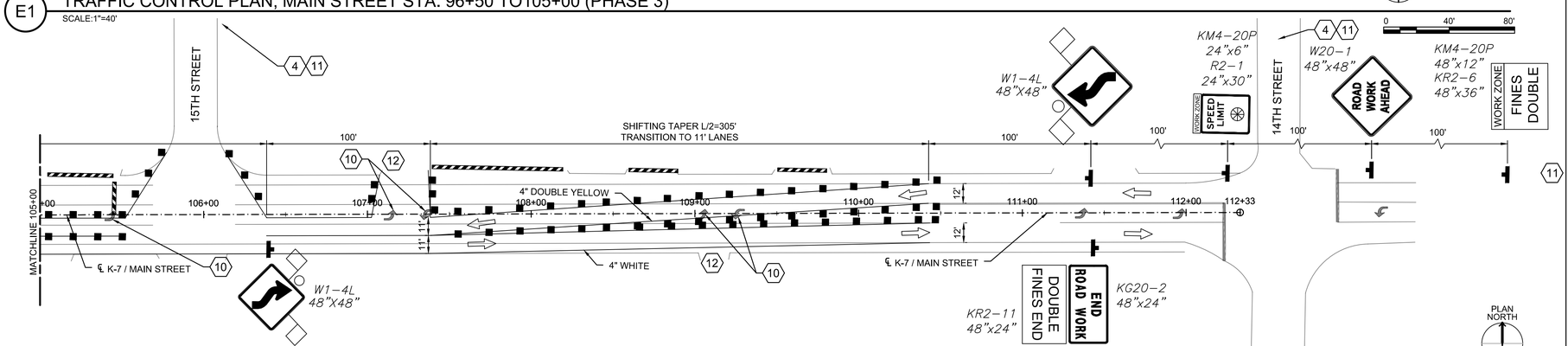
K1 TRAFFIC CONTROL PLAN, MAIN STREET STA. 87+50 TO 96+50 (PHASE 3)

SCALE:1"=40'



E1 TRAFFIC CONTROL PLAN, MAIN STREET STA. 96+50 TO105+00 (PHASE 3)

SCALE:1"=40'



A1 TRAFFIC CONTROL PLAN, MAIN STREET STA. 105+00 TO114+00 (PHASE 3)

SCALE:1"=40'

SHEET NOTES

1. CONSTRUCT AND CONNECT NORTH HALF OF STORM SEWER ON MAIN STREET.
2. CONSTRUCT STORM SEWER PIPE, INLETS AND JUNCTION BOXES ALONG 16TH STREET INTERSECTION.
3. 16TH STREET SHALL BE CLOSED AT MAIN STREET INTERSECTION. SEE KEYNOTE #2.
4. CONTRACTOR SHALL PROVIDE ACCESS TO ALL RESIDENTIAL PROPERTIES THROUGH THE DURATION OF CONSTRUCTION.
5. TEMPORARY CONCRETE SAFETY BARRIER SHALL BE PINNED.
6. ALIGNMENT ALONG MAIN ST FOR NOTES PURPOSES ONLY.
7. SEE SHEETS C-527-C-538 FOR TRAFFIC CONTROL DETAILS.
8. ALIGNMENT FOR REFERENCE ONLY TO AID IN CONSTRUCTION.
9. CONSTRUCTION OF PHASE 4 STORM PIPE TO BEGIN AT SOUTH END NEAR 16TH STREET INTERSECTION WHILE PHASE 3 TRAFFIC CONTROL IS IN PLACE.

SHEET KEYNOTES

1. CONSTRUCT STORM SEWER NORTH SIDE OF MAIN STREET.
2. CONSTRUCT STORM SEWER PIPE AND STRUCTURES AT MAIN AND 16TH STREET.
3. ACCESS MUST REMAIN OPEN FROM 16TH STREET FOR 1535 MAIN STREET PROPERTY DURING PHASE 3 CONSTRUCTION.
4. INSTALL W20-1 (ROAD CLOSED AHEAD) & KG20-2a (END ROAD WORK) 100' FROM INTERSECTION.
5. WEST ENTRANCE TO REMAIN OPEN AT 1642 MAIN ST. EAST ENTRANCE TO BE CLOSED DURING PHASE 2 AND 3.
6. ACCESS TO MAIN STREET TO BE CLOSED DURING PHASE 2 AND 3 AT 1629 MAIN STREET. BUSINESS STILL HAS ACCESS FROM 17TH STREET.
7. ENTRANCE AT 1531 MAIN STREET TO BE CLOSED. ACCESS TO PROPERTY FROM ADJACENT CONNECTED DRIVE.
8. ENTRANCE ON MAIN STREET TO BE CLOSED FOR STORAGE UNITS. ACCESS FROM 17TH STREET.
9. SIDEWALK TO REMAIN OPEN. IF CLOSURE IS NEEDED, REFER TO GC103 FOR PEDESTRIAN ROUTING.
10. REMOVE CONFLICTING PAVEMENT MARKINGS
11. INSTALL KR5-17 100' AHEAD OF KR2-6 OR W20-1 SIGNS. SEE KDOT SPECIAL PROVISION 15-08029.
12. INSTALL R3-2 AT ALL RIGHT-OUT ONLY ACCESS POINTS.

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- CHANNELIZING DEVICE
- TYPE 3 BARRICADES
- TRAFFIC CONTROL SIGN
- TRAFFIC CONTROL PHASE 3

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

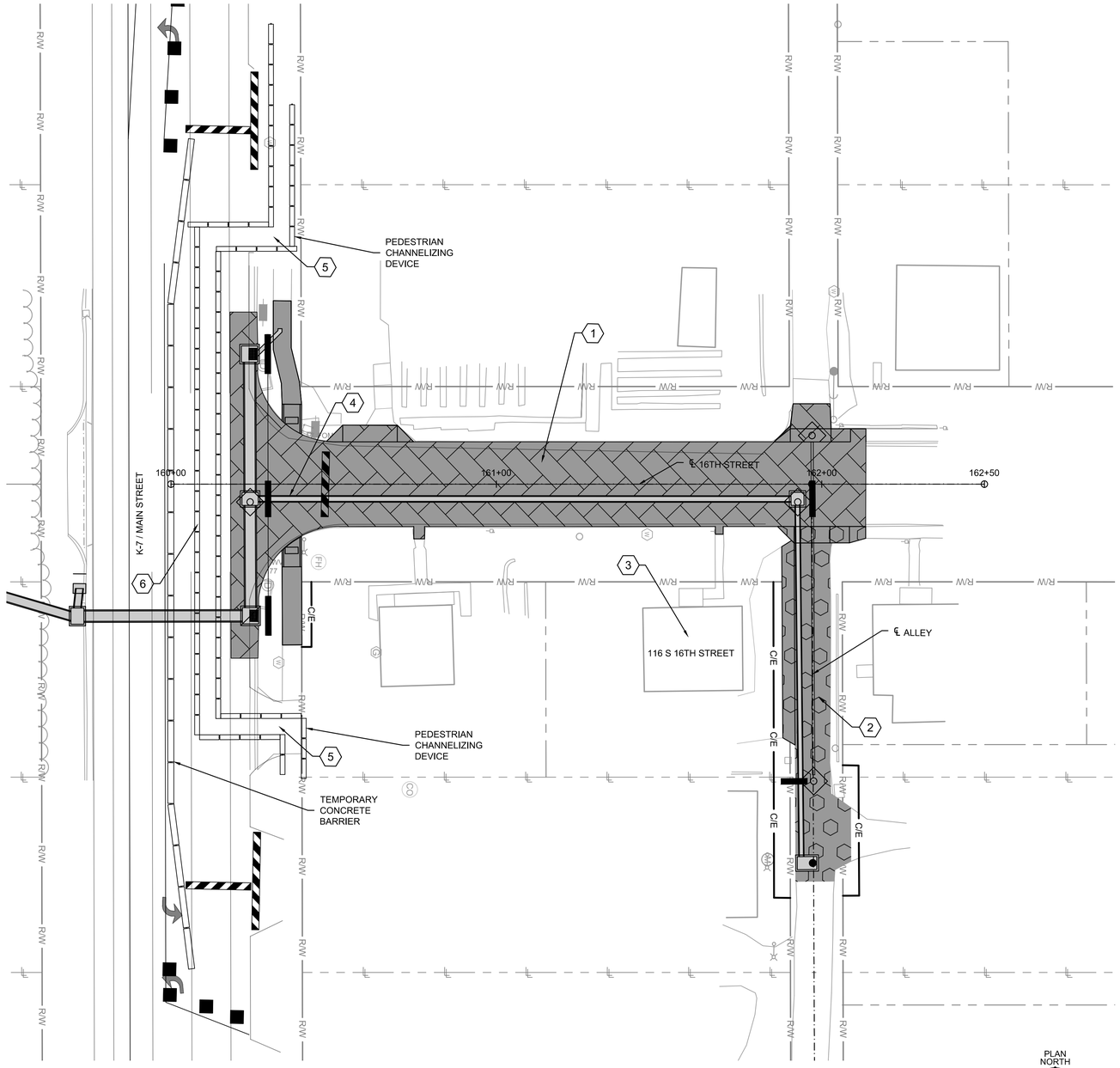
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

K-7 / MAIN STREET NORTH
TRAFFIC CONTROL PLAN (PHASE 3)

SHEET ID
GC102

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC102_MAIN_N.dwg
9/9/2026 3:44:09 PM

READY TO ADVERTISE



C3 TRAFFIC CONTROL PLAN, 16TH STREET STA. 160+00 TO 162+50 (PHASE 4 & PHASE 5)

SCALE:1"=20'

SHEET NOTES

1. CONSTRUCTION ALONG 16TH STREET TO BE COMPLETED BEFORE CONSTRUCTION ON ALLEY CAN BEGIN. FULL ACCESS TO ALLEY TO REMAIN OPEN DURING PHASE 4.
2. CONSTRUCTION ON PHASE 4 TO BEGIN AT SOUTH END AT 16TH STREET INTERSECTION WHILE PHASE 3 TRAFFIC CONTROL IS IN PLACE.
3. CONSTRUCTION ON ALLEY MAY BE DONE UNDER FULL CLOSURE.
4. CONSTRUCT SIDEWALK AND RAMPS ONCE STORM SEWER CONSTRUCTION IS COMPLETED.
5. CONTRACTOR SHALL PROVIDE ACCESS TO ALL RESIDENTIAL PROPERTIES THROUGH THE DURATION OF CONSTRUCTION.
6. PHASE 6 (SHOWN ON SHEETS GC104 AND GC105) CAN BE CONSTRUCTED CONCURRENTLY.
7. REFER TO TYPICAL APPLICATION 19 AND TYPICAL APPLICATION 20 (FIGURES 6P-19, 6P-20) OF THE MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES (MUTCD) FOR WHEN 16TH STREET AND THE ALLEY ARE CLOSED TO TRAFFIC.
8. REFER TO SHEET GC102 FOR PHASE 3 TRAFFIC CONTROL SETUP.
9. REFER TO SHEETS C-527-C-538 FOR TRAFFIC CONTROL DETAILS.

SHEET KEYNOTES

1. CONSTRUCT PHASE 4 STORM SEWER & ROAD RECONSTRUCTION ALONG 16TH ST.
2. CONSTRUCT PHASE 5 STORM SEWER ALONG ALLEY.
3. ACCESS MUST REMAIN OPEN FROM 16TH ST FOR 116 S 16TH STREET PROPERTY DURING PHASE 5 CONSTUCTION.
4. PHASE 3 TRAFFIC CONTROL TO REMAIN IN PLACE WHEN SIDE WALK IS CLOSED; OR WHEN CONSTRUCTION IS NEAR INTERSECTION.
5. DRIVEWAYS TO BE USED AS SIDEWALK RAMP DETOUR. REFER TO C-531 FOR SIDE WALK CLOSURE DETAILS.
6. PHASE 3 TRAFFIC CONTROL AND PEDESTRIAN DETOUR MAY BE REMOVED ONCE WORK ENCROACHING ON MAIN STREET IS COMPLETE AND SIDEWALKS ARE REOPENED.

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- PEDESTRIAN CHANNELIZING DEVICE
- TRAFFIC CONTROL PHASE 4
- TRAFFIC CONTROL PHASE 5

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

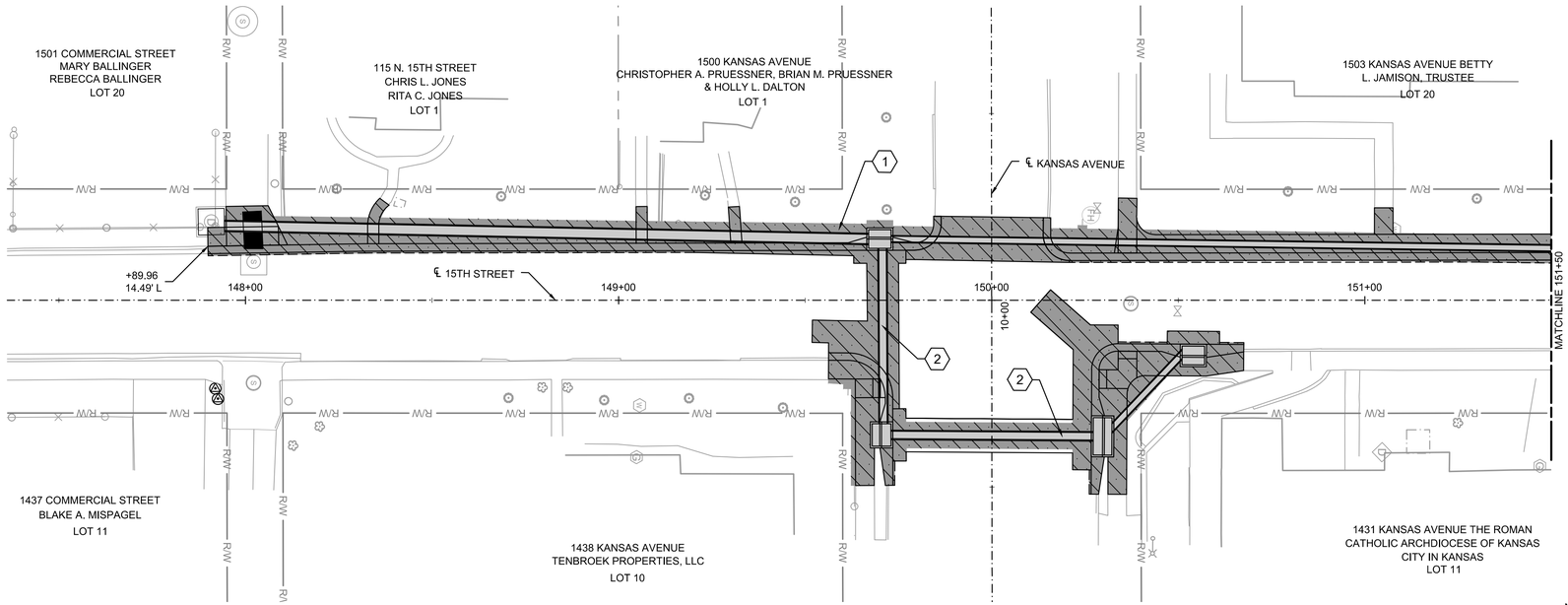
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
TRAFFIC CONTROL PLAN (PHASE 4 & 5)

SHEET ID
GC103

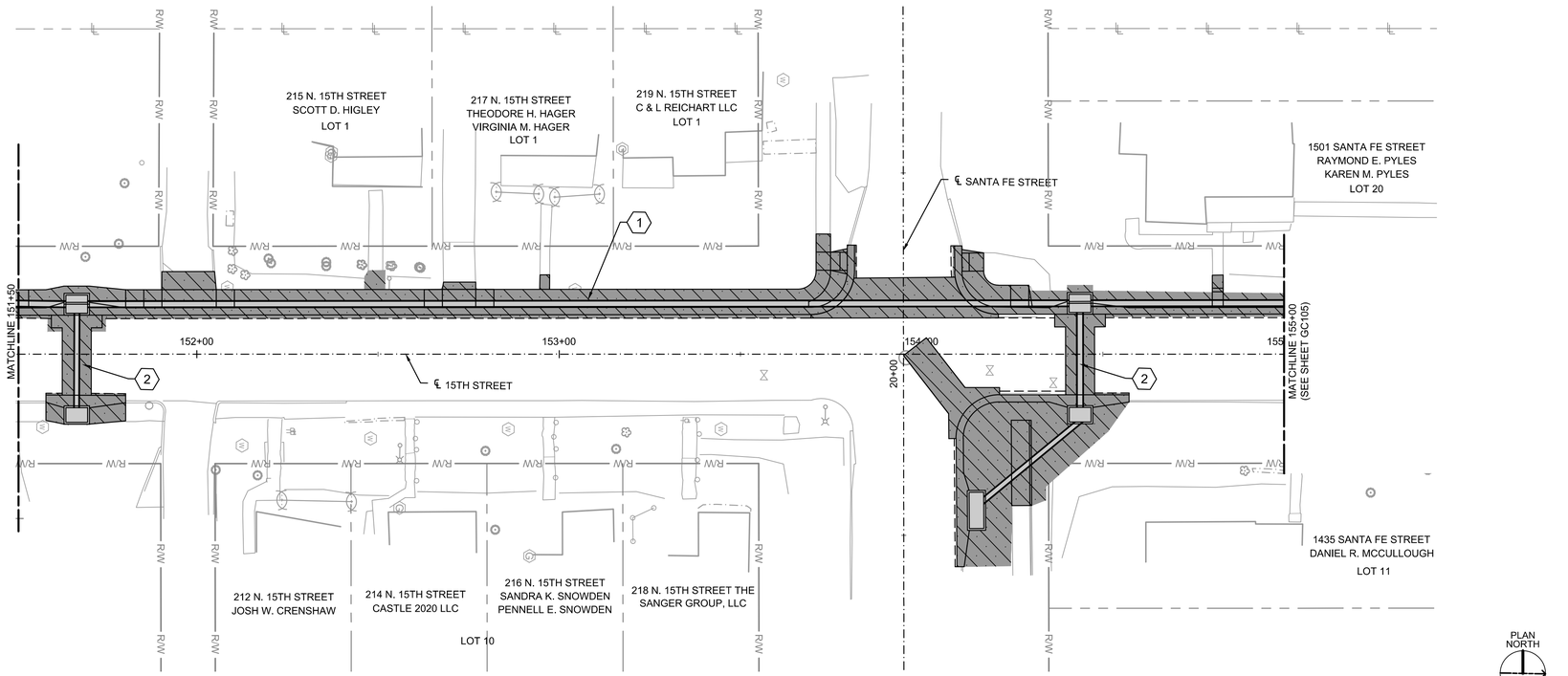
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC103_16TH.dwg
9/9/2026 3:46:00 PM

READY TO ADVERTISE



H1 TRAFFIC CONTROL PLAN, 15TH STREET STA. 147+90 TO 151+50 (PHASE 6)

SCALE:1"=20'



A1 TRAFFIC CONTROL PLAN, 15TH STREET STA. 151+50 TO 155+00 (PHASE 6)

SCALE:1"=20'

SHEET NOTES

1. PHASE 6 CAN BE CONSTRUCTED SIMULTANEOUSLY DURING PHASES 4 AND 5 (SHOWN ON SHEET GC103).
2. CONSTRUCT SIDEWALK AND RAMPS AFTER STORM SEWER IS COMPLETED.
3. CONTRACTOR SHALL PROVIDE ACCESS TO ALL RESIDENTIAL PROPERTIES THROUGH THE DURATION OF CONSTRUCTION.
4. CONTRACTOR SHALL START WORK AT DOWNSTREAM (SOUTH) END OF PROJECT.
5. IF CONTRACTOR IS UTILIZING FULL ROAD AND INTERSECTION CLOSURES, ONLY A SINGLE INTERSECTION AND SINGLE ROADWAY SEGMENT MAY BE CLOSED AT A TIME. ADJACENT INTERSECTIONS MAY NOT BE CLOSED AT THE SAME TIME.
6. REFER TO TYPICAL APPLICATION 19 AND TYPICAL APPLICATION 20 (FIGURES 6P-19, 6P-20) OF THE MUTCD FOR TYPICAL TEMPORARY TRAFFIC CONTROL SETUP.
7. REFER TO SHEET C-527-C-538 FOR TRAFFIC CONTROL DETAILS.

SHEET KEYNOTES

1. CONSTRUCT PHASE 6 STORM SEWER IMPROVEMENTS ALONG 15TH STREET.
2. STREET CROSSINGS MAY BE CONSTRUCTED WHILE THAT PORTION OF ROADWAY OR INTERSECTION IS ALREADY CLOSED DUE TO 15TH STREET MAINLINE STORM SEWER CONSTRUCTION. ALTERNATIVELY, CROSSINGS MAY BE CONSTRUCTED SEPARATELY ONCE THE MAINLINE CONSTRUCTION HAS BEEN COMPLETED.

LEGEND

- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- TRAFFIC CONTROL PHASE 6

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

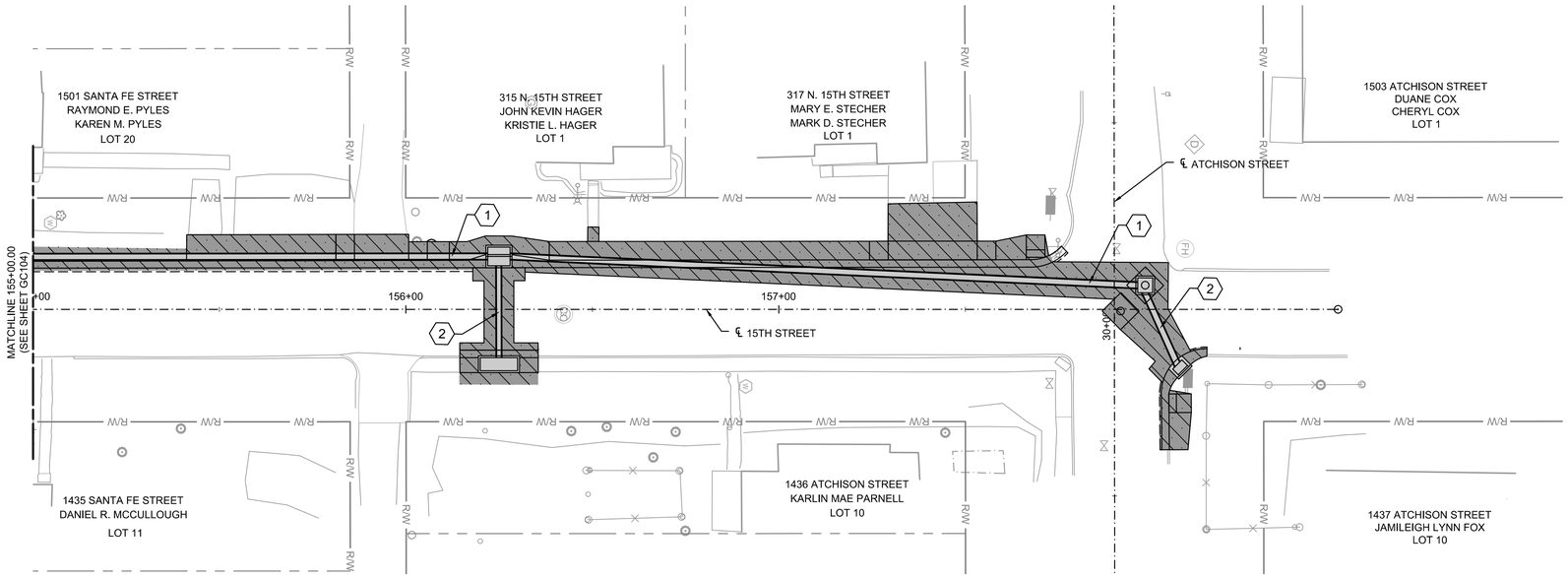
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
TRAFFIC CONTROL PLAN (PHASE 6)

SHEET ID
GC104

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC104-GC105_15TH.dwg
9/9/2026 3:47:05 PM

READY TO ADVERTISE



H1 TRAFFIC CONTROL PLAN, 15TH STREET STA 155+00 TO 158+37 (PHASE 6)

SCALE:1"=20'

SHEET NOTES

1. PHASE 6 CAN BE CONSTRUCTED SIMULTANEOUSLY DURING PHASES 4 AND 5 (SHOWN ON SHEET GC103).
2. CONSTRUCT SIDEWALK AND RAMPS AFTER STORM SEWER IS COMPLETED.
3. CONTRACTOR SHALL PROVIDE ACCESS TO ALL RESIDENTIAL PROPERTIES THROUGH THE DURATION OF CONSTRUCTION.
4. CONTRACTOR SHALL START WORK AT DOWNSTREAM (SOUTH) END OF PROJECT.
5. IF CONTRACTOR IS UTILIZING FULL ROAD AND INTERSECTION CLOSURES, ONLY A SINGLE INTERSECTION AND SINGLE ROADWAY SEGMENT MAY BE CLOSED AT A TIME. ADJACENT INTERSECTIONS MAY NOT BE CLOSED AT THE SAME TIME.
6. REFER TO TYPICAL APPLICATION 19 AND TYPICAL APPLICATION 20 (FIGURES 6P-19, 6P-20) OF THE MUTCD FOR TYPICAL TEMPORARY TRAFFIC CONTROL SETUP.
7. REFER TO SHEET C-527-C-538 FOR TRAFFIC CONTROL DETAILS.

SHEET KEYNOTES

1. CONSTRUCT PHASE 6 STORM SEWER IMPROVEMENTS ALONG 15TH STREET.
2. STREET CROSSINGS MAY BE CONSTRUCTED WHILE THAT PORTION OF ROADWAY OR INTERSECTION IS ALREADY CLOSED DUE TO 15TH STREET MAINLINE STORM SEWER CONSTRUCTION. ALTERNATIVELY, CROSSINGS MAY BE CONSTRUCTED SEPARATELY ONCE THE MAINLINE CONSTRUCTION HAS BEEN COMPLETED.

LEGEND

PROPOSED DRAINAGE STRUCTURE

PROPOSED STORM DRAINAGE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

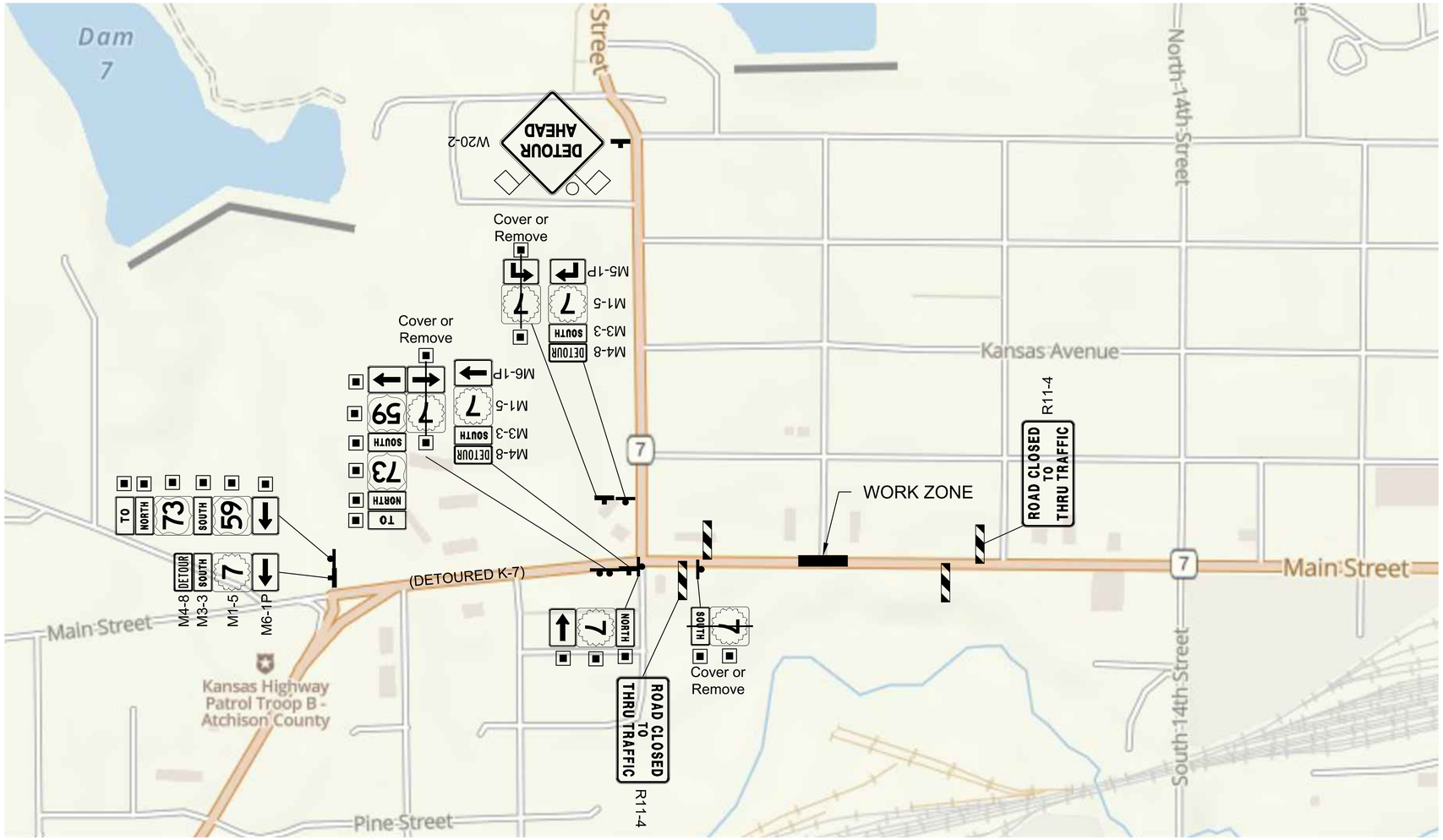
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
TRAFFIC CONTROL PLAN (PHASE 6)

SHEET ID
GC105

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC104-GC105_15TH.dwg
9/9/2026 3:47:20 PM

READY TO ADVERTISE



B1 TRAFFIC CONTROL DETOUR PLAN (SHEET 1 OF 3)

NOT TO SCALE

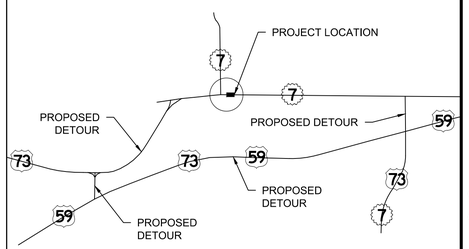
SHEET NOTES

1. ALL CONFLICTING ROUTE MARKERS ALONG THE DETOUR ROUTE SHALL BE COVERED OR REMOVED.
2. EXISTING SIGNS NOT SHOWN AT THESE JUNCTIONS SHALL REMAIN IN PLACE UNLESS OTHERWISE DIRECTED BY THE ENGINEER.
3. ERECT DETOUR CONFIRMING ASSEMBLIES IN CONJUNCTION WITH EXISTING ROUTE MARKER ASSEMBLIES ALONG DETOUR ROUTE.
4. EXISTING SIGN POSTS SHOULD NOT BE DISTURBED. ANY SIGNS NEEDED FOR DETOUR SHOULD BE PLACED ALONG SIDE OF EXISTING SIGN POSTS, WITH IT'S OWN POST.
5. D1-X SIGNS SHOULD BE SHOWN ON TOP OF EXISTING SIGNS BUT, BE PLACED ON THEIR OWN SIGN STAND DIRECTLY ABOVE AND BEHIND.

LEGEND

- EXISTING SIGN SYMBOL
- DETOUR SIGN SYMBOL
- EXISTING SIGN FACE
- TYPE 3 BARRICADES

DETOUR KEY MAP



NOTE: CIRCLE ON MAP DENOTES LOCATION OF DETOUR.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
DATE: ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012
SCALE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

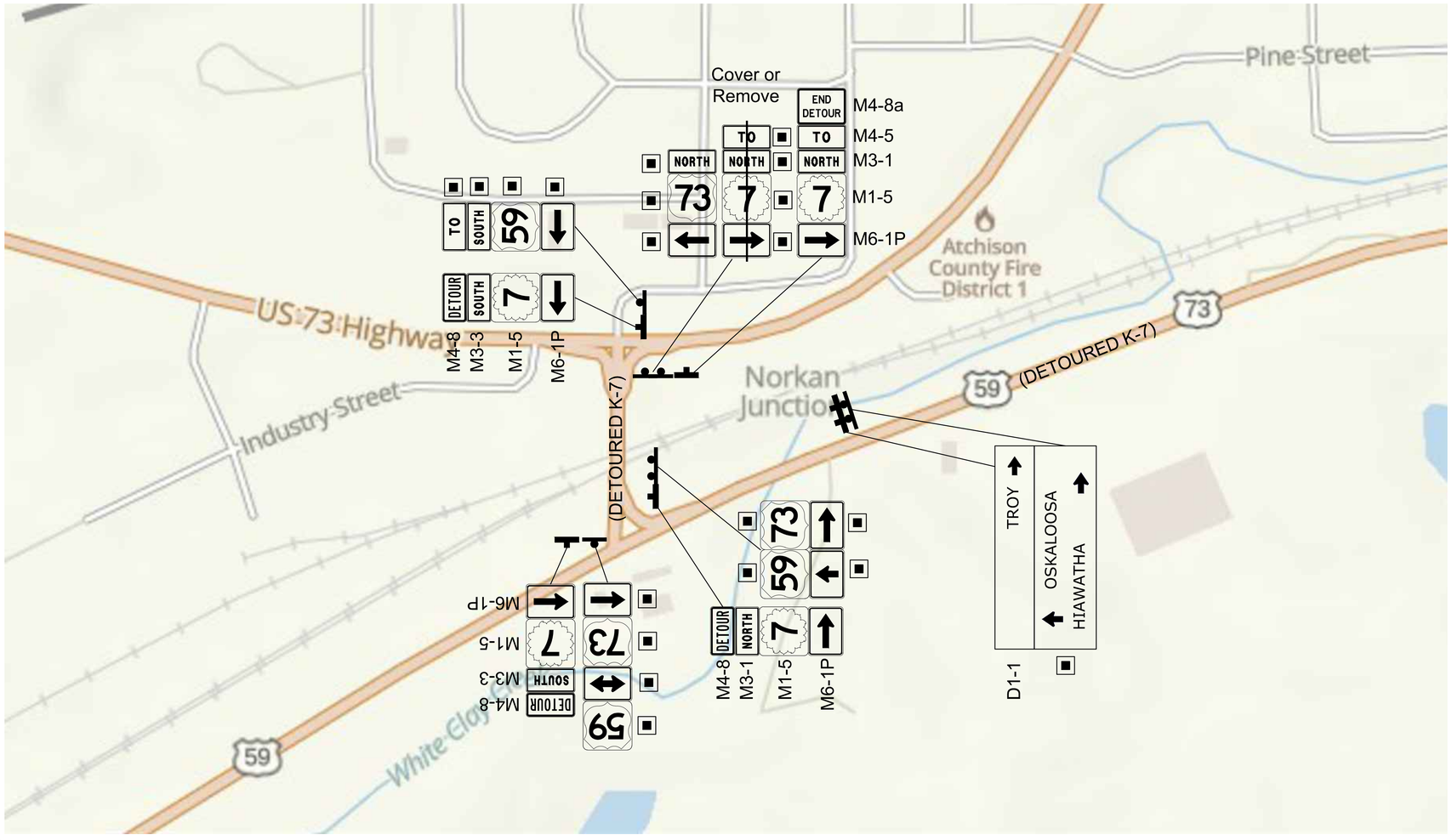
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

TRAFFIC CONTROL
K-7 DETOUR PLAN

SHEET ID
GC106

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC106-GC108_DTR.dwg
9/9/2026 3:47:43 PM

READY TO ADVERTISE



PLAN NORTH

C1 TRAFFIC CONTROL DETOUR PLAN (SHEET 2 OF 3)

NOT TO SCALE

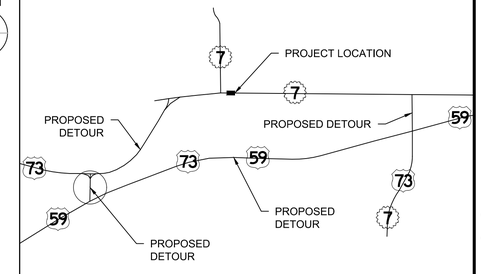
SHEET NOTES

1. ALL CONFLICTING ROUTE MARKERS ALONG THE DETOUR ROUTE SHALL BE COVERED OR REMOVED.
2. EXISTING SIGNS NOT SHOWN AT THESE JUNCTIONS SHALL REMAIN IN PLACE UNLESS OTHERWISE DIRECTED BY THE ENGINEER.
3. ERECT DETOUR CONFIRMING ASSEMBLIES IN CONJUNCTION WITH EXISTING ROUTE MARKER ASSEMBLIES ALONG DETOUR ROUTE.
4. EXISTING SIGN POSTS SHOULD NOT BE DISTURBED. ANY SIGNS NEEDED FOR DETOUR SHOULD BE PLACED ALONG SIDE OF EXISTING SIGN POSTS, WITH IT'S OWN POST.
5. D1-X SIGNS SHOULD BE SHOWN ON TOP OF EXISTING SIGNS BUT, BE PLACED ON THEIR OWN SIGN STAND DIRECTLY ABOVE AND BEHIND.

LEGEND

- EXISTING SIGN SYMBOL
- DETOUR SIGN SYMBOL
- EXISTING SIGN FACE
- TYPE 3 BARRICADES

DETOUR KEY MAP



NOTE: CIRCLE ON MAP DENOTES LOCATION OF DETOUR.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPAITE WOODS BUILDING 6
OVERLAND PARK, KS 66210

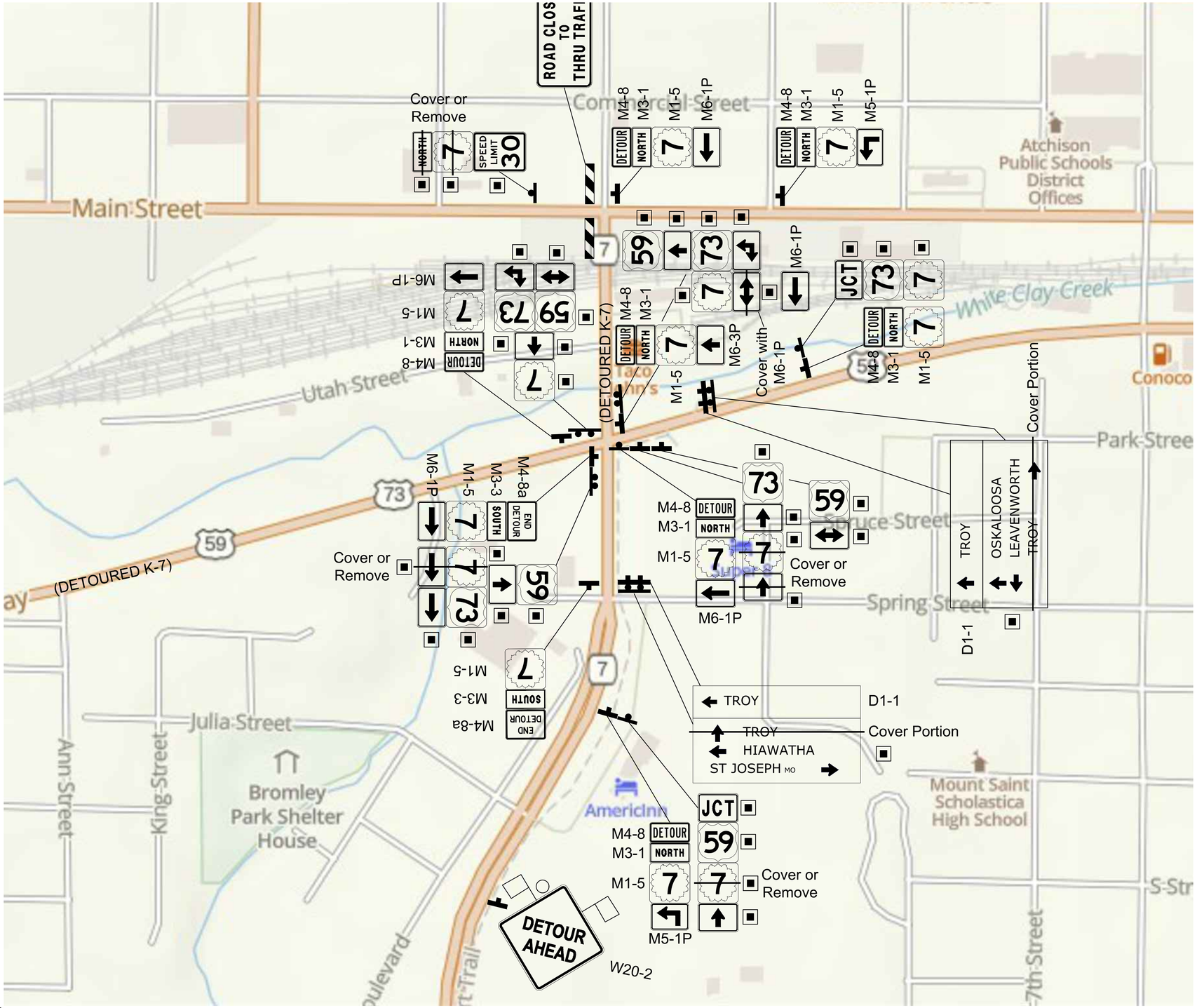
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

TRAFFIC CONTROL
K-7 DETOUR PLAN

SHEET ID
GC107

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC106-GC108_DTR.dwg
9/9/2026 3:47:49 PM

READY TO ADVERTISE



A1 TRAFFIC CONTROL DETOUR PLAN (SHEET 3 OF 3)
NOT TO SCALE

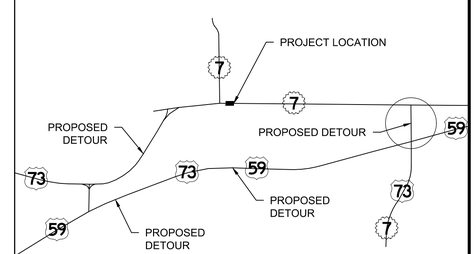
SHEET NOTES

1. ALL CONFLICTING ROUTE MARKERS ALONG THE DETOUR ROUTE SHALL BE COVERED OR REMOVED.
2. EXISTING SIGNS NOT SHOWN AT THESE JUNCTIONS SHALL REMAIN IN PLACE UNLESS OTHERWISE DIRECTED BY THE ENGINEER.
3. ERECT DETOUR CONFIRMING ASSEMBLIES IN CONJUNCTION WITH EXISTING ROUTE MARKER ASSEMBLIES ALONG DETOUR ROUTE.
4. EXISTING SIGN POSTS SHOULD NOT BE DISTURBED. ANY SIGNS NEEDED FOR DETOUR SHOULD BE PLACED ALONG SIDE OF EXISTING SIGN POSTS, WITH IT'S OWN POST.
5. D1-X SIGNS SHOULD BE SHOWN ON TOP OF EXISTING SIGNS BUT, BE PLACED ON THEIR OWN SIGN STAND DIRECTLY ABOVE AND BEHIND.

LEGEND

- EXISTING SIGN SYMBOL
- DETOUR SIGN SYMBOL
- EXISTING SIGN FACE
- TYPE 3 BARRICADES

DETOUR KEY MAP



NOTE: CIRCLE ON MAP DENOTES LOCATION OF DETOUR.

PLAN NORTH

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

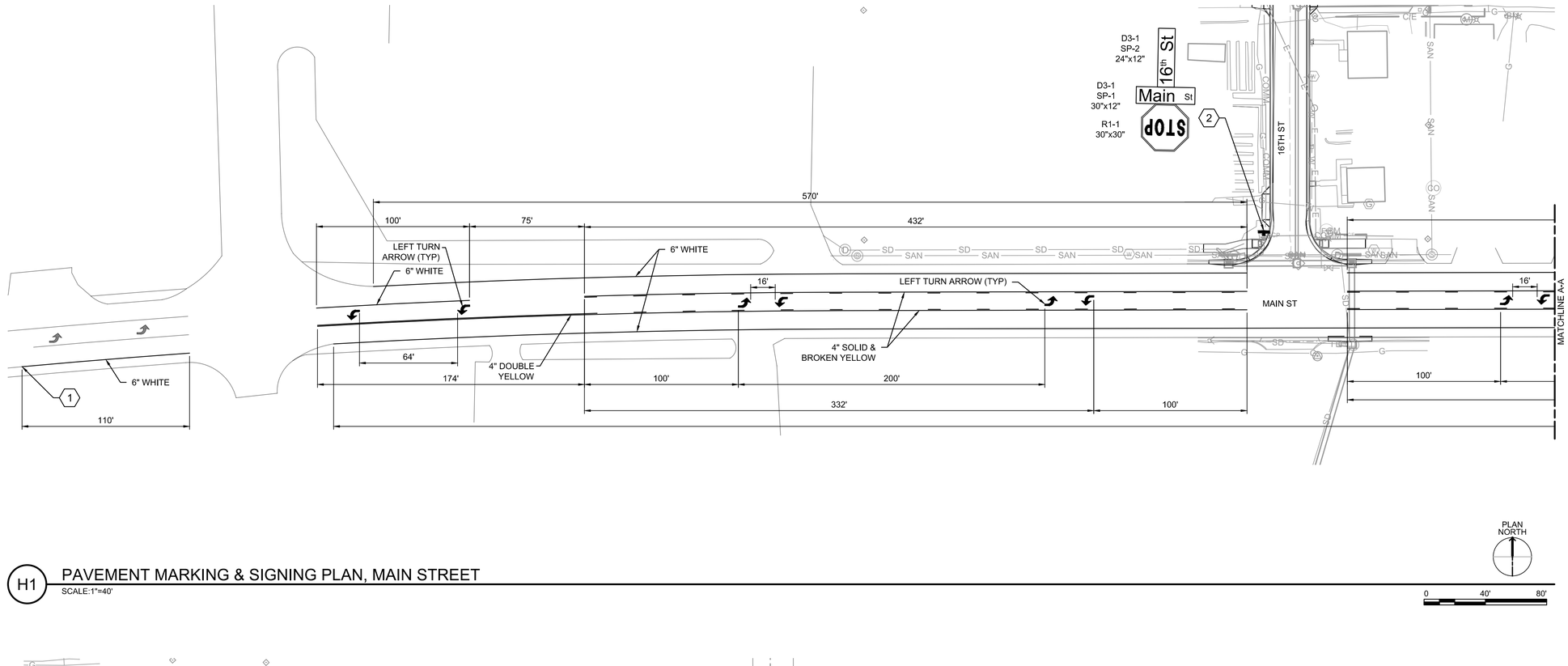
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

TRAFFIC CONTROL
K-7 DETOUR PLAN

SHEET ID
GC108

READY TO ADVERTISE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC106-GC108_DTR.dwg
9/9/2026 3:47:57 PM

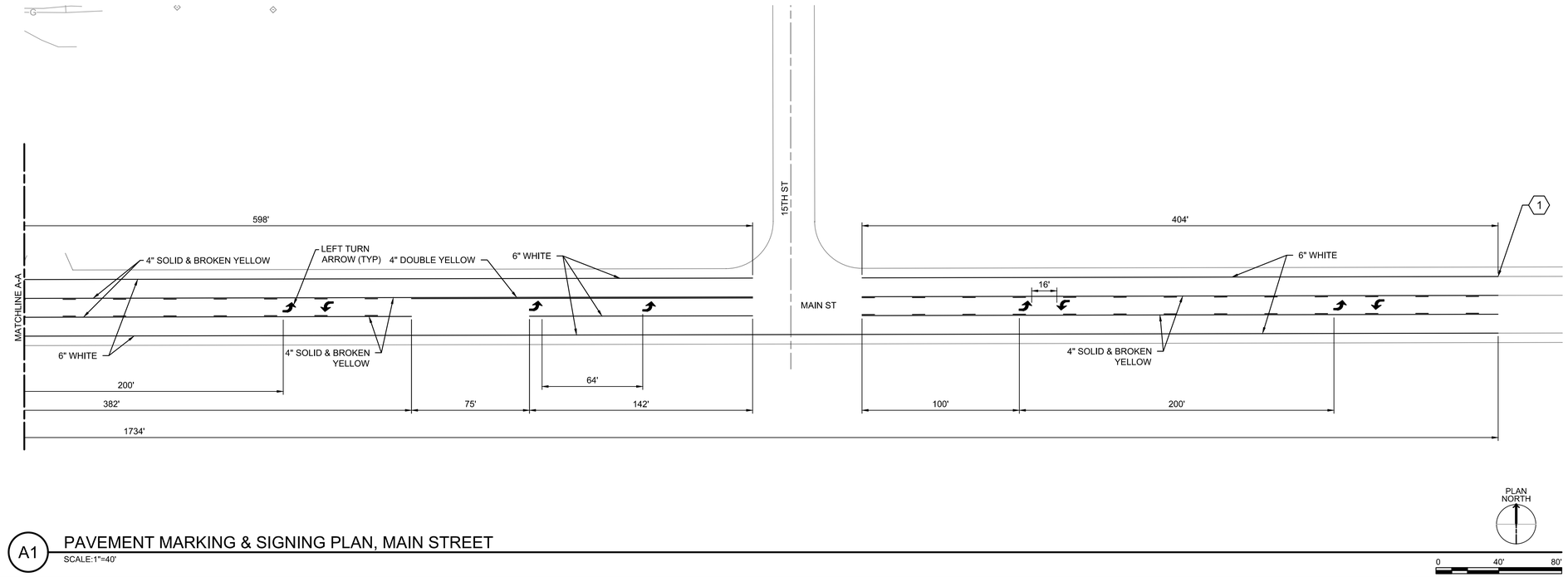


PLAN NORTH

0 40' 80'

H1 PAVEMENT MARKING & SIGNING PLAN, MAIN STREET

SCALE:1"=40'



PLAN NORTH

0 40' 80'

A1 PAVEMENT MARKING & SIGNING PLAN, MAIN STREET

SCALE:1"=40'

SHEET NOTES

1. NEW STREET NAME SIGNS ARE PLACED BACK TO BACK ON SIGN POST (QUANTITY 2 EACH).
2. SEE SHEETS C-539-C-542 FOR PAVEMENT MARKING AND SIGNING DETAILS.

SHEET KEYNOTES

1. MATCH EXISTING
2. REMOVE AND REPLACE WITH NEW SIGN ASSEMBLY

LEGEND

SIGN POST

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

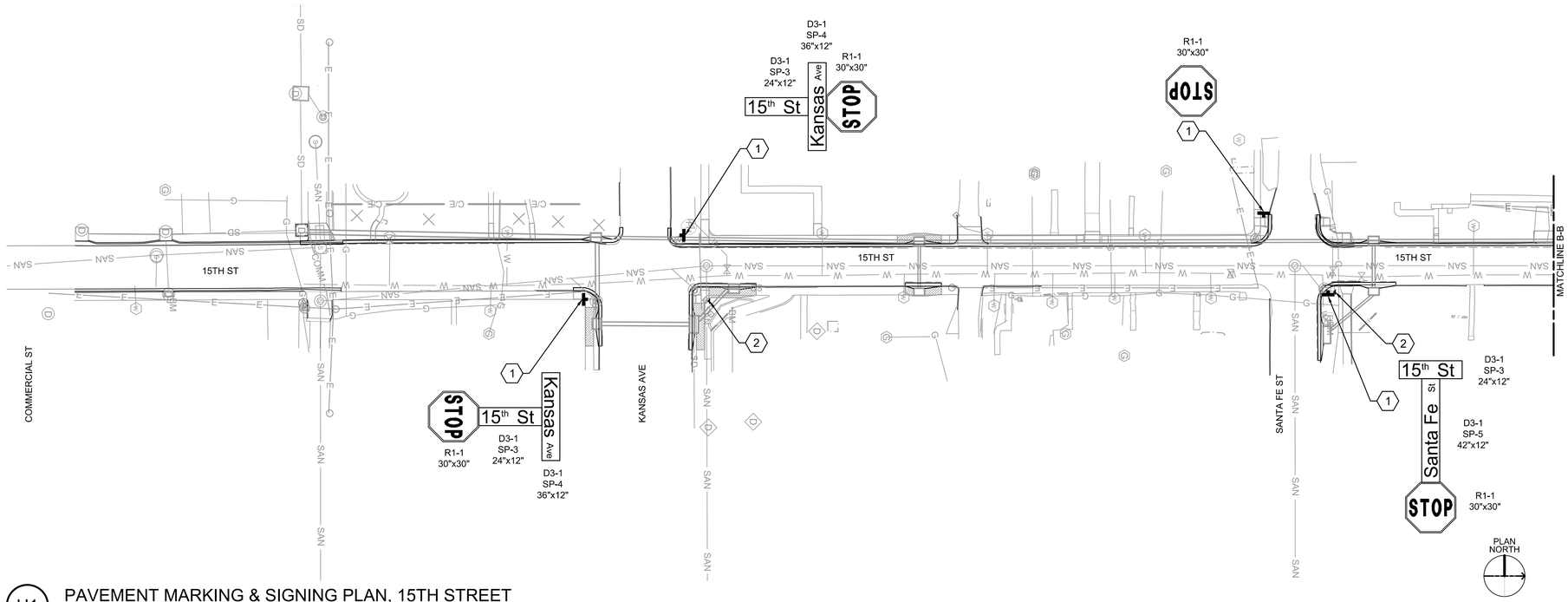
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

K-7 / MAIN STREET
PAVEMENT MARKING & SIGNING PLAN

SHEET ID
GC109

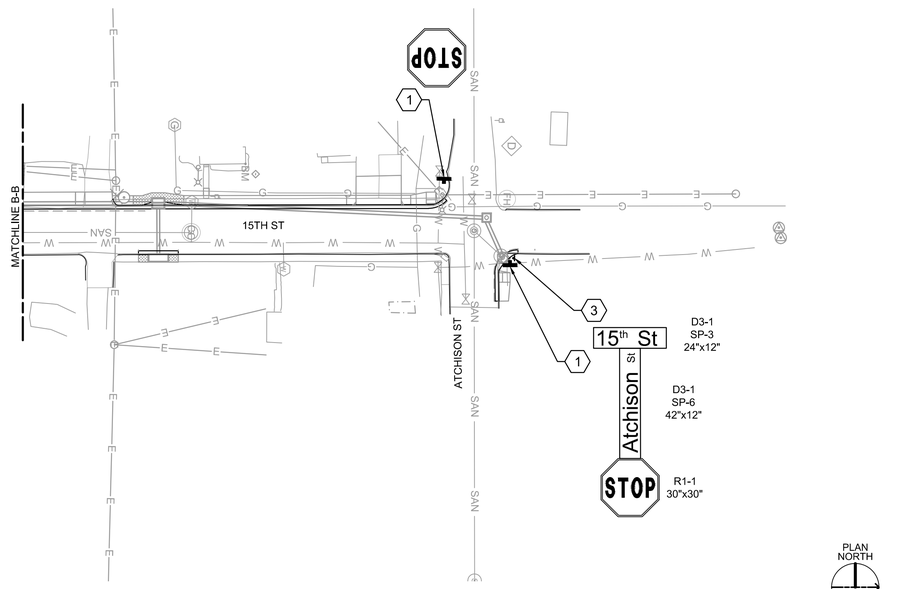
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC109-GC110_PM-SGN.dwg
9/9/2026 3:49:07 PM

READY TO ADVERTISE



H1 PAVEMENT MARKING & SIGNING PLAN, 15TH STREET

SCALE:1"=40'



A1 PAVEMENT MARKING & SIGNING PLAN, 15TH STREET

SCALE:1"=40'

SHEET NOTES

1. NEW STREET NAME SIGNS ARE PLACED BACK TO BACK ON SIGN POST (QUANTITY 2 EACH).
2. SEE SHEETS C-539-C-542 FOR PAVEMENT MARKING AND SIGNING DETAILS.

SHEET KEYNOTES

1. REMOVE AND REPLACE WITH NEW SIGN ASSEMBLY
2. REMOVE AND RESET STONE SIGN (NO DIRECT PAY)
3. REMOVE STONE SIGN (NO DIRECT PAY)

LEGEND

SIGN POST

LIMESTONE STREET NAME SIGN

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROWL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
SIZE:	ANSI D
ISSUE DATE:	23 FEBRUARY 2026
SOLICITATION NO.:	
TASK ORDER # W912D24F1019	
CONTRACT NO.:	W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

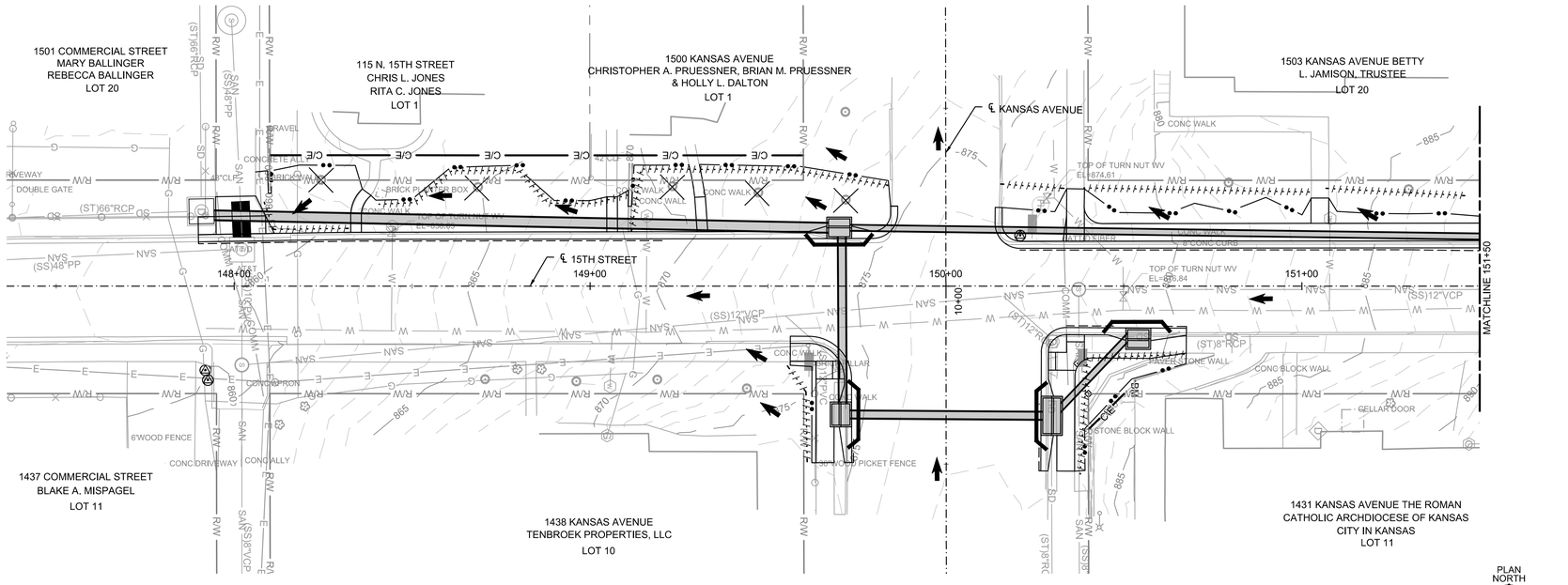
15TH STREET
PAVEMENT MARKING & SIGNING PLAN

SHEET ID

GC110

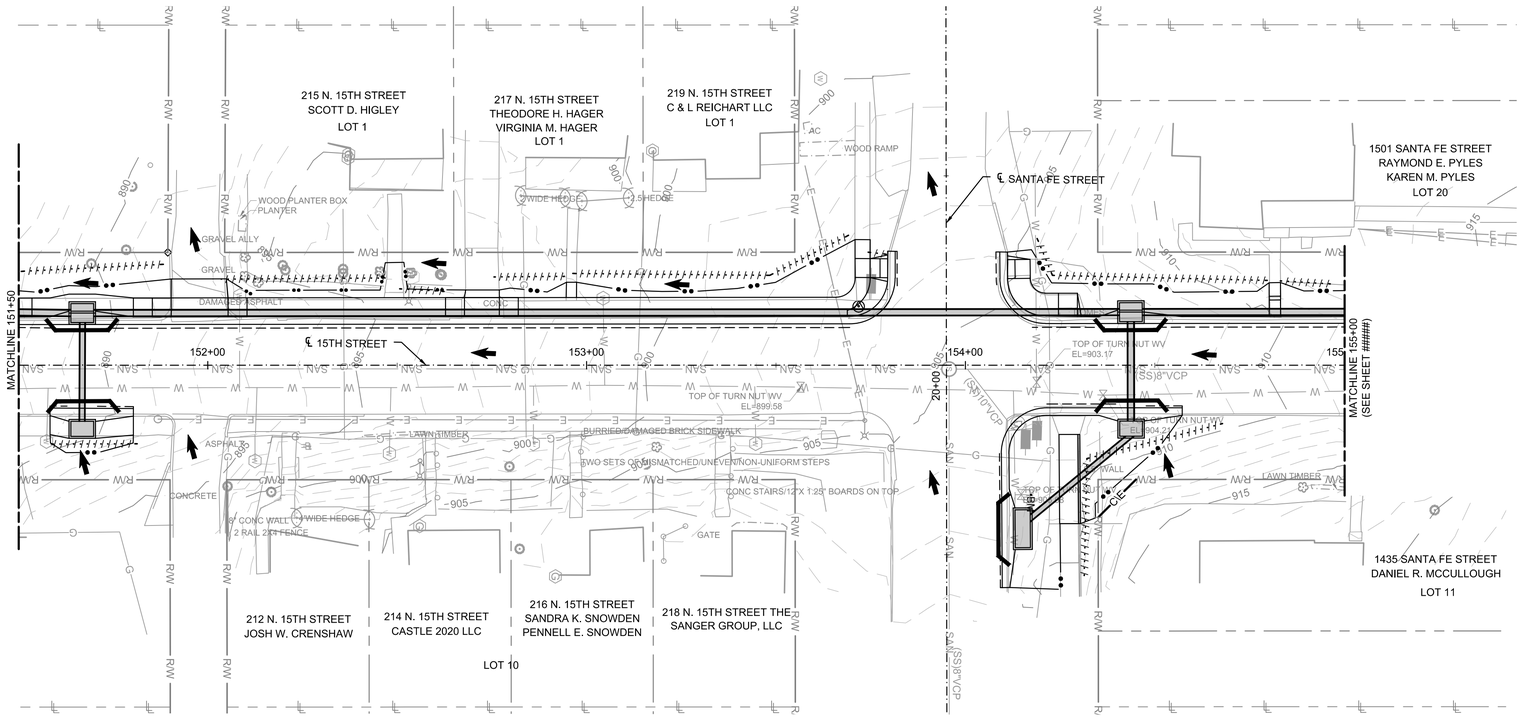
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC109-GC110_PM-SGN.dwg
9/9/2026 3:49:22 PM

READY TO ADVERTISE



H1 EROSION CONTROL PLAN, 15TH STREET STA. 147+90 TO 151+50

SCALE:1"=20'



A1 EROSION CONTROL PLAN, 15TH STREET STA. 151+50 TO 155+00

SCALE:1"=20'

SHEET NOTES

1. SEE SHEET C-523, C-524, AND C-525 FOR EROSION CONTROL DETAILS.

LEGEND

- FLOW DIRECTION
- INLET BARRIER (CURB INLET) *
- BIODEGRADABLE LOG
- INLET PROTECTION (GRATE INLET) *

* THE CONFIGURATION OF THE TEMPORARY INLET SEDIMENT BARRIERS, AS SHOWN, ARE FOR STORM SEWER STRUCTURES WITHOUT INSTALLED TOPS. SEE EROSION CONTROL DETAILS FOR OTHER CONFIGURATIONS.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

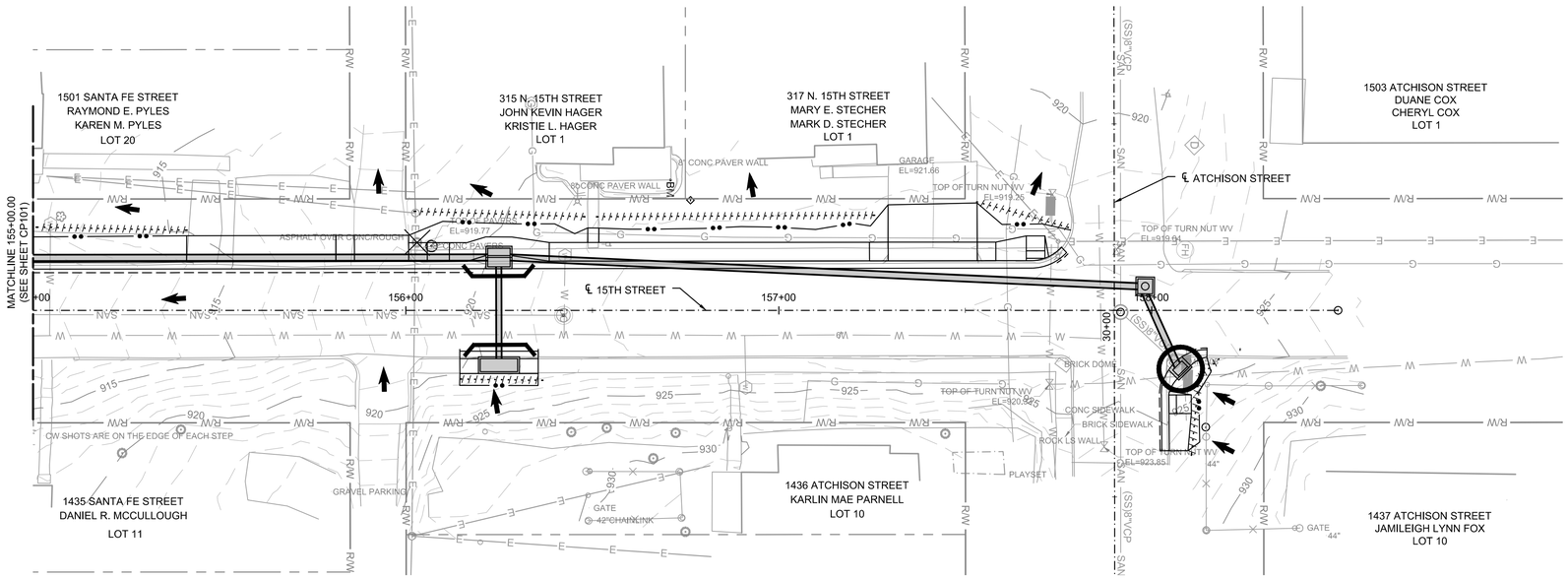
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
EROSION CONTROL PLAN

SHEET ID
GC111

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC111-GC112_15TH-EC.dwg
9/9/2026 3:50:32 PM

READY TO ADVERTISE



H1 EROSION CONTROL PLAN, 15TH STREET STA 155+00 TO 158+37

SCALE:1"=20'

SHEET NOTES

1. SEE SHEET C-523, C-524, AND C-525 FOR EROSION CONTROL DETAILS.

LEGEND

- FLOW DIRECTION
- INLET BARRIER (CURB INLET) *
- BIODEGRADABLE LOG
- INLET PROTECTION (GRATE INLET) *

* THE CONFIGURATION OF THE TEMPORARY INLET SEDIMENT BARRIERS, AS SHOWN, ARE FOR STORM SEWER STRUCTURES WITHOUT INSTALLED TOPS. SEE EROSION CONTROL DETAILS FOR OTHER CONFIGURATIONS.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

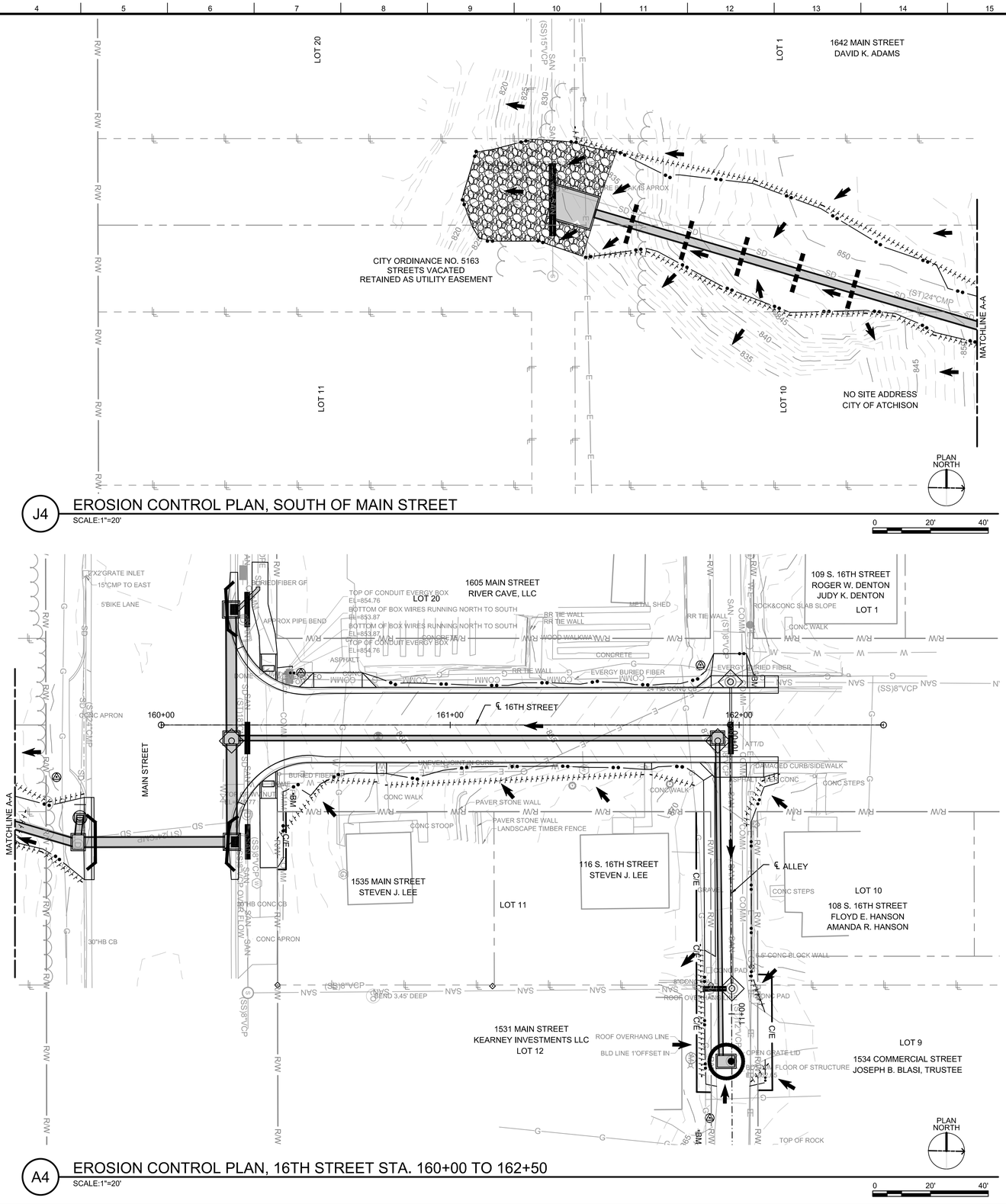
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
EROSION CONTROL PLAN

SHEET ID
GC112

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC111-GC112_15TH-EC.dwg
9/9/2026 3:50:47 PM

READY TO ADVERTISE



J4 EROSION CONTROL PLAN, SOUTH OF MAIN STREET
SCALE:1"=20'

A4 EROSION CONTROL PLAN, 16TH STREET STA. 160+00 TO 162+50
SCALE:1"=20'

SHEET NOTES

1. SEE SHEET C-523, C-524, AND C-525 FOR EROSION CONTROL DETAILS

LEGEND

- FLOW DIRECTION
- DITCH CHECK
- INLET BARRIER (CURB INLET) *
- BIODEGRADABLE LOG
- INLET PROTECTION (GRATE INLET) *

* THE CONFIGURATION OF THE TEMPORARY INLET SEDIMENT BARRIERS, AS SHOWN, ARE FOR STORM SEWER STRUCTURES WITHOUT INSTALLED TOPS. SEE EROSION CONTROL DETAILS FOR OTHER CONFIGURATIONS.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

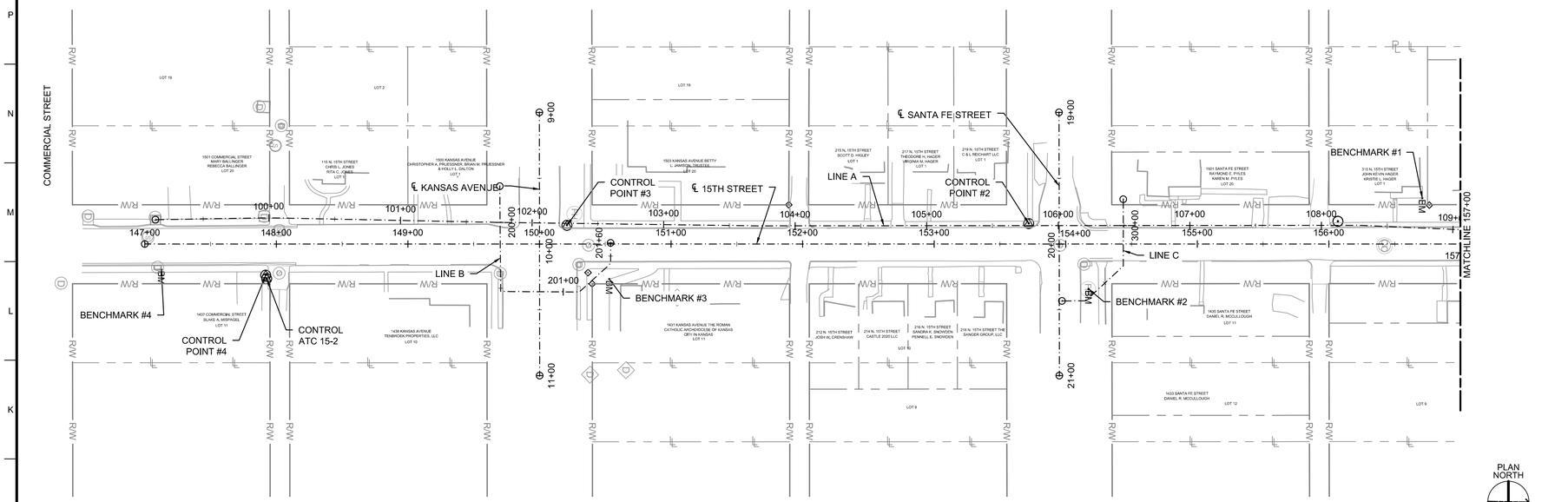
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
EROSION CONTROL PLAN

SHEET ID
GC113

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_GC113_16TH-EC.dwg
9/9/2026 3:51:43 PM

READY TO ADVERTISE

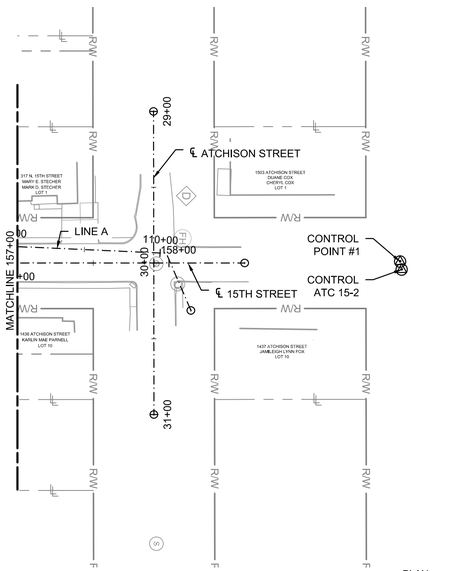


PLAN NORTH

0 50' 100'

J1 PROJECT CONTROL, 15TH STREET
SCALE:1"=50'

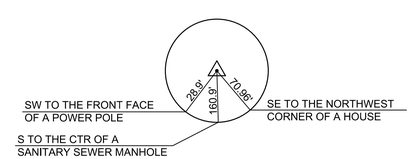
PROJECT CONTROL



PLAN NORTH

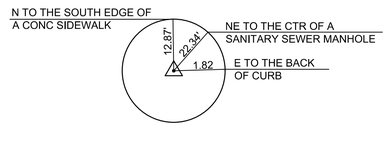
A1 PROJECT CONTROL, 15TH STREET
SCALE:1"=50'

0 50' 100'



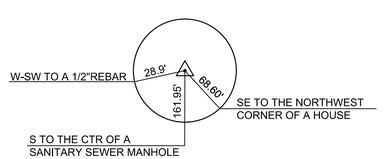
CP#1: SET A 1/2"REBAR W/ A CONTROL POINT CAP AT THE DEAD END STREET OF N. 15TH STREET NORTH OF THE INTERSECTION OF N. 15TH STREET AND ATCHISON STREET

N=461,605.142 E=2,119,123.527



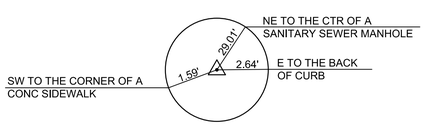
CP#3: SET A 1/2"REBAR W/ A CONTROL POINT CAP AT THE NORTHWEST CORNER OF THE INTERSECTION OF KANSAS AVENUE AND N. 15TH STREET

N=460,673.379 E=2,119,129.120



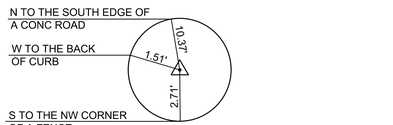
ATC 15-1: SET A 3-1/2" DOMED BRONZE DISK IN A 10"Ø , 36" DEEP CONCRETE BASE AT THE DEAD END STREET OF N. 15TH STREET NORTH OF THE INTERSECTION OF N. 15TH STREET AND ATCHISON STREET

N=461,606.274 E=2,119,128.754
ELEV= 934.12'



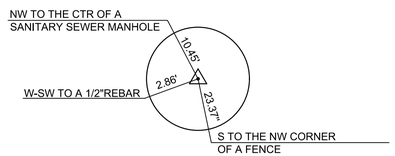
CP#2: SET A MAG NAIL AND SHINER IN SIDEWALK AT THE SOUTHWEST CORNER OF THE INTERSECTION OF SANTA FE STREET AND N. 15TH STREET

N=461,024.255 E=2,119,120.715



CP#4: SET A 1/2"REBAR W/ A CONTROL POINT CAP AT THE SOUTHEAST CORNER OF THE INTERSECTION OF N. 15TH STREET AND THE ALLEY NORTH OF COMMERCIAL STREET

N=460,445.245 E=2,119,171.618



ATC 15-2: SET A 3-1/2" DOMED BRONZE DISK IN A 10"Ø , 36" DEEP CONCRETE BASE N. 15TH STREET AND THE ALLEY NORTH OF COMMERCIAL STREET

N=460,446.017 E=2119174.370
ELEV= 859.95'

BENCHMARK #1:
NORTHEAST CORNER OF A CONCRETE PORCH AT HOUSE# 315 N. 15TH STREET.
ELEVATION=922.37

BENCHMARK #2:
SET A TRIPLE PUNCH MARK IN THE NORTHWEST BONNET BOLT OF A FIRE HYDRANT AT THE NORTHEAST CORNER OF THE INTERSECTION OF SANTA FE STREET AND N. 15TH STREET
ELEVATION=910.37

BENCHMARK #3:
SOUTHWEST CORNER OF A CONCRETE SIDEWALK WITH HANDRAIL AT THE SOUTHWEST CORNER OF THE CHURCH BUILDING AT THE NORTHEAST CORNER OF THE INTERSECTION OF KANSAS AVENUE AND N. 15TH STREET.
ELEVATION=883.56

BENCHMARK #4:
SET A CUT SQUARE ON THE NORTHEAST CORNER OF A CURB INLET ON THE EAST SIDE OF N. 15TH STREET, NORTH OF COMMERCIAL STREET.
ELEVATION=858.66

SHEET NOTES

1. THIS SURVEY IS A TOPOGRAPHIC SURVEY ONLY, NO RECORD PROPERTY RESEARCH WAS PERFORMED BY THE SURVEYOR OR DIRECTORATE OF PUBLIC WORKS (DPW)
2. THE UNDERGROUND UTILITY LINES AS SHOWN ARE AS LOCATED IN THE FIELD AND MAPPING PROVIDED BY THE DPW, SANITARY AND STORM LINES SHOWN REPRESENT A "STRAIGHT-LINE CONNECTION" BETWEEN OBSERVABLE STRUCTURES, PIPELINE ELEVATIONS ARE AS MEASURED FROM THE FLOWLINE OF THE STRUCTURES.

3.1 THE PROJECT SITE IS ALL ZONED SINGLE FAMILY RESIDENTIAL. THE PROJECT SITE IS ON 15TH STREET AND EXTENDS FROM COMMERCIAL STREET ON THE SOUTH TO ATCHISON STREET ON THE NORTH.

3.2 HORIZONTAL DATA SHOWN ON THIS PROJECT IS BASED ON THE KANSAS STATE PLANE COORDINATE SYSTEM (NORTH ZONE). NAD83 (2011), US SURVEY FEET. ANY BEARINGS SHOWN ARE GRID NORTH.

VERTICAL DATA SHOWN ON THIS PROJECT IS NORTH AMERICAN VERTICAL DATUM, 1988 (NAVD 88) AND ARE SHOWN IN US SURVEY FEET.

BOTH HORIZONTAL AND VERTICAL DATA WERE ESTABLISHED USING NATIONAL GEODETIC SURVEY (NGS) POINT DESIGNATED AS W 281, PID KE0493, AS A BASE CONTROL POINT.

THE PUBLISHED VALUES FOR W 281 ARE:
N: 463,982.13' E: 2,116,796.88' ELEV: 998.64'

3. 3.3 ON-SITE CONTROL FOR THIS PROJECT IS BASED ON NGS POINT W 281 BEING THE ORIGINATING BASE POINT. PROJECT CONTROL WAS ESTABLISHED BY OCCUPYING NGS W 281 WITH A GPS BASE UNIT AND USING GPS ROVER UNITS TO MEASURE TO SEMI-PERMANENT CONTROL POINTS ON-SITE FOR SURVEY PURPOSES AND CONTRACTORS USE. THE CONTROL WAS MEASURED FOUR INDIVIDUAL TIMES AND THE VALUES AVERAGED TO DETERMINE FINAL VALUES. A CLOSED LEVEL LOOP WAS PERFORMED THROUGH ALL SEMI-PERMANENT CONTROL POINTS AND BENCHMARKS. TOPOGRAPHIC SURVEY WORK WAS PERFORMED USING REAL TIME KINEMATIC (RTK) SURVEY METHODS WITH A BASE AND ROVER SETUP AND THE USE OF ROBOTIC TOTAL STATIONS. THE SEMI-PERMANENT CONTROL POINTS SET ARE 1/2" BY 18" REBARS WITH CONTROL POINT CAPS IN GRASSY AREAS OR 1-1/2"MAG NAILS WITH SHINERS IN PAVED AREAS AND ARE IDENTIFIED AS CP#1 THRU CP#4. SEE PLAN SHEETS FOR LOCATION AND TYPES OF SEMI-PERMANENT MARKERS SET. STANDARD TYPE C 3-1/2" BRONZE USACE SURVEY MARKERS ARE IDENTIFIED AS ATC 15-1 AND ATC 15-2. THIS SURVEY WAS PERFORMED BETWEEN NOVEMBER 25, 2024 AND FEBRUARY 4, 2025, BY AFFINIS CORP.

ATTENTION: JONATHAN AUBREY MEYER, KS PLS #1579
8900 INDIAN CREEK PARKWAY, SUITE 450
OVERLAND PARK, KS 66210
TELEPHONE (913)-239-1100

LEGEND

- PROPERTY PIN
- CONTROL POINT
- •BM BENCH MARK
- R/W EXISTING RIGHT-OF-WAY
- EXISTING PROPERTY LINE
- EXISTING LOT LINE
- EXISTING PROPERTY/LOT LINE
- CENTER LINE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D224F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

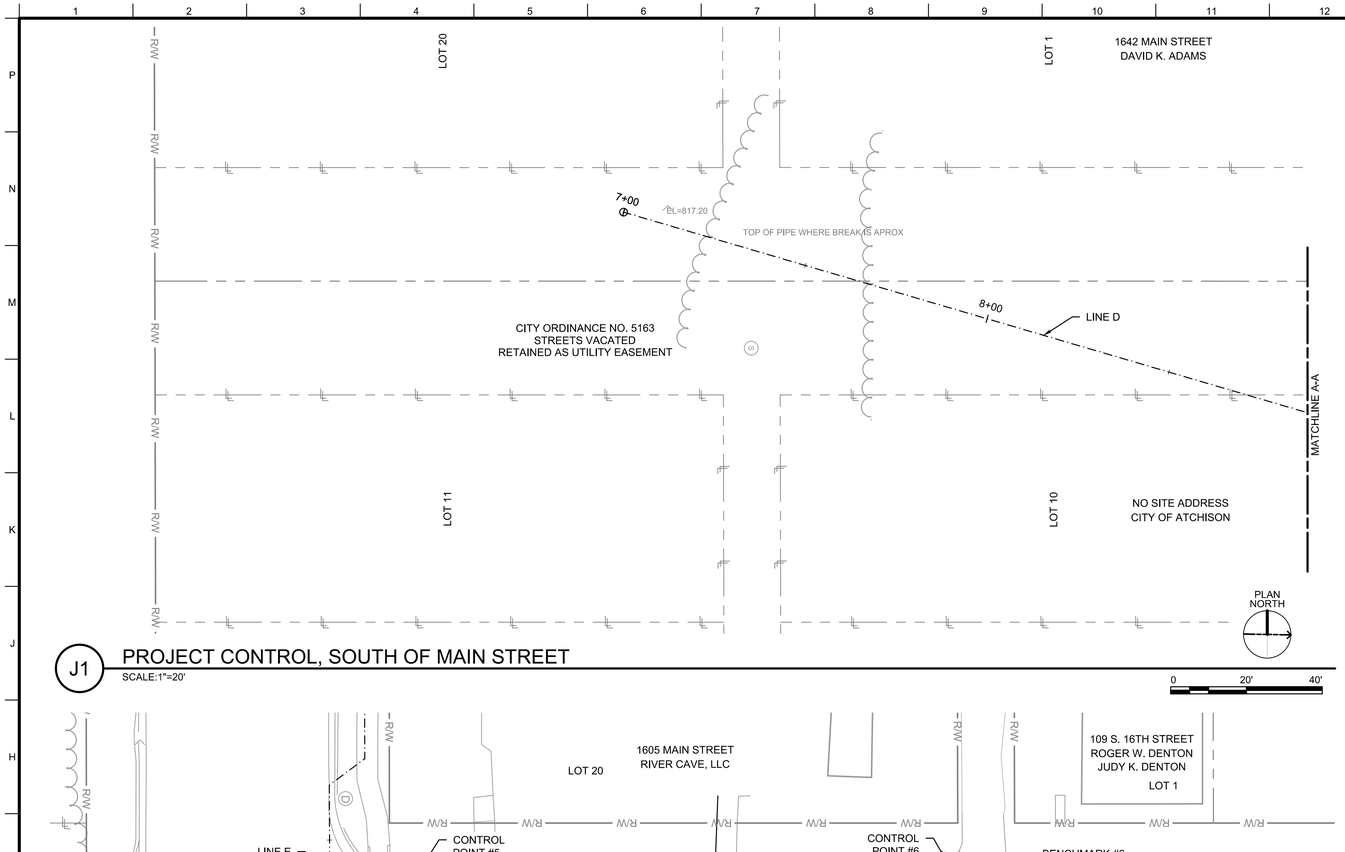
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
PROJECT CONTROL

SHEET ID
V-101

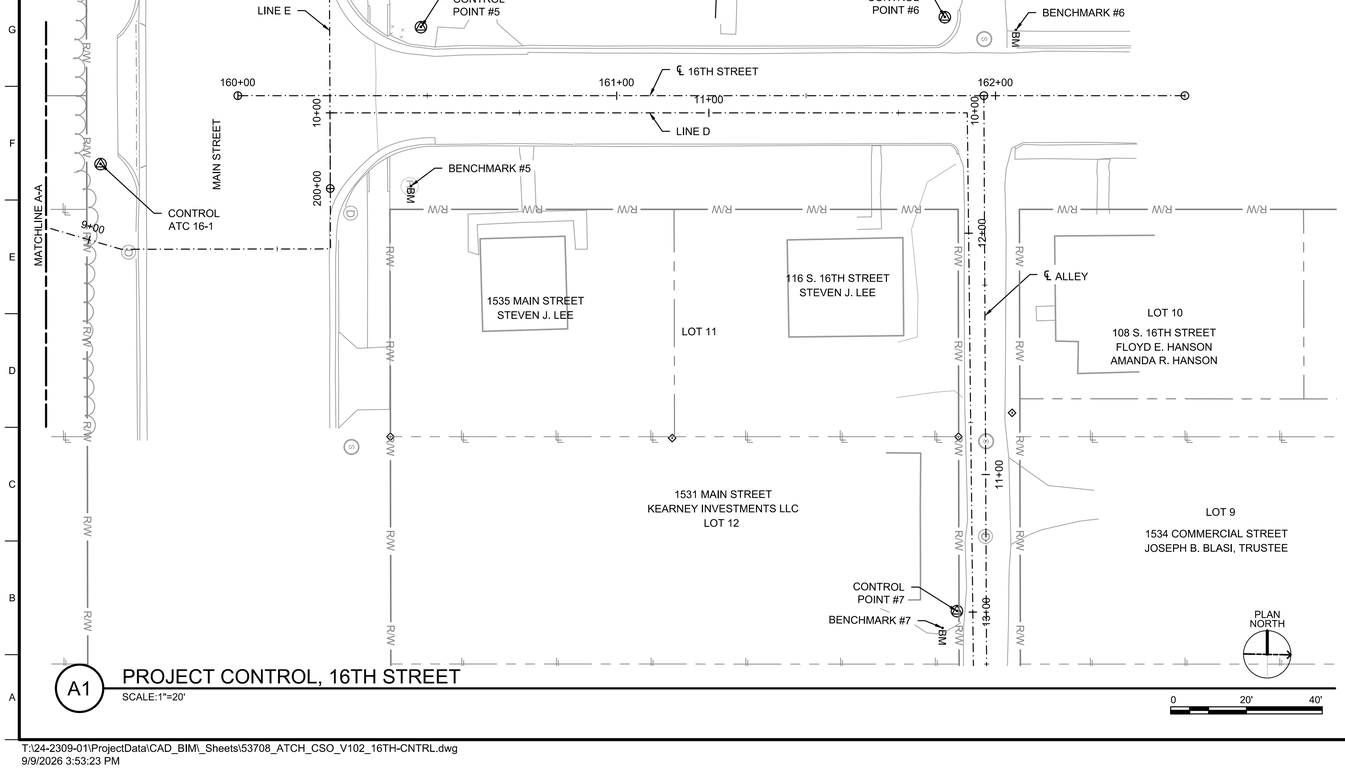
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_V101_15TH-CNTRL.dwg
9/9/2026 3:52:46 PM

READY TO ADVERTISE



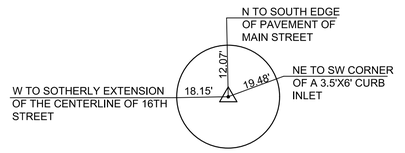
J1 PROJECT CONTROL, SOUTH OF MAIN STREET

SCALE:1"=20'



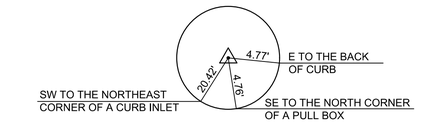
A1 PROJECT CONTROL, 16TH STREET

SCALE:1"=20'



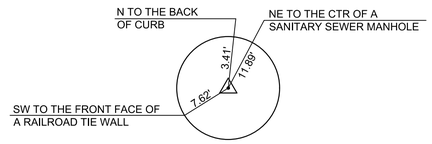
16-1: SET A 3-1/2" DOMED BRONZE DISK IN A 10"Ø , 36" DEEP CONCRETE BASE ON THE SOUTH SIDE OF MAIN STREET, EAST OF 16TH STREET

N=459,811.576 E=2,118,518.544
ELEV= 855.52'



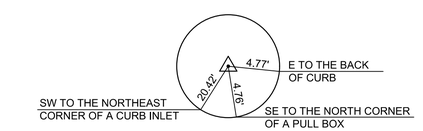
CP#5: SET A MAG NAIL AND SHINER IN EAST EDGE OF SIDEWALK AT THE NORTHWEST CORNER OF THE INTERSECTION OF MAIN STREET AND N. 16TH STREET

N=459,895.429 E=2,118,480.779



CP#6: SET A 1/2"REBAR W/ A CONTROL POINT CAP AT THE SOUTHWEST CORNER OF THE INTERSECTION OF N. 16TH STREET AND THE ALLEY NORTH OF MAIN STREET

N=460,033.644 E=2,118,475.448



CP#7: SET A 1/2"REBAR W/ A CONTROL POINT CAP ON THE SOUTH SIDE OF THE EASTERLY ALLEY WAY NORTH OF THE INTERSECTION OF MAIN STREET AND N. 16TH STREET

N=460,039.891 E=2,118,632.093

BENCHMARK #5:
SET TRIPLE PUNCH MARK IN THE NORTHWEST TOP FLANGE BOLT OF A FIRE HYDRANT AT THE NORTHEAST CORNER OF MAIN STREET AND N. 16TH STREET.
ELEVATION=859.09'

BENCHMARK #6:
SOUTHEAST CORNER OF STONE CAP ON SOUTH END OF BLOCK WALL AT THE NORTHWEST CORNER OF THE INTERSECTION OF N. 16TH STREET AND FIRST ALLEY NORTH OF MAIN STREET.
ELEVATION=874.70'

BENCHMARK #7:
SET STAINLESS STEEL BENCH TIE IN THE NORTH FACE OF A POWER POLE ON THE SOUTH SIDE OF THE FIRST ALLEY NORTH OF MAIN STREET, 135'+/- EAST OF 16TH STREET.
ELEVATION=867.10'

SHEET NOTES

1. THIS SURVEY IS A TOPOGRAPHIC SURVEY ONLY, NO RECORD PROPERTY RESEARCH WAS PERFORMED BY THE SURVEYOR OR DIRECTORATE OF PUBLIC WORKS (DPW)
2. THE UNDERGROUND UTILITY LINES AS SHOWN ARE AS LOCATED IN THE FIELD AND MAPPING PROVIDED BY THE DPW, SANITARY AND STORM LINES SHOWN REPRESENT A "STRAIGHT-LINE CONNECTION" BETWEEN OBSERVABLE STRUCTURES, PIPELINE ELEVATIONS ARE AS MEASURED FROM THE FLOWLINE OF THE STRUCTURES.

3.1 THE PROJECT SITE IS MIXED BETWEEN GENERAL BUSINESS USE AND SINGLE FAMILY RESIDENTIAL. THE PROPERTIES ALONG MAIN STREET AND SOUTH OF THE FIRST ALLEY TO THE NORTH ARE ZONED GENERAL BUSINESS USE. PROPERTIES NORTH OF THE ALLEY ARE ZONED SINGLE FAMILY RESIDENTIAL. THE PROJECT SITE IS ON 16TH STREET AND EXTENDS FROM MAIN STREET TO THE FIRST ALLEY TO THE NORTH, AND THEN BENDS TO THE EAST ON THE ALLEY FOR APPROXIMATELY 120 FEET.

3.2 HORIZONTAL DATA SHOWN ON THIS PROJECT IS BASED ON THE KANSAS STATE PLANE COORDINATE SYSTEM (NORTH ZONE). NAD83 (2011), US SURVEY FEET. ANY BEARINGS SHOWN ARE GRID NORTH.

VERTICAL DATA SHOWN ON THIS PROJECT IS NORTH AMERICAN VERTICAL DATUM, 1988 (NAVD 88) AND ARE SHOWN IN US SURVEY FEET.

BOTH HORIZONTAL AND VERTICAL DATA WERE ESTABLISHED USING NATIONAL GEODETIC SURVEY (NGS) POINT DESIGNATED AS W 281, PID KE0493, AS A BASE CONTROL POINT.

THE PUBLISHED VALUES FOR W 281 ARE:
N: 463,982.13' E: 2,116,796.88' ELEV: 998.64'

3.3 ON-SITE CONTROL FOR THIS PROJECT IS BASED ON NGS POINT W 281 BEING THE ORIGINATING BASE POINT. PROJECT CONTROL WAS ESTABLISHED BY OCCUPYING NGS W 281 WITH A GPS BASE UNIT AND USING GPS ROVER UNITS TO MEASURE TO SEMI-PERMANENT CONTROL POINTS ON-SITE FOR SURVEY PURPOSES AND CONTRACTORS USE. THE CONTROL WAS MEASURED FOUR INDIVIDUAL TIMES AND THE VALUES AVERAGED TO DETERMINE FINAL VALUES. A CLOSED LEVEL LOOP WAS PERFORMED THROUGH ALL SEMI-PERMANENT CONTROL POINTS AND BENCHMARKS. TOPOGRAPHIC SURVEY WORK WAS PERFORMED USING REAL TIME KINEMATIC (RTK) SURVEY METHODS WITH A BASE AND ROVER SETUP AND THE USE OF ROBOTIC TOTAL STATIONS. THE SEMI-PERMANENT CONTROL POINTS SET ARE 1/2" BY 18" REBARS WITH CONTROL POINT CAPS IN GRASSY AREAS OR 1-1/2"MAG NAILS WITH SHINERS IN PAVED AREAS AND ARE IDENTIFIED AS CP#5 THRU CP#7. SEE PLAN SHEETS FOR LOCATION AND TYPES OF SEMI-PERMANENT MARKERS SET. STANDARD TYPE C 3-1/2" BRONZE USACE SURVEY MARKER IS IDENTIFIED AS ATC 16-1. THIS SURVEY WAS PERFORMED BETWEEN NOVEMBER 25, 2024 AND FEBRUARY 4, 2025, BY AFFINIS CORP.

A FUTURE SUBMITTAL.

4. THIS SURVEY WAS PERFORMED BETWEEN NOVEMBER 25, 2024 AND FEBRUARY 4, 2025, BY AFFINIS CORP.

ATTENTION: JONATHAN AUBREY MEYER, KS PLS #1579
8900 INDIAN CREEK PARKWAY, SUITE 450
OVERLAND PARK, KS 66210
TELEPHONE (913)-239-1100

LEGEND

- PROPERTY PIN
- CONTROL POINT
- •BM BENCH MARK
- R/W EXISTING RIGHT-OF-WAY
- EXISTING PROPERTY LINE
- EXISTING LOT LINE
- EXISTING PROPERTY/LOT LINE
- CENTER LINE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D024F1019
CONTRACT NO.:
W912DQ19D1012
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

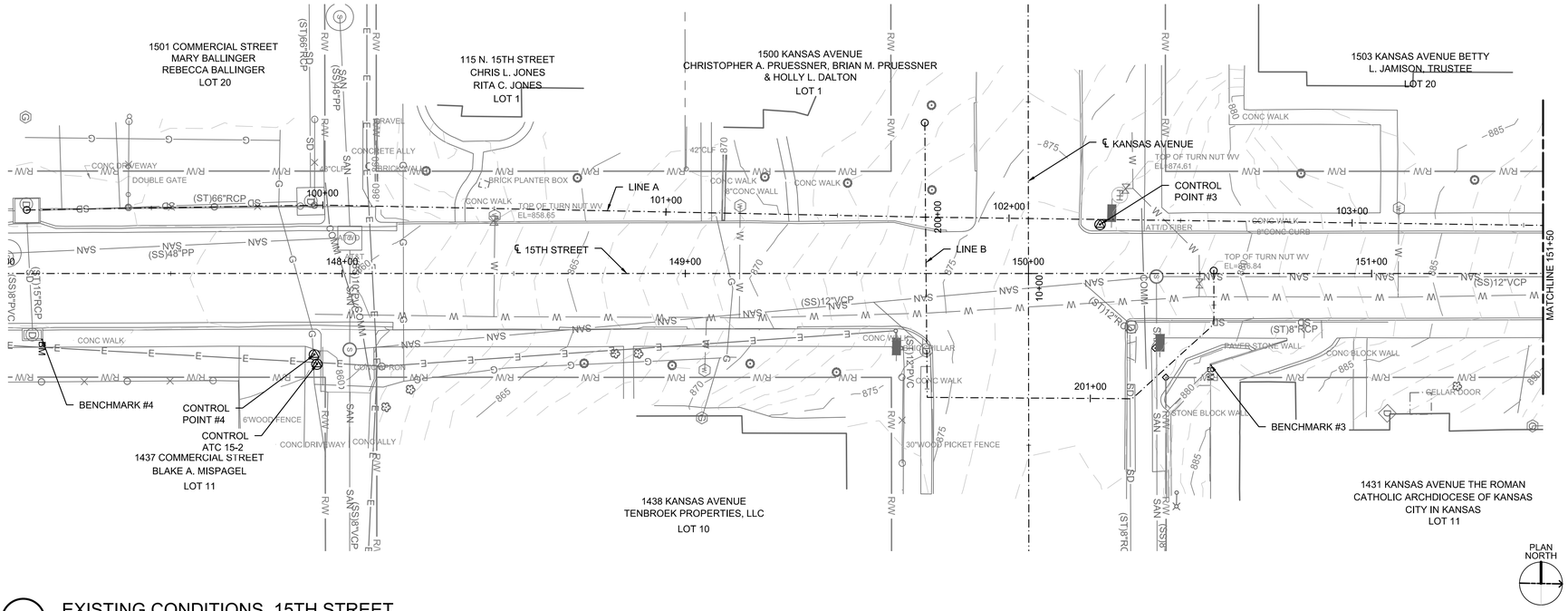
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
PROJECT CONTROL

SHEET ID
V-102

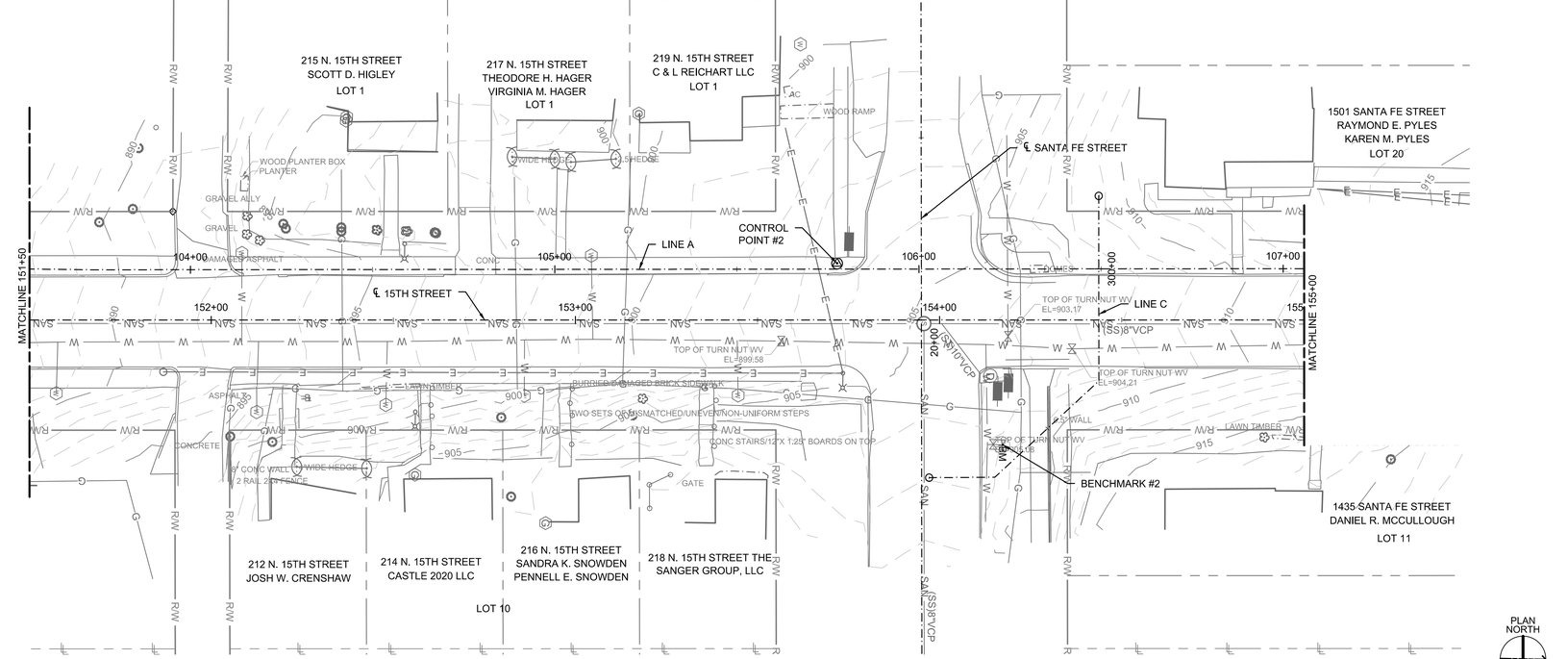
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_V102_16TH-CNTRL.dwg
9/9/2026 3:53:23 PM

READY TO ADVERTISE



H1 EXISTING CONDITIONS, 15TH STREET

SCALE:1"=20'



A1 EXISTING CONDITIONS, 15TH STREET

SCALE:1"=20'

SHEET NOTES

1. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

LEGEND

- EXISTING SANITARY SEWER PIPE
- EXISTING STORM SEWER PIPE
- EXISTING BURIED COMMUNICATION LINE
- EXISTING OVERHEAD ELECTRICAL LINE
- EXISTING WATERLINE
- EXISTING GAS
- MAJOR CONTOUR
- MINOR CONTOUR
- EXISTING EDGE OF PAVEMENT
- EXISTING CHAIN LINK FENCE
- EXISTING SANITARY SEWER STRUCTURE
- EXISTING SANITARY SEWER CLEANOUT
- EXISTING STORM SEWER STRUCTURE
- EXISTING COMMUNICATION POLE
- EXISTING LIGHT POLE
- EXISTING POWER METER
- EXISTING WATER METER
- EXISTING FIRE HYDRANT
- EXISTING WATER VALVE
- EXISTING GAS METER
- EXISTING STREET SIGN
- EXISTING MAILBOX
- EXISTING BORE HOLE
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- EXISTING SHRUBLINE
- PROPERTY PIN
- CONTROL POINT
- BENCH MARK
- EXISTING RIGHT-OF-WAY
- EXISTING PROPERTY LINE
- EXISTING LOT LINE
- EXISTING PROPERTY/LOT LINE
- CENTER LINE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

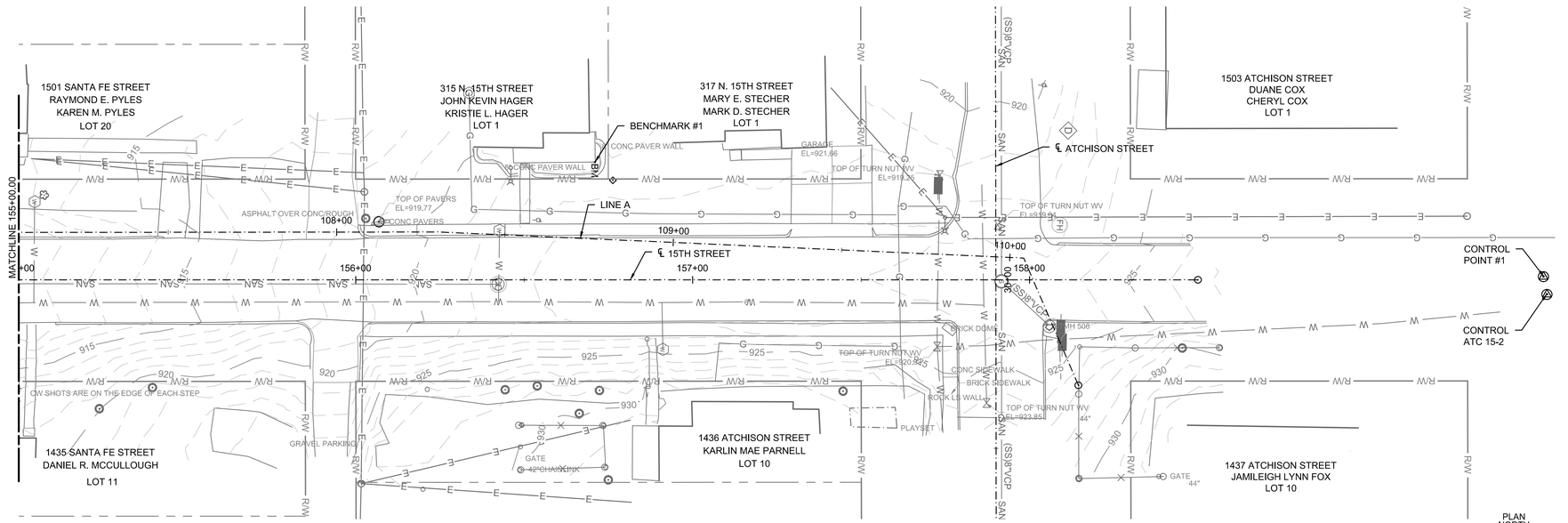
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
EXISTING CONDITIONS

SHEET ID
V-103

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_V103-V104_15TH-EXIST.dwg
9/9/2026 3:54:05 PM

READY TO ADVERTISE



H1 EXISTING CONDITIONS, 15TH STREET

SCALE:1"=20'

SHEET NOTES

1. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

LEGEND

- (SS)8"VCP / SAN — EXISTING SANITARY SEWER PIPE
- (ST)15"HDPE / SD — EXISTING STORM SEWER PIPE
- COMM — EXISTING BURIED COMMUNICATION LINE
- E — EXISTING OVERHEAD ELECTRICAL LINE
- W — EXISTING WATERLINE
- G — EXISTING GAS
- MAJOR CONTOUR
- MINOR CONTOUR
- EXISTING EDGE OF PAVEMENT
- EXISTING CHAIN LINK FENCE
- S — EXISTING SANITARY SEWER STRUCTURE
- CO — EXISTING SANITARY SEWER CLEANOUT
- D — EXISTING STORM SEWER STRUCTURE
- EXISTING COMMUNICATION POLE
- EXISTING LIGHT POLE
- M — EXISTING POWER METER
- W — EXISTING WATER METER
- FH — EXISTING FIRE HYDRANT
- EXISTING WATER VALVE
- G — EXISTING GAS METER
- EXISTING STREET SIGN
- EXISTING MAILBOX
- EXISTING BORE HOLE
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- EXISTING SHRUBLINE
- PROPERTY PIN
- CONTROL POINT
- •BM — BENCH MARK
- R/W — EXISTING RIGHT-OF-WAY
- EXISTING PROPERTY LINE
- ℄ — EXISTING LOT LINE
- ℄ ℄ — EXISTING PROPERTY/LOT LINE
- CENTER LINE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D224F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

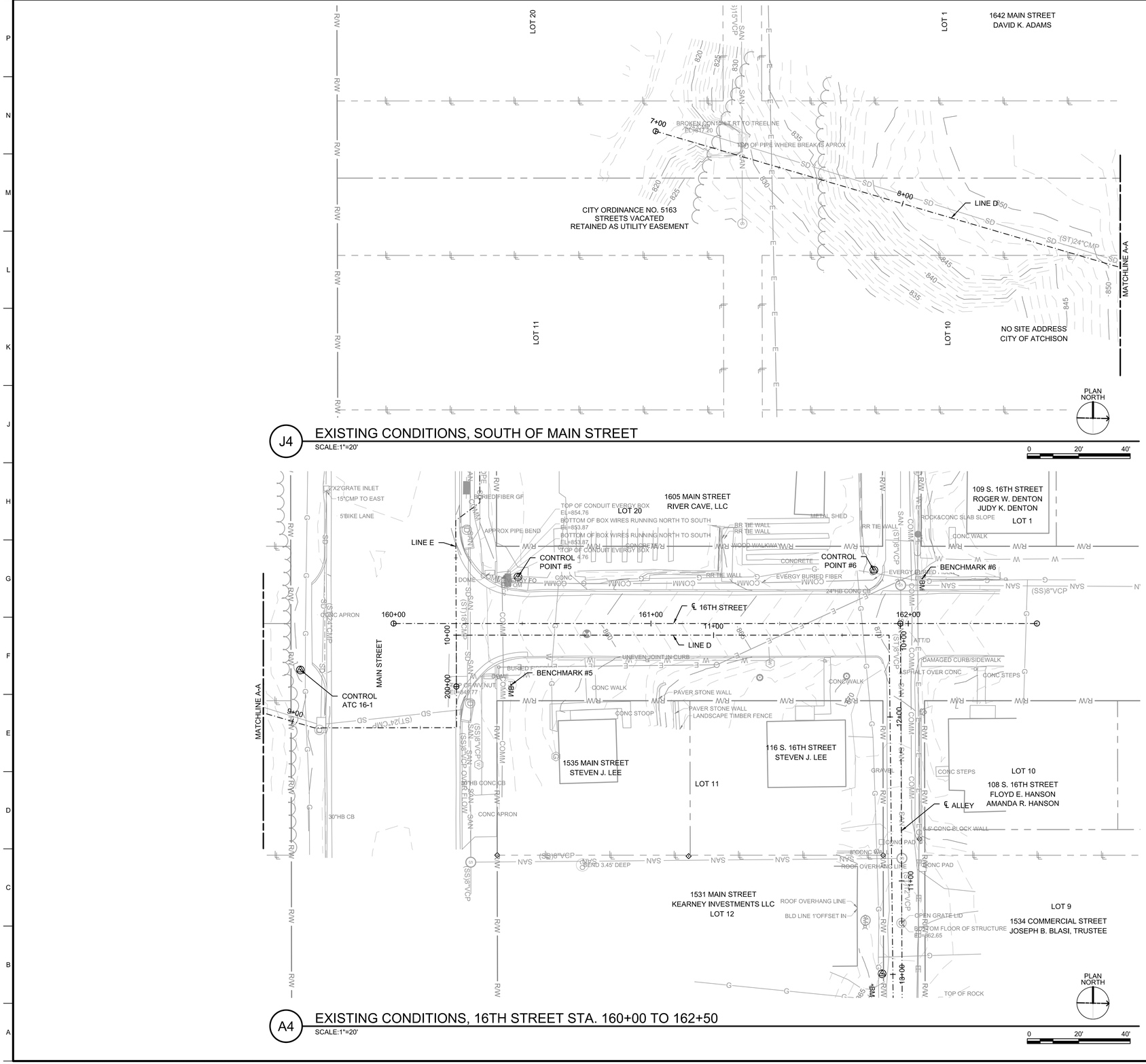
15TH STREET
EXISTING CONDITIONS

SHEET ID

V-104

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_V103-V104_15TH-EXIST.dwg
9/9/2026 3:54:14 PM

READY TO ADVERTISE



J4 EXISTING CONDITIONS, SOUTH OF MAIN STREET

SCALE:1"=20'

A4 EXISTING CONDITIONS, 16TH STREET STA. 160+00 TO 162+50

SCALE:1"=20'

SHEET NOTES

1. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

LEGEND

- EXISTING SANITARY SEWER PIPE
- EXISTING STORM SEWER PIPE
- EXISTING BURIED COMMUNICATION LINE
- EXISTING OVERHEAD ELECTRICAL LINE
- EXISTING WATERLINE
- EXISTING GAS
- MAJOR CONTOUR
- MINOR CONTOUR
- EXISTING EDGE OF PAVEMENT
- EXISTING CHAIN LINK FENCE
- EXISTING SANITARY SEWER STRUCTURE
- EXISTING SANITARY SEWER CLEANOUT
- EXISTING STORM SEWER STRUCTURE
- EXISTING COMMUNICATION POLE
- EXISTING LIGHT POLE
- EXISTING POWER METER
- EXISTING WATER METER
- EXISTING FIRE HYDRANT
- EXISTING WATER VALVE
- EXISTING GAS METER
- EXISTING STREET SIGN
- EXISTING MAILBOX
- EXISTING BORE HOLE
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- EXISTING SHRUBLINE
- PROPERTY PIN
- CONTROL POINT
- BENCH MARK
- EXISTING RIGHT-OF-WAY
- EXISTING PROPERTY LINE
- EXISTING LOT LINE
- EXISTING PROPERTY/LOT LINE
- CENTER LINE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D224F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

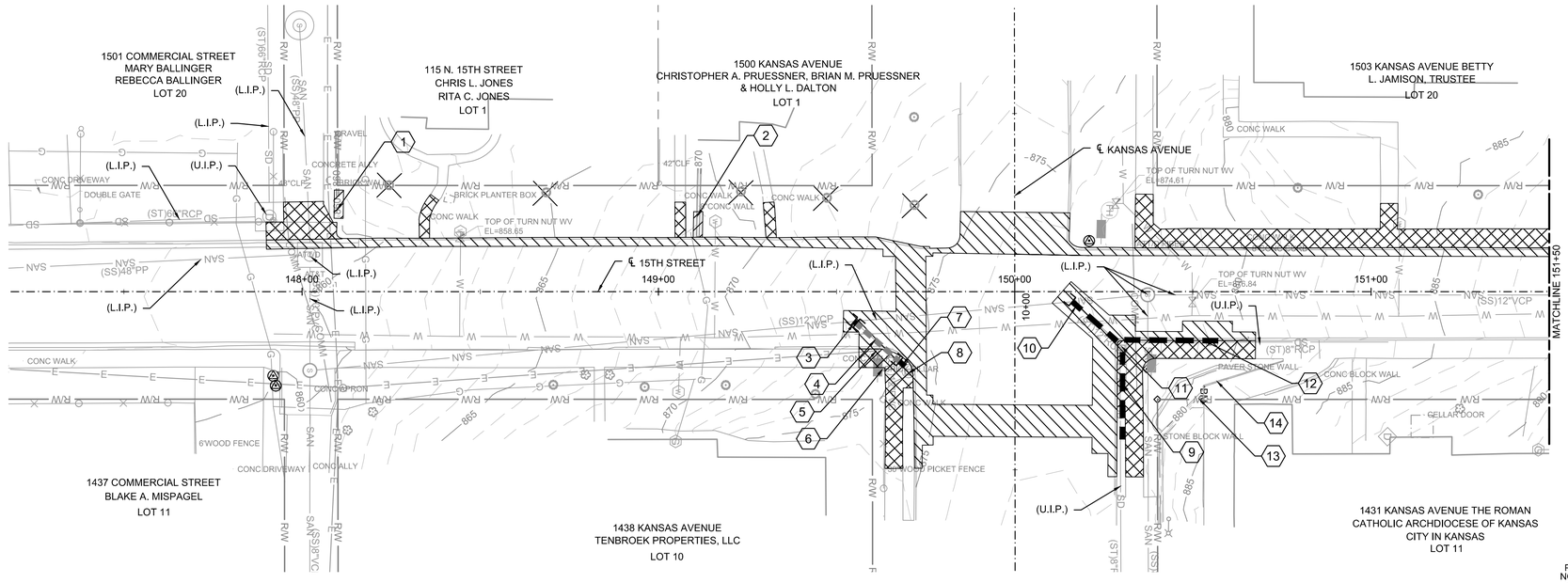
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
EXISTING CONDITIONS

SHEET ID
V-105

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_V105_16TH-EXIST.dwg
9/9/2026 3:54:52 PM

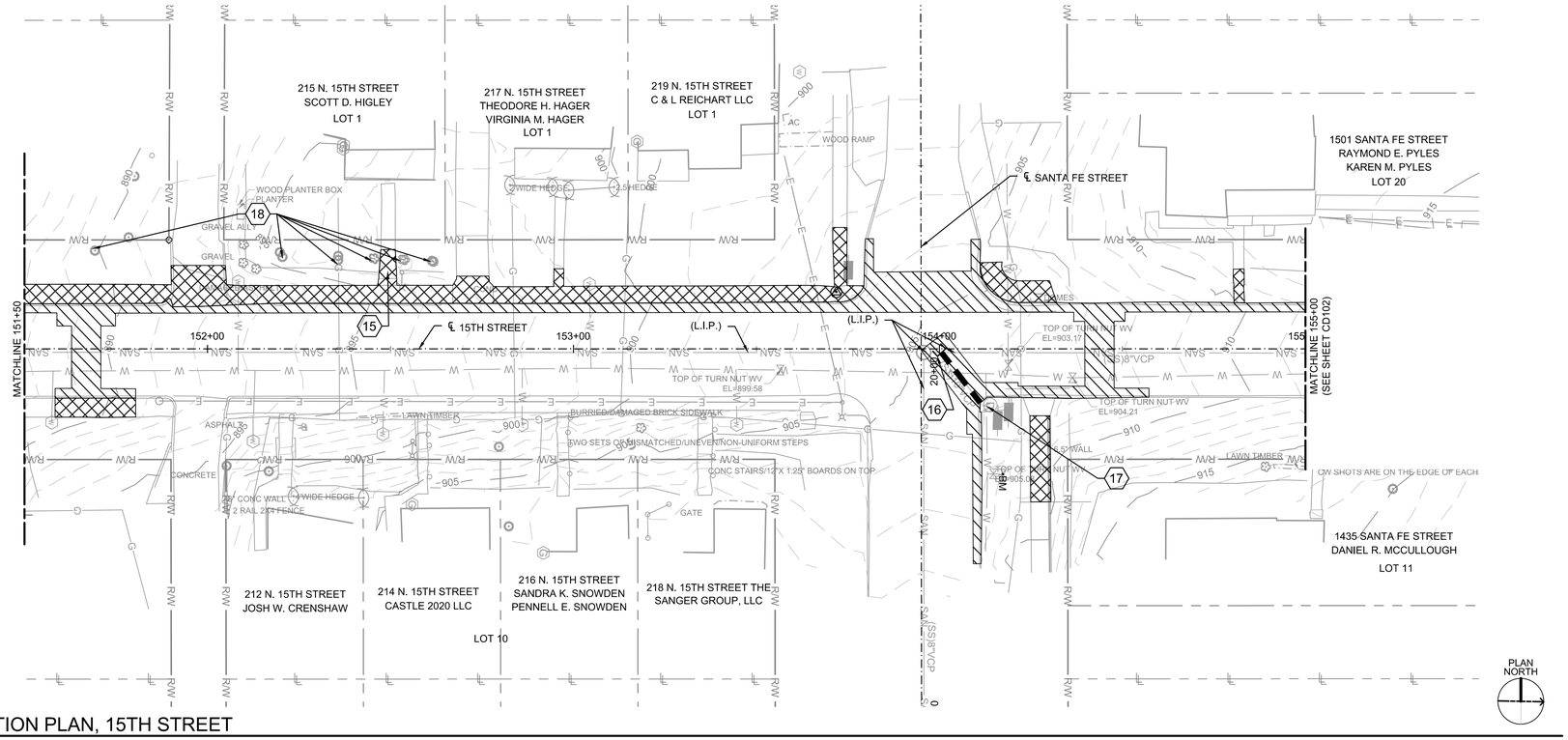
READY TO ADVERTISE



H1 DEMOLITION PLAN, 15TH STREET

SCALE:1"=20'

0 20' 40'



A1 DEMOLITION PLAN, 15TH STREET

SCALE:1"=20'

0 20' 40'

SHEET NOTES

1. SEE SHEET C-506 FOR GROUT END CAP AND SHEET C-522 FOR MANHOLE AND TEE CAP-OFF DETAILS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

SHEET KEYNOTES

1. SAWCUT AND REMOVE 5 L.F. BRICK WALL. CONSTRUCT GROUT END CAP.
2. SAWCUT AND REMOVE 7 L.F. CONCRETE WALL. CONSTRUCT GROUT END CAP.
3. TEE CAP-OFF AND REMOVE APPROXIMATELY 3 L.F. EXISTING 12" PVC (NE).
4. CAP-OFF AND FLOWFILL APPROXIMATELY 16 L.F. EXISTING 12" PVC (NE).
5. PROTECT EXISTING LIGHT POLE DURING CONSTRUCTION (NO DIRECT PAY).
6. PROTECT EXISTING BRICK PILLAR AND WOOD FENCES DURING CONSTRUCTION (NO DIRECT PAY).
7. REMOVE APPROXIMATELY 3 L.F. EXISTING 12" RCP (SW)
8. REMOVE EXISTING STRUCTURE
9. REMOVE 33.5 L.F. EXISTING 8" RCP (W)
10. TEE CAP-OFF AND REMOVE 16.7 L.F. EXISTING 12" RCP (NE)
11. REMOVE EXISTING STRUCTURE
12. REMOVE APPROXIMATELY 25 L.F. EXISTING 8" RCP (S)
13. PROTECT EXISTING STATUE DURING CONSTRUCTION (NO DIRECT PAY).
14. PROTECT EXISTING WALL DURING CONSTRUCTION (NO DIRECT PAY).
15. REMOVE EXISTING CONCRETE STEPS AND SIDEWALK. SALVAGE TRELLIS AND HANDRAIL TO CITY. TO BE REPLACED BY CITY.
16. TEE CAP-OFF AND REMOVE APPROXIMATELY 25L.F. EXISTING 10" VCP.
17. REMOVE EXISTING STRUCTURE.
18. TRIM TREES AS NEEDED FOR CONSTRUCTION. COORDINATE WITH CITY AND HOME OWNERS. (NO DIRECT PAY)

LEGEND

- PAVEMENT & CURB REMOVAL
- SIDEWALK, DRIVEWAY, STEPS & INTEGRAL SIDEWALK RETAINING WALL REMOVAL
- EXISTING PIPE REMOVAL
- EXISTING PIPE ABANDONED
- TREE REMOVAL
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- TEE AND MANHOLE CAP-OFF

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D2024F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

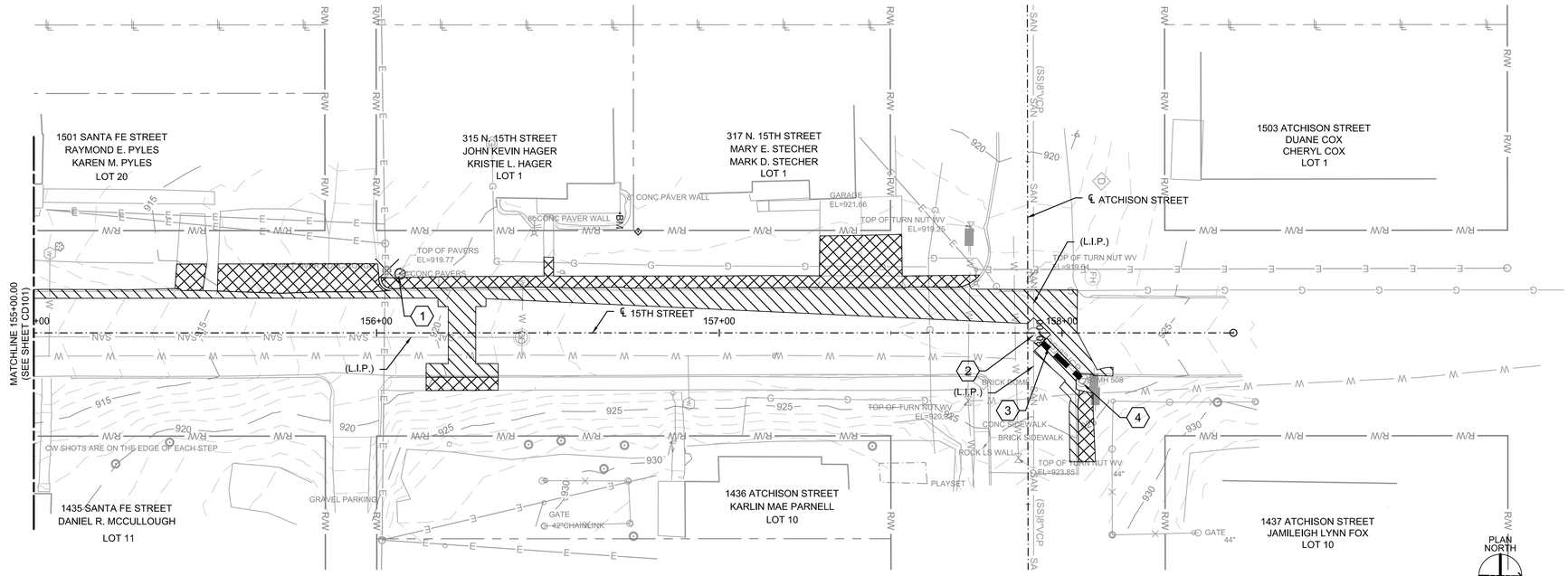
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
DEMOLITION PLAN

SHEET ID
CD101

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CD101-CD102_15TH-DEMO.dwg
9/9/2026 3:55:43 PM

READY TO ADVERTISE



H1 DEMOLITION PLAN, 15TH STREET

SCALE:1"=20'

SHEET NOTES

1. SEE SHEET C-522 FOR MANHOLE AND TEE CAP-OFF DETAILS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

SHEET KEYNOTES

1. REMOVE EXISTING BRICK PAVERS (NO DIRECT PAY).
2. PROTECT EXISTING SSMH (U.I.P.)
3. MANHOLE CAP-OFF AND REMOVE APPROXIMATELY 15 L.F. EXISTING 8" VCP (NE)
4. REMOVE EXISTING STRUCTURE

LEGEND

- PAVEMENT & CURB REMOVAL
- SIDEWALK, DRIVEWAY, STEPS & INTEGRAL SIDEWALK RETAINING WALL REMOVAL
- EXISTING PIPE REMOVAL
- TREE REMOVAL
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- TEE AND MANHOLE CAP-OFF

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROWL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
SIZE:	ANSI D
ISSUE DATE:	23 FEBRUARY 2026
SOLICITATION NO.:	
TASK ORDER # W912D24F1019	
CONTRACT NO.:	W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

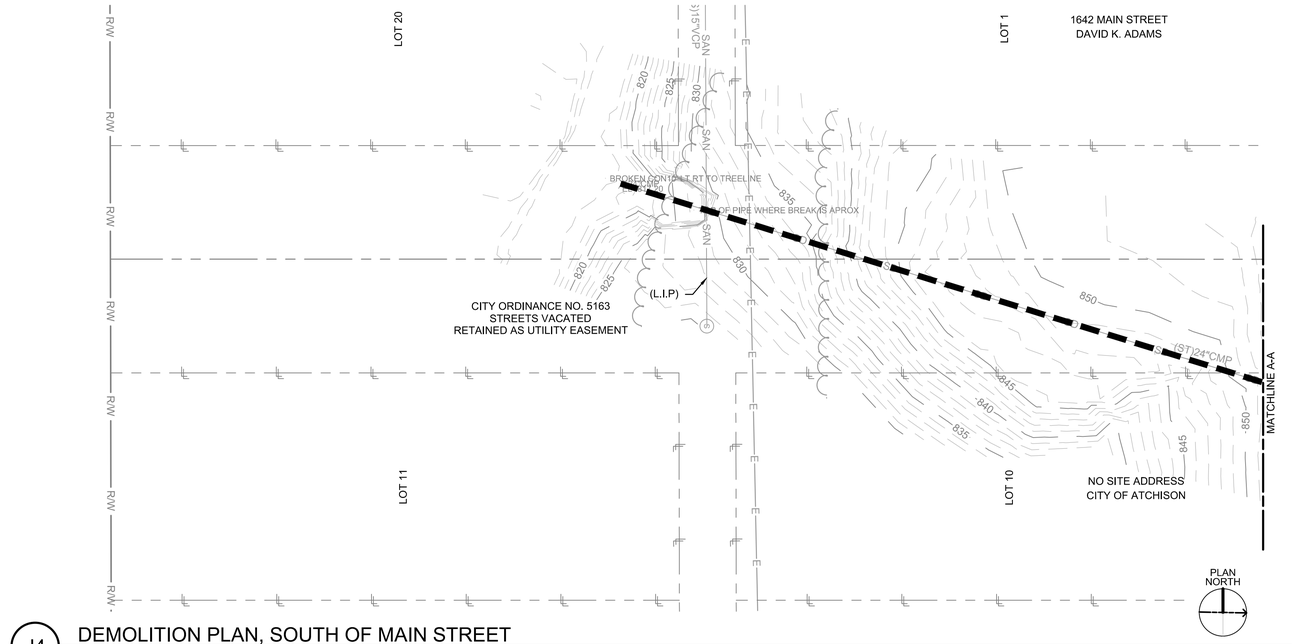
15TH STREET
DEMOLITION PLAN

SHEET ID

CD102

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CD101-CD102_15TH-DEMO.dwg
9/9/2026 3:55:56 PM

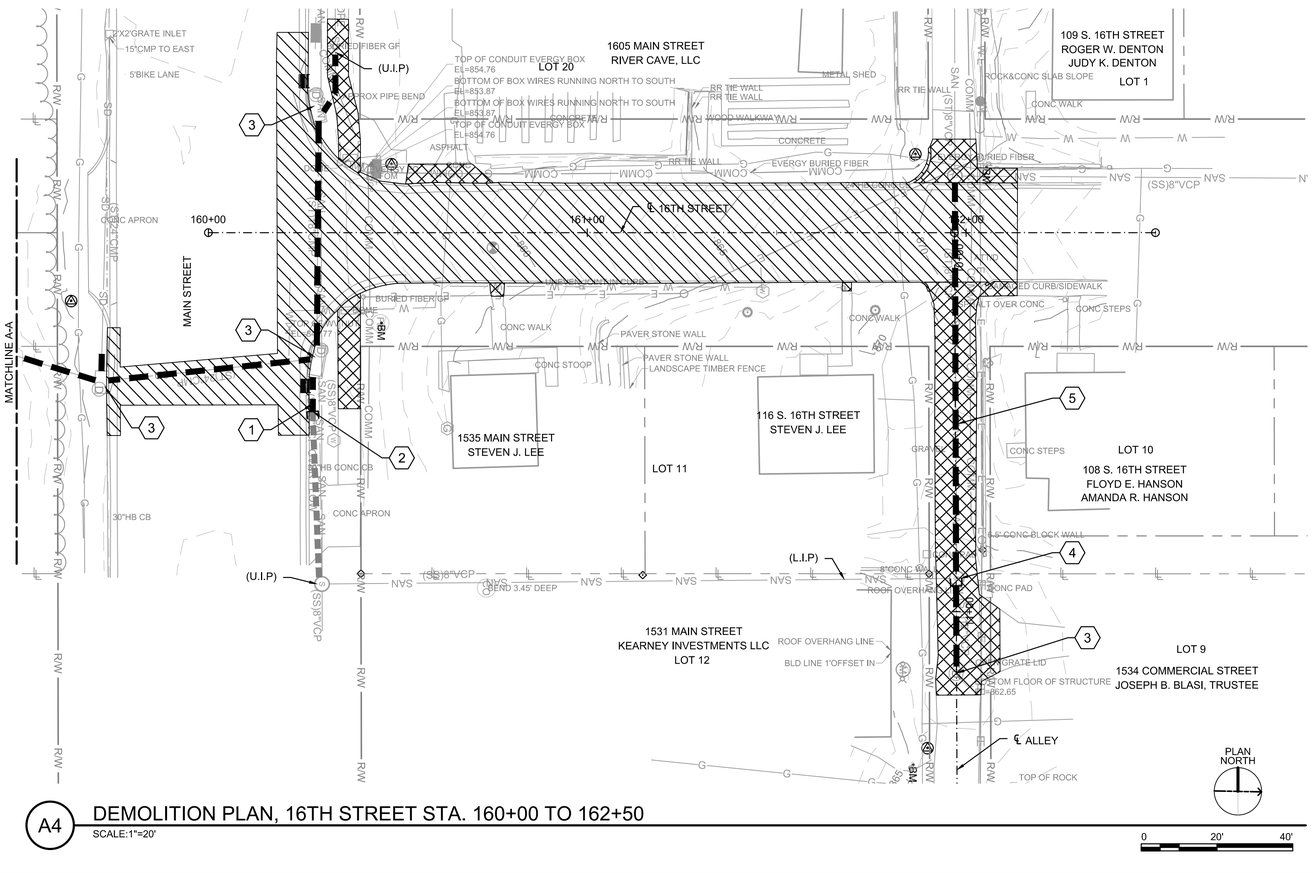
READY TO ADVERTISE



J4 DEMOLITION PLAN, SOUTH OF MAIN STREET

SCALE:1"=20'

0 20' 40'



A4 DEMOLITION PLAN, 16TH STREET STA. 160+00 TO 162+50

SCALE:1"=20'

0 20' 40'

SHEET NOTES

1. SEE SHEET C-522 FOR MANHOLE AND TEE CAP-OFF DETAILS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.

SHEET KEYNOTES

1. REMOVE APPROXIMATELY 10 L.F. EXISTING 8" VCP (E) AT CURB INLET
2. CAP-OFF AND FLOWFILL APPROXIMATELY 48 L.F. EXISTING 8" VCP TO SSMH
3. REMOVE EXISTING STRUCTURE
4. PROTECT EXISTING SSMH (U.I.P.) CAP-OFF AND REMOVE APPROXIMATELY 21 L.F. EXISTING 12" PVC (NE)
5. REMOVE APPROXIMATELY 103 L.F. EXISTING 8" VCP

LEGEND

- PAVEMENT & CURB REMOVAL
- SIDEWALK & DRIVEWAY REMOVAL
- EXISTING PIPE REMOVAL
- EXISTING PIPE ABANDONED
- EXISTING TREE
- EXISTING SMALL TREE OR BUSH
- TEE AND MANHOLE CAP-OFF

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

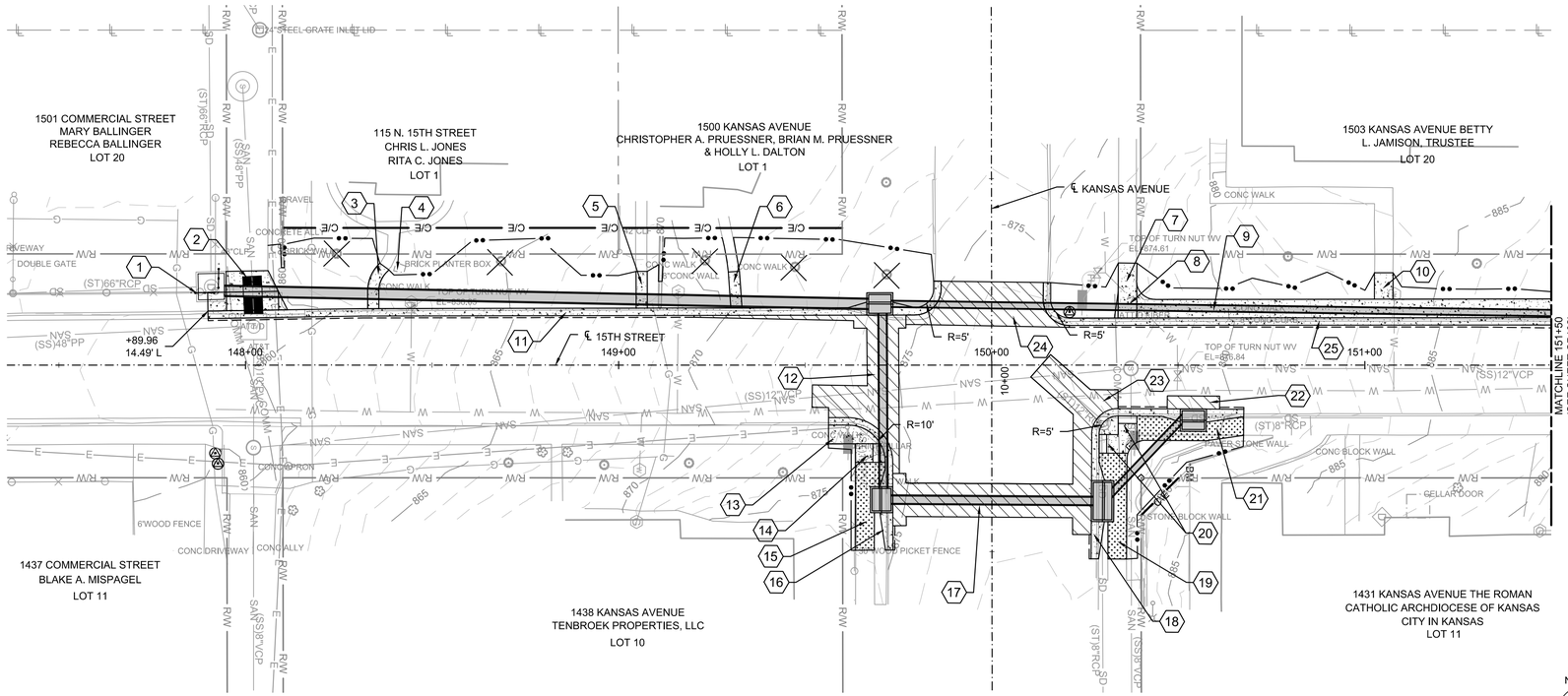
16TH STREET
DEMOLITION PLAN

SHEET ID

CD103

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CD103_16TH-DEMO.dwg
9/9/2026 3:56:38 PM

READY TO ADVERTISE



D1 CIVIL PLAN, 15TH STREET STA. 147+90 TO 151+50

SCALE:1"=20'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.
6. SEE SHEET CP106 FOR INTERSECTION DETAILS.

SHEET KEYNOTES

1. REMOVE AND REPLACE 15 L.F. 48" BLACK CHAIN LINK FENCE IN KIND
2. CONSTRUCT ALLEY DRIVE
   SEE DETAIL A1 ON SHEET C-505
3. CONSTRUCT 12 L.F. SIDEWALK (3' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
4. PROTECT BRICK
5. CONSTRUCT 10 L.F. SIDEWALK (3' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
6. CONSTRUCT 10 L.F. SIDEWALK (3' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
7. CONSTRUCT SIDEWALK (5' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
8. CONSTRUCT SIDEWALK RAMP
   SEE SHEET CP110 AND DETAIL A1 ON SHEET C-503
9. CONSTRUCT SIDEWALK (5' WIDTH)
   SEE SECTION L1 ON SHEET C-501
   SEE DETAIL A1 ON SHEET C-503
10. CONSTRUCT 10 L.F. SIDEWALK (5' WIDTH)
    SEE DETAIL A1 ON SHEET C-503
11. CONSTRUCT TYPE I CURB AND GUTTER
    SEE L1 ON SHEET C-501
    SEE DETAIL A1 ON SHEET C-502
12. CONSTRUCT TYPE I STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
13. CONSTRUCT SIDEWALK (5' WIDTH)
    SEE DETAIL A1 ON SHEET C-503
14. CONSTRUCT SIDEWALK RAMP
    SEE SHEET CP110 AND DETAIL A1 ON SHEET C-503
15. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH)
    SEE DETAIL A1 ON SHEET C-511
16. CONSTRUCT TYPE I CURB AND GUTTER
    SEE DETAIL A1 ON SHEET C-502
17. CONSTRUCT TYPE I STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
18. CONSTRUCT TYPE I CURB AND GUTTER
    SEE DETAIL A1 ON SHEET C-502
19. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH)
    SEE DETAIL A1 ON SHEET C-511
20. CONSTRUCT (2) SIDEWALK RAMPS
    SEE SHEET CP110 AND DETAIL A1 ON SHEET C-503
21. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH)
    SEE DETAIL A1 ON SHEET C-511
22. CONSTRUCT TYPE I STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
23. CONSTRUCT TYPE I STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
24. CONSTRUCT TYPE I STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
25. CONSTRUCT TYPE I CURB AND GUTTER
    SEE L1 ON SHEET C-501
    SEE DETAIL A1 ON SHEET C-502

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E CONSTRUCTION EASEMENT
- 6" REINFORCED SIDEWALK

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

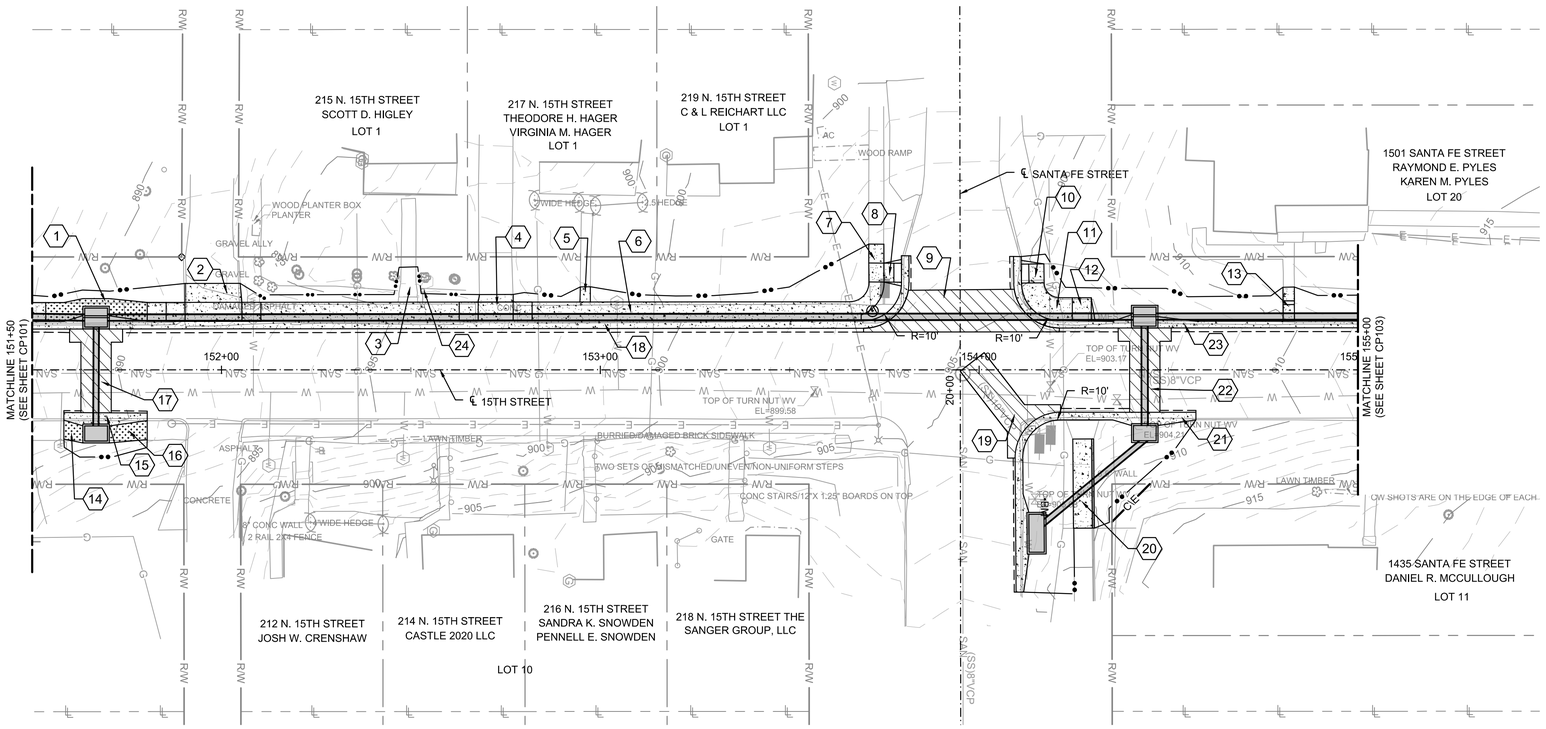
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
CIVIL PLAN

SHEET ID
CP101

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP101-CP103_15TH.dwg
9/9/2026 3:57:55 PM

READY TO ADVERTISE



D1 CIVIL PLAN, 15TH STREET STA. 151+50 TO 155+00

SCALE:1"=20'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.
6. SEE SHEET CP107 FOR INTERSECTION DETAILS.

SHEET KEYNOTES

1. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH) SEE DETAIL A1 ON SHEET C-511
2. CONSTRUCT ALLEY DRIVE SEE DETAIL A1 ON SHEET C-505
3. REPLACEMENT BY OTHERS (CITY).
4. CONSTRUCT RESIDENTIAL DRIVE SEE DETAIL A1 ON SHEET C-504
5. CONSTRUCT SIDEWALK (3' WIDTH) SEE DETAIL A1 ON SHEET C-503
6. CONSTRUCT SIDEWALK (5' WIDTH) SEE L1 ON SHEET C-501 SEE DETAIL A1 ON SHEET C-503
7. CONSTRUCT SIDEWALK (4' WIDTH) SEE DETAIL A1 ON SHEET C-503
8. CONSTRUCT SIDEWALK RAMP SEE SHEET CP111 AND DETAIL A1 ON SHEET C-503
9. CONSTRUCT TYPE I STREET REPAIR SEE DETAIL J8 ON SHEET C-508
10. CONSTRUCT SIDEWALK RAMP SEE SHEET CP111 AND DETAIL A1 ON SHEET C-503
11. CONSTRUCT SIDEWALK (6' WIDTH) SEE L1 ON SHEET C-501 SEE DETAIL A1 ON SHEET C-503
12. CONSTRUCT SIDEWALK RAMP SEE SHEET CP111 AND DETAIL A1 ON SHEET C-503
13. CONSTRUCT STEP & SIDEWALK (3' WIDTH) SEE DETAIL A1 ON SHEET C-507
14. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH) SEE DETAIL A1 ON SHEET C-511
15. CONSTRUCT TYPE I CURB AND GUTTER SEE DETAIL A1 ON SHEET C-502
16. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH) SEE DETAIL A1 ON SHEET C-511
17. CONSTRUCT TYPE I STREET REPAIR SEE DETAIL J8 ON SHEET C-508
18. CONSTRUCT TYPE I CURB AND GUTTER SEE L1 ON SHEET C-501 SEE DETAIL A1 ON SHEET C-502
19. CONSTRUCT TYPE I STREET REPAIR SEE DETAIL J8 ON SHEET C-508
20. CONSTRUCT INTEGRAL SIDEWALK RETAINING WALL SEE DETAIL A1 ON SHEET C-506
21. CONSTRUCT TYPE I CURB AND GUTTER SEE DETAIL A1 ON SHEET C-502
22. CONSTRUCT TYPE I STREET REPAIR SEE DETAIL J8 ON SHEET C-508
23. CONSTRUCT TYPE I CURB AND GUTTER SEE L1 ON SHEET C-501 SEE DETAIL A1 ON SHEET C-502
24. EXISTING LIGHT POLE (D.N.D.)

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E — CONSTRUCTION EASEMENT
- 6" REINFORCED SIDEWALK

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

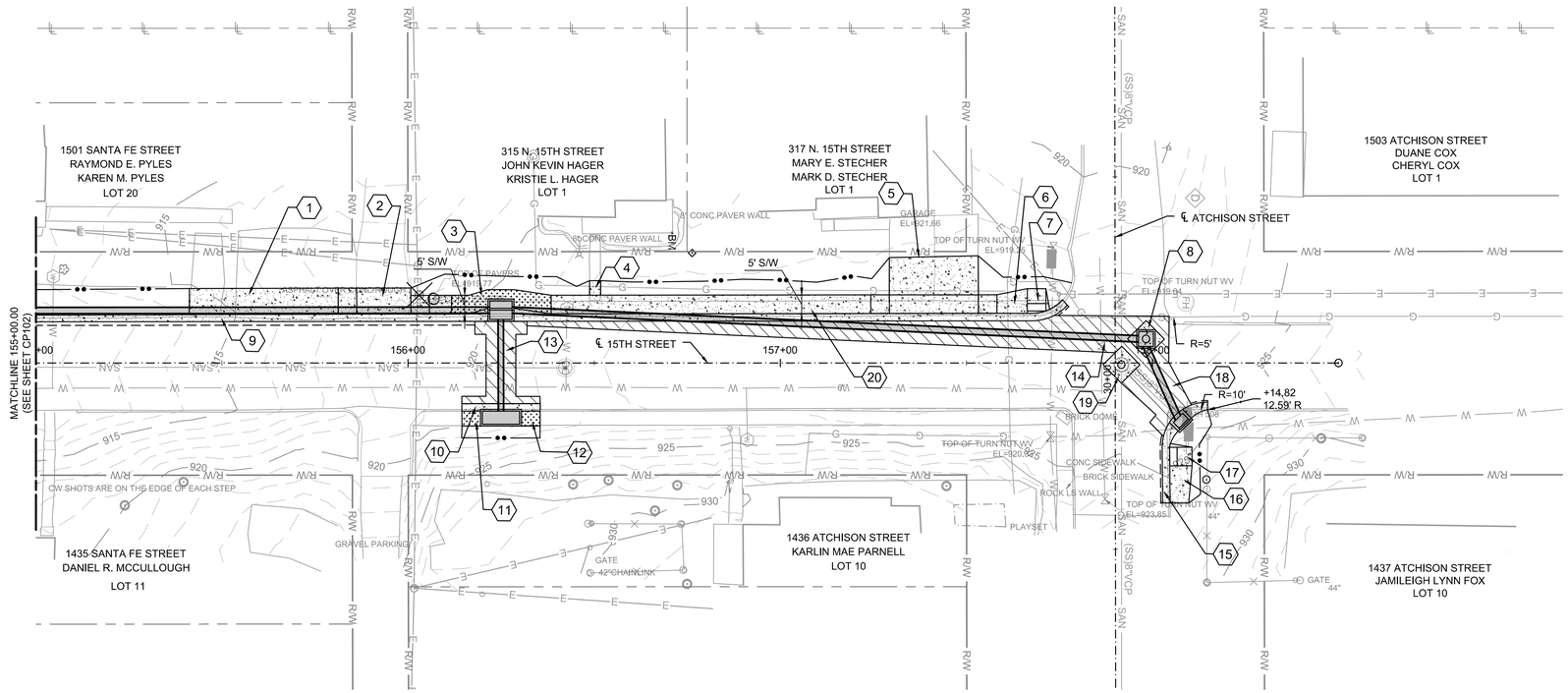
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
CIVIL PLAN

SHEET ID
CP102

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP101-CP103_15TH.dwg
9/9/2026 3:58:22 PM

READY TO ADVERTISE



D1 CIVIL PLAN, 15TH STREET STA 155+00 TO 158+37

SCALE:1"=20'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.
6. SEE SHEET CP108 FOR INTERSECTION DETAILS.

SHEET KEYNOTES

1. CONSTRUCT RESIDENTIAL DRIVE
   SEE DETAIL A1 ON SHEET C-504
2. CONSTRUCT ALLEY DRIVE
   SEE DETAIL A1 ON SHEET C-505
3. CONSTRUCT 6" REINFORCED SIDEWALK (5' WIDTH)
   SEE DETAIL A1 ON SHEET C-511
4. CONSTRUCT SIDEWALK (3' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
5. CONSTRUCT RESIDENTIAL DRIVE
   SEE DETAIL A1 ON SHEET C-504
6. CONSTRUCT SIDEWALK (5' TO 6.7' TRANSITION)
   SEE L1 ON SHEET C-501
   SEE DETAIL A1 ON SHEET C-503
7. CONSTRUCT SIDEWALK RAMP
   SEE SHEET CP112 AND DETAIL A1 ON SHEET C-503
8. CONSTRUCT CONCRETE BLOCKOUT
   SEE DETAIL H2 ON SHEET C-508
9. CONSTRUCT TYPE I CURB AND GUTTER
   SEE L1 ON SHEET C-501
   SEE DETAIL A1 ON SHEET C-502
10. CONSTRUCT TYPE I CURB AND GUTTER
    SEE DETAIL A1 ON SHEET C-502
11. CONSTRUCT 6" REINFORCED SIDEWALK (4' WIDTH)
    SEE DETAIL A1 ON SHEET C-511
12. CONSTRUCT 6" REINFORCED SIDEWALK (4' WIDTH)
    SEE DETAIL A1 ON SHEET C-511
13. CONSTRUCT TYPE 1 STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
14. CONSTRUCT TYPE 1 STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
15. CONSTRUCT TYPE I CURB AND GUTTER
    SEE DETAIL A1 ON SHEET C-502
16. CONSTRUCT SIDEWALK (5' TO 6' TRANSITION)
    SEE DETAIL A1 ON SHEET C-503
17. CONSTRUCT SIDEWALK RAMP
    SEE SHEET CP112 AND DETAIL A1 ON SHEET C-503
18. CONSTRUCT TYPE 1 STREET REPAIR
    SEE DETAIL J8 ON SHEET C-508
19. CONSTRUCT CONCRETE BLOCKOUT
    SEE DETAIL H2 ON SHEET C-508
20. CONSTRUCT SIDEWALK (5' WIDTH)
    SEE L1 ON SHEET C-501
    SEE DETAIL A1 ON SHEET C-503

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- CONSTRUCTION EASEMENT (C\E)
- 6" REINFORCED SIDEWALK

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

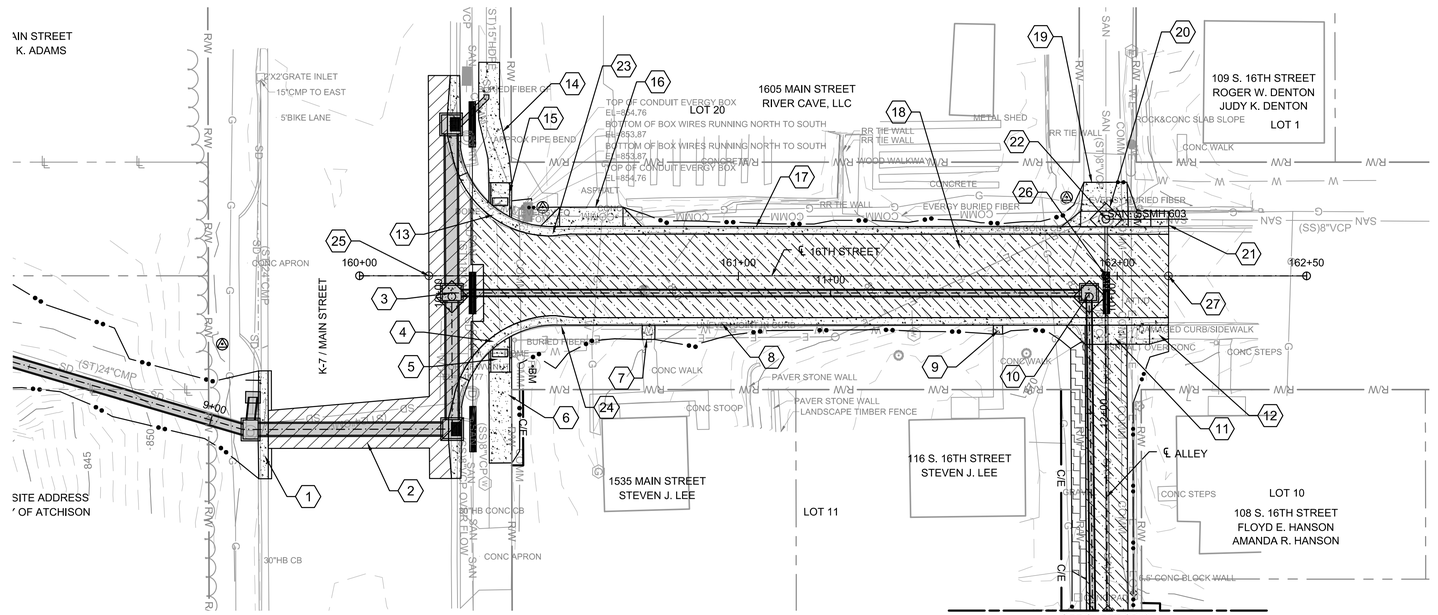
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

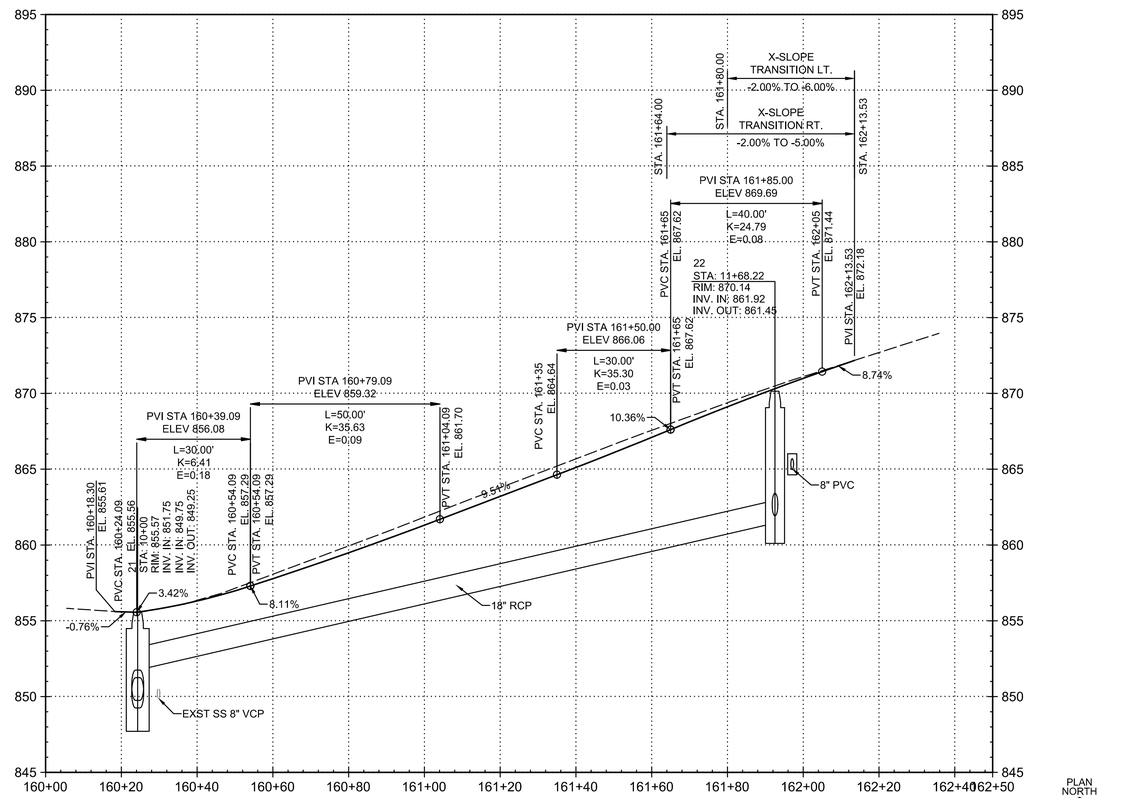
15TH STREET
CIVIL PLAN

SHEET ID
CP103

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP101-CP103_15TH.dwg
9/9/2026 3:58:49 PM

READY TO ADVERTISE





SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. SEE SHEET CU107 FOR PIPE PROFILE INFORMATION.
5. SEE SHEET CP301 THRU CP302 FOR CROSS SECTIONS
6. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
7. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.
8. SEE SHEET CP109 FOR INTERSECTION DETAILS

SHEET KEYNOTES

1. CONSTRUCT TYPE III CURB AND GUTTER (2'-6" WIDTH)
   SEE DETAIL A1 ON SHEET C-502
2. ASPHALTIC CONCRETE STREET REPAIR TYPE 1
   SEE DETAIL J8 ON SHEET C-508
3. CONSTRUCT CONCRETE BLOCKOUT
   SEE DETAIL H2 ON SHEET C-508
4. CONSTRUCT TYPE III CURB AND GUTTER (2'-6" WIDTH)
   SEE DETAIL A1 ON SHEET C-502
5. CONSTRUCT SIDEWALK RAMP
   SEE SHEET CP113 AND DETAIL A1 ON SHEET C-503
6. CONSTRUCT SIDEWALK (5' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
7. CONSTRUCT SIDEWALK (3' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
8. CONSTRUCT TYPE I CURB AND GUTTER
   SEE G1 ON SHEET C-501
   SEE DETAIL A1 ON SHEET C-502
9. CONSTRUCT SIDEWALK (3' WIDTH)
   SEE DETAIL A1 ON SHEET C-503
10. CONSTRUCT CONCRETE BLOCKOUT
    SEE DETAIL H2 ON SHEET C-508
11. CONSTRUCT ALLEY DRIVE
    SEE SECTION A1 ON SHEET C-505
12. CONSTRUCT SIDEWALK (III.6' TO 4' TRANSITION)
    SEE DETAIL A1 ON SHEET C-503
13. CONSTRUCT TYPE III CURB AND GUTTER (2'-6" WIDTH)
    SEE DETAIL A1 ON SHEET C-502
14. CONSTRUCT SIDEWALK (5.5' WIDTH)
    SEE DETAIL A1 ON SHEET C-503
15. CONSTRUCT SIDEWALK RAMP
    SEE SHEET CP113 AND DETAIL A1 ON SHEET C-503
16. CONSTRUCT COMMERCIAL DRIVE
    SEE DETAIL A1 ON SHEET C-505
17. CONSTRUCT TYPE I CURB AND GUTTER
    SEE G1 ON SHEET C-501
    SEE DETAIL A1 ON SHEET C-502
18. FULL DEPTH ASPHALT CONSTRUCTION
    SEE SECTION G1 ON SHEET C-501
19. CONSTRUCT ALLEY DRIVE
    SEE DETAIL A1 ON SHEET C-505
20. CONSTRUCT CONCRETE BLOCKOUT
    SEE DETAIL H2 ON SHEET C-508
21. CONSTRUCT SIDEWALK (4' WIDTH)
    SEE DETAIL A1 ON SHEET C-503
22. ADJUST SSMH 603 TO GRADE, EXISTING ELEV.=870.38, PROPOSED ELEV.=870.30
23. 5' CURB TRANSITION TYPE III TO TYPE 1
24. 5' CURB TRANSITION TYPE III TO TYPE 1
25. BEGIN FULL DEPTH ASPHALT RECONSTRUCTION
    STA. 160+18.30
    N:459865.74 E:2118499.43
    SAWCUT EXISTING FULL DEPTH
    (NO DIRECT PAY)
26. STA. 161+96.94, 16TH STREET=
    STA. 10+00.00, ALLEY
    N:460044.34, E:2118495.95
27. END FULL DEPTH ASPHALT RECONSTRUCTION
    STA. 162+13.53
    N:460060.94 E:2118495.63
    SAWCUT EXISTING FULL DEPTH
    (NO DIRECT PAY)

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E — CONSTRUCTION EASEMENT
- FULL DEPTH ASPHALT REPLACEMENT
- GRAVEL DRIVE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
CIVIL PLAN AND PROFILE

SHEET ID
CP104

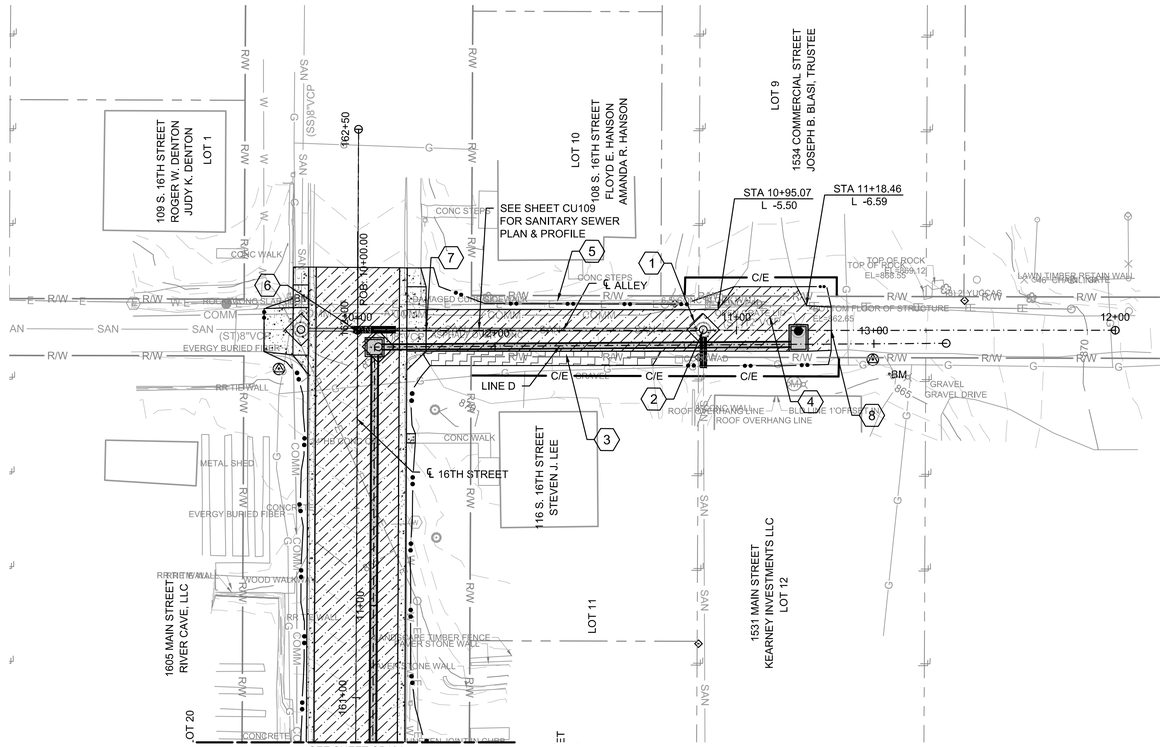
A1 CIVIL PLAN & PROFILE, 16TH STREET STA. 160+00 TO 162+50
SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

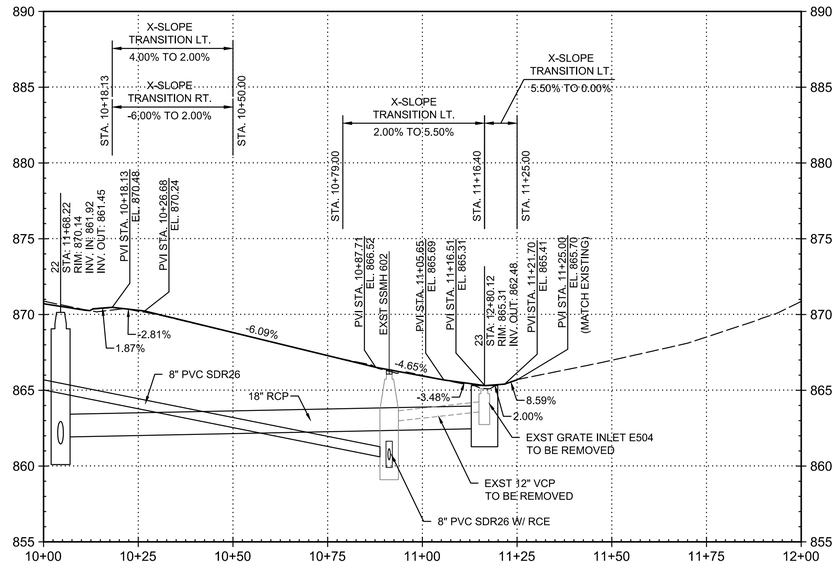
PLAN NORTH

0 20' 40'

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP104_16TH.dwg
9/9/2026 4:00:02 PM

READY TO ADVERTISE





SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET CU109 FOR SANITARY PLAN AND PROFILE.
4. SEE SHEET CP303 FOR CROSS SECTIONS
5. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
6. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.

SHEET KEYNOTES

1. CONSTRUCT CONCRETE BLOCKOUT
   SEE DETAIL H2 ON SHEET C-508
2. ADJUST SSMH 602 TO GRADE, EXISTING ELEV.=866.07, PROPOSED ELEV.=866.36
3. CONSTRUCT GRAVEL DRIVE
   SEE A1 ON SHEET C-504
4. FULL DEPTH ASPHALT CONSTRUCTION
   SEE A3 ON SHEET C-501
5. EXISTING RETAINING WALL (D.N.D.)
6. STA. 10+00, ALLEY=
   STA. 161+96.94, 16TH ST
   N:460044.34, E:2118495.95
7. BEGIN FULL DEPTH
   ASPHALT
   CONSTRUCTION
   STA. 10+18.13
   N:460044.78, E:2118514.07
8. END FULL DEPTH
   ASPHALT
   CONSTRUCTION
   STA. 11+25
   N:460047.37, E:2118620.91
   SAWCUT EXISTING
   PAVEMENT FULL DEPTH
   (NO DIRECT PAY)

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- CONSTRUCTION EASEMENT
- FULL DEPTH ASPHALT REPLACEMENT
- GRAVEL DRIVE

A2 CIVIL PLAN & PROFILE, ALLEY STA. 10+00 TO 12+00

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

PLAN NORTH

0 20' 40'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D024F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

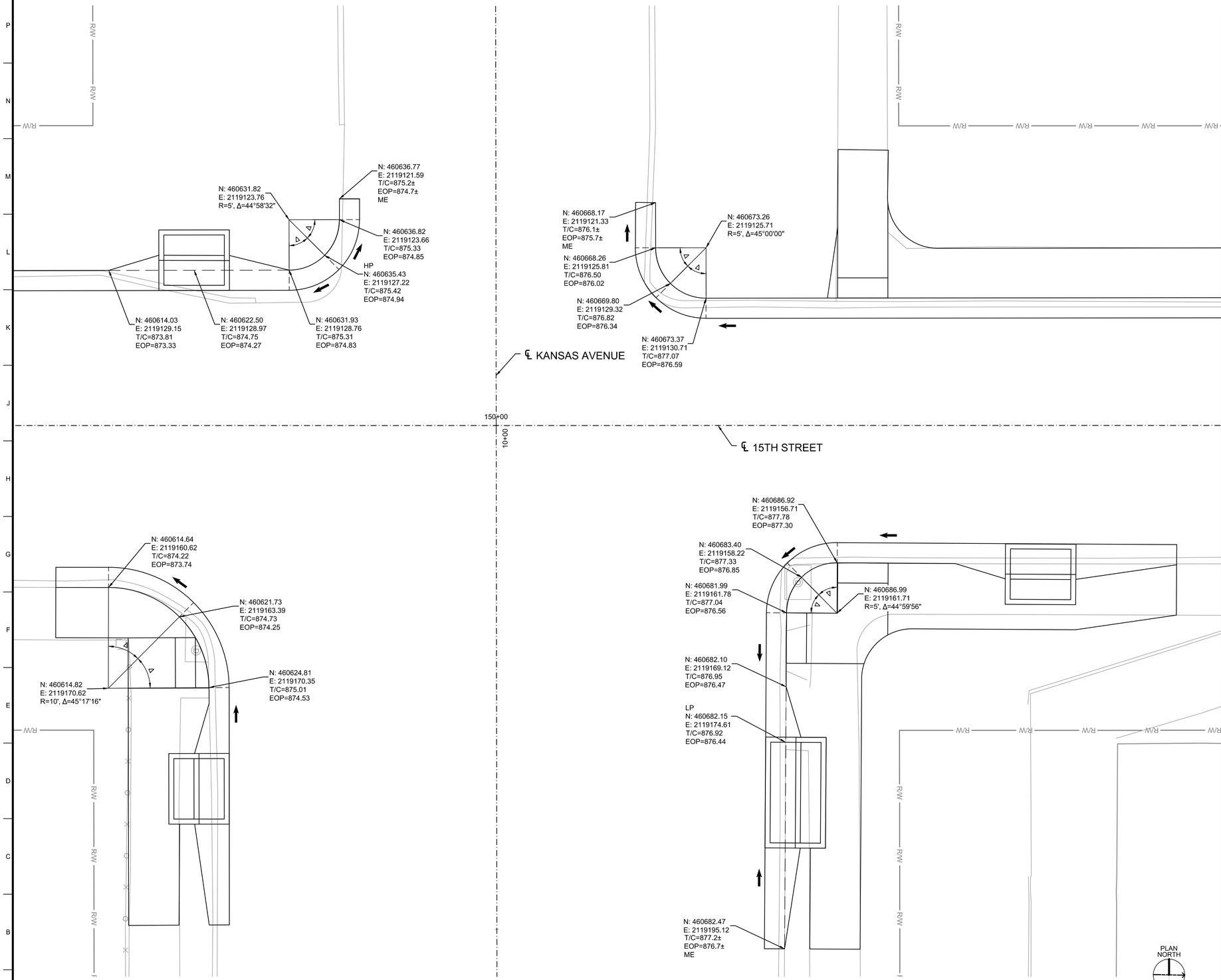
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

ALLEY
CIVIL PLAN AND PROFILE

SHEET ID
CP105

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP105_ALLEY.dwg
9/9/2026 4:01:07 PM

READY TO ADVERTISE



A1 INTERSECTION DETAILS, 15TH STREET & KANSAS AVENUE

SCALE:1"=5'

SHEET NOTES

1. SEE SHEETS CP101, CP102, AND CP103 FOR 15TH STREET CIVIL PLANS.
2. ALL CURB & GUTTER SHALL BE APWA TYPE I CURB & GUTTER UNLESS OTHERWISE STATED ON PLANS. CURB ELEVATIONS SHOWN ARE THE THEORETICAL TOP OF CURB. SEE SHEET C-502 FOR DETAILS.
3. CONTRACTOR IS RESPONSIBLE FOR POSITIVE DRAINAGE AT ALL INTERSECTIONS.

LEGEND

EOP	EDGE OF PAVEMENT
T/C	TOP OF CONCRETE
HP	HIGH POINT
LP	LOW POINT
ME	MATCH EXISTING

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:: W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY:
J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

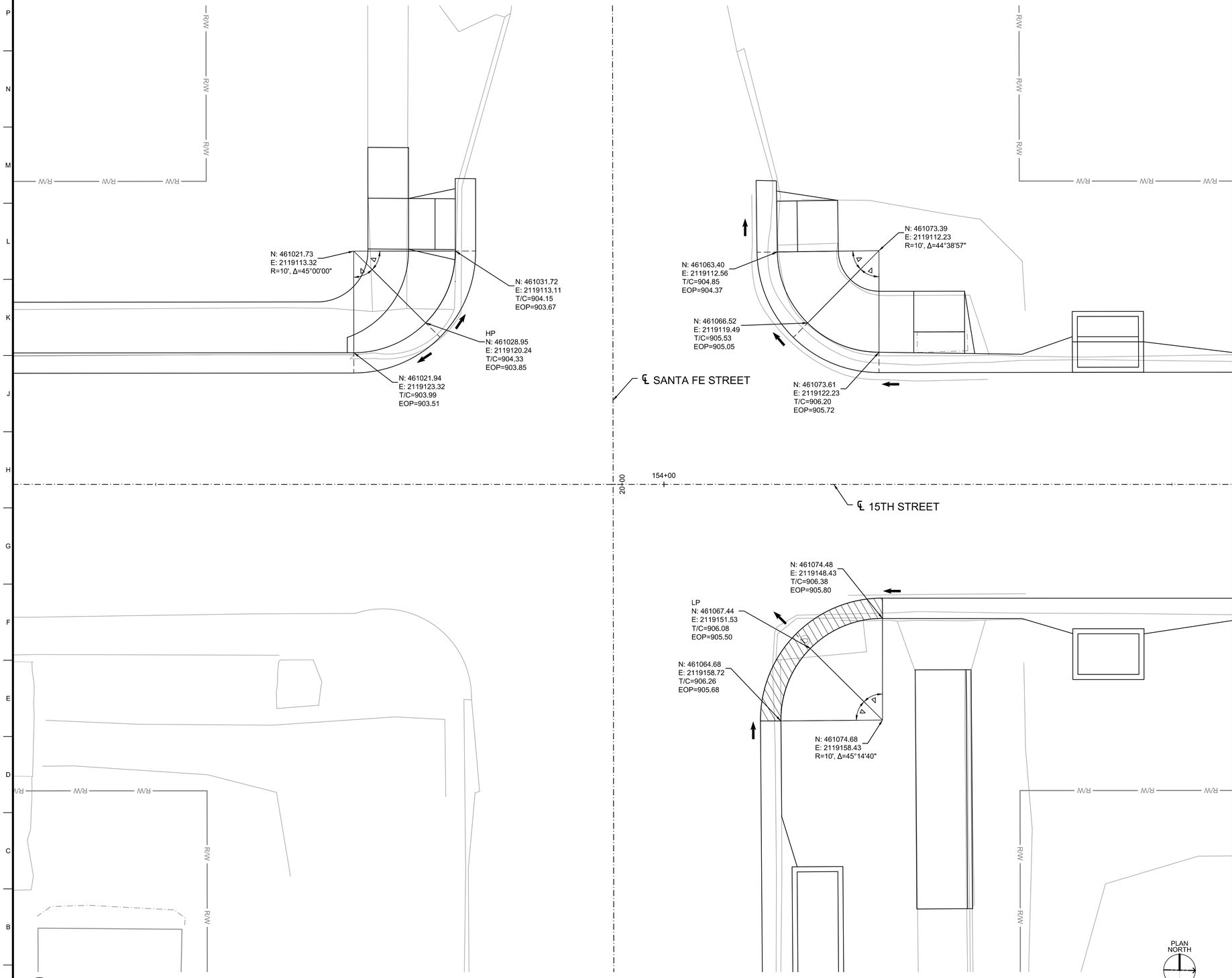
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
INTERSECTION DETAILS

SHEET ID
CP106

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP106-CP108_15TH.dwg
9/9/2026 4:02:04 PM

READY TO ADVERTISE



A1 INTERSECTION DETAILS, 15TH STREET & SANTA FE STREET

SCALE:1"=5'

SHEET NOTES

1. SEE SHEETS CP101, CP102, AND CP103 FOR 15TH STREET CIVIL PLANS.
2. ALL CURB & GUTTER SHALL BE APWA TYPE I CURB & GUTTER UNLESS OTHERWISE STATED ON PLANS. CURB ELEVATIONS SHOWN ARE THE THEORETICAL TOP OF CURB. SEE SHEET C-502 FOR DETAILS.
3. CONTRACTOR IS RESPONSIBLE FOR POSITIVE DRAINAGE AT ALL INTERSECTIONS.

LEGEND

EOP	EDGE OF PAVEMENT
T/C	TOP OF CONCRETE
HP	HIGH POINT
LP	LOW POINT
ME	MATCH EXISTING

DRY CURB, SEE SHEET C-502 FOR TYPE I (DRY) CURB AND GUTTER

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

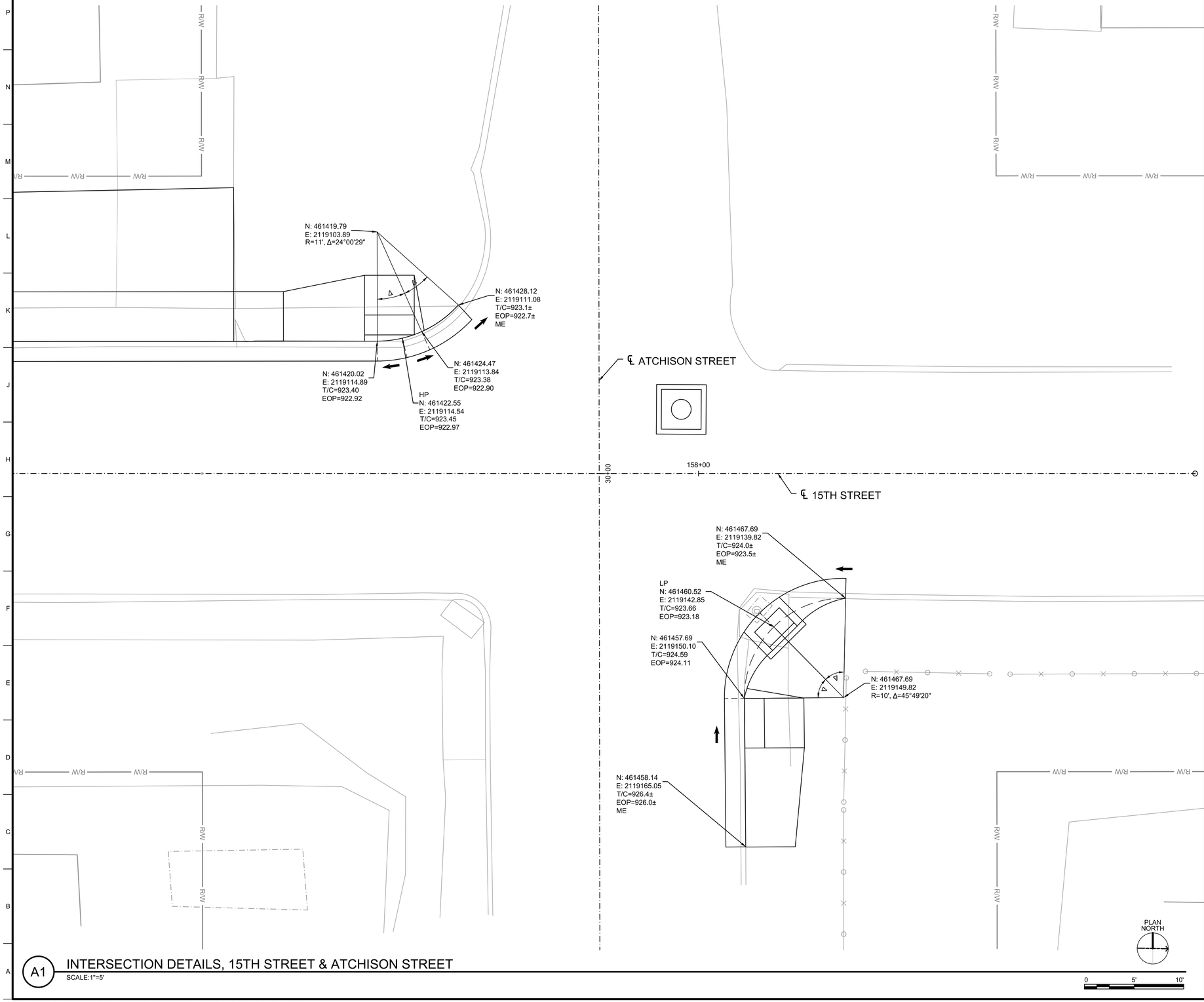
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
INTERSECTION DETAILS

SHEET ID
CP107

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP106-CP108_15TH.dwg
9/9/2026 4:02:24 PM

READY TO ADVERTISE



SHEET NOTES

1. SEE SHEETS CP101, CP102, AND CP103 FOR 15TH STREET CIVIL PLANS.
2. ALL CURB & GUTTER SHALL BE APWA TYPE I CURB & GUTTER UNLESS OTHERWISE STATED ON PLANS. CURB ELEVATIONS SHOWN ARE THE THEORETICAL TOP OF CURB. SEE SHEET C-502 FOR DETAILS.
3. CONTRACTOR IS RESPONSIBLE FOR POSITIVE DRAINAGE AT ALL INTERSECTIONS.

LEGEND

EOP	EDGE OF PAVEMENT
T/C	TOP OF CONCRETE
HP	HIGH POINT
LP	LOW POINT
ME	MATCH EXISTING

A1 INTERSECTION DETAILS, 15TH STREET & ATCHISON STREET
SCALE:1"=5'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

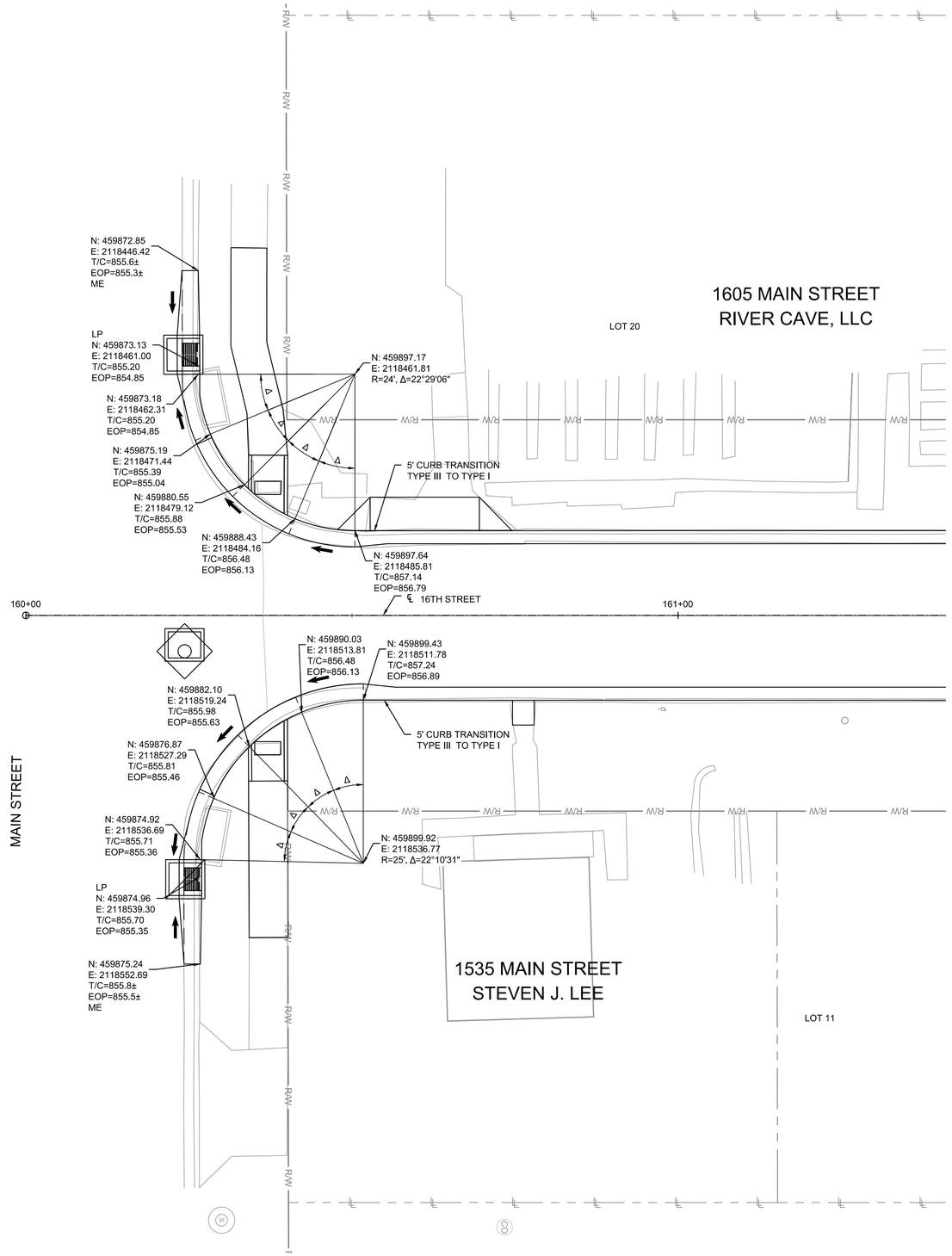
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
INTERSECTION DETAILS

SHEET ID
CP108

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP106-CP108_15TH.dwg
9/9/2026 4:02:40 PM

READY TO ADVERTISE



A1 INTERSECTION DETAILS, 16TH STREET & MAIN STREET

SCALE:1"=5'

SHEET NOTES

1. SEE SHEET CP104 FOR 16TH STREET CIVIL PLANS.
2. ALL CURB & GUTTER SHALL BE TYPE III CURB & GUTTER UNLESS OTHERWISE STATED ON PLANS. CURB ELEVATIONS SHOWN ARE THE THEORETICAL TOP OF CURB. SEE SHEET C-502 FOR DETAILS.
3. CONTRACTOR IS RESPONSIBLE FOR POSITIVE DRAINAGE AT ALL INTERSECTIONS.

LEGEND

EOP	EDGE OF PAVEMENT
T/C	TOP OF CONCRETE
HP	HIGH POINT
LP	LOW POINT

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

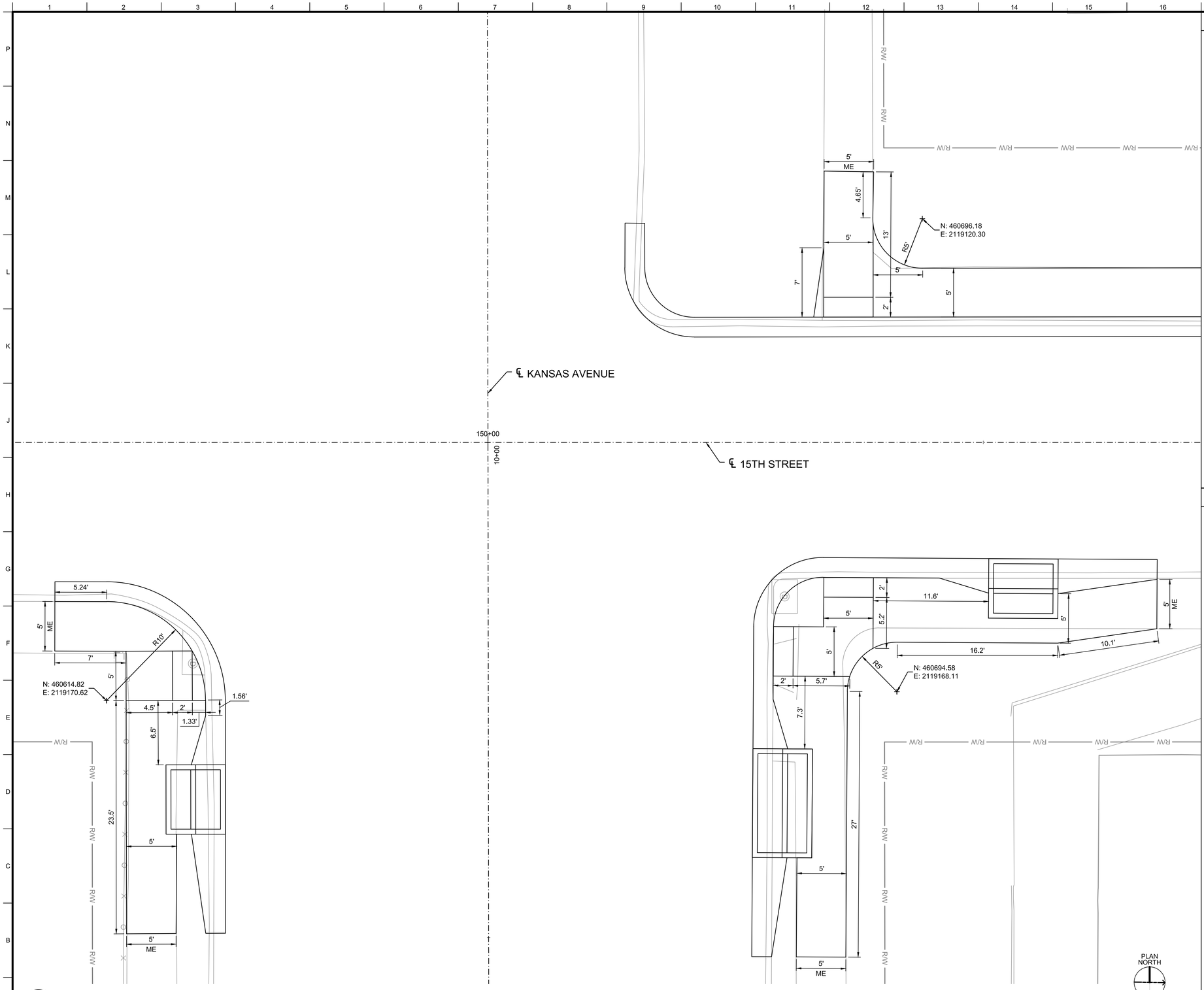
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
INTERSECTION DETAILS

SHEET ID
CP109

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP109_16TH.dwg
9/9/2026 4:03:38 PM

READY TO ADVERTISE



SHEET NOTES

1. SEE SHEETS CP101, CP102, AND CP103 FOR 15TH STREET CIVIL PLANS.
2. CONSTRUCT SIDEWALK AND RAMPS AFTER STORM SEWER IS COMPLETED.
3. SEE SHEET C-503 FOR ADDITIONAL RAMP DETAILS.

LEGEND

ME MATCH EXISTING

A1 RAMP DETAILS, 15TH STREET & KANSAS AVENUE

SCALE:1"=5'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

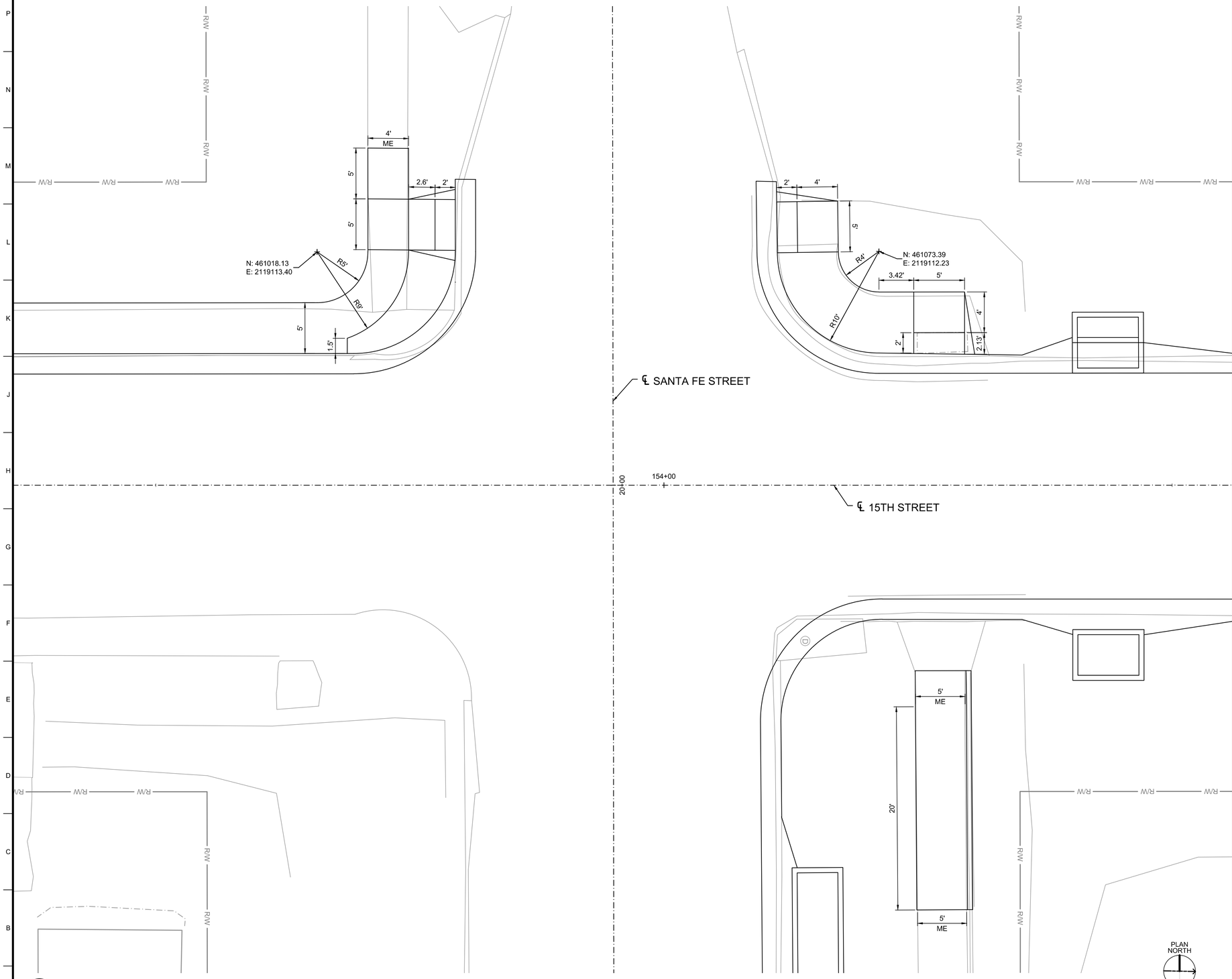
15TH STREET
RAMP DETAILS

SHEET ID

CP110

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP110-CP112_15TH.dwg
9/9/2026 4:04:37 PM

READY TO ADVERTISE



SHEET NOTES

1. SEE SHEETS CP101, CP102, AND CP103 FOR 15TH STREET CIVIL PLANS.
2. CONSTRUCT SIDEWALK AND RAMPS AFTER STORM SEWER IS COMPLETED.
3. SEE SHEET C-503 FOR ADDITIONAL RAMP DETAILS.

LEGEND

ME	MATCH EXISTING

A1 RAMP DETAILS, 15TH STREET & SANTA FE STREET

SCALE:1"=5'

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP110-CP112_15TH.dwg
9/9/2026 4:04:57 PM

READY TO ADVERTISE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

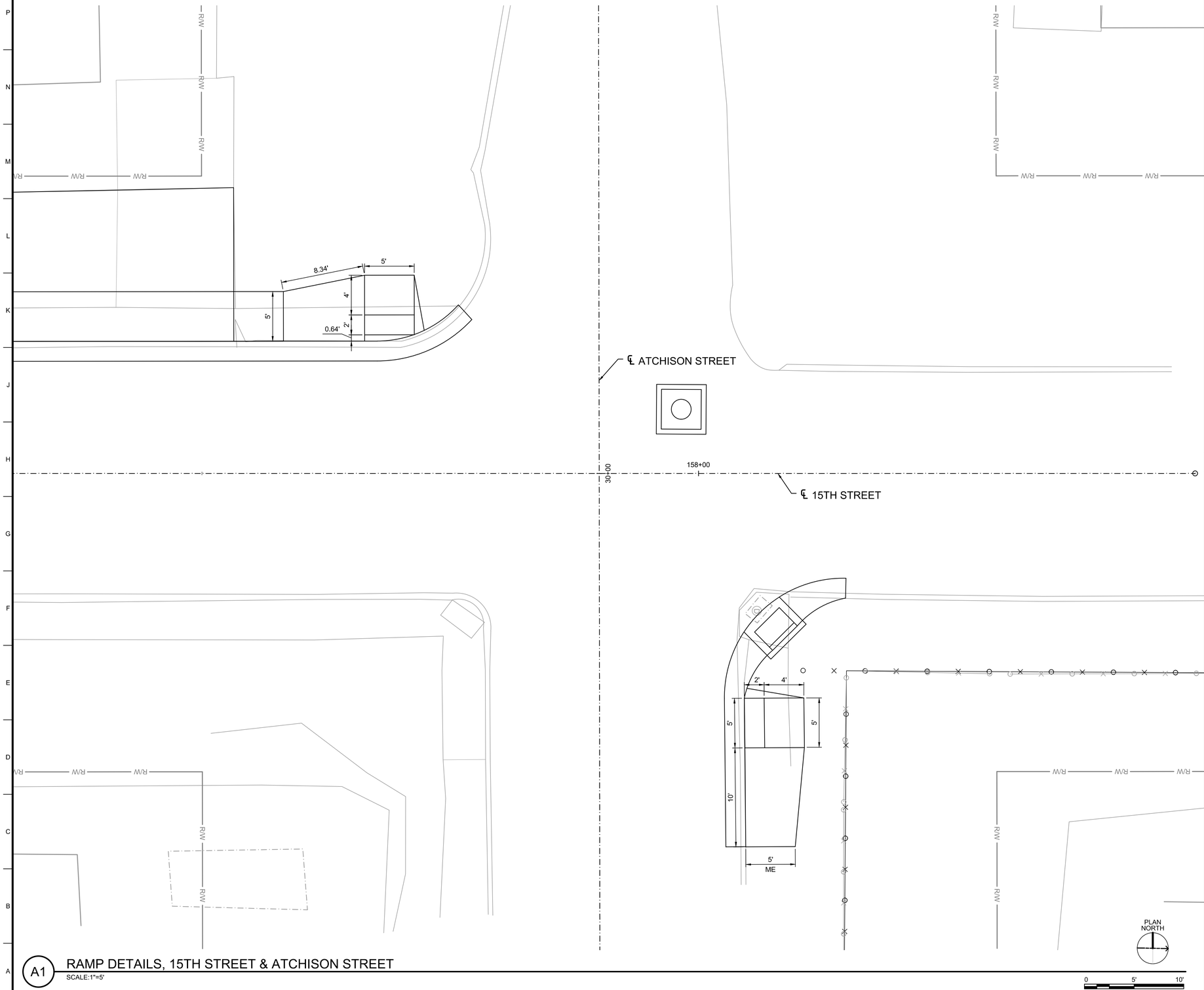
U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
RAMP DETAILS

SHEET ID
CP111



SHEET NOTES

1. SEE SHEETS CP101, CP102, AND CP103 FOR 15TH STREET CIVIL PLANS.
2. CONSTRUCT SIDEWALK AND RAMPS AFTER STORM SEWER IS COMPLETED.
3. SEE SHEET C-503 FOR ADDITIONAL RAMP DETAILS.

LEGEND

ME	MATCH EXISTING

A1 RAMP DETAILS, 15TH STREET & ATCHISON STREET

SCALE:1"=5'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

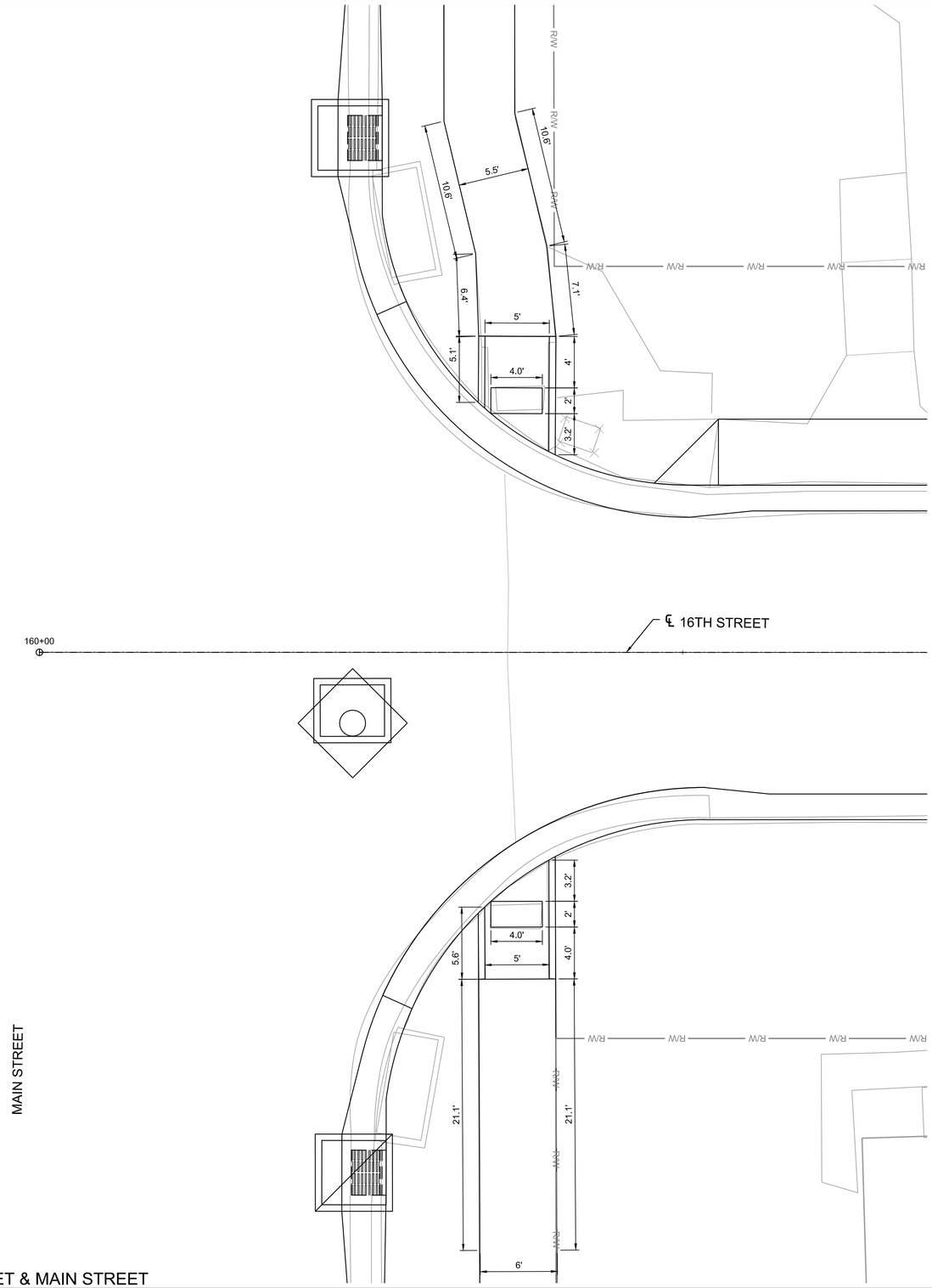
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
RAMP DETAILS

SHEET ID
CP112

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP110-CP112_15TH.dwg
9/9/2026 4:05:16 PM

READY TO ADVERTISE



A1 RAMP DETAILS, 16TH STREET & MAIN STREET

SCALE:1"=5'

SHEET NOTES

1. SEE SHEET CP104 FOR 16TH STREET CIVIL PLANS.
2. CONSTRUCT SIDEWALK AND RAMPS AFTER STORM SEWER IS COMPLETED.
3. SEE SHEET C-503 FOR ADDITIONAL RAMP DETAILS.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

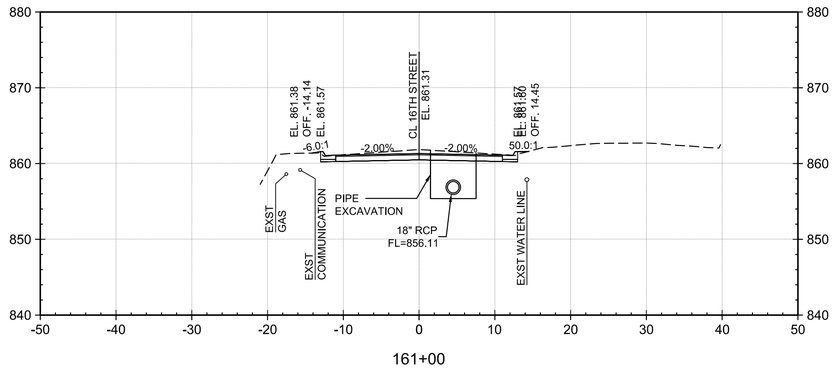
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
RAMP DETAILS

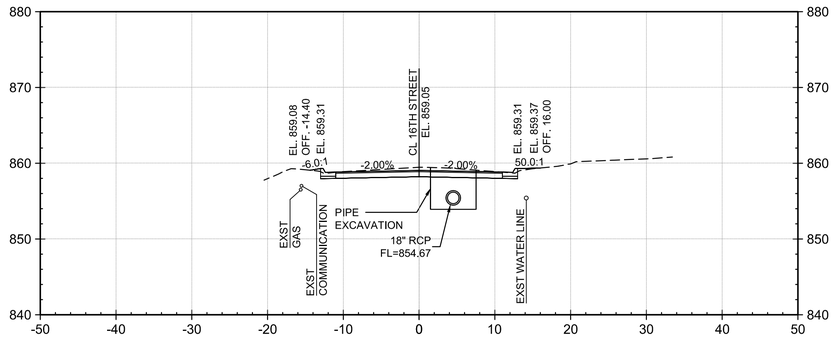
SHEET ID
CP113

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP113_16TH.dwg
9/9/2026 4:06:07 PM

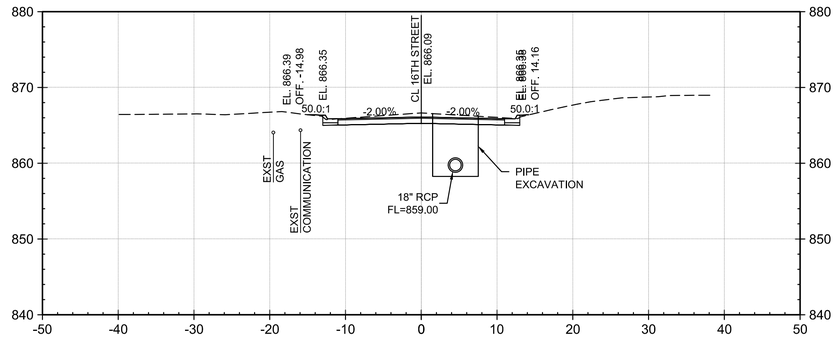
READY TO ADVERTISE



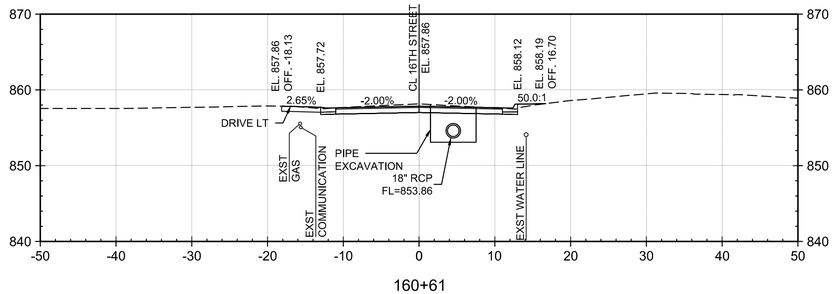
161+00



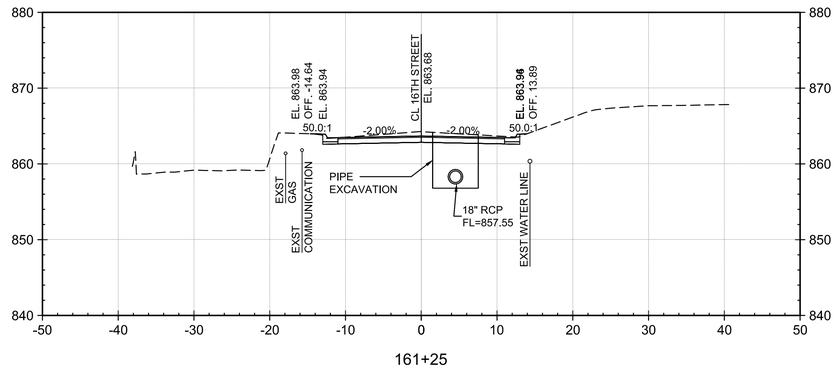
160+75



161+50



160+61



161+25

A1 CIVIL CROSS SECTIONS, 16TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR
4. SEE SHEET CP104 FOR 16TH STREET PLAN AND PROFILE.
5. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
6. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
CIVIL CROSS SECTIONS

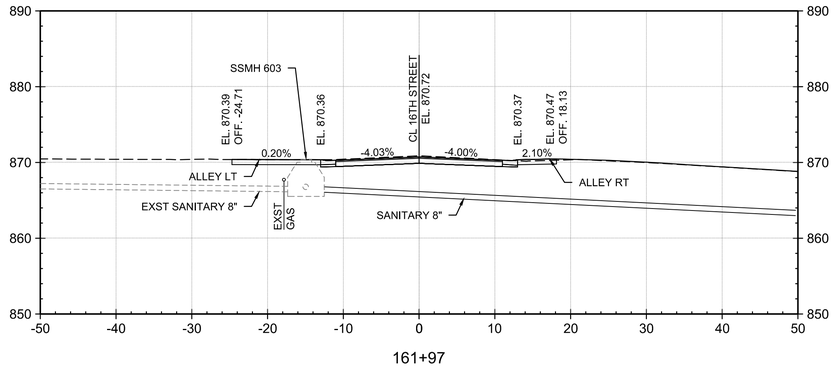
SHEET ID
CP301

READY TO ADVERTISE

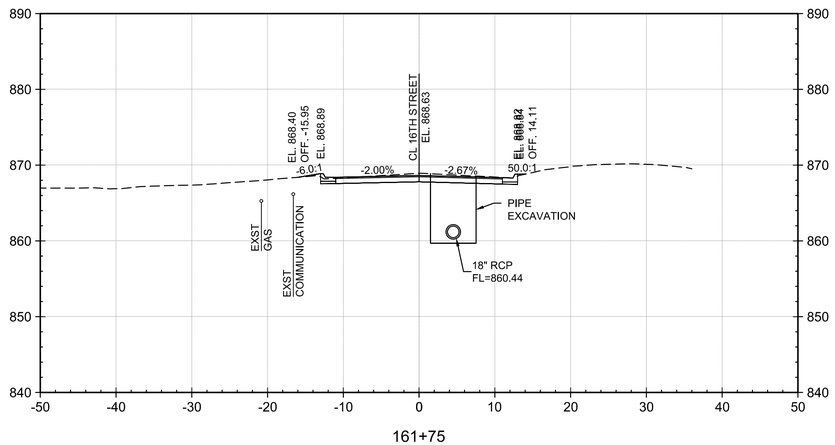
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP301-CP302_16TH.dwg
9/9/2026 4:06:59 PM

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR
4. SEE SHEET CP104 FOR 16TH STREET PLAN AND PROFILE.
5. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
6. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



161+97



161+75

A1 CIVIL CROSS SECTIONS, 16TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

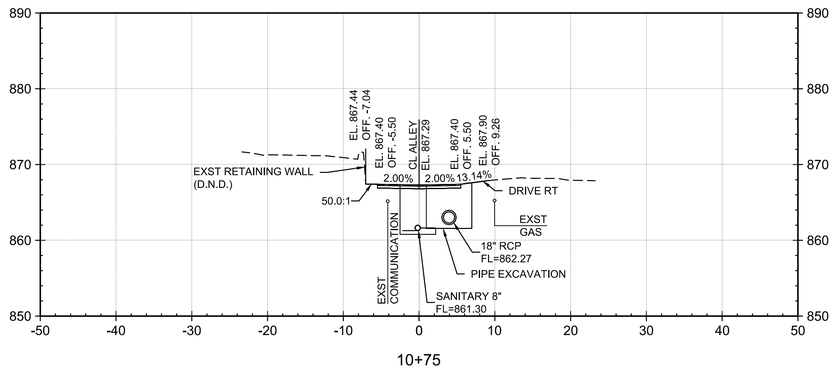
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
CIVIL CROSS SECTIONS

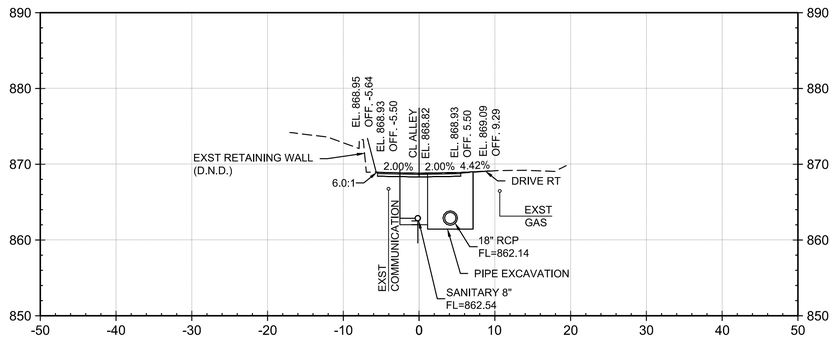
SHEET ID
CP302

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP301-CP302_16TH.dwg
9/9/2026 4:07:10 PM

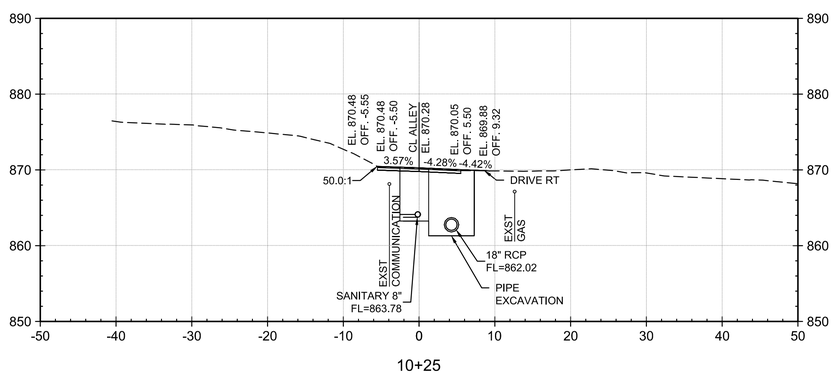
READY TO ADVERTISE



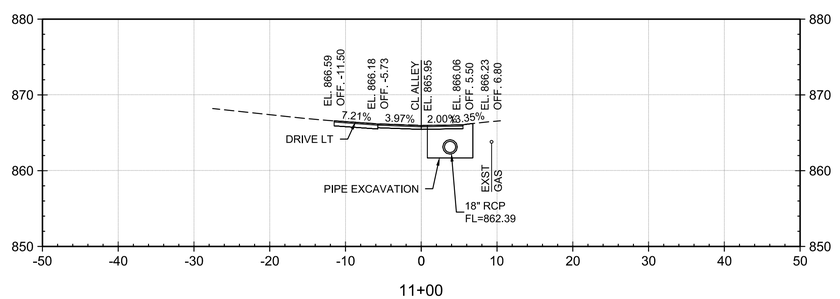
10+75



10+50



10+25



11+00

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
4. SEE SHEET CP105 FOR ALLEY PLAN AND PROFILE
5. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
6. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.

A1 CIVIL CROSS SECTIONS, ALLEY

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BINSCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

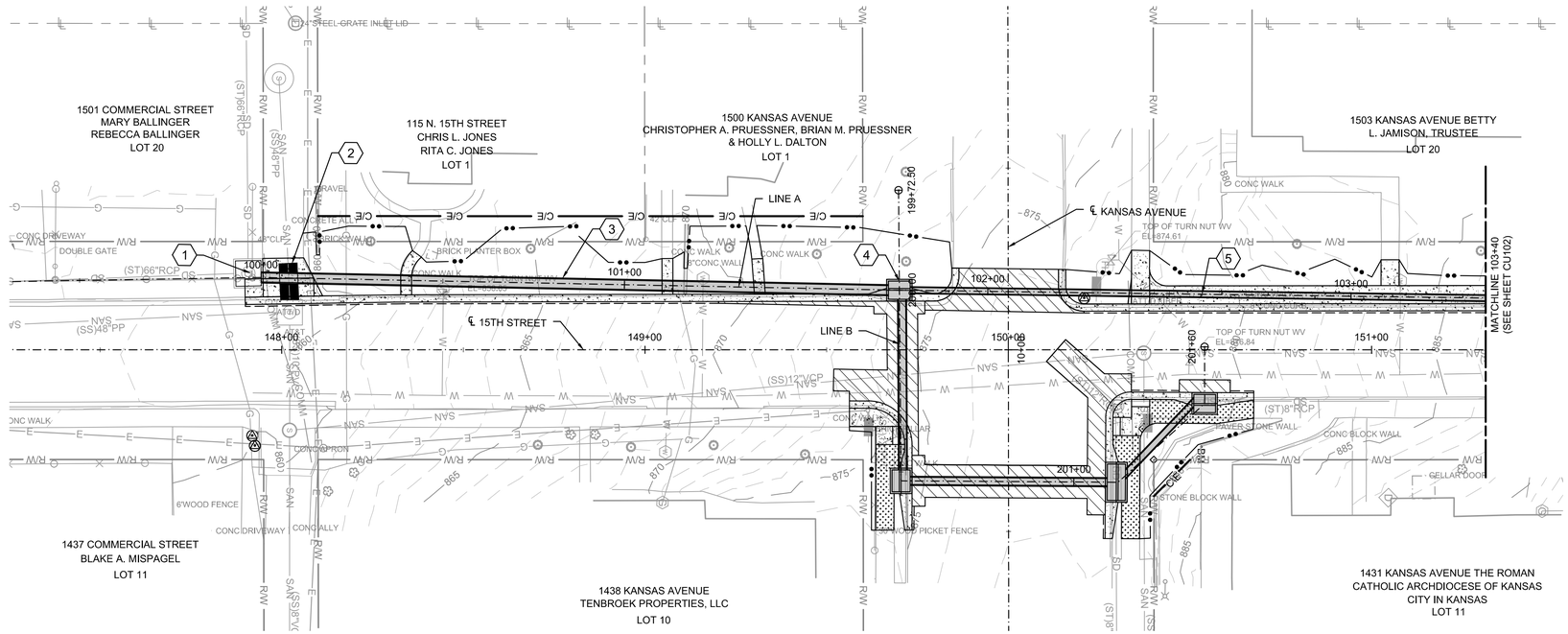
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

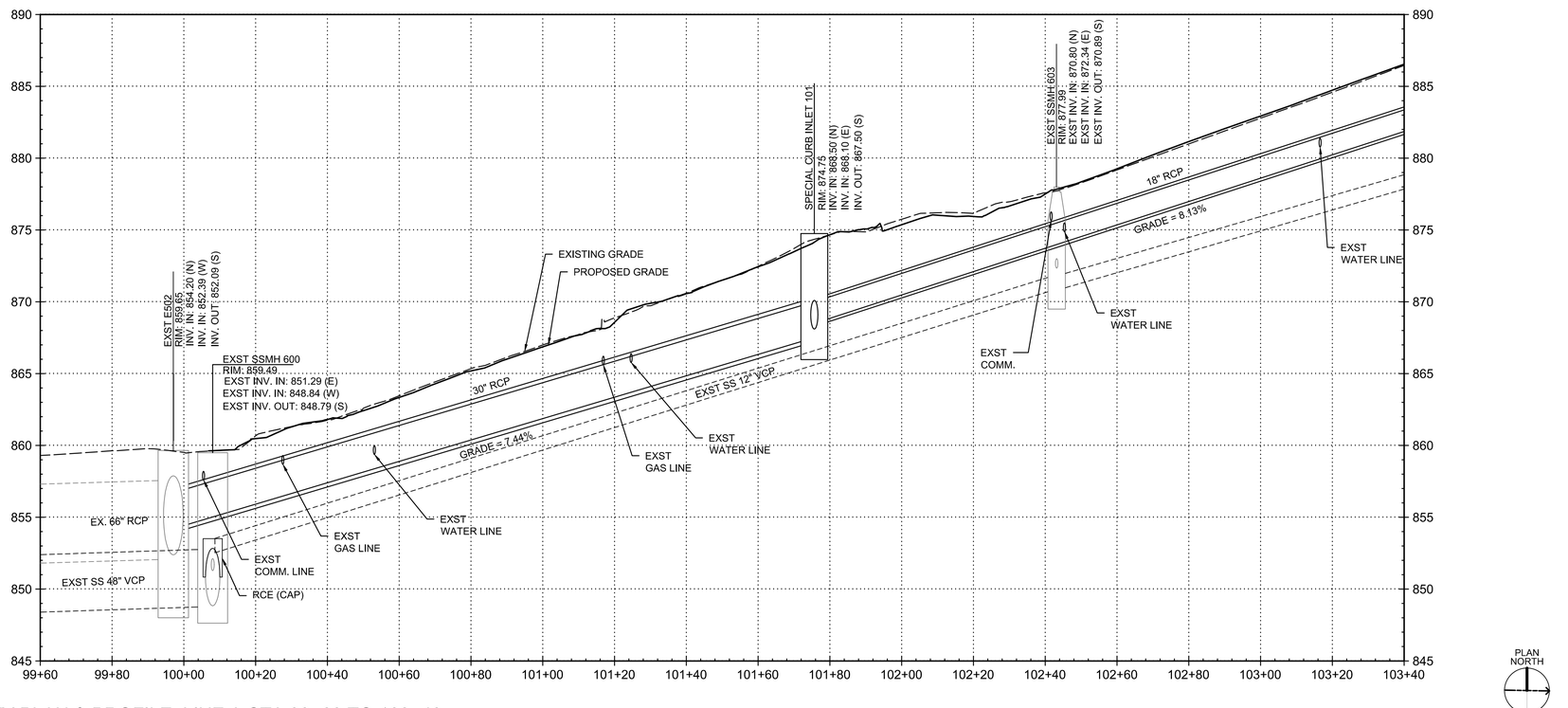
ALLEY
CIVIL CROSS SECTIONS

SHEET ID
CP303

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CP303_ALLEY.dwg
9/9/2026 4:07:53 PM

READY TO ADVERTISE





PLAN NORTH

0 20' 40'

A1 UTILITY PLAN & PROFILE, LINE A STA 99+60 TO 103+40

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. SEE SHEET CU104 FOR LINE B UTILITY PLAN AND PROFILE
5. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
6. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.

SHEET KEYNOTES

1. EXISTING 7'X7' JUNCTION BOX E502 (U.I.P.) STA 99+97.07, LINE A= STA 147+91.42, 21.11' L ℄ 15TH ST, N:460443.83, E:2119126.89 CONNECT 30" RCP (N), PLACE 9" FROM EAST INSIDE FACE SEE DETAIL D6 ON SHEET C-521
2. CONSTRUCT 10 L.F. RCE (CAP) SEE DETAIL L7 ON SHEET C-521
3. CONSTRUCT 30" RCP
4. CONSTRUCT 6'X5' SPECIAL CURB INLET 101 STA 101+75.67, LINE A= STA 200+00.00, LINE B= STA 149+70.01, 16.41' L ℄ 15TH ST, N:460622.47, E:2119127.97 CONNECT 24" RCP (E), PLACE 12" FROM NORTH INSIDE FACE SEE SHEET CU104 FOR ADDITIONAL UTILITY PLAN AND PROFILE SEE DETAIL A1 ON SHEET C-514
5. CONSTRUCT 18" RCP

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E CONSTRUCTION EASEMENT
- 6" REINFORCED SIDEWALK

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D224F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

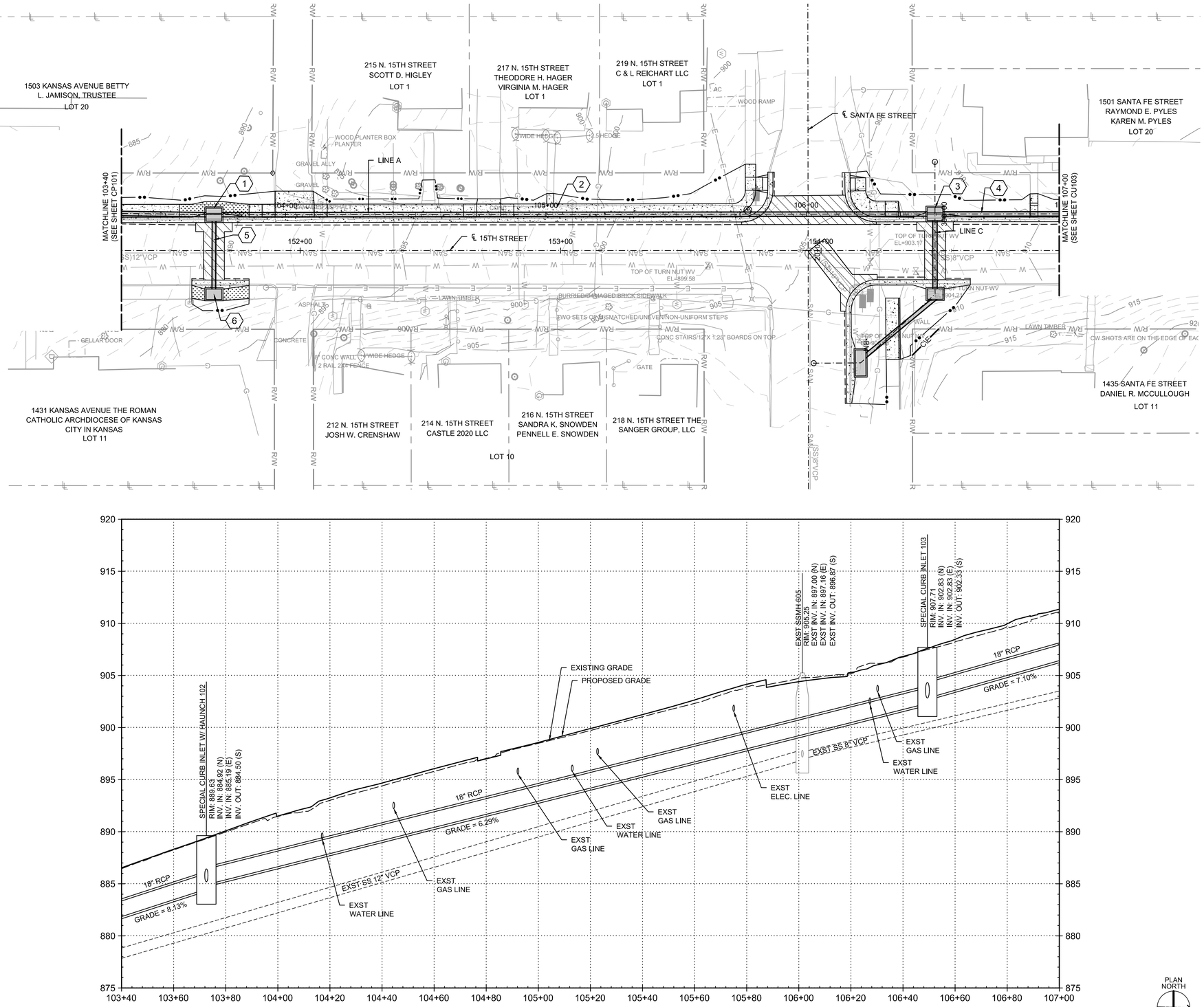
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
UTILITY PLAN AND PROFILE

SHEET ID
CU101

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU101-CU103_15TH.dwg
9/9/2026 4:09:18 PM

READY TO ADVERTISE



A1 UTILITY PLAN & PROFILE, LINE A STA. 103+40 TO 107+00

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. SEE SHEET CU104 FOR LINE C UTILITY PLAN AND PROFILE.
5. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
6. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.

SHEET KEYNOTES

1. CONSTRUCT 6'X5' SPECIAL CURB INLET W/ HAUNCH 102 STA 103+72.57, LINE A= STA 151+66.89, 13.78' L ℄ 15TH ST, N:460819.36, E:2119126.61
   SEE SHEET CU105 FOR ADDITIONAL UTILITY PLAN AND PROFILE
   SEE DETAIL A1 ON SHEET C-514
2. CONSTRUCT 18" RCP
3. CONSTRUCT 6'X5' SPECIAL CURB INLET 103 STA 106+49.37, LINE A= STA 300+00.00, LINE C=
   STA 154+43.69, 14.03' L ℄ 15TH ST, N:461096.10, E:2119120.75
   SEE SHEET CU104 FOR ADDITIONAL UTILITY PLAN AND PROFILE
   SEE DETAIL A1 ON SHEET C-514
4. CONSTRUCT 18" RCP
5. CONSTRUCT 15" RCP
   SEE SHEET CU105 FOR UTILITY PROFILE
6. CONSTRUCT 6'X4' CURB INLET W/ HAUNCH 110 STA 103+72.57, 30.58' R LINE A= STA 151+66.89, 16.80' R ℄ 15TH ST, N:460819.99, E:2119157.18
   SEE SHEET CU105 FOR UTILITY PROFILE
   SEE DETAIL A1 ON SHEET C-510

CALLOUTS

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E — CONSTRUCTION EASEMENT
- 6" REINFORCED SIDEWALK

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D224F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

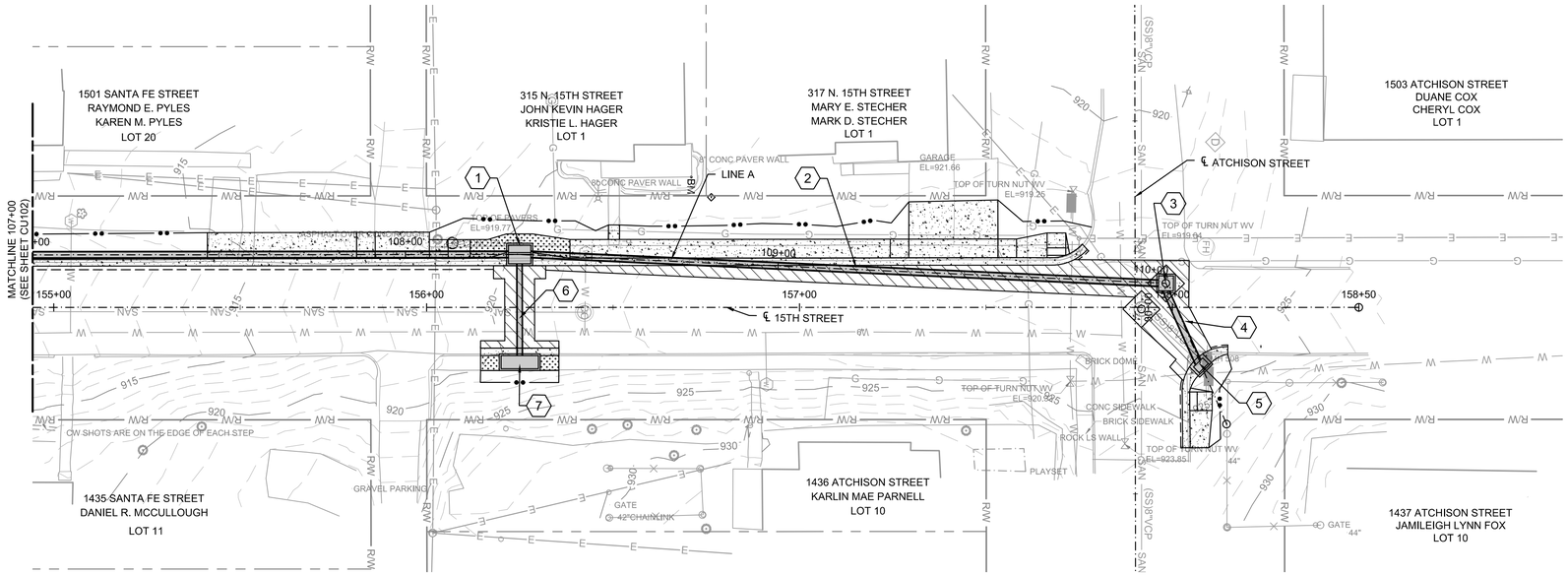
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

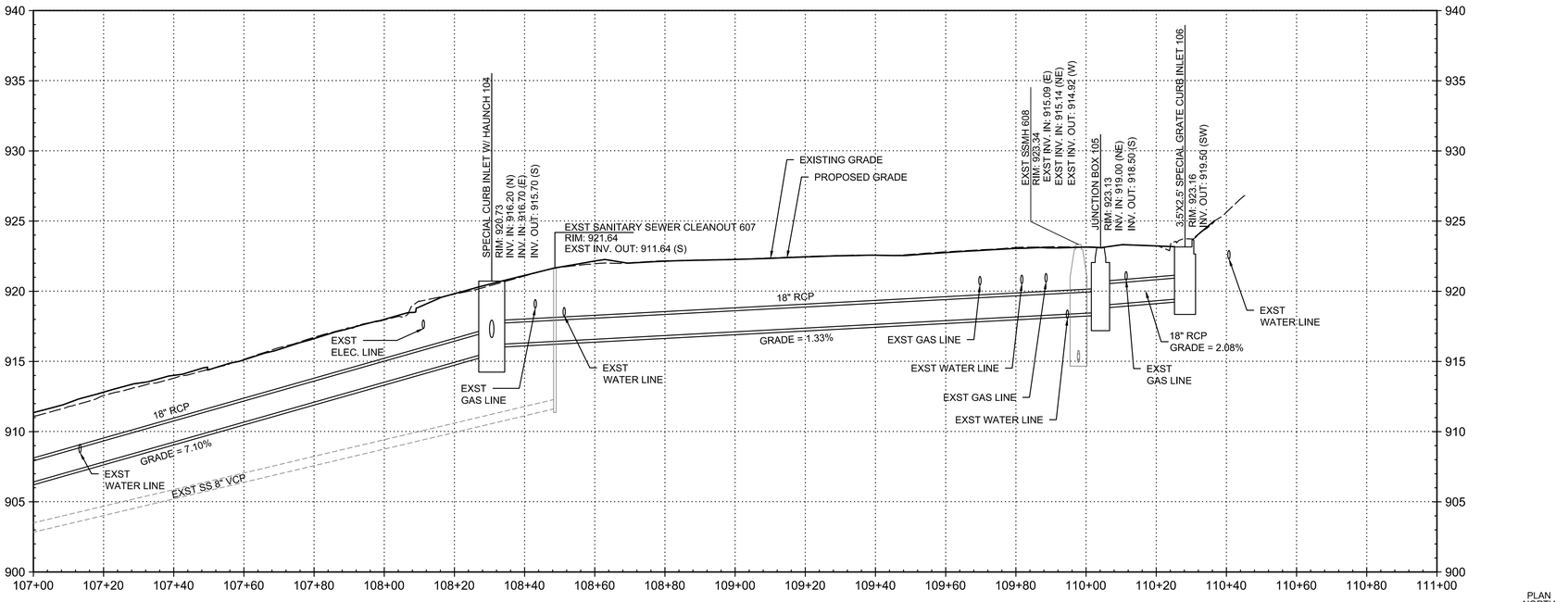
15TH STREET
UTILITY PLAN AND PROFILE

SHEET ID
CU102

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU101-CU103_15TH.dwg
9/9/2026 4:09:52 PM

READY TO ADVERTISE





PLAN NORTH

0 20' 40'

A1 UTILITY PLAN & PROFILE, LINE A STA 107+00 TO 111+00

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.

SHEET KEYNOTES

1. CONSTRUCT 6'X5' SPECIAL CURB INLET W/ HAUNCH 104 STA 108+30.62, LINE A= STA 299+99.87, LINE C=
   STA 156+24.94, 14.20' L ℄ 15TH ST, N:461277.32, E:2119116.91
   SEE SHEET CU105 FOR ADDITIONAL UTILITY PLAN AND PROFILE
   SEE DETAIL A1 ON SHEET C-514
2. CONSTRUCT 18" RCP
3. CONSTRUCT 4'X4' JUNCTION BOX 105 STA 110+04.12, LINE A=
   STA 157+98.27, 6.46' L ℄ 15TH ST, N:461450.76, E:2119121.14
   SEE DETAIL A1 ON SHEET C-516
4. CONSTRUCT 18" RCP
5. CONSTRUCT 3.5'X2.5' SPECIAL GRATE CURB INLET 106 STA 110+28.21, LINE A=
   STA 158+07.79, 15.67' R ℄ 15TH ST, N:461460.74, E:2119143.06
   SEE DETAIL A1 ON SHEET C-513
6. CONSTRUCT 15" RCP
   SEE SHEET CU105 FOR UTILITY PROFILE
7. CONSTRUCT 10'X3.5' NON-SETBACK CURB INLET W/ HAUNCH 113
   STA 108+31.93, 28.85' R LINE A=
   STA 156+24.96, 14.68' R ℄ 15TH ST, N:461277.92, E:2119145.78
   SEE SHEET CU105 FOR UTILITY PROFILE
   SEE DETAIL A1 ON SHEET C-512

CALLOUTS

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- 6" REINFORCED SIDEWALK

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D224F1019
CONTRACT NO.:: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

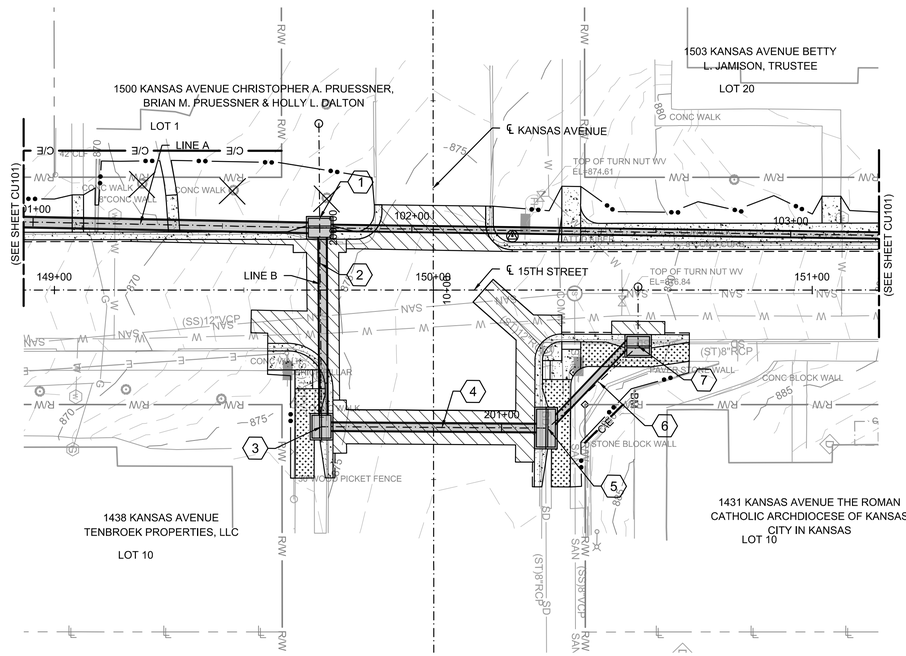
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

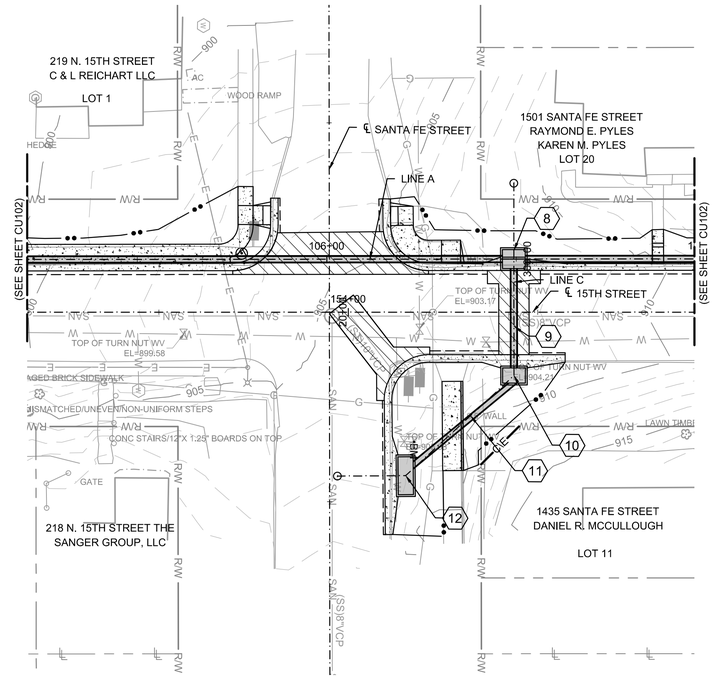
15TH STREET
UTILITY PLAN AND PROFILE

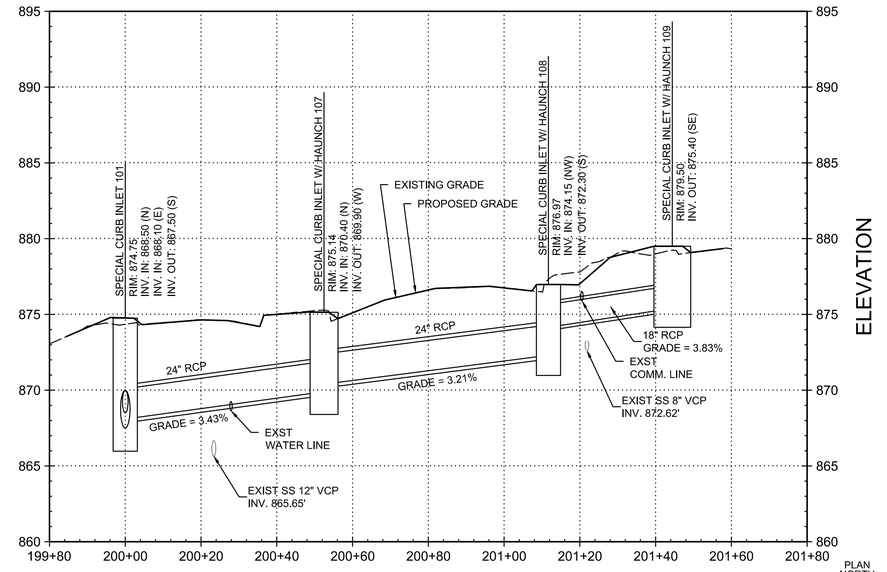
SHEET ID
CU103

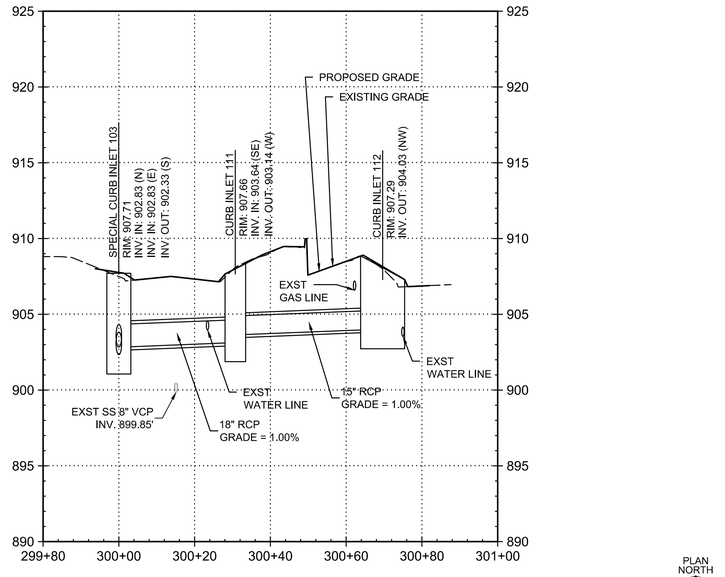
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU101-CU103_15TH.dwg
9/9/2026 4:10:21 PM

READY TO ADVERTISE









SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STEPS, INLET, AND MANHOLE DETAILS.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE DETAIL J8 ON SHEET C-508.

SHEET KEYNOTES

1. CONSTRUCT 5'X6' SPECIAL CURB INLET 101 STA 101+75.67, LINE B= STA 101+75.67, LINE A= STA 149+70.01, 16.41' L ℄ 15TH ST, N:460622.47, E:2119127.97 SEE SHEET CU101 FOR ADDITIONAL UTILITY PLAN AND PROFILE SEE DETAIL A1 ON SHEET C-514
2. CONSTRUCT 24" RCP
3. CONSTRUCT 6'X5' SPECIAL CURB INLET W/ HAUNCH 107 LINE B= STA 10+36.02, 29.55' R ℄ KANSAS AVE, N:460624.02, E:2119180.42 CONNECT 24" RCP (W), PLACE 9" FROM NORTH INSIDE FACE SEE DETAIL A1 ON SHEET C-514
4. CONSTRUCT 24" RCP
5. CONSTRUCT 10'X5' SPECIAL CURB INLET W/ HAUNCH 108 LINE B= STA 10+36.48, 29.67' L ℄ KANSAS AVE, N:460683.24, E:2119179.60 CONNECT 18" RCP (N), PLACE 16" FROM WEST INSIDE FACE CONNECT EXISTING 8" RCP (E) FIELD VERIFY PIPE LOCATION AND ℄ PRIOR TO CONSTRUCTION SEE DETAIL A1 ON SHEET C-515
6. CONSTRUCT 18" RCP
7. CONSTRUCT 6'X5' SPECIAL CURB INLET W/ HAUNCH 109 LINE B= STA 150+54.05, 14.77' R ℄ 15TH ST, N:460707.13, E:2119157.44 CONNECT EXISTING 8" RCP (N) FIELD VERIFY PIPE LOCATION AND ℄ PRIOR TO CONSTRUCTION SEE DETAIL A1 ON SHEET C-514
8. CONSTRUCT 5'X6' SPECIAL CURB INLET 103 STA 106+49.37, LINE C= STA 106+49.37, LINE A= STA 154+43.69, 14.03' L ℄ 15TH ST, N:461096.10, E:2119120.75 SEE SHEET CU102 FOR ADDITIONAL UTILITY PLAN AND PROFILE SEE DETAIL A1 ON SHEET C-514
9. CONSTRUCT 18" RCP
10. CONSTRUCT 6'X4' CURB INLET 111 STA 300+30.74, LINE C= STA 154+43.73, 16.71' R ℄ 15TH ST, N:461096.77, E:2119151.48 CONNECT 18" RCP (E), PLACE 17" FROM SOUTH INSIDE FACE SLOPE TOP OF STRUCTURE 4:1 SEE DETAIL A1 ON SHEET C-510
11. CONSTRUCT 15" RCP
12. CONSTRUCT 10'X4' CURB INLET 112 LINE C= STA 20+43.11, 20.11' L ℄ SANTE FE ST, N:461068.75, E:2119178.43 CONNECT 18" RCP (N), PLACE 14" FROM WEST INSIDE FACE SLOPE TOP OF STRUCTURE 6:1 SEE DETAIL A1 ON SHEET C-510

CALLOUTS

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- CONSTRUCTION EASEMENT
- 6" REINFORCED SIDEWALK

A1 UTILITY PLAN & PROFILE, LINE B STA 199+80 TO 201+80
SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'
0 20' 40'

A10 UTILITY PLAN & PROFILE, LINE C STA 299+80 TO 301+00
SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'
0 20' 40'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
UTILITY PLAN AND PROFILE

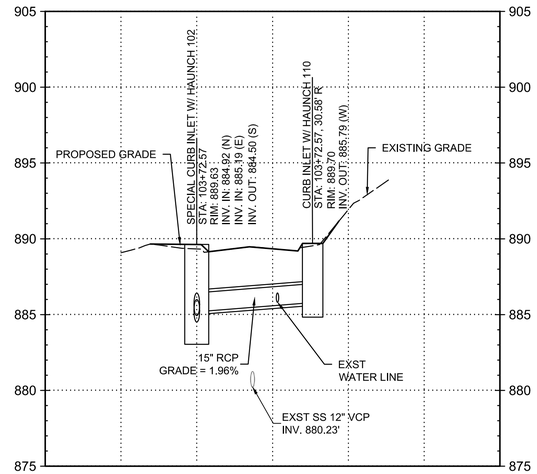
SHEET ID
CU104

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU104_15TH.dwg
9/9/2026 4:11:35 PM

READY TO ADVERTISE

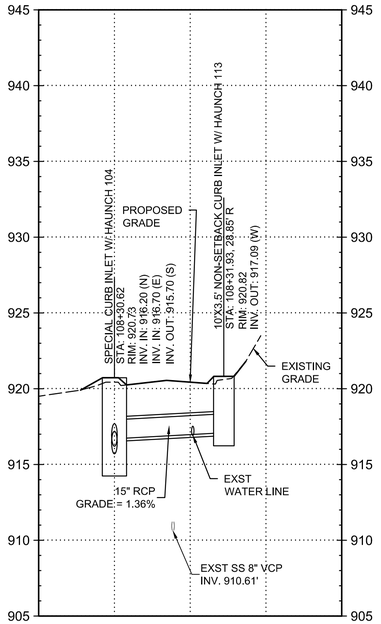
SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STEPS, INLET, AND MANHOLE DETAILS.



E5 UTILITY PROFILE, LINE A STA. 103+72.57 TO 103+72.57, 30.58' R

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'



E12 UTILITY PROFILE, LINE A STA. 108+30.62 TO 108+31.93, 28.85' R

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

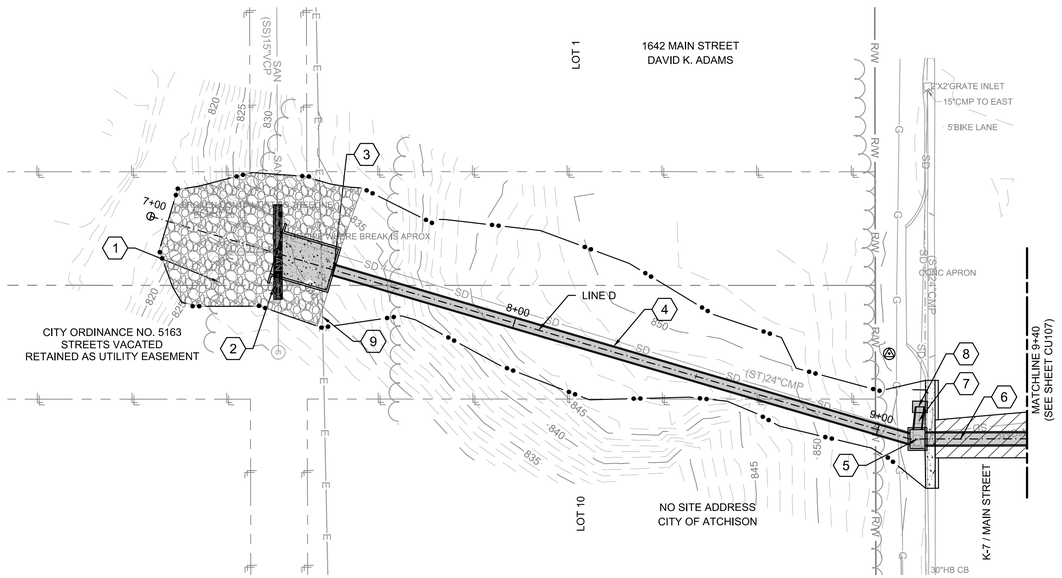
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

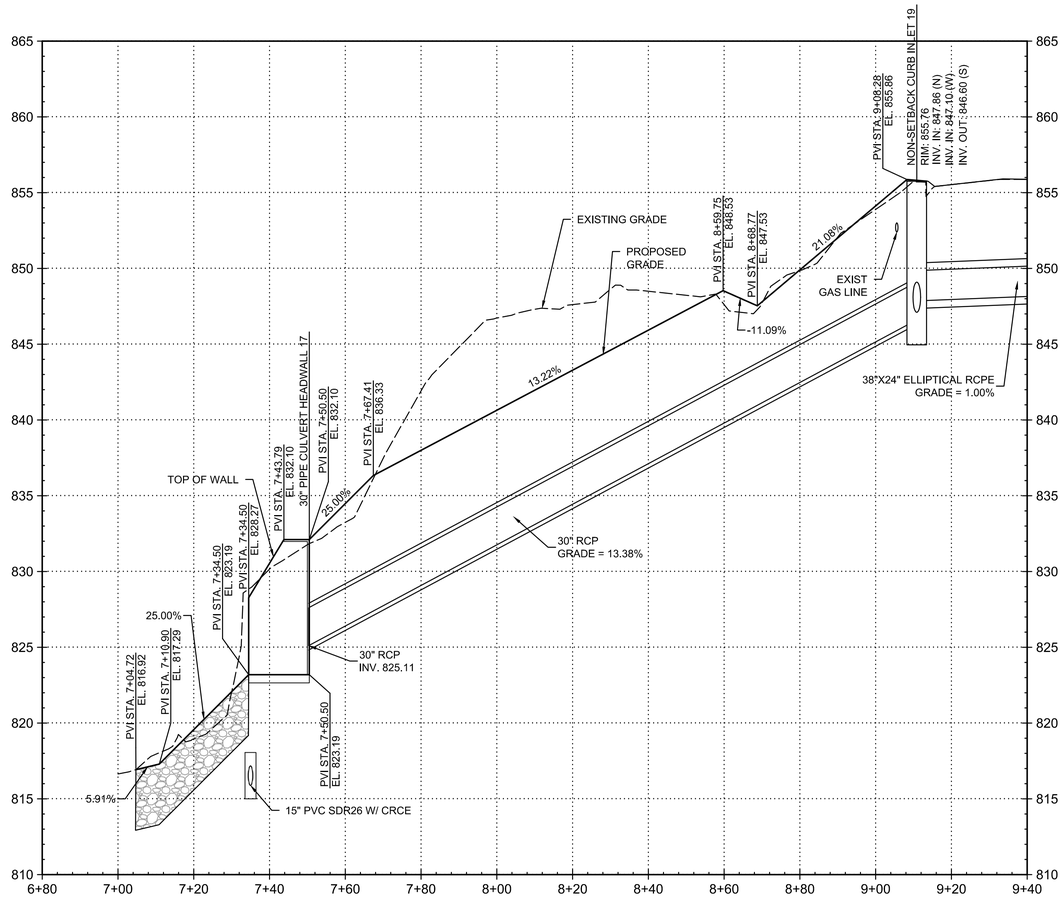
15TH STREET
UTILITY PROFILE

SHEET ID
CU105

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU105_15TH.dwg
9/9/2026 4:12:13 PM

READY TO ADVERTISE





A2 UTILITY PLAN AND PROFILE, LINE D STA. 7+00 TO 9+40

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

PLAN NORTH

0 20' 40'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.

2 SEE SHEET CG102 FOR CIVIL GRADING PLAN.

3. SEE SHEET C-526 FOR FOR TYPICAL SECTIONS, CROSS SECTION SHEETS CU306 & CU307.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE SHEET C-508 FOR DETAIL.

SHEET KEYNOTES

1. APWA RIPRAP (HEAVY STONE). SEE SPECIFICATION 33 40 00. SEE SHEET C-526
2. CONSTRUCT 25 L.F. 15" PVC SDR26 W/ CRCE SEE DETAIL J14 ON SHEET C-522
3. CONSTRUCT PIPE CULVERT HEADWALL 17 STA 7+50.50, LINE D= N:459664.68, E:2118499.36 SEE DETAIL A1 ON SHEET C-520
4. CONSTRUCT 30" RCP SEE DETAIL B2 ON SHEET C-508
5. CONSTRUCT 6'X4' NON-SETBACK CURB INLET 19 STA 9+10.86, LINE D= N:459819.48, E:2118540.89 SEE DETAIL A1 ON SHEET C-512
6. CONSTRUCT 38"X24" RCPE
7. CONSTRUCT 24" RCP
8. CONSTRUCT CONCRETE COLLAR 26 STA 9+11.50, 8.48 ' L LINE D= N:459819.95, E:2118532.39 CONNECT EXISTING 24" CMP SEE DETAIL A8 ON SHEET C-508
9. COORDINATE WITH UTILITIES FOR OVERHEAD SAFETY.

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- RIPRAP

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROWL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
SIZE:	ANSI D
ISSUE DATE:	23 FEBRUARY 2026
SOLICITATION NO.:	TASK ORDER # W912D24F1019
CONTRACT NO.:	W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

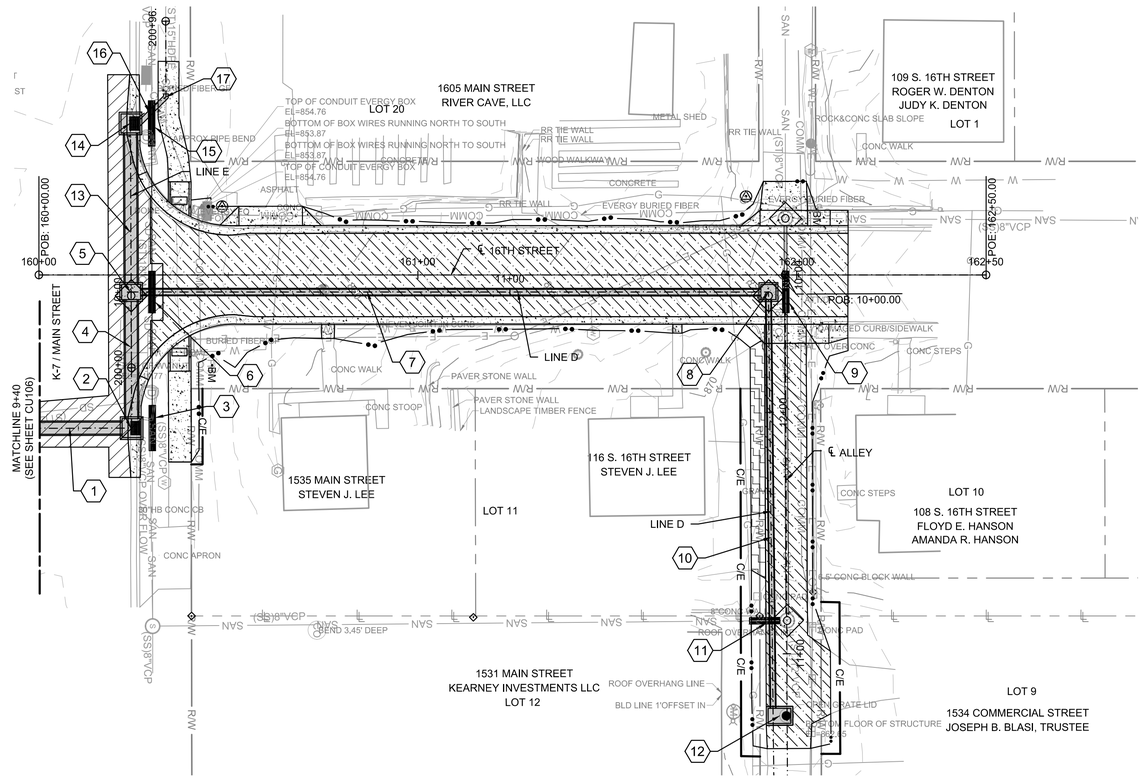
16TH STREET
UTILITY PLAN AND PROFILE

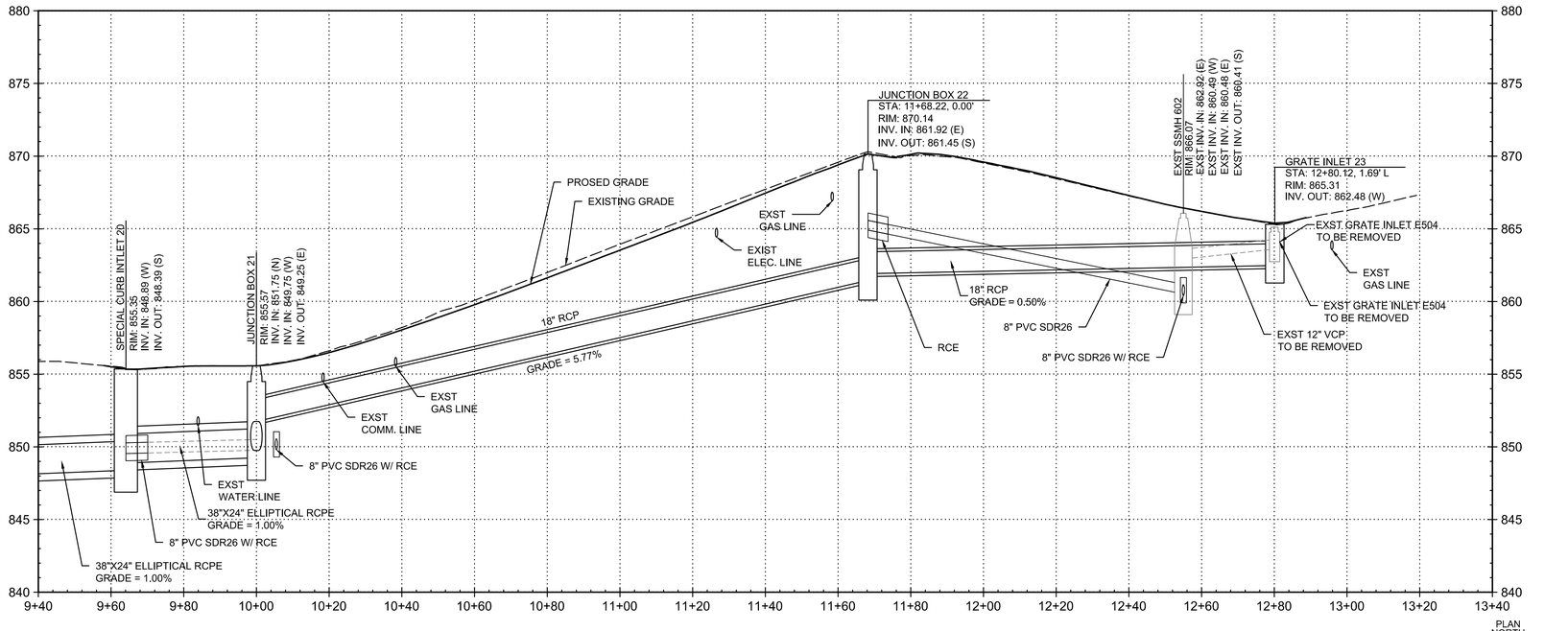
SHEET ID

CU106

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU106-CU108_16TH.dwg
9/9/2026 4:13:17 PM

READY TO ADVERTISE





A2 UTILITY PLAN AND PROFILE, LINE D STA. 9+40 TO 13+40

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

0 20' 40'

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.

2 SEE SHEET CG102 FOR CIVIL GRADING PLAN.

3. SEE SHEET C-526 FOR FOR TYPICAL SECTIONS, CROSS SECTION SHEETS CU306 & CU307.
4. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
5. AREAS DEPICTING PAVEMENT PATCH INCLUDE 12" MINIMUM BENCHING. SEE SHEET C-508 FOR DETAIL.

SHEET KEYNOTES

1. CONSTRUCT 38"X24" RCPE
2. CONSTRUCT 5'X5' SPECIAL CURB INTLET 20 STA 9+64.09, LINE D= STA 160+24.45, 40.43' R ℄ 16TH ST, N:459872.68, E:2118539.74
3. CONSTRUCT 12 L.F. 8" PVC SDR26 W/ RCE SEE DETAIL L7 ON SHEET C-521
4. CONSTRUCT 38"X24" RCPE
5. CONSTRUCT 5'X4' JUNCTION BOX 21 STA 10+00.00, LINE D= STA 200+19.98, LINE E= STA 160+24.32, 4.52' R ℄ 16TH ST, N:459871.85, E:2118503.84 SEE DETAIL A1 ON SHEET C-516 SEE SHEET CU108 FOR ADDITIONAL UTILITY PROFILE
6. CONSTRUCT 12 L.F. 8" PVC SDR26 W/ RCE SEE DETAIL L7 ON SHEET C-521
7. CONSTRUCT 18" RCP
8. CONSTRUCT 4'X4' JUNCTION BOX 22 STA 11+68.22, LINE D= STA 161+92.55, 4.51' R ℄ 16TH ST, N:460040.04, E:2118500.55 SEE DETAIL A1 ON SHEET C-516
9. CONSTRUCT 8" PVC SDR26 W/ RCE SEE PLAN & PROFILE SHEET CU109
10. CONSTRUCT 18" RCP
11. CONSTRUCT 8" PVC SDR26 W/ RCE SEE PLAN & PROFILE SHEET CU109
12. CONSTRUCT 4'X6' GRATE INLET 23 STA 12+80.12, 1.69 L, STA 11+16.38, 2.00' R ℄ ALLEY, N:460045.16, E:2118612.35 CONNECT 18" RCP (W) 6" FROM SOUTH INSIDE FACE SEE DETAIL A1 ON SHEET C-517
13. CONSTRUCT 38"X24" RCPE SEE SHEET CU108 FOR UTILITY PROFILE
14. CONSTRUCT 5'X5' SPECIAL CURB INTLET 24 STA 200+64.48, LINE E= STA 160+24.15, 39.97' L ℄ 16TH ST, N:459870.81, E:2118459.35 SEE SHEET CU108 FOR UTILITY PROFILE SEE DETAIL A1 ON SHEET C-513
15. CONSTRUCT 12 L.F. 8" PVC SDR26 W/ RCE SEE DETAIL L7 ON SHEET C-521
16. CONSTRUCT 15" HDPE SEE SHEET CU108 FOR UTILITY PROFILE
17. CONSTRUC PIPE CONNECTION 25 STA 10+09.18, LINE E= STA 160+33.50, 46.97' L ℄ 16TH ST, N:459880.02, E:2118452.17 CONNECT EXISTING 15 HDPE FIELD VERIFY PIPE LOCATION AND ELEVATION PRIOR TO CONSTRUCTION

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E CONSTRUCTION EASEMENT
- FULL DEPTH ASPHALT REPLACEMENT
- GRAVEL DRIVE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D024F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
UTILITY PLAN AND PROFILE

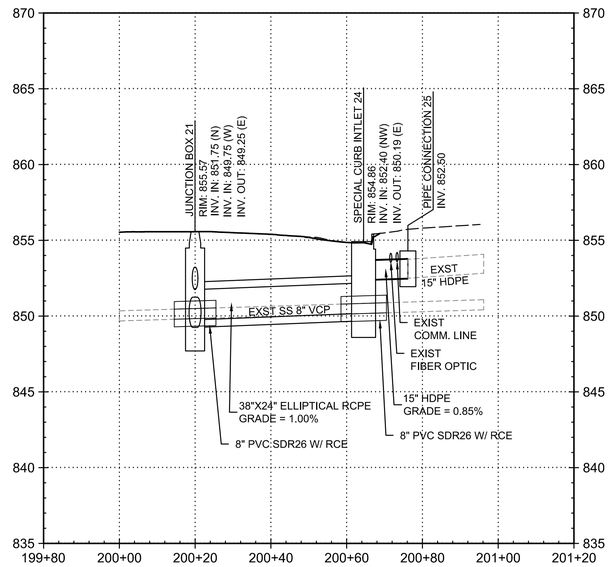
SHEET ID
CU107

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU106-CU108_16TH.dwg
9/9/2026 4:13:41 PM

READY TO ADVERTISE

SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.



E8 UTILITY PROFILE, LINE E STA. 200+20.00 TO 200+76.32

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

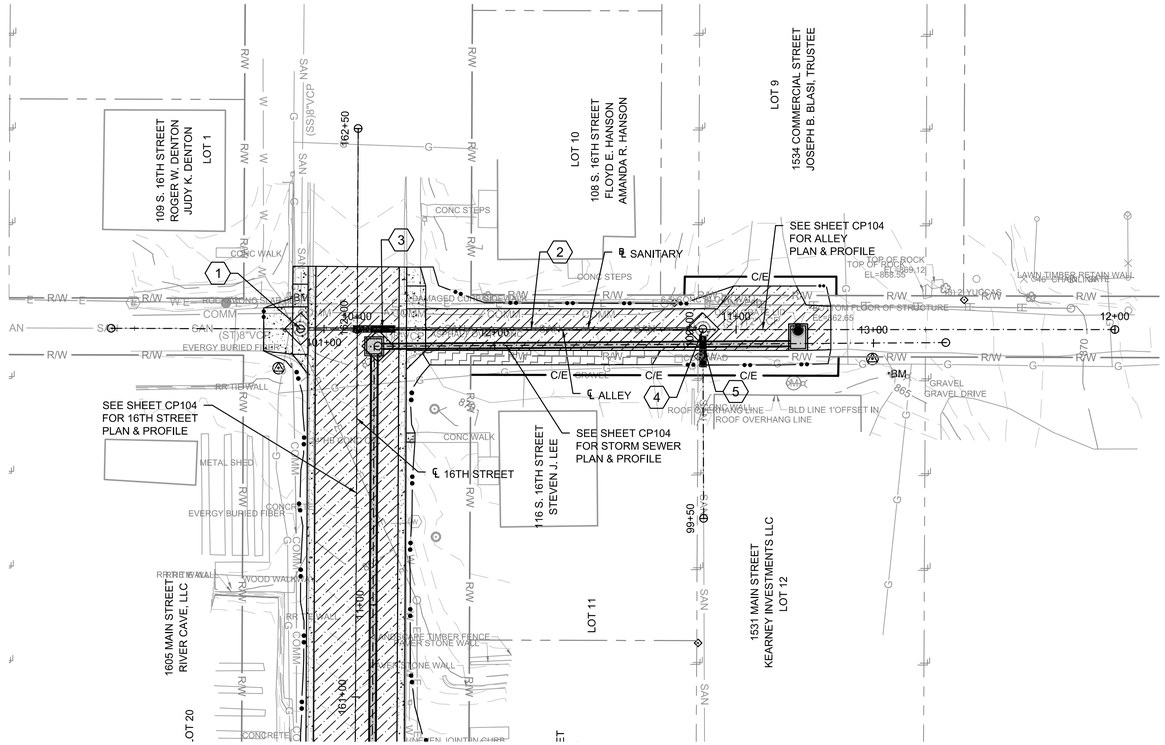
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

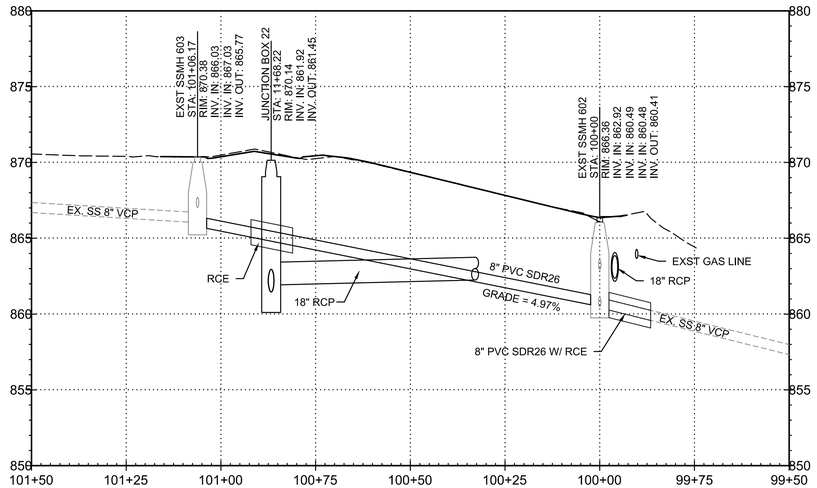
16TH STREET
UTILITY PROFILE

SHEET ID
CU108

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU106-CU108_16TH.dwg
9/9/2026 4:13:53 PM

READY TO ADVERTISE





SHEET NOTES

1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. SEE SHEET CP302- CP303 FOR CROSS SECTIONS
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.

SHEET KEYNOTES

1. EXISTING 4' DIA. SSMH 603 (U.I.P.) STA 101+06.17, LINE = STA 161+97.04, 14.94' L ℄ 16TH ST, N:460044.15, E:2118481.01
   CONNECT 8" PVC SDR26 (E)
   SEE DETAIL D6 ON SHEET C-521
2. CONSTRUCT 8" PVC SDR26
3. CONSTRUCT 12 L.F. RCE
   SEE DETAIL L7 ON SHEET C-521
4. EXISTING 4' DIA. SSMH 602 (U.I.P.) STA 100+00.00, LINE = STA 10+91.23, 0.16' L ℄ ALLEY, N:460046.71, E:2118587.15
   ADJUST SSMH 602 TO GRADE, EXISTING ELEV.=866.07, PROPOSED ELEV.=866.36
   CONNECT 8" PVC SDR26 (W)
   CONNECT 8" PVC SDR26 W/RCE (S)
   SEE DETAILS D1 & D6 ON SHEET C-521
5. CONSTRUCT 8 L.F. 8" PVC SDR26 W/ RCE
   SEE DETAIL L7 ON SHEET C-521

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- CONSTRUCTION EASEMENT
- FULL DEPTH ASPHALT REPLACEMENT
- GRAVEL DRIVE

PLAN NORTH

0 20' 40'

A2 UTILITY PLAN & PROFILE, STA. 101+06.17 TO 100+00

SCALE: HORIZONTAL 1"=20', VERTICLE 1"=5'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

ALLEY
UTILITY PLAN AND PROFILE

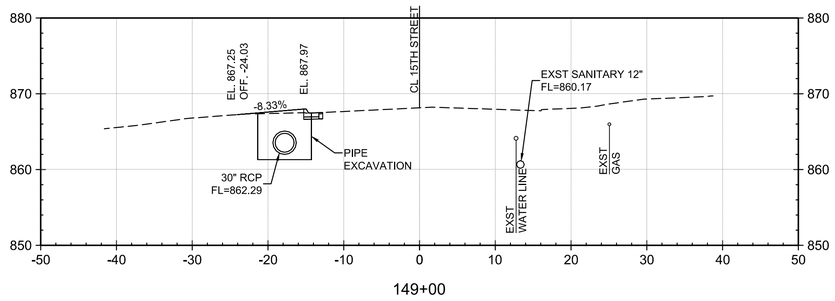
SHEET ID
CU109

READY TO ADVERTISE

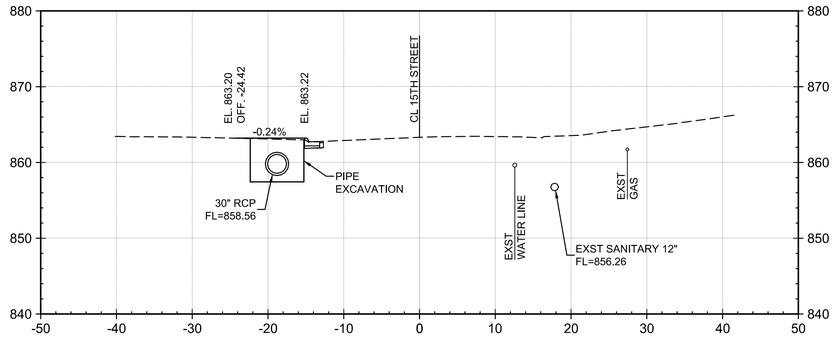
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU109_ALLEY SAN.dwg
9/9/2026 4:14:59 PM

SHEET NOTES

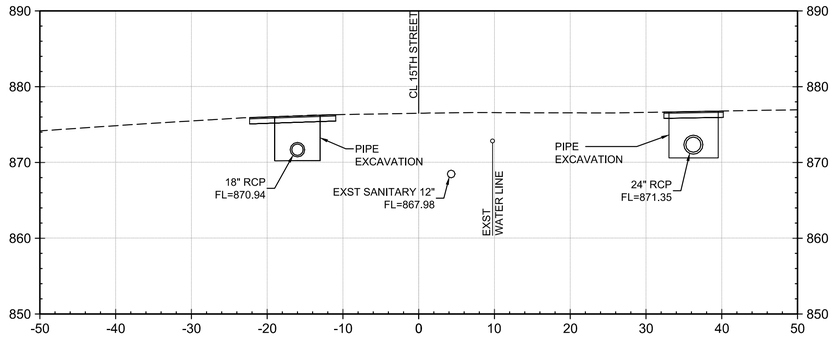
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



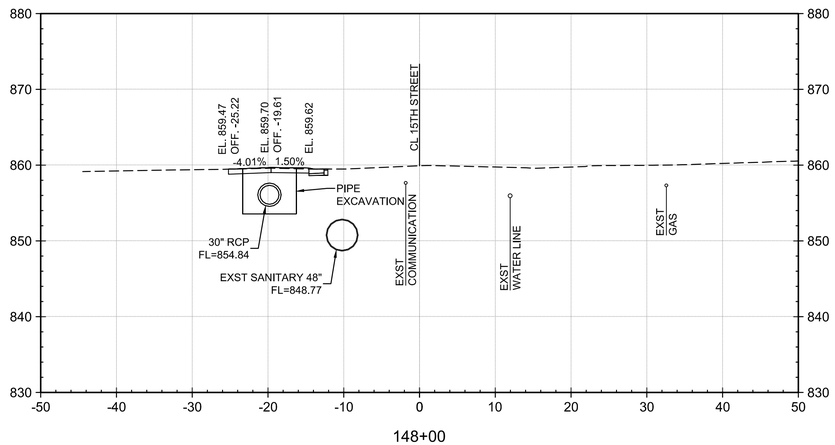
149+00



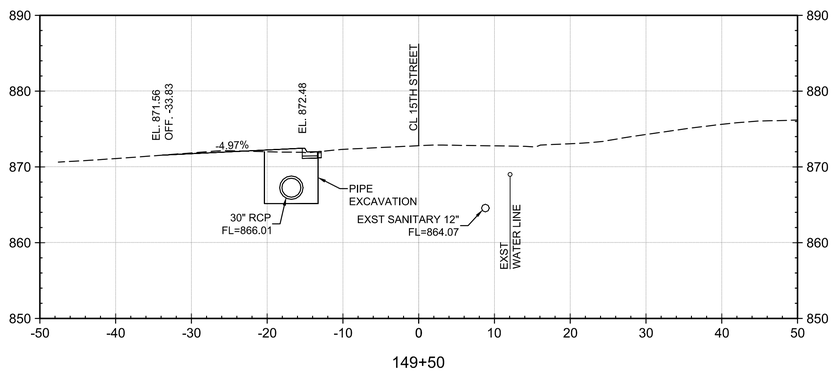
148+50



150+00



148+00



149+50

A1 UTILITY CROSS SECTIONS, 15TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	ISSUE DATE:
M. CROWL	23 FEBRUARY 2026
DRAWN BY:	SOLICITATION NO.:
T. ROUSH	TASK ORDER # W912D24F1019
CHECKED BY:	CONTRACT NO.:
J. DAVIS	W912D0219D1012
SUBMITTED BY:	
S. BRISCOE	
SIZE:	
ANSI D	

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
UTILITY CROSS SECTIONS

SHEET ID

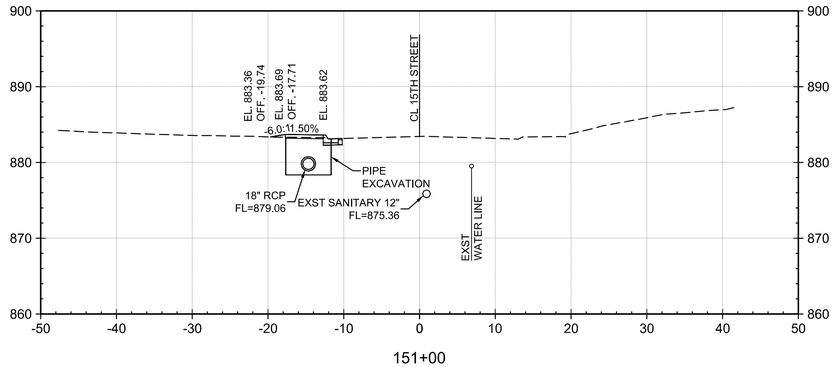
CU301

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU301-CU305_15TH.dwg
9/9/2026 4:15:46 PM

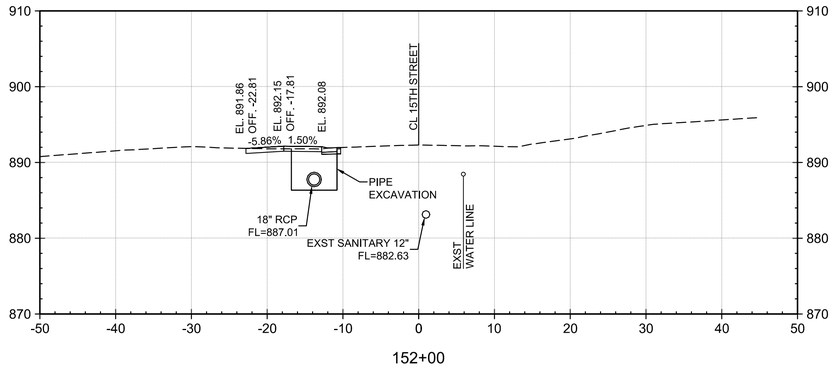
READY TO ADVERTISE

SHEET NOTES

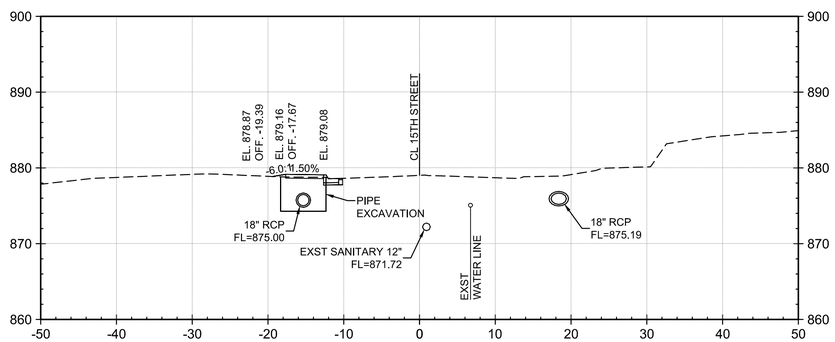
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



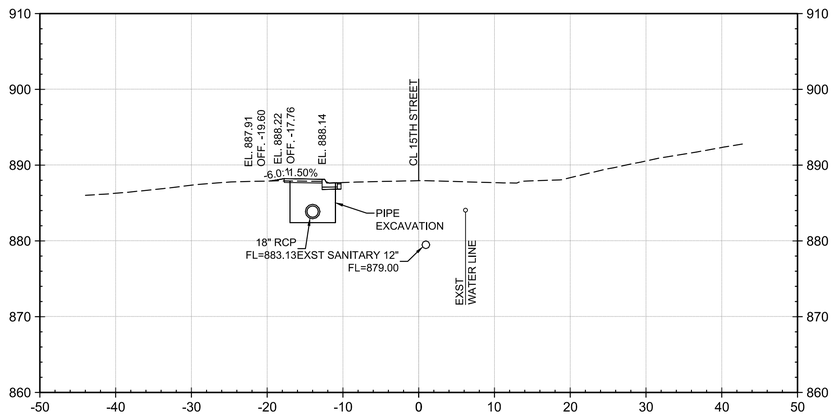
151+00



152+00



150+50



151+50

A1 UTILITY CROSS SECTIONS, 15TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	ISSUE DATE:
M. CROWL	23 FEBRUARY 2026
DRAWN BY:	SOLICITATION NO.:
T. ROUSH	TASK ORDER # W912D24F1019
CHECKED BY:	CONTRACT NO.:
J. DAVIS	W912DQ19D1012
SUBMITTED BY:	
S. BRISCOE	
SIZE:	
ANSI D	

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
UTILITY CROSS SECTIONS

SHEET ID

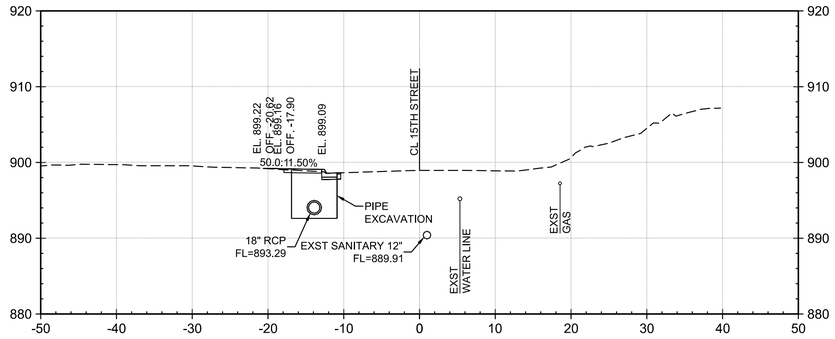
CU302

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU301-CU305_15TH.dwg
9/9/2026 4:15:58 PM

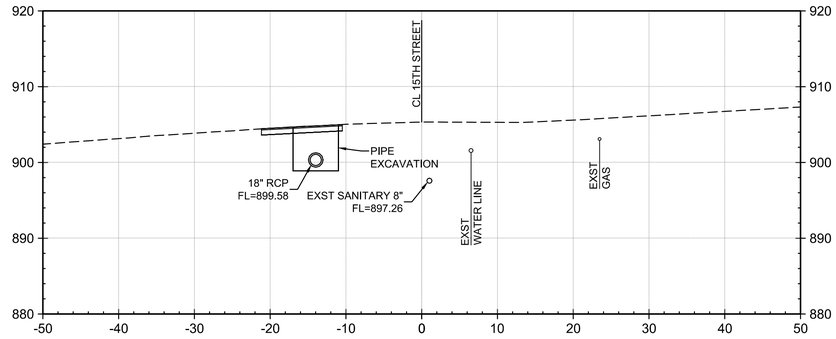
READY TO ADVERTISE

SHEET NOTES

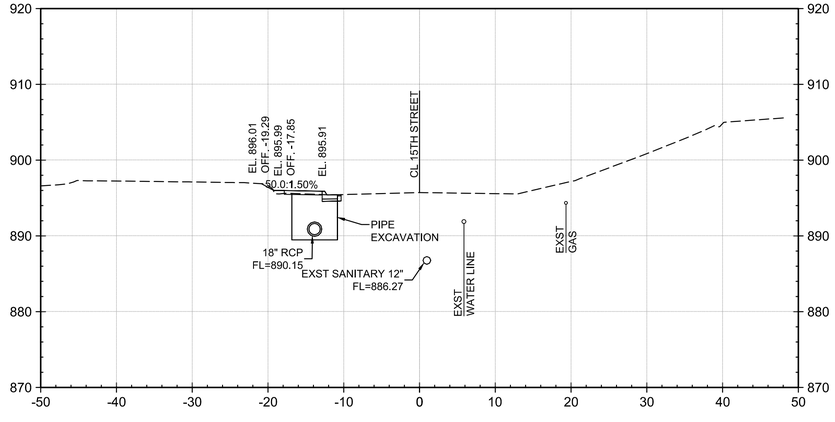
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



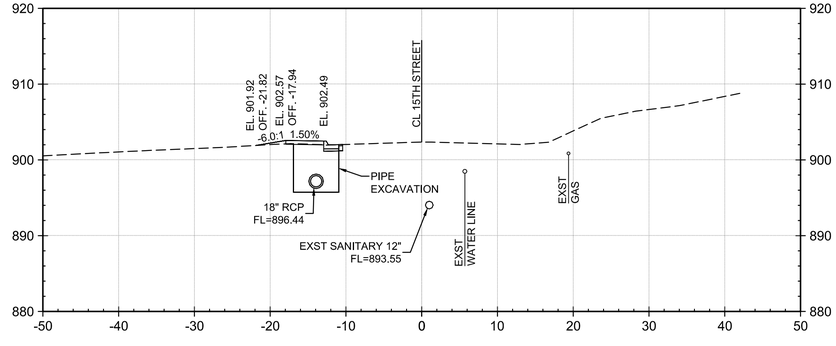
153+00



154+00



152+50



153+50

A1 UTILITY CROSS SECTIONS, 15TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU301-CU305_15TH.dwg
9/9/2026 4:16:10 PM

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

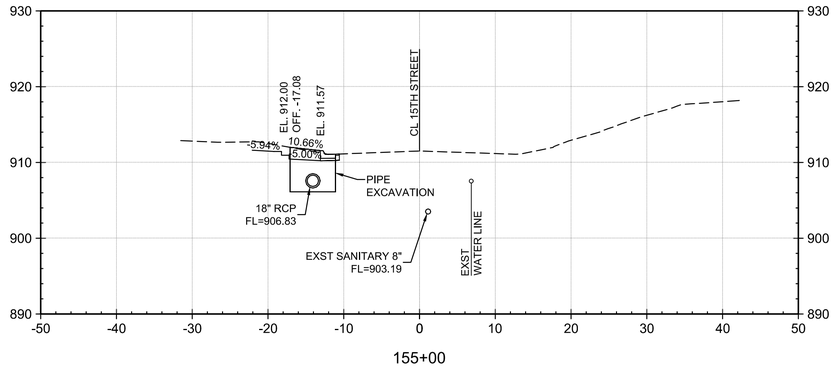
15TH STREET
UTILITY CROSS SECTIONS

SHEET ID
CU303

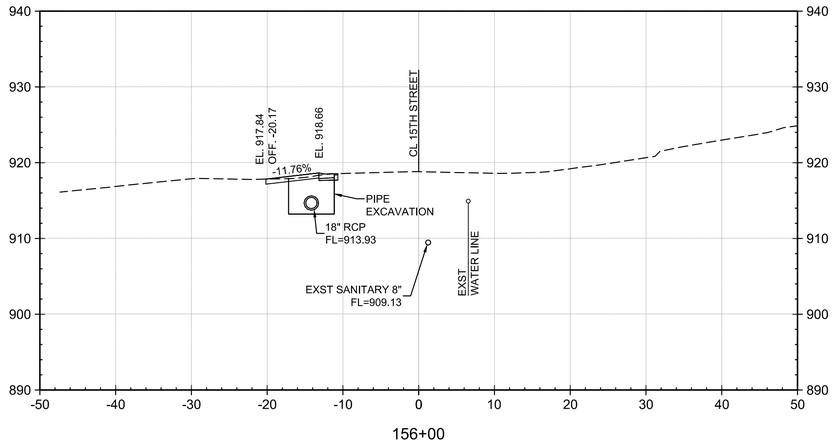
READY TO ADVERTISE

SHEET NOTES

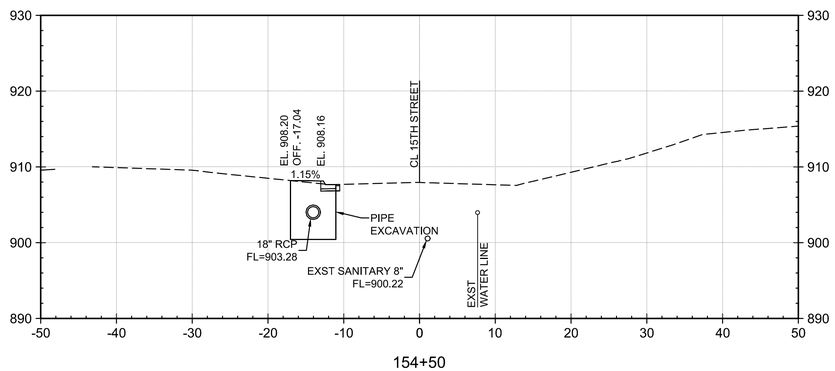
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



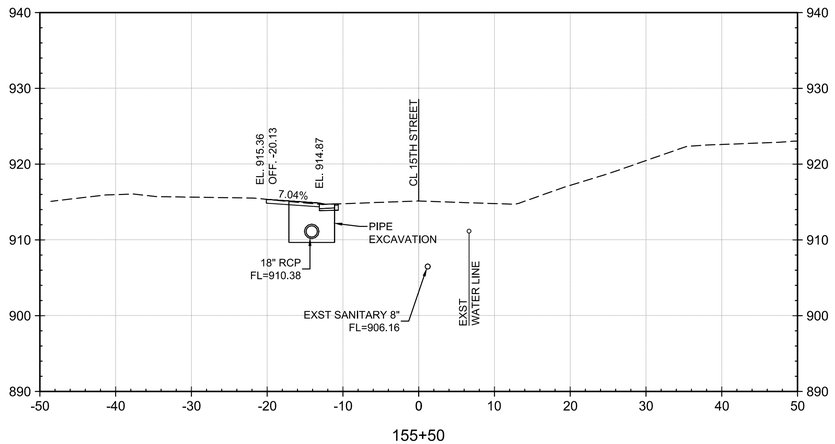
155+00



156+00



154+50



155+50

A1 UTILITY CROSS SECTIONS, 15TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU301-CU305_15TH.dwg
9/9/2026 4:16:20 PM

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

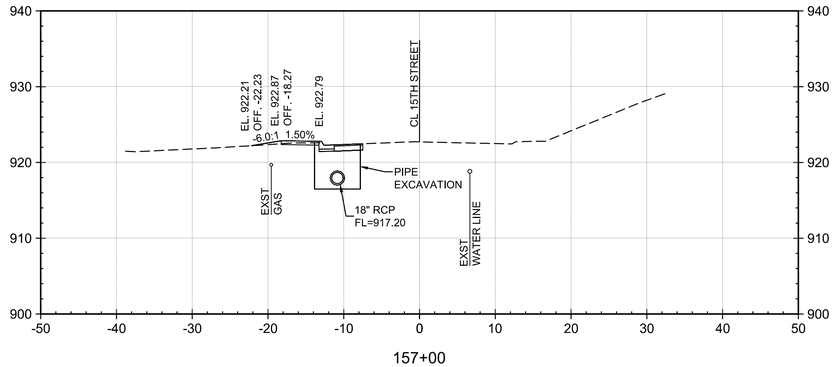
15TH STREET
UTILITY CROSS SECTIONS

SHEET ID
CU304

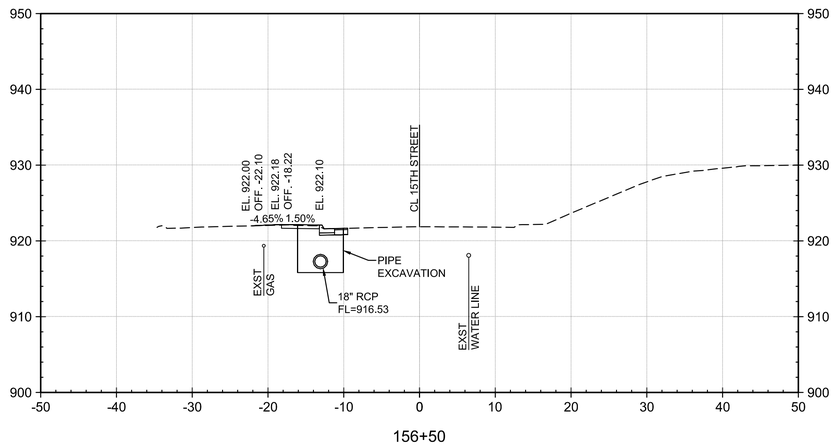
READY TO ADVERTISE

SHEET NOTES

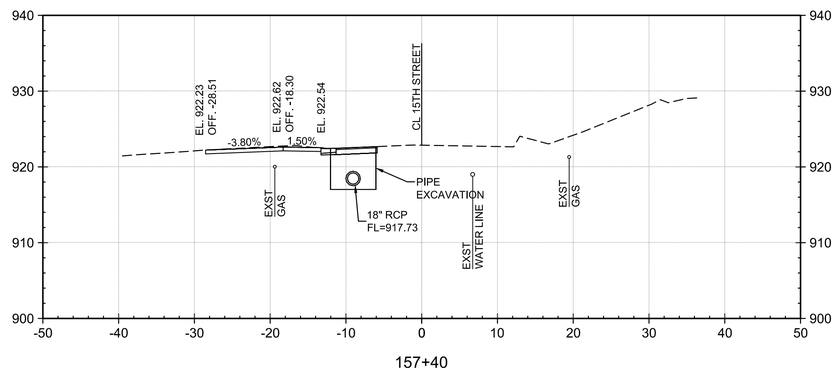
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR.
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



157+00



156+50



157+40

A1 UTILITY CROSS SECTIONS, 15TH STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU301-CU305_15TH.dwg
9/9/2026 4:16:32 PM

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

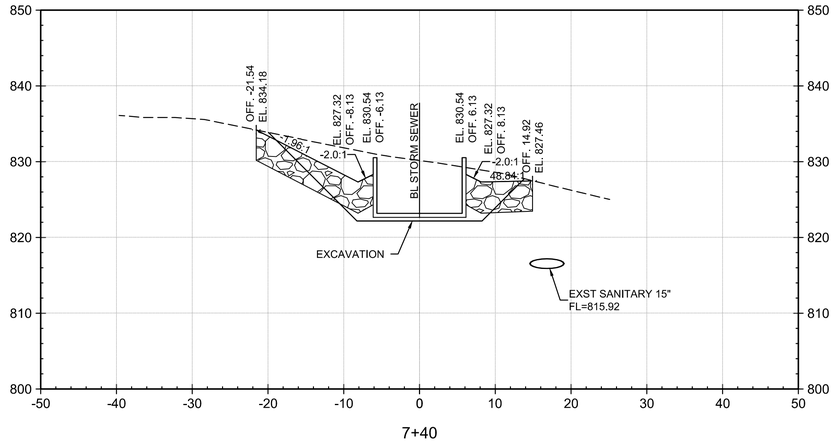
15TH STREET
UTILITY CROSS SECTIONS

SHEET ID
CU305

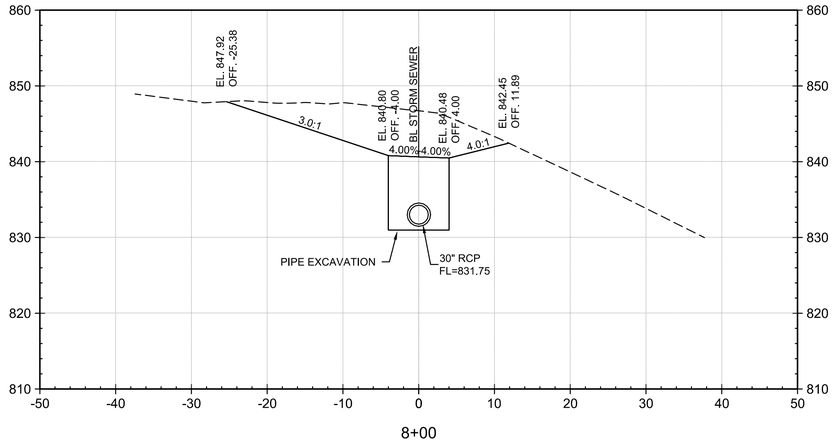
READY TO ADVERTISE

SHEET NOTES

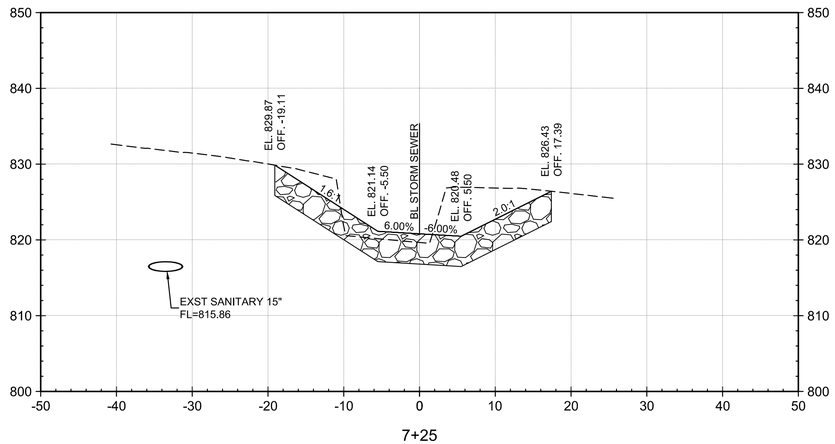
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



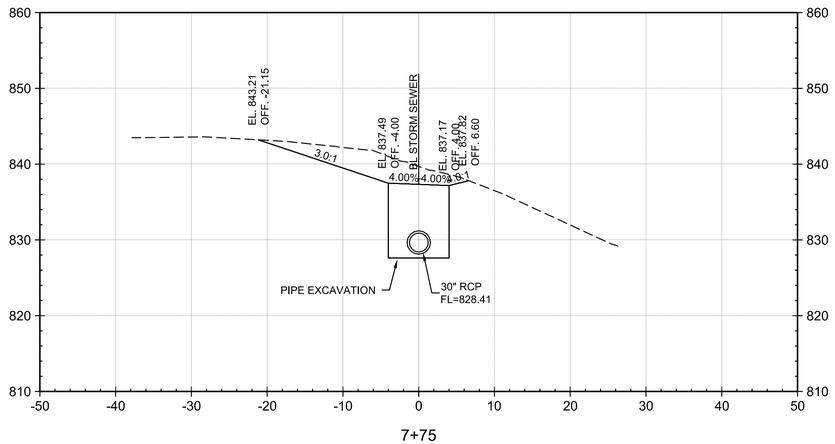
7+40



8+00



7+25



7+75

A1 UTILITY CROSS SECTIONS, SOUTH OF K7 \ MAIN STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

SOUTH OF K7 \ MAIN STREET
UTILITY CROSS SECTIONS

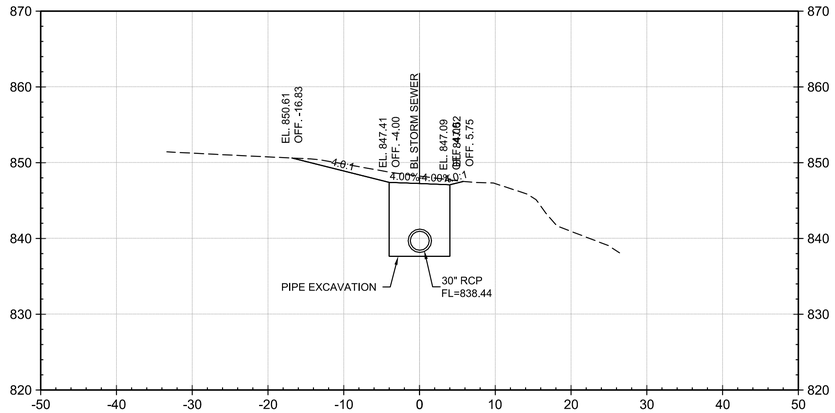
SHEET ID
CU306

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU306-CU307_16TH.dwg
9/9/2026 4:17:17 PM

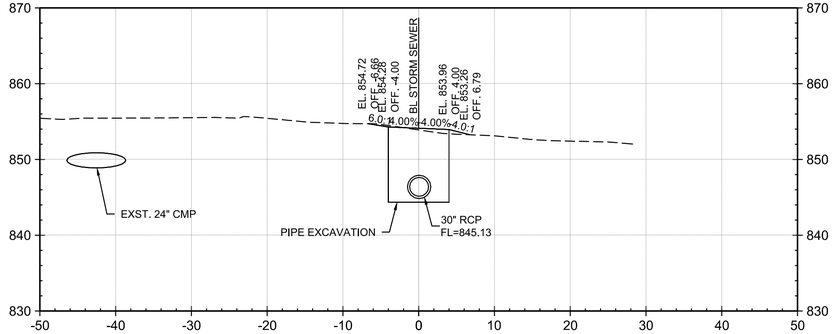
READY TO ADVERTISE

SHEET NOTES

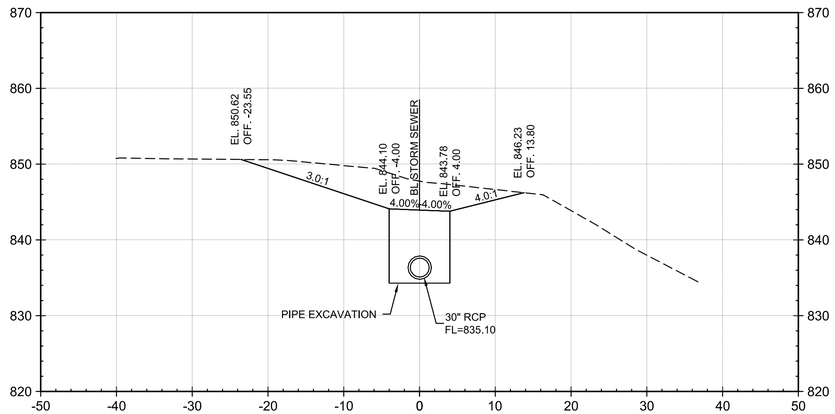
1. SEE SPECIFICATION 33 40 00 FOR PIPE CLASS REQUIREMENTS.
2. CONTRACTOR IS RESPONSIBLE FOR UTILITY COORDINATION ACTIVITIES WITH ALL UTILITIES IN THE PROJECT AREA.
3. CROSS SECTIONS DEPICTING "PIPE EXCAVATION" ASSUME TRENCH BOX CONSTRUCTION. SEE SPECIFICATION 31 10 00 FOR INSTRUCTIONS ON EXCAVATION AND TRENCHING PLAN TO BE PROVIDED BY THE CONTRACTOR
4. SEE SHEET C-501 THRU C-522 FOR PIPE, CURB AND GUTTER, PAVEMENT, SIDEWALK, STAIRS, INLET, AND MANHOLE DETAILS.
5. SEE SHEET C-509 FOR STANDARD RCP INSTALLATION AND BACK FILL DETAILS.



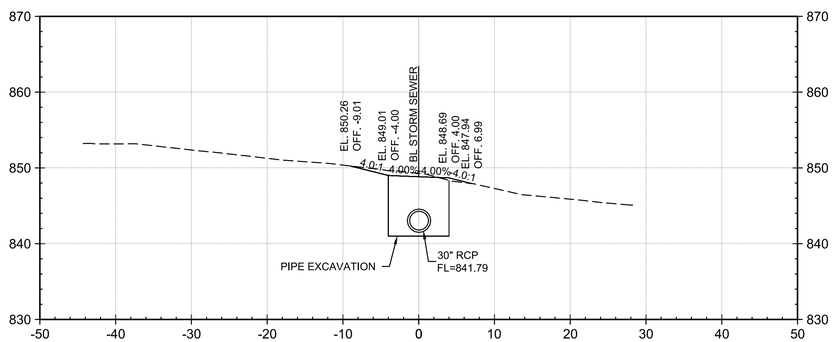
8+50



9+00



8+25



8+75

A1 UTILITY CROSS SECTIONS, SOUTH OF K7 \ MAIN STREET

SCALE: HORIZONTAL 1"=10', VERTICLE 1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	ISSUE DATE:
M. CROWL	23 FEBRUARY 2026
DRAWN BY:	SOLICITATION NO.:
T. ROUSH	TASK ORDER # W912D24F1019
CHECKED BY:	CONTRACT NO.:
J. DAVIS	W912D0219D1012
SUBMITTED BY:	
S. BRISCOE	
SIZE:	
ANSI D	

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

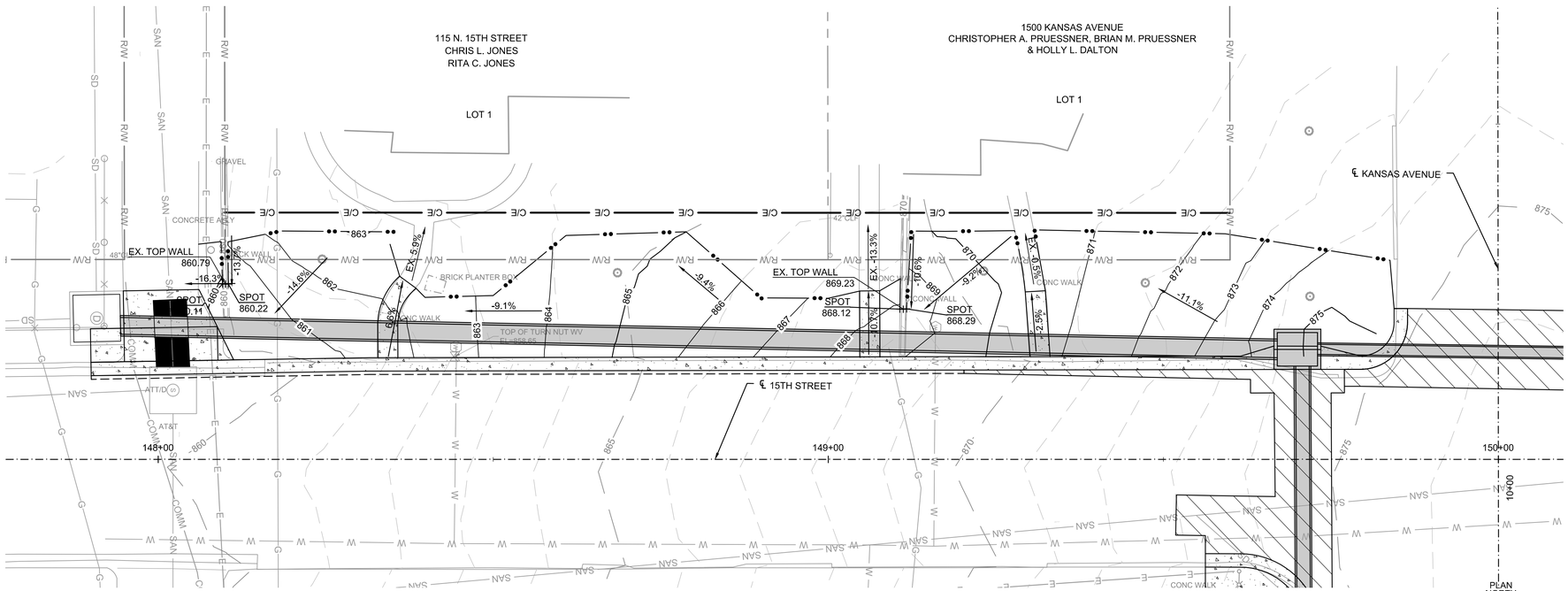
SOUTH OF K7 \ MAIN STREET
UTILITY CROSS SECTIONS

SHEET ID

CU307

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CU306-CU307_16TH.dwg
9/9/2026 4:17:27 PM

READY TO ADVERTISE



E1 CIVIL GRADING, 15TH STREET

SCALE:1"=10'

SHEET NOTES

SHEET KEYNOTES

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- C\E CONSTRUCTION EASEMENT
- 875 PROPOSED INDEX CONTOURS AT 5 FT. INTERVAL
- 874 PROPOSED INTERMEDIATE CONTOURS AT 1 FT. INTERVAL
- 875 EXISTING INDEX CONTOURS AT 5 FT. INTERVAL
- EXISTING INTERMEDIATE CONTOURS AT 1 FT. INTERVAL

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

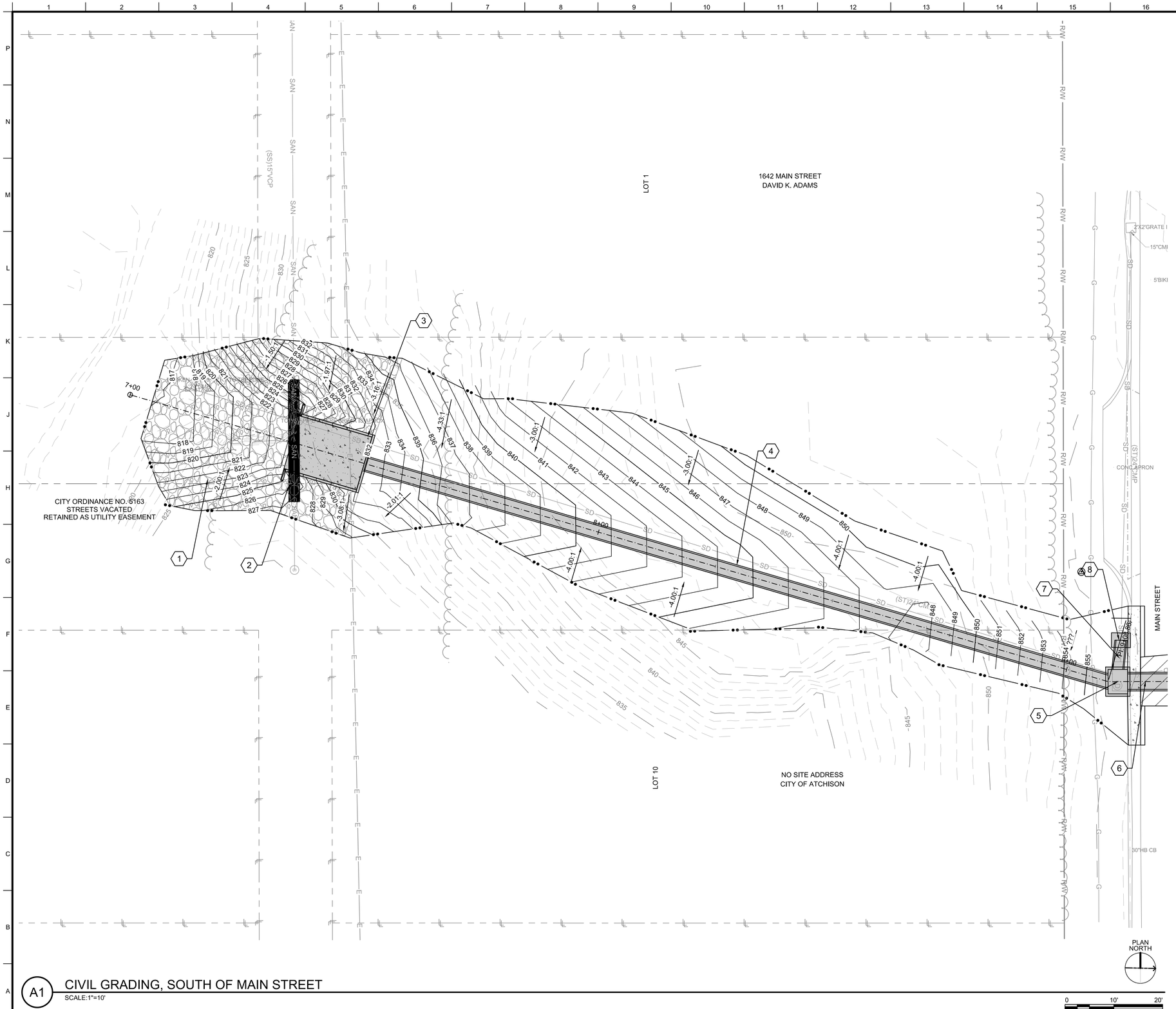
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
CIVIL GRADING PLAN

SHEET ID
CG101

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CG101_15TH.dwg
9/9/2026 4:19:05 PM

READY TO ADVERTISE



SHEET NOTES

1. SEE SHEET C-526 FOR FOR TYPICAL SECTIONS, CROSS SECTION SHEETS CU306 & CU307.

SHEET KEYNOTES

1. APWA RIPRAP (HEAVY STONE). SEE SPECIFICATION 33 40 00. SEE SHEET C-526
2. CONSTRUCT 15" PVC SDR26 w/CRCE
3. CONSTRUCT PIPE CULVERT HEADWALL 17 SEE SHEET C-520
4. CONSTRUCT 30" RCP
5. CONSTRUCT 6'X4' NON-SETBACK CURB INTLET 19
6. CONSTRUCT 38" X 24" RCPE CONSTRUCT 24" RCP
7. CONSTRUCT CONCRETE COLLAR 26

LEGEND

- PROPOSED CONCRETE DRIVE, SIDEWALK, AND CURB
- PAVEMENT PATCHING
- PROPOSED DRAINAGE STRUCTURE
- PROPOSED STORM DRAINAGE
- APPROXIMATE GRADING LIMITS
- CENTER LINE
- RIPRAP
- 875 — PROPOSED INDEX CONTOURS AT 5 FT. INTERVAL
- 874 — PROPOSED INTERMEDIATE CONTOURS AT 1 FT. INTERVAL
- 875 — EXISTING INDEX CONTOURS AT 5 FT. INTERVAL
- EXISTING INTERMEDIATE CONTOURS AT 1 FT. INTERVAL

A1 CIVIL GRADING, SOUTH OF MAIN STREET
SCALE:1"=10'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL	ISSUE DATE: 23 FEBRUARY 2026
DRAWN BY: T. ROUSH	SOLICITATION NO.: TASK ORDER # W912D24F1019
CHECKED BY: J. DAVIS	CONTRACT NO.: W912DQ19D1012
SUBMITTED BY: S. BRISCOE	
SIZE: ANSI D	

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

SOUTH OF MAIN STREET
CIVIL GRADING PLAN

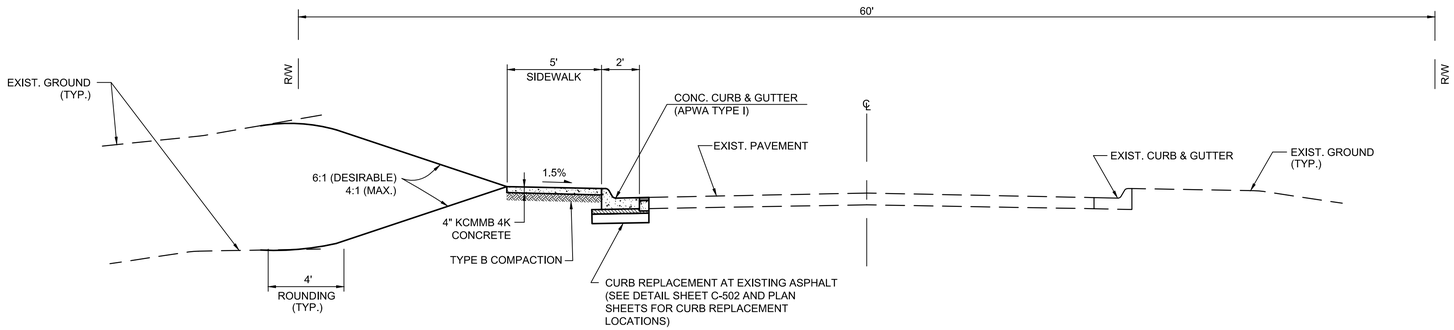
SHEET ID
CG102

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_CG102_16TH.dwg
9/9/2026 4:20:33 PM

READY TO ADVERTISE

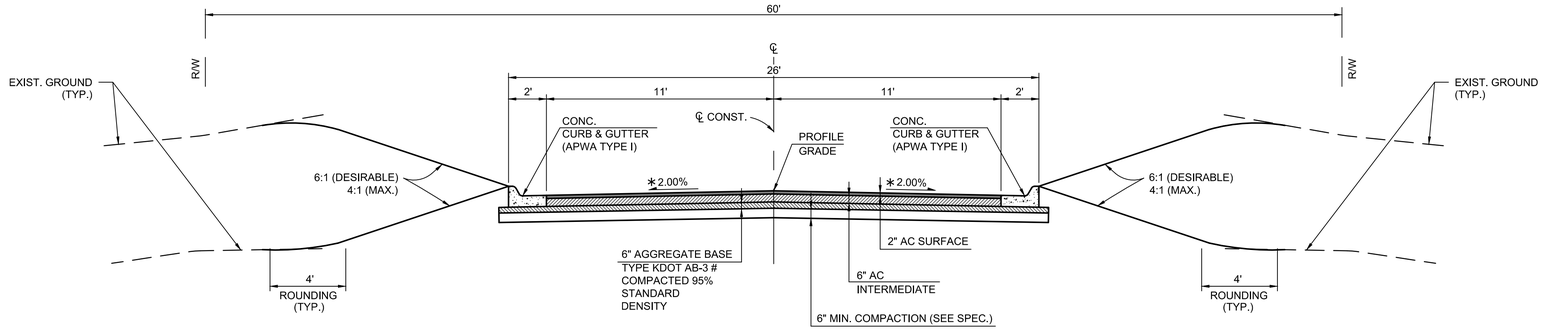
SHEET NOTES

1. NEW SIDEWALK MAY BE ADJUSTED TO AVOID CONFLICTS WITH EXISTING UTILITIES, TREES OR LANDSCAPING, AS APPROVED BY THE CITY OF ATCHISON AND THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
2. SEE PLAN SHEETS FOR CURB REPLACEMENT AND SIDEWALK.



L1 SECTION - 15TH STREET (KANSAS AVE. TO ATCHISON ST.) FROM STA. 147+89.96 TO 158+14.86

SCALE: NOT TO SCALE

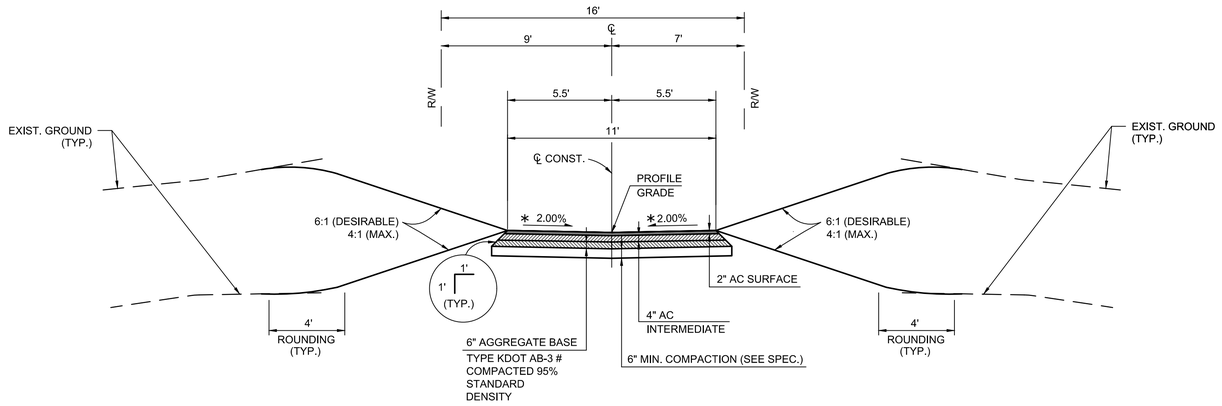


G1 SECTION - 16TH STREET (MAIN ST. TO ALLEY) FROM STA. 160+18.12 TO 162+13.53

SCALE: NOT TO SCALE

∗ = CROSS SLOPE MAY VARY, SEE CROSS SECTIONS

= INCLUDES 2" TEMPORARY SURFACING MATERIAL TO REMAIN IN PLACE AS SPECIFIED.



A3 SECTION - ALLEY (EAST OF 16TH ST.) FROM STA. 10+18.05 TO 11+22.02

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

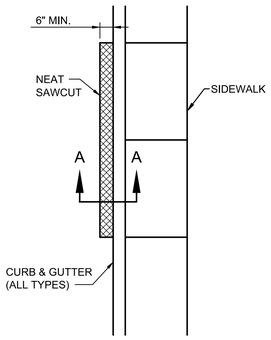
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

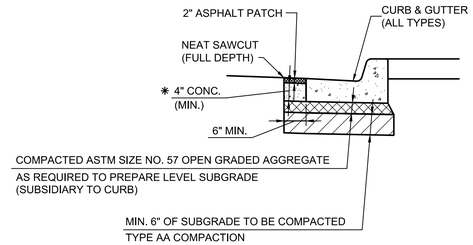
SHEET ID
C-501

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C501-TYP.dwg
9/9/2026 4:21:00 PM

READY TO ADVERTISE



PLAN VIEW



SECTION A-A
(CURB & GUTTER)

✳ CONCRETE PATCH SHALL BE A MINIMUM OF 4" THICK OR MATCH THE THICKNESS OF THE ADJACENT PAVEMENT. CONCRETE AND ASPHALT ARE SUBSIDIARY TO THE CURB.

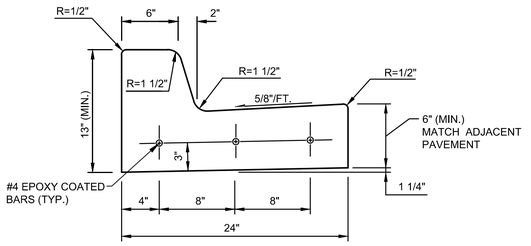
NOTE:

THIS DETAIL SHALL NOT BE USED FOR CURB REPLACEMENT ADJACENT TO FULL DEPTH ASPHALT PATCH, CONCRETE PAVEMENT OR FULL PAVEMENT RECONSTRUCTION.

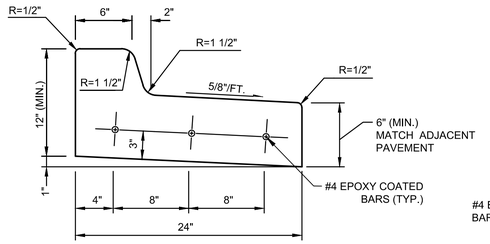
CURB REPLACEMENT AT EXISTING ASPHALT

CURB REPLACEMENT NOTES:

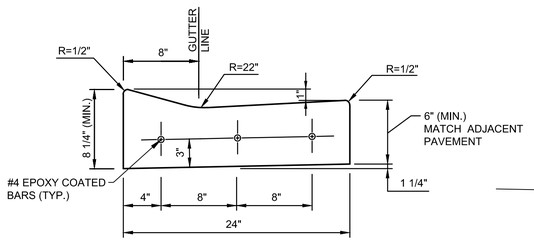
1. 3/4" PREMOLDED EXPANSION JOINTS SHALL BE PLACED AT POINTS OF CURVATURE, CURB RETURNS, CURB INLETS, AND AT 150' CENTERS. THREE 24" EPOXY COATED SMOOTH #4 BARS AT 8" ON CENTER SHALL BE USED ACROSS THESE JOINTS AND SUPPORTED ON FABRICATED BAR SUPPORTS. THE #4 BAR SHALL BE PREGREASED. CONTRACTION JOINTS SHALL BE 1-1/2" DEEP, AND PLACED AT APPROXIMATELY 10' INTERVALS EQUALLY SPACED BETWEEN EXPANSION JOINTS.
2. THREE 24" EPOXY COATED SMOOTH #4 BARS WITH ½" PREMOLDED EXPANSION SHALL BE PLACED WHERE NEW CURB MATCHES EXISTING CURB.
3. STEEL REINFORCING NOT REQUIRED WHEN CURB AND GUTTER IS PLACED ON A MINIMUM 4" THICK ASPHALT BASE.
4. ALL REINFORCING STEEL SHALL BE SUPPORTED ON FABRICATED STEEL BAR SUPPORTS AT 3'-0" MAXIMUM SPACING.



APWA TYPE I CURB AND GUTTER (24")
(NOT TO SCALE)



APWA TYPE I (DRY) CURB AND GUTTER (24")
(NOT TO SCALE)

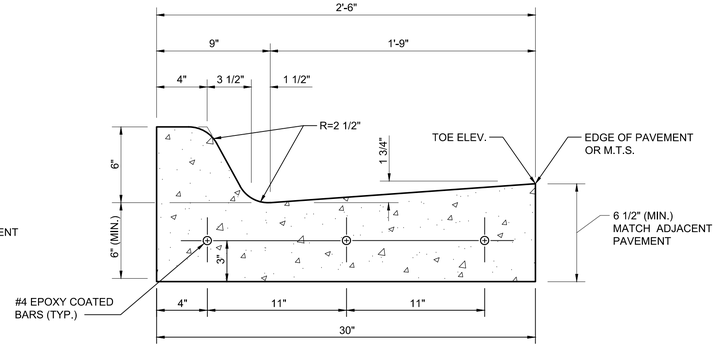


APWA TYPE II CURB AND GUTTER (24")
(NOT TO SCALE)

CURB AND GUTTER NOTES:

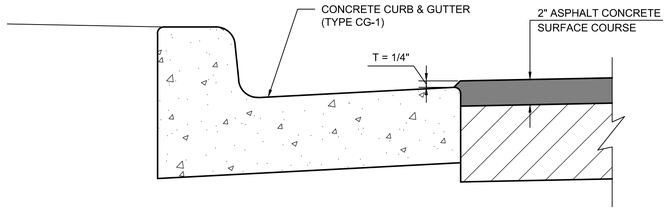
1. JOINTS IN THE CURB AND GUTTER SHALL BE FILLED WITH THE SAME MATERIAL USED FOR THE PAVEMENT JOINT.
2. WHERE CURB AND GUTTER ABUTS ASPHALT PAVEMENT, CONTRACTION JOINTS SHALL BE PLACED AT 10' CENTERS UNLESS DIRECTED OTHERWISE BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
3. WHERE CURB AND GUTTER ABUTS CONCRETE PAVEMENT, CONTRACTION JOINTS IN THE CONCRETE PAVEMENT SHALL BE CONTINUED THROUGH CURB AND GUTTER. NO OTHER PLANES OF WEAKNESS SHALL BE REQUIRED.
4. INSTALL #5 X 30" REBARS AT 30" CTRS. WHERE CURB AND GUTTER ABUTS NEW OR EXISTING CONCRETE PAVEMENT.
5. PROVIDE A 4 FT. TRANSITION FROM FULL CURB HEIGHT TO NO CURB HEIGHT AT LOCATIONS WHERE THE END OF THE CURB AND GUTTER DOES NOT CONNECT TO AN EXISTING CURB OR A STRUCTURE.

(NOT TO SCALE)



T/C ELEV. = TOE ELEV. + 0.125'

COMBINED CURB & GUTTER - TYPE III (2'-6" WIDTH)
(NOT TO SCALE)



EDGE OF PAVEMENT AT GUTTER LIP DETAIL
(NOT TO SCALE)

A1 CURB AND GUTTER DETAILS

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

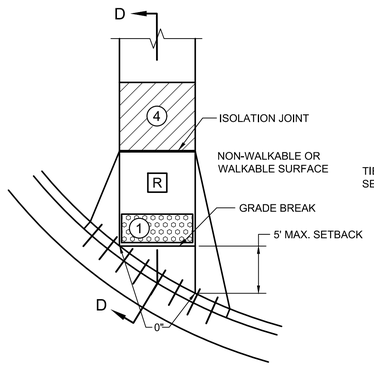
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

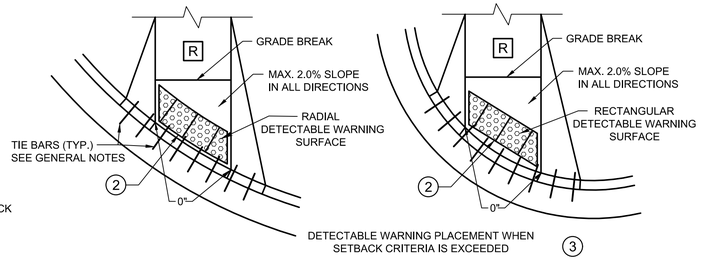
SHEET ID
C-502

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C502-CURB.dwg
9/9/2026 4:21:26 PM

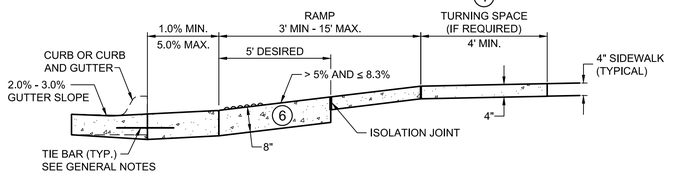
READY TO ADVERTISE



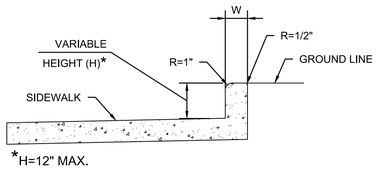
STANDARD ONE-WAY DIRECTIONAL



ONE-WAY DIRECTIONAL
WITH DETECTABLE WARNING AT BACK OF CURB



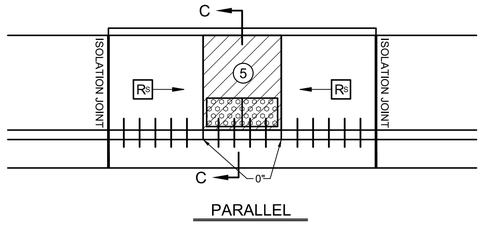
SECTION D-D



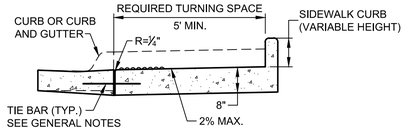
SIDEWALK CURB	
CURB HEIGHT	CURB WIDTH
*(H)	(W)
<6"	4"
>6"	6"

SIDEWALK CURB DETAIL

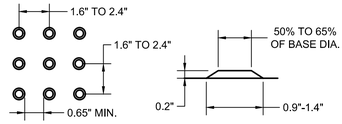
(NOT TO SCALE)



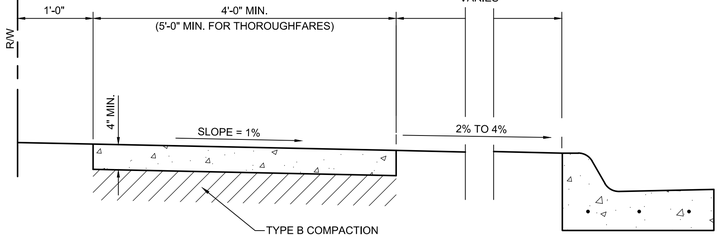
PARALLEL



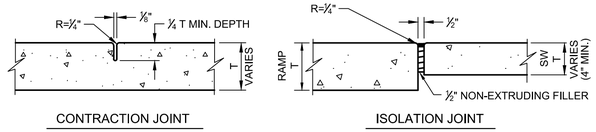
SECTION C-C
PARALLEL/ DEPRESSED CORNER



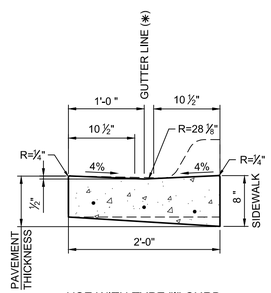
TRUNCATED DOME DIMENSIONS



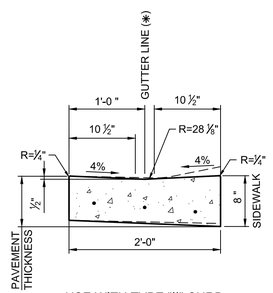
STANDARD SIDEWALK



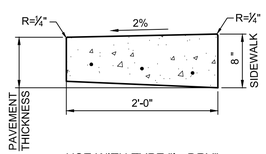
JOINT DETAILS



USE WITH TYPE "I" CURB



USE WITH TYPE "II" CURB



USE WITH TYPE "I - DRY"
OR TYPE "II - DRY" CURB

✱ - 0.5%MINIMUM POSITIVE SLOPE IN GUTTER LINE SHALL BE MAINTAINED THROUGH TRANSITION FROM STANDARD CURB TO RAMP CURB

STREET CURB DETAIL AT RAMP

NOTE: SEE STANDARD CURB DETAIL FOR ADDITIONAL REQUIREMENTS

SIDEWALK RAMP ELEMENTS GENERAL REQUIREMENTS:

RAMP (REQUIRED TO TRANSITION ELEVATION): MAX. RUNNING SLOPE - 8.33%
MAX. CROSS SLOPE - 2% (1.5% PREFERRED)
MIN. WIDTH - 4'

TURNING SPACE (REQUIRED TO CHANGE DIRECTION OF TRAVEL): MAX. RUNNING SLOPE - 2%
MAX. CROSS SLOPE - 2% (1% PREFERRED)
MIN. WIDTH - 4'

DETECTABLE WARNING SURFACE: WIDTH EQUALS RAMP OPENING WIDTH @ CURB
MIN. LENGTH - 2'
DOMES SHOULD BE ALIGNED WITH
DIRECTION OF TRAVEL.

CAST IRON DETECTABLE WARNING PLATE NOTES:

1. SEE PRE-APPROVED MATERIALS LIST FOR PRODUCTS.
2. DETECTABLE WARNING SURFACE TO EXTEND A MINIMUM OF 2' IN DIRECTION OF TRAVEL.
3. DETECTABLE WARNING PLATES PLACED RADIALLY BEHIND THE CURB SHALL BE RADIUS PLATES.
4. DETECTABLE WARNING PLATES SHALL BE INSTALLED PER MANUFACTURER'S RECOMMENDATIONS AND CLEANED OF EXCESS MATERIAL AFTER INSTALLATION.
5. VOID IN VENT HOLES SHALL BE FILLED WITH JOINT SEALANT (ASTM C 920 CLASS 35)

GENERAL SIDEWALK & SIDEWALK RAMP NOTES:

1. KDOT GRADE 4.0 CONCRETE SHALL BE USED THROUGHOUT. SEE SPECIFICATION 32 16 19.
2. SIDEWALK RAMP LOCATION DETERMINED FROM THE INTERSECTION OF THE EXTENSION OF BACK OF SIDEWALK AND BACK OF CURB & GUTTER.
3. POSITIVE FLOW LINE DRAINAGE SHALL BE MAINTAINED THROUGH THE PEDESTRIAN ACCESS ROUTE (PAR). NO PONDING SHALL BE PRESENT IN THE PAR. ANY VERTICAL LIP THAT OCCURS AT THE FLOW LINE SHALL NOT BE GREATER THAN 1/4 INCH.
4. TURNING SPACES SHALL BE LOCATED ANYWHERE THE PAR CHANGES DIRECTION, AND IF THE APPROACHING WALK IS INVERSE GRADE.
5. THE MAXIMUM CROSS SLOPE REQUIREMENTS FOR PERPENDICULAR CURB RAMPS AND BLENDED TRANSITIONS ADJACENT TO PEDESTRIAN STREET CROSSINGS ARE AS FOLLOWS:
   AT YIELD OR STOP CONTROL - 2%;
   WITHOUT YIELD OR STOP CONTROL, OR WITH TRAFFIC SIGNALS - 5%
   AT MIDBLOCK - NO GREATER THAN THE STREET GRADE.
6. PAR AND RAMP CROSS-SLOPE, 1% DESIRED (2% MAX), WHEN NOT ADJACENT TO PEDESTRIAN STREET CROSSING.
7. CONTRACTION JOINTS SHALL BE CONSTRUCTED ALONG ALL GRADE BREAKS AND AT THE TOPS OF CONCRETE FLARES ADJACENT TO WALKABLE SURFACES.
8. ALL GRADE BREAKS WITHIN THE PAR SHALL BE PERPENDICULAR TO THE PATH OF TRAVEL.
9. ALL RAMP TYPES SHOULD HAVE A MINIMUM OF 3' RAMP LENGTH.
10. DETECTABLE WARNINGS SHALL CONTINUOUSLY EXTEND FOR A MINIMUM OF 24" IN THE PATH OF TRAVEL. DETECTABLE WARNING TO COVER THE ENTIRE WIDTH OF SIDEWALK AND SHARED-USE PATHS. ARC LENGTH OF RADIAL DETECTABLE WARNINGS SHOULD NOT BE GREATER THAN 20 FEET.
11. RECTANGULAR DETECTABLE WARNINGS SHALL BE SETBACK 2" MINIMUM TO 9" MAXIMUM FROM THE BACK OF CURB. SEE (2) FOR INFORMATION REGARDING RECTANGULAR DETECTABLE WARNING PLACEMENT.
    RADIAL DETECTION WARNINGS SHALL BE SETBACK 2" MINIMUM TO 6" MAXIMUM FROM THE BACK OF CURB.
13. LONGITUDINAL JOINT SPACING TO MATCH WIDTH OF SIDEWALK (4' MIN.)
14. ISOLATION JOINTS SHALL BE PLACED WHERE WALK ABUTS DRIVEWAYS AND SIMILAR STRUCTURES, AND 250' CENTERS MAX.
15. INSTALL 18" TIE BARS #4 EPOXY COATED @ 18" O.C. AND THROUGH WINGS AS SHOWN.
16. SIDEWALK RAMP SHALL BE LENGTHENED AS NEEDED TO PROVIDE COMPLIANT SLOPE (8.3% MAX.) BUT NEED NOT EXCEED 15' REGARDLESS OF RESULTING SLOPE.
17. CURB DEPRESSION AT RAMP OPENING SHALL BE STAKED PRIOR TO CURB CONSTRUCTION.
18. SEE CURB REPLACEMENT DETAILS FOR ADDITIONAL REQUIREMENTS.
19. NO CASTING OR UTILITY BOXES SHALL BE ALLOWED IN RAMPS OR TURNING SPACES. CONTRACTOR SHALL BE RESPONSIBLE FOR ADJUSTING UTILITY BOXES AND COORDINATING WITH UTILITIES TO OBTAIN RAMP AND SIDEWALK COMPLIANCE.
20. DO NOT SCALE THESE DRAWINGS FOR DIMENSIONS.

RAMP ELEMENT NOTES:

1. PLACE DOMES AT THE BACK OF CURB WHEN ALLOWABLE SETBACK CRITERIA IS EXCEEDED.
2. RECTANGULAR DETECTABLE WARNINGS MAY BE SETBACK UP TO 9" FROM THE BACK OF CURB WITH CORNERS SET 2" FROM BACK OF CURB. IF 9" SETBACK IS EXCEEDED, USE RADIAL DETECTABLE WARNINGS. RADIAL DETECTABLE WARNINGS SHALL BE SETBACK UP TO 6" MAXIMUM FROM BACK OF CURB.
3. FOR DIRECTIONAL RAMPS WITH THE DETECTABLE WARNINGS PLACED AT THE BACK OF CURB, THE DETECTABLE WARNINGS SHALL COVER THE ENTIRE WIDTH OF THE WALK/PATH.
4. A TURNING SPACE IS NOT REQUIRED AT THE TOP OF A PERPENDICULAR OR STANDARD ONE-WAY DIRECTIONAL CURB RAMP IF THERE IS NO CHANGE IN DIRECTION AT THE TOP OF THE RAMP.
5. DETECTABLE WARNINGS MAY BE PART OF THE 4' X 4' MIN. LANDING AREA IF IT IS NOT FEASIBLE TO CONSTRUCT THE LANDING OUTSIDE OF THE DETECTABLE WARNING AREA.
6. ONLY 8" THICK PANELS WILL BE PAID FOR AS "RAMP". SIDEWALK BEHIND ISOLATION JOINT WILL BE PAID FOR AS "4" SIDEWALK".

LEGEND

- R — CURB RAMP
- TURNING SPACE - 4' X 4' MIN. (5' X 4' MIN. WHEN CONSTRAINED). MAX 2.0% SLOPE IN ALL DIRECTIONS. TURNING SPACES SHALL BE FULL WIDTH OF INCOMING PARS.
- DETECTABLE WARNING PLATES

A1 SIDEWALK AND SIDEWALK RAMP DETAILS

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

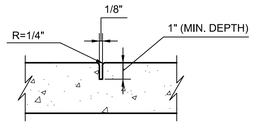
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

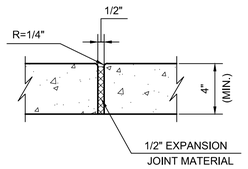
SHEET ID
C-503

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C503-RAMP.dwg
9/9/2026 4:21:52 PM

READY TO ADVERTISE



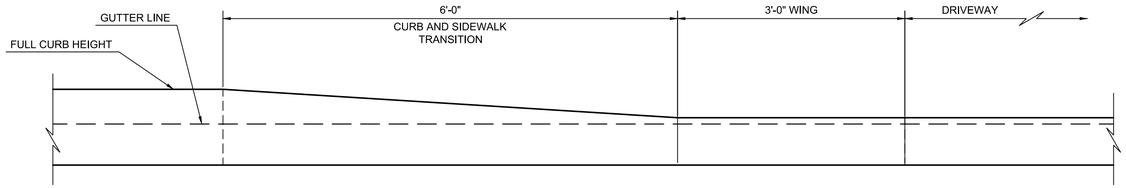
TYPE A



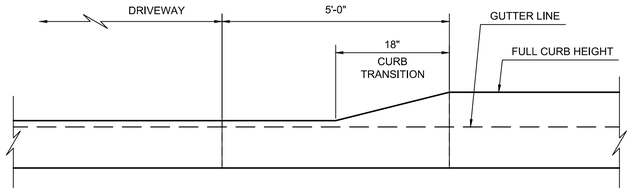
TYPE B

JOINT DETAILS

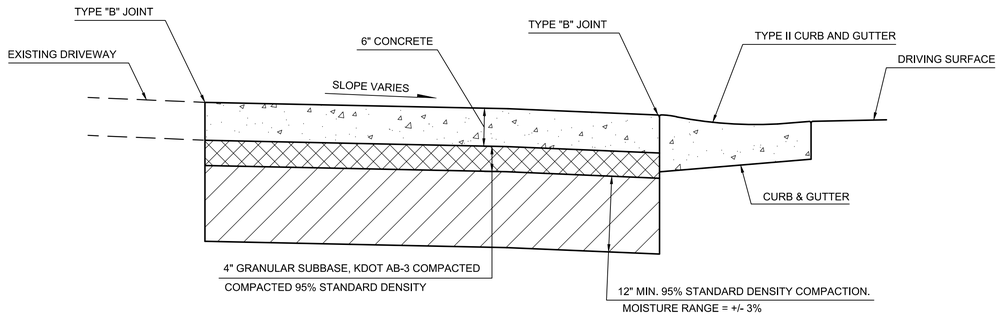
NOT TO SCALE



SECTION E-E



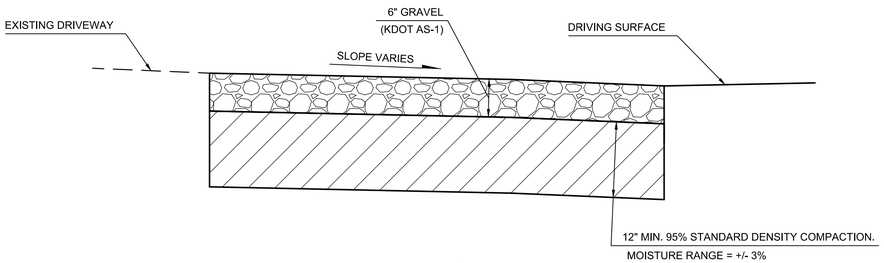
SECTION B-B



SECTION A-A

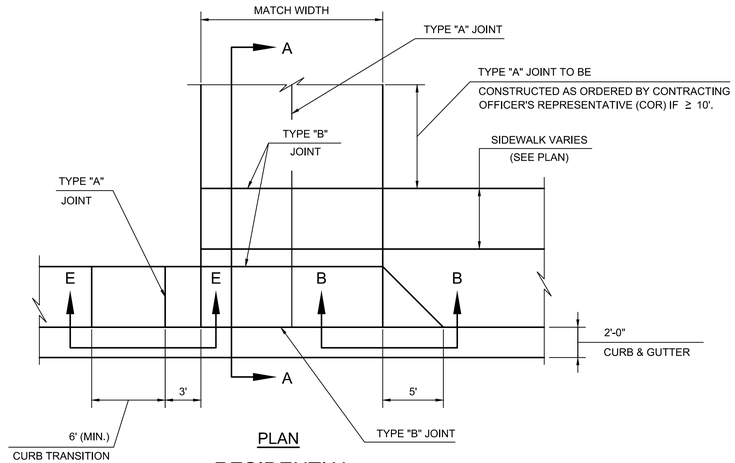
RESIDENTIAL
PCC CONCRETE DRIVE

NOT TO SCALE



RESIDENTIAL
GRAVEL DRIVE SECTION

NOT TO SCALE



PLAN

RESIDENTIAL
PCC CONCRETE DRIVE

NOT TO SCALE

A1 RESIDENTIAL DRIVEWAY DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

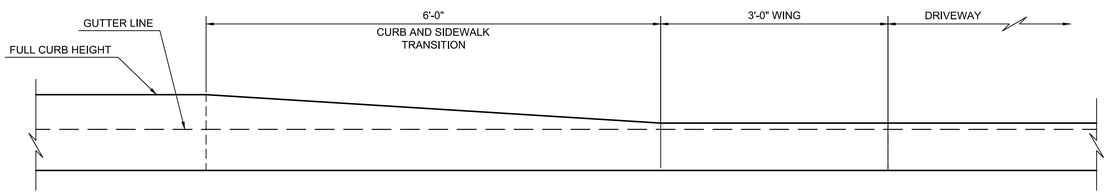
CIVIL DETAILS

SHEET ID

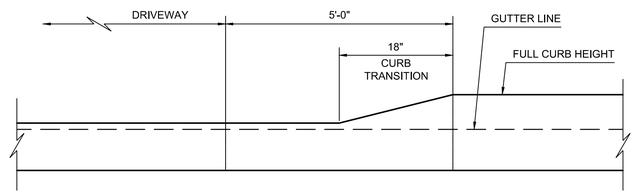
C-504

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C504-505_DRIVE.dwg
9/9/2026 4:22:12 PM

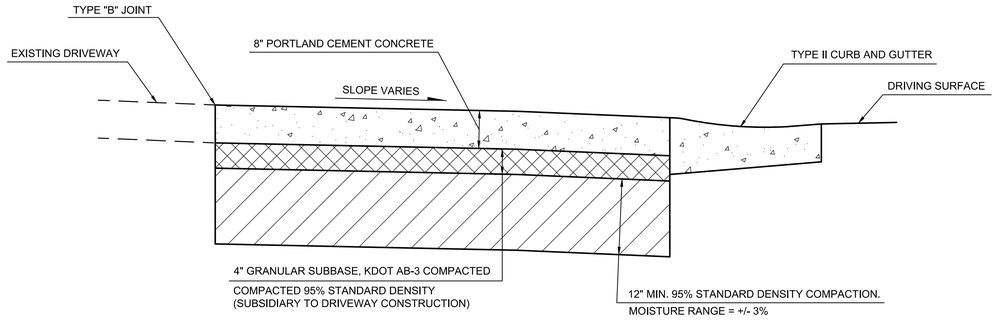
READY TO ADVERTISE



SECTION E-E



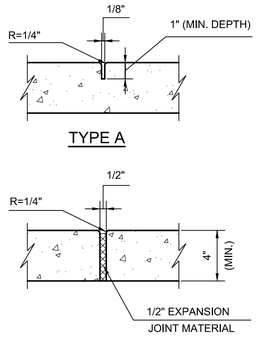
SECTION B-B



SECTION A-A

ALLEY PCC
CONCRETE DRIVE

NOT TO SCALE

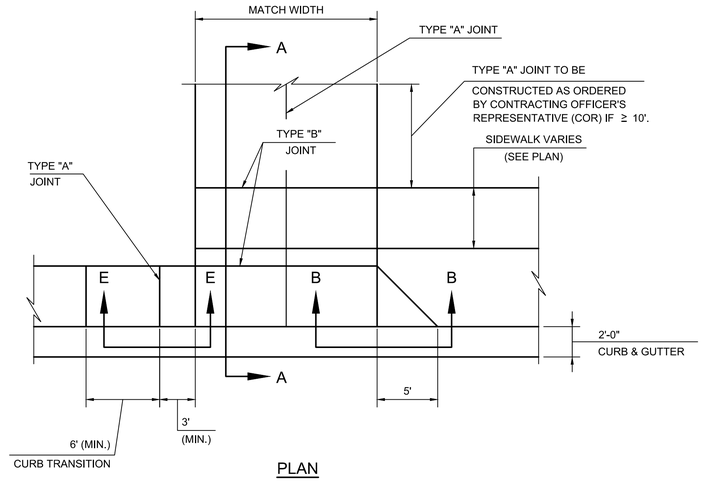


TYPE A

TYPE B

JOINT DETAILS

NOT TO SCALE



PLAN

COMMERCIAL/ALLEY
PCC CONCRETE DRIVE

NOT TO SCALE

A1 COMMERCIAL / ALLEY DRIVEWAY DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-505

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C504-505_DRIVE.dwg
9/9/2026 4:22:16 PM

READY TO ADVERTISE

INTEGRAL SIDEWALK AND RETAINING WALL NOTES:

GENERAL

1. ROCK BACKFILL AND TOPSOIL PLACEMENT SHALL BE CONSIDERED SUBSIDIARY TO INTEGRAL SIDEWALK AND RETAINING WALL CONSTRUCTION.
2. DO NOT SCALE THESE DRAWINGS FOR DIMENSIONS OR CLEARANCES. ANY QUESTIONS REGARDING DIMENSIONS SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER PRIOR TO CONSTRUCTION.
3. SEE PLAN VIEW FOR WEEPHOLE LOCATIONS.
4. ALL FILTER FABRIC USED FOR PIPE UNDERDRAIN CONSTRUCTION SHALL MEET THE REQUIREMENTS OF AASHTO M288 CLASS 1. SEE SPECIFICATION 32 16 19.

CONCRETE

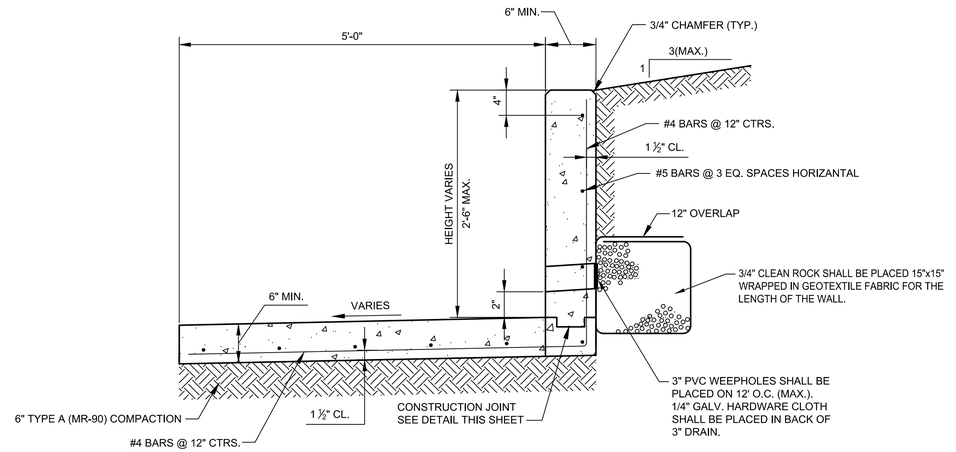
5. ALL EXPOSED WALL FACES SHALL BE HAND FINISHED WITH AN APPROVED GROUT WHILE THE WALL IS GREEN. FORM MARKS SHALL BE REMOVED PRIOR TO FINISHING WALL.
6. ALL FORM TIES SHALL BE RECESSED TYPE.
7. TIES AND SPREADERS USED INSIDE FORMS SHALL BE SO CONSTRUCTED THAT AFTER REMOVAL OF FORMS, THE METAL MAYBE REMOVED TO A DEPTH OF ONE INCH BELOW THE SURFACE OF THE CONCRETE. SURFACE FINISH SHALL BEGIN IMMEDIATELY FOLLOWING THE REMOVAL OF THE FORMS. FINS AND IRREGULAR PROJECTIONS SHALL BE REMOVED. FORM TIE CAVITIES, HOLES AND HONEYCOMB SPOTS SHALL BE THOROUGHLY CLEANED, SATURATED WITH WATER AND CAREFULLY POINTED WITH A MORTAR MATCHING THE SHADING OF THE CONCRETE.

JOINTS

8. * TYPE A JOINT SPACING SHALL BE 8'.
9. ** WHERE WALK ABUTS JUNCTION OF EXISTING WALK, CONCRETE CURBS, DRIVEWAYS, AND SIMILAR STRUCTURES, AND 100' CENTERS MAXIMUM.
10. *** DOWELED CONSTRUCTION JOINT WITH NEOPRENE RUBBER SEAL SHALL BE INSTALLED AT THE LOCATIONS WHERE NEW WALL ABUTS EXISTING.
11. TYPE B HORIZONTAL EXPANSION JOINTS SHALL HAVE 5/8" SMOOTH DOWEL ASSEMBLIES WITH GREASED CAPS @ 12" CTRS. MAX. SEE SPECIFICATION 32 16 19.
12. KEY ALL CONSTRUCTION JOINTS.

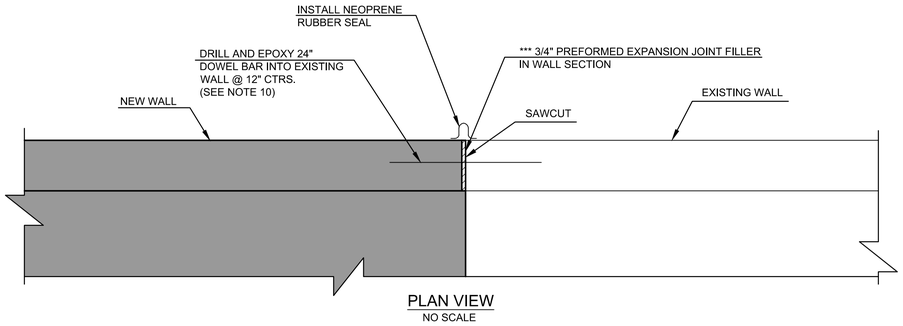
REINFORCING

13. REINFORCING STEEL SHALL BE NEW BILLET, MINIMUM GRADE 40 AS PER ASTM A615, AND SHALL BE BENT COLD AS PER SPECIFICATION 32 16 19.
14. ALL DIMENSIONS RELATIVE TO REINFORCING STEEL ARE TO CENTERLINE OF BARS. 2" CLEARANCE SHALL BE PROVIDED THROUGHOUT UNLESS NOTED OTHERWISE. TOLERANCE OF +/- 1/8" SHALL BE PERMITTED.
15. ALL LAP SPLICES NOT SHOWN SHALL BE A MINIMUM OF 40 BAR DIAMETERS IN LENGTH.
16. ALL DOWELS SHALL BE ACCURATELY PLACED AND SECURELY TIED IN PLACE PRIOR TO PLACEMENT OF BOTTOM SLAB CONCRETE. STICKING OF DOWELS INTO FRESH OR PARTIALLY HARDENED CONCRETE WILL NOT BE ACCEPTABLE.
17. ALL REINFORCING STEEL SHALL BE SUPPORTED ON FABRICATED STEEL BAR SUPPORTS @ 3'-0" MAXIMUM SPACING.



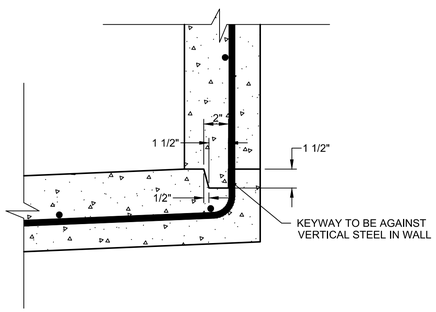
TYPICAL SECTION VIEW

NO SCALE



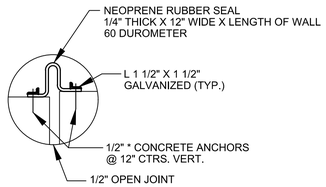
PLAN VIEW

NO SCALE



CONSTRUCTION JOINT DETAIL

NO SCALE



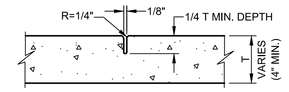
RETAINING WALL CONNECTION

NO SCALE

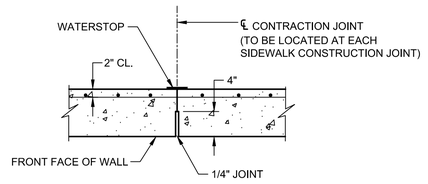
NOTES:

1. INSTALL 3/4" PREFORMED EXPANSION JOINT FILLER BETWEEN WALL EXTENSION AND NEW WALL CONSTRUCTION. DISCONTINUE WALL REINFORCING THROUGH JOINT.
2. PAID FOR AS INTEGRAL SIDEWALK AND RETAINING WALL.

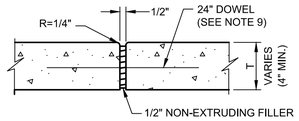
* BONDING AGENT TO BE SPECIFIED BY CONCRETE SUPPLIER FOR COMPATIBILITY.



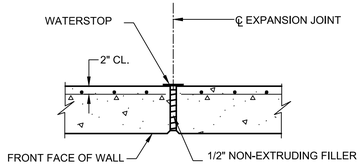
TYPE A* (HORIZONTAL)



VERTICAL CONTRACTION JOINT DETAIL



TYPE B** (HORIZONTAL)

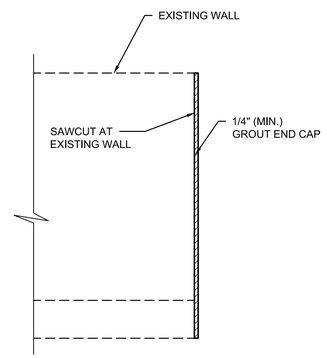


VERTICAL EXPANSION JOINT DETAIL

NOTE: VERTICAL EXPANSION JOINTS REQUIRED EVERY 100'

CONCRETE JOINT DETAILS

NO SCALE



GROUT END CAP AT EXISTING WALL

NO SCALE

A17

A1 INTEGRAL SIDEWALK AND RETAINING WALL DETAILS

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

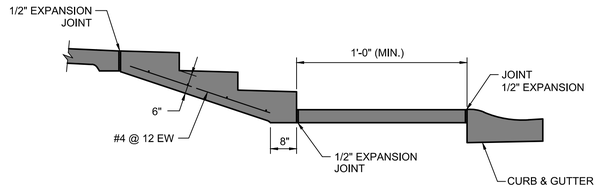
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

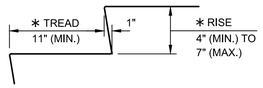
CIVIL DETAILS

SHEET ID
C-506

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C506-ISRW.dwg
9/9/2026 4:22:39 PM

READY TO ADVERTISE



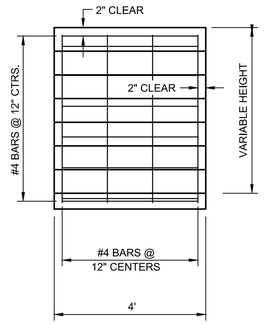


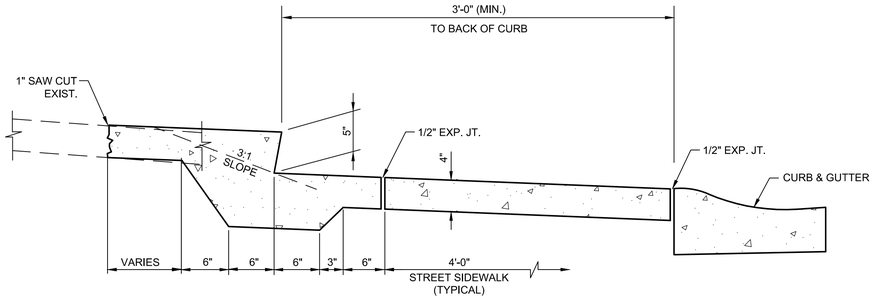
STEP DETAIL

∗ TREAD & RISER DIMESIONS MAY VARY TO MEET EXISTING CONDITIONS PROVIDED :

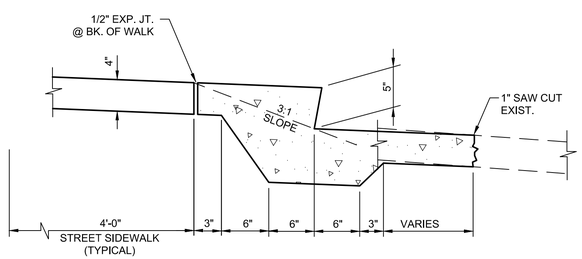
1. EACH RISER IN ANY SET OF STEPS SHALL BE OF EQUAL HEIGHT.
2. EACH TREAD IN ANY SET OF STEPS SHALL BE OF EQUAL WIDTH.
3. (R) X (T) (IN INCHES) SHALL NOT BE LESS THAN 70 OR GREATER THAN 75.

REINFORCED CONCRETE STEPS





SECTION UP



SECTION DOWN

SINGLE STEP DETAILS

A1 CONCRETE STEP DETAILS
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

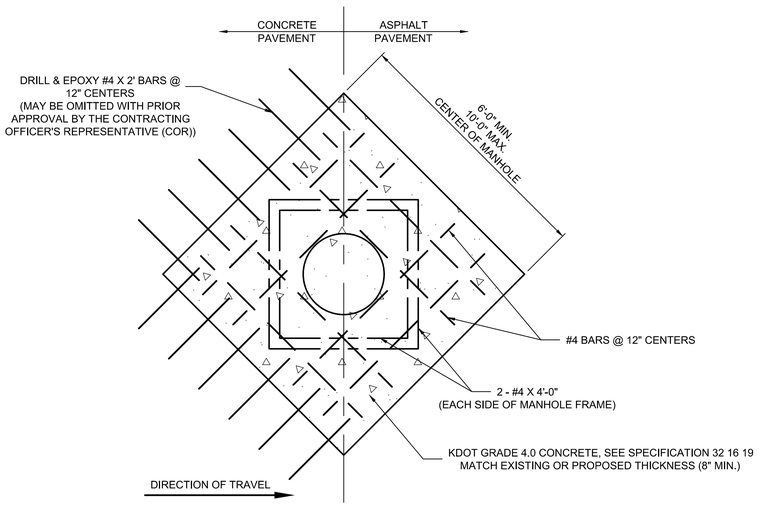
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-507

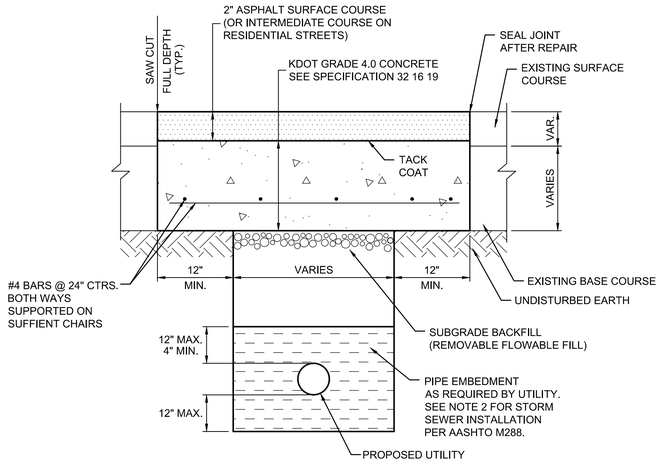
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C507-STEPS.dwg
9/9/2026 4:22:58 PM

READY TO ADVERTISE



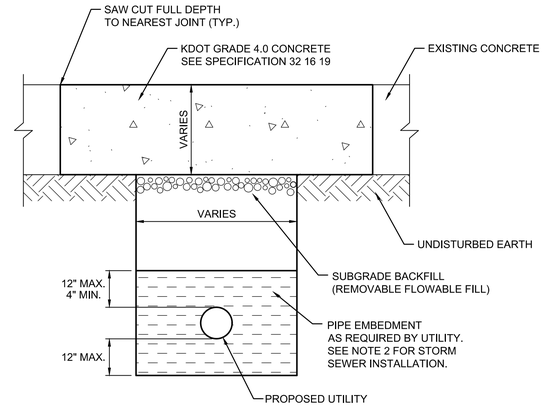
H2 MANHOLE IN PAVED AREA

- DETAIL APPLIES TO IMPROVED THOROUGHFARES AND ALL IMPROVED CONCRETE STREETS. FOR EXISTING CONCRETE STREETS, COORDINATE WITH EXISTING JOINTS.



J8 ASPHALTIC CONCRETE STREET REPAIR TYPE 1

(CONCRETE BASE WITH ASPHALTIC CONCRETE SURFACE)



J14 REPAIR UNDER CURB, SIDEWALK, SHARED USE PATH OR DRIVEWAY

REPAIR UNDER CURB/SIDEWALK/DRIVEWAY NOTES:

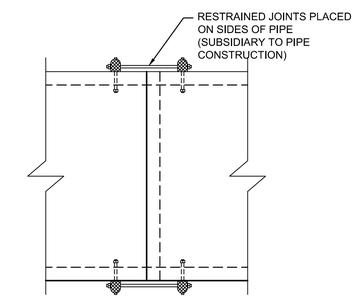
1. CURBS, SIDEWALKS AND DRIVEWAYS SHALL BE SAWED FULL DEPTH, REMOVED AND REPLACED TO NEAREST CONTRACTION OR EXPANSION JOINT.
2. IF ADJACENT SECTION OF CURB, SIDEWALK OR DRIVEWAY IS CRACKED, CHIPPED OR OTHERWISE DAMAGED, IT SHALL ALSO BE REMOVED AND REPLACED TO NEAREST JOINT.
3. ALL CONCRETE USED IN THIS WORK SHALL BE KDOT GRADE 4.0.

STREET REPAIR

NOT TO SCALE

STREET REPAIR NOTES:

1. ANY EXCAVATION LEFT OVERNIGHT IN ANY ROADWAY SHALL BE SECURELY PLATED. PLATES SHALL BE PROPERLY ANCHORED AND ALL EDGES OF THE PLATE SHALL BE RAMPED WITH ASPHALT SURFACE MIX TO PREVENT RATTLING.
2. FOR STORM SEWER INSTALLATION PIPE EMBEDMENT SHALL BE FILLED TO 4" ABOVE THE TOP OF PIPE. SEE RCP INSTALLATION DETAIL.
3. PERMANENT PAVEMENT MARKINGS SHALL BE REPLACED WITH LIKE MATERIALS WITHIN FOURTEEN DAYS AFTER THE PAVEMENT SURFACE HAS BEEN REPLACED, UNLESS OTHERWISE AUTHORIZED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).



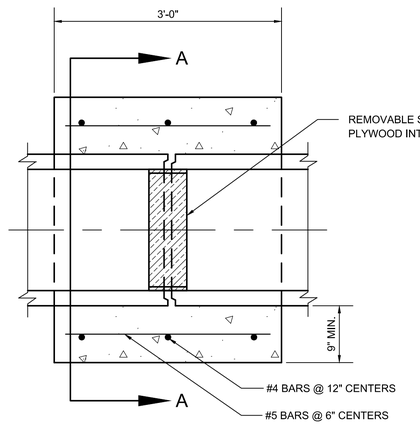
PLAN VIEW

RESTRAINED JOINT NOTES:

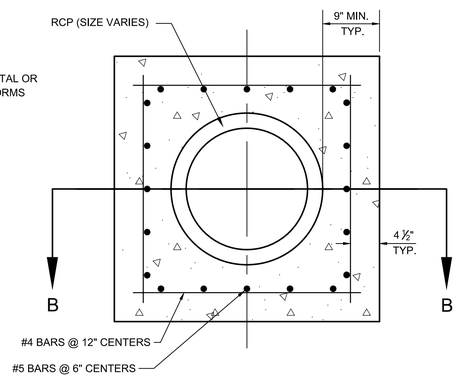
1. THE LAST THREE JOINTS OF RCP SHALL BE MECHANICALLY RESTRAINED.
2. THE RESTRAINED JOINT HARDWARE SHALL BE INSTALLED SO THE BOLT PROJECTION INSIDE THE PIPE DOES NOT EXCEED 2".
3. THE RESTRAINED JOINT HARDWARE SHALL BE HOT DIP ZINC COATED IN ACCORDANCE WITH ASTM A-123.

B2 RESTRAINED JOINT

SCALE: NOT TO SCALE



SECTION B-B



SECTION A-A

CONCRETE COLLAR NOTES

1. USE 4,000 PSI CONCRETE THROUGHOUT.
2. REINFORCING STEEL SHALL BE NEW BILLET, MINIMUM GRADE 40 AS PER ASTM A615, AND SHALL BE BENT COLD.
3. ALL DIMENSIONS RELATIVE TO REINFORCING STEEL ARE TO CENTERLINE OF BARS. 2" CLEARANCE SHALL BE PROVIDED THROUGHOUT UNLESS NOTED OTHERWISE. TOLERANCE OF +/- 1/8" SHALL BE PERMITTED.
4. ALL LAP SPLICES NOT SHOWN SHALL BE A MINIMUM OF 40 BAR DIAMETERS IN LENGTH.
5. ALL DOWELS SHALL BE ACCURATELY PLACED AND SECURELY TIED IN PLACE PRIOR TO PLACEMENT OF BOTTOM SLAB CONCRETE. STICKING OF DOWELS INTO FRESH OR PARTIALLY HARDENED CONCRETE WILL NOT BE ACCEPTABLE.
6. ALL REINFORCING STEEL SHALL BE SUPPORTED ON FABRICATED STEEL BAR SUPPORTS @ 3'-0" MAXIMUM SPACING.
7. DO NOT SCALE THESE DRAWINGS FOR DIMENSIONS OR CLEARANCES. ANY QUESTIONS REGARDING DIMENSIONS SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) PRIOR TO CONSTRUCTION.

A8 CONCRETE COLLAR DETAIL

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

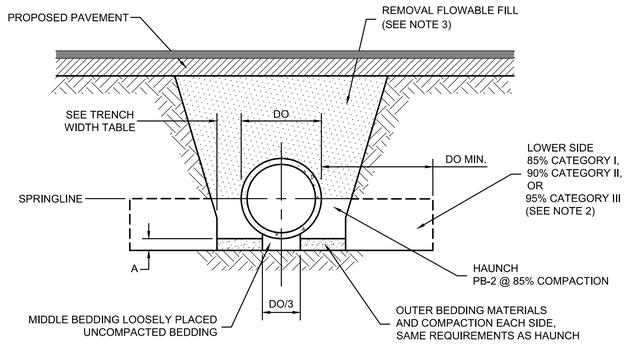
SHEET ID
C-508

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C508-STR REPAIR.dwg
9/9/2026 4:23:19 PM

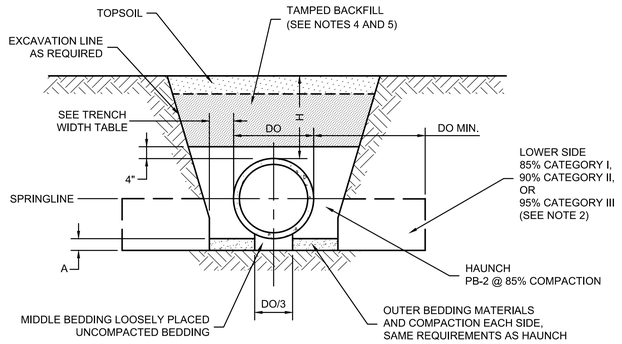
READY TO ADVERTISE

TRENCH WIDTH TABLE

PIPE SIZE (INCHES)	MINIMUM SIDE WALL CLEARANCE (INCHES)
12 - 18	6
21	6.5
24	7
27	8
30	8.5
33	9
36	10
42	11
48	12.5
54	13.5
60	15
66	15
72	15.5
78	15.5
84	16
90	16.5
96	17
102	17.5
108	18



TRENCH INSTALLATION UNDER PAVED SURFACE



TRENCH INSTALLATION OUTSIDE PAVED SURFACE

TABLE OF FILL DEPTHS BELOW PIPE

D	"A" MIN. IN SOIL	"A" MIN. IN ROCK
60" & SMALLER	3"	6"
66" & LARGER	6"	12"

RCP INSTALLATION

NOT TO SCALE

RCP INSTALLATION NOTES:

1. CATEGORY I MATERIAL SHALL BE CRUSHED STONE OR SILICEOUS GRAVEL MEETING THE REQUIREMENTS OF KDOT SECTION 1107 OF THE STANDARD SPECIFICATIONS, AGGREGATE DESIGNATION PB-2.
2. COMPACTION AND SOIL SYMBOLS - I.E. "95% CATEGORY I" - REFERS TO CATEGORY I SOIL MATERIAL WITH MINIMUM STANDARD PROCTOR COMPACTION OF 95%
3. REMOVABLE FLOWABLE FILL IS REQUIRED AS SHOWN ON THE TRENCH INSTALLATION DETAIL IN ANY PAVED PORTION OF THE RIGHT OF WAY, INCLUDING THE AREA WITHIN EXISTING OR FUTURE PUBLIC STREET PAVEMENT, UNDER EXISTING OR FUTURE CURB AND GUTTER, MEDIAN, SHARED USE PATH, PARKING LOTS, AND CONCRETE SIDEWALK. ALSO UNDER EXISTING CONCRETE, ASPHALT, OR GRAVEL DRIVEWAY APPROACH AND UNDER ANY FUTURE DRIVEWAY APPROACH, WHEN THE LOCATION IS KNOWN.
4. TAMPED BACKFILL SHALL BE USED AT ALL OTHER LOCATIONS UNLESS OTHERWISE NOTED. TAMPED BACKFILL SHALL BE FINELY DIVIDED JOB EXCAVATED MATERIAL FREE FROM DEBRIS, ORGANIC MATERIAL, AND STONES, COMPACTED TO 95% MAXIMUM DENSITY AS DETERMINED BY AASHTO STANDARD, METHOD T-99.
5. GRANULAR FILL MAY BE SUBSTITUTED FOR ALL OR PART OF TAMPED BACKFILL. GRANULAR FILL SHALL BE KDOT AB-3, AS SPECIFIED IN THE STANDARD SPECIFICATIONS, COMPACTED TO A MINIMUM OF 95% OF OPTIMUM DENSITY, WITHIN 0 TO MINUS 2 PERCENT OF OPTIMUM MOISTURE PLACED UP TO AN ELEVATION 6" BELOW SURFACE. TOPSOIL SHALL BE USED IN THE TOP 6".
6. WHEN THE EXISTING GRADE IS NOT AT LEAST 18" ABOVE THE PROPOSED TOP OF PIPE PROFILE, INSTALL COMPACTED FILL (95% STD. PROCTOR) TO 18" ABOVE TOP OF PIPE PRIOR TO EXCAVATION FOR STORM SEWER INSTALLATION, OR AS APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
7. SOIL IN OUTER BEDDING AND HAUNCH ZONES SHALL BE COMPACTED TO AT LEAST THE SAME COMPACTION AS SPECIFIED FOR THE MAJORITY OF SOIL IN THE BACKFILL ZONE.
8. THE TRENCH WIDTH SHALL BE WIDER THAN SHOWN IF REQUIRED FOR ADEQUATE SPACE TO ATTAIN THE SPECIFIED COMPACTION IN THE HAUNCH AND BEDDING ZONES.
9. FOR TRENCH WALLS THAT ARE WITHIN 10 DEGREES OF VERTICAL, THE COMPACTION OF FIRMNESS OF THE SOIL IN THE TRENCH WALLS AND LOWER SIDE ZONE NEED NOT BE CONSIDERED.
10. FOR TRENCH WALLS WITH GREATER THAN 10 DEGREE SLOPES THAT CONSIST OF EMBANKMENT, THE LOWER SIDE SHALL BE COMPACTED TO AT LEAST THE SAME COMPACTION AS SPECIFIED FOR THE SOIL IN THE OVERFILL ZONE.
11. SEE STREET REPAIR DETAILS FOR ADDITIONAL BACKFILL REQUIREMENTS UNDER STREETS , ROADWAYS, CURBS, SIDEWALKS, AND OTHER STRUCTURES IN RIGHT-OF-WAY.

A1 RCP INSTALLATION DETAILS
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

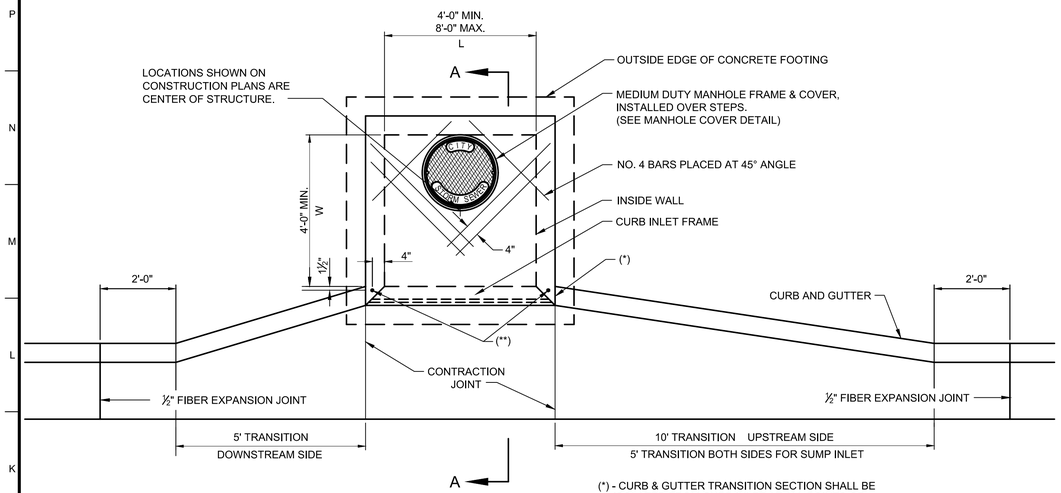
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-509

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C509-RCP INSTL.dwg
9/9/2026 4:23:38 PM

READY TO ADVERTISE



PLAN

NOT TO SCALE

(*) - CURB & GUTTER TRANSITION SECTION SHALL BE ISOLATED FROM CURB INLET AT ALL POINTS OF CONTACT. ISOLATION MATERIAL SHALL BE 30 LB. ROOFING FELT OR APPROVED EQUAL.

(**) - NO. 4 VERTICAL BARS SHALL BE PLACED IN WALL 1 ½" CLEAR FROM FINISHED TOP OF SLAB.

CURB INLET NOTES

GENERAL

1. ALL STORM SEWER STRUCTURES SHALL BE PRE-CAST OR POURED IN PLACE. IF PRE-CAST STRUCTURES ARE USED FOR PUBLICLY FINANCED, MAINTAINED OR ADMINISTERED CONSTRUCTION, THE TOPS SHALL BE POURED IN PLACE AND THE WALL STEEL SHALL BE LEFT EXPOSED TO A HEIGHT 2" BELOW THE FINISH TOP ELEVATION, OR AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
2. PRE-CAST SHOP DRAWINGS ARE TO BE APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) FOR PUBLICLY FINANCED OR ADMINISTERED PROJECTS.
3. DO NOT SCALE THESE DRAWINGS FOR DIMENSIONS OR CLEARANCES. ANY QUESTIONS REGARDING DIMENSIONS SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) PRIOR TO CONSTRUCTION.
4. ON-GRADE INLETS SHALL CONFORM TO THE STREET GRADE AND SUMP INLETS SHALL BE LEVEL.
5. THE FIRST DIMENSION LISTED IN THE CONSTRUCTION NOTES IS THE "L" DIMENSION. THE SECOND DIMENSION IS THE "W" DIMENSION. THE CONCRETE THICKNESS AND REINFORCEMENT SHOWN IS FOR BOXES WITH ("L"+"H") AND ("W"+"H") LESS THEN OR EQUAL TO 20. FOR BOXES WITH EITHER OF THESE CALCULATIONS GREATER THAN 20, A SPECIAL DESIGN IS REQUIRED.

CONCRETE

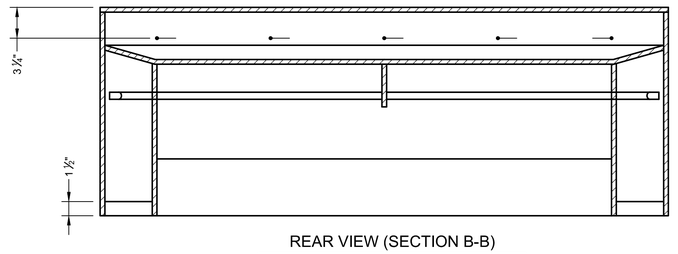
6. CONCRETE USED IN THIS WORK SHALL BE KDOT GRADE 4.0. SEE SPECIFICATION 32 16 19.
7. CONCRETE CONSTRUCTION SHALL MEET THE APPLICABLE REQUIREMENTS OF STANDARD SPECIFICATIONS FOR STATE ROAD AND BRIDGE CONSTRUCTION, KANSAS DEPARTMENT OF TRANSPORTATION, LATEST EDITION, EXCEPT AS MODIFIED IN THE ATCHISON MUNICIPAL CODE.
8. INLET FLOORS SHALL BE SHAPED WITH NON-REINFORCED CONCRETE INVERTS TO PROVIDE SMOOTH FLOW.
9. BEVEL ALL EXPOSED EDGES WITH 3/4" TRIANGULAR MOLDING.

REINFORCING STEEL

10. REINFORCING STEEL SHALL BE NEW BILLET, MINIMUM GRADE 40 AS PER ASTM A615M, AND SHALL BE BENT COLD.
11. ALL DIMENSIONS RELATIVE TO REINFORCING STEEL ARE TO CENTERLINE OF BARS. 2" CLEARANCE SHALL BE PROVIDED THROUGHOUT UNLESS NOTED OTHERWISE. TOLERANCE OF +/- 1/8" SHALL BE PERMITTED.
12. ALL LAP SPLICES NOT SHOWN SHALL BE A MINIMUM OF 40 BAR DIAMETERS IN LENGTH.
13. ALL REINFORCING STEEL SHALL BE SUPPORTED ON FABRICATED STEEL BAR SUPPORTS @ 3'-0" MAXIMUM SPACING.
14. ALL DOWELS SHALL BE ACCURATELY PLACED AND SECURELY TIED IN PLACE PRIOR TO PLACEMENT OF BOTTOM SLAB CONCRETE. STICKING OF DOWELS INTO FRESH OR PARTIALLY HARDENED CONCRETE WILL NOT BE ACCEPTABLE.

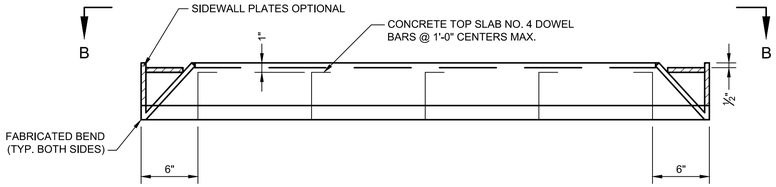
CONSTRUCTION

15. THE BOTTOM SLAB SHALL BE AT LEAST 24 HOURS OLD BEFORE PLACING SIDEWALL CONCRETE. ALL SIDEWALL FORMS SHALL REMAIN IN PLACE A MINIMUM OF 24 HOURS AFTER SIDEWALLS ARE POURED BEFORE REMOVAL, AND AFTER REMOVAL SHALL BE IMMEDIATELY TREATED WITH MEMBRANE CURING COMPOUND.
16. ALL CURB INLET TOPS ARE TO BE CONSTRUCTED AFTER FINAL CURB STRING LINE HAS BEEN APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) AND PRIOR TO CURB CONSTRUCTION, OR AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
17. PIPE CONNECTIONS TO PRE-CAST STRUCTURES SHALL HAVE A MINIMUM OF 6" OF CONCRETE AROUND THE ENTIRE PIPE WITHIN 2' OF THE STRUCTURE.
18. MATERIAL SELECTION AND COMPACTION REQUIREMENTS FOR BACKFILL AROUND STRUCTURES SHALL BE AS SPECIFIED IN THE MANUAL OF INFRASTRUCTURE STANDARDS, AS PROMULGATED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).



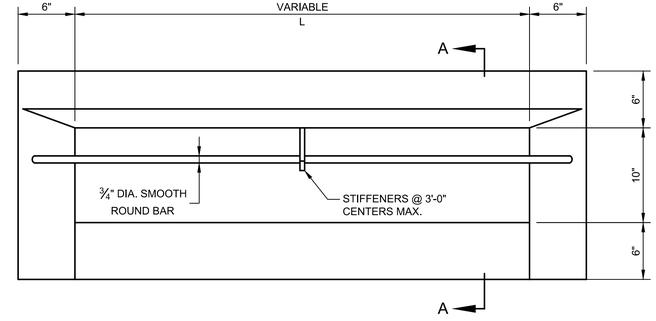
REAR VIEW (SECTION B-B)

NOT TO SCALE



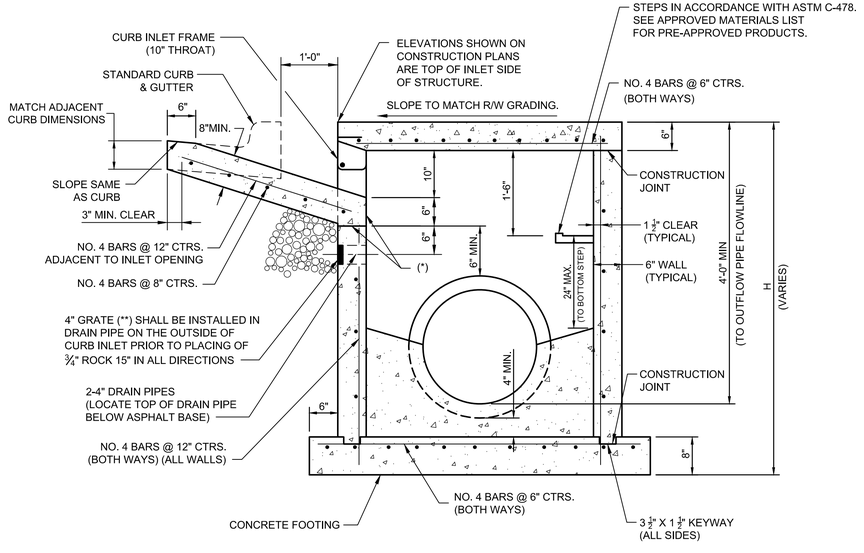
TOP VIEW

NOT TO SCALE



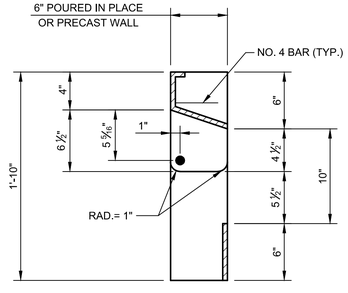
FRONT VIEW

NOT TO SCALE



SECTION A-A

NOT TO SCALE



SIDE VIEW (SECTION A-A)

NOT TO SCALE

CURB INLET FRAME NOTES

1. ALL WELDS SHALL BE PERFORMED IN ACCORDANCE WITH APPROPRIATE AWS SPECIFICATIONS AND PROCEDURES.
2. ALL WELDS ON EXPOSED SURFACES SHALL BE DRESSED SO AS TO PROVIDE A PLEASING FINISHED APPEARANCE.
3. ALL FLAT STEEL SHALL BE 7 GAGE OR 3/16" THICK.
4. THE ENTIRE FRAME SHALL BE HOT DIP ZINC COATED IN ACCORDANCE WITH ASTM A-123.

CURB INLET FRAME

A1 CURB INLET DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

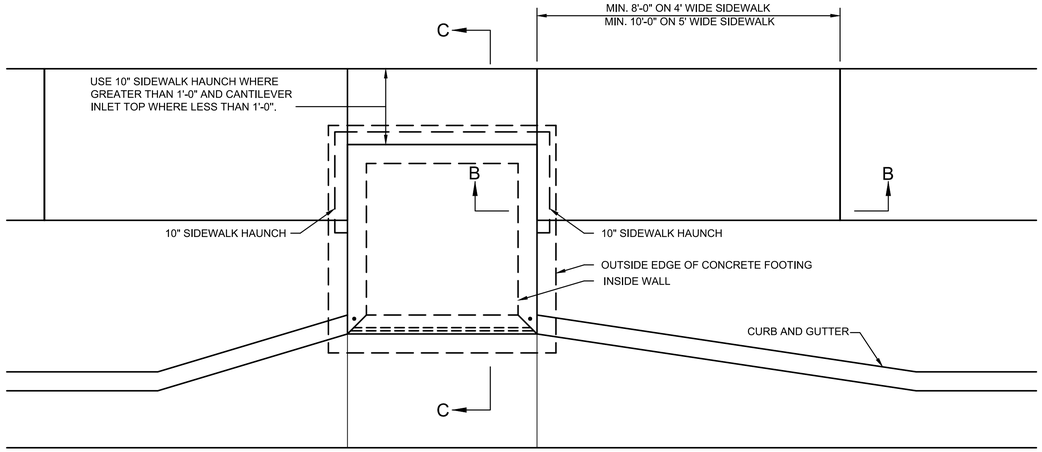
CIVIL DETAILS

SHEET ID

C-510

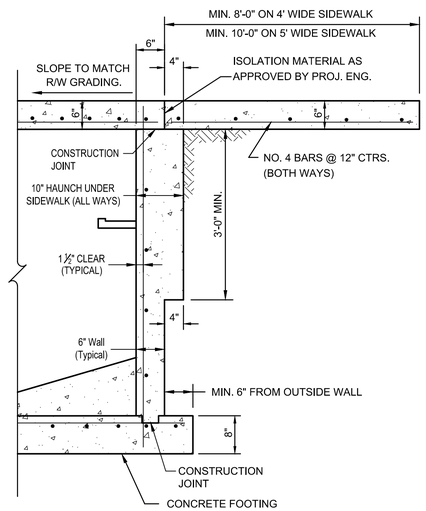
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C510-CINLET.dwg
9/9/2026 4:24:00 PM

READY TO ADVERTISE



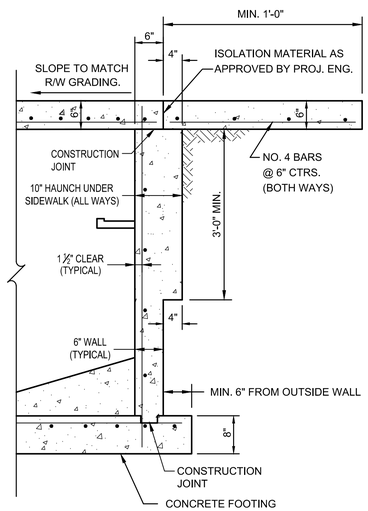
PLAN

NOT TO SCALE



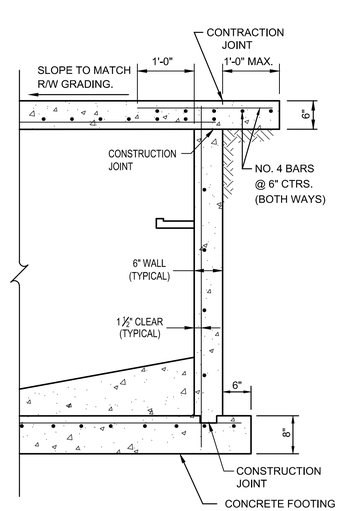
SECTION B-B

WITH SIDEWALK HAUNCH



SECTION C-C

WITH SIDEWALK HAUNCH WHERE SIDEWALK WIDTH > 1' ALONG BACK OF INLET



SECTION C-C

WITH CANTILEVER INLET TOP WHERE SIDEWALK WIDTH < 1' ALONG BACK OF INLET

A1 CURB INLET WITH ADJOINING SIDEWALK DETAILS

SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C511-CI-SW.dwg
9/9/2026 4:24:19 PM

READY TO ADVERTISE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

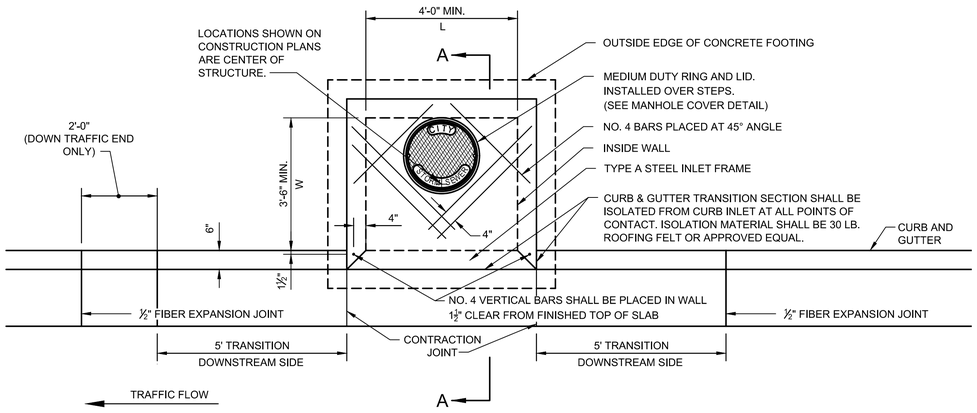
AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

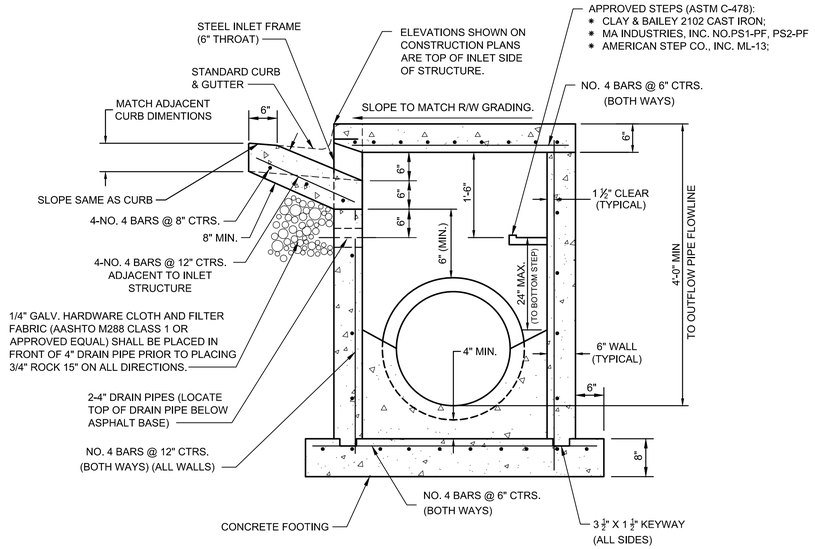
SHEET ID

C-511



PLAN

NOT TO SCALE



SECTION A-A

NOT TO SCALE

CURB INLET (6" THROAT)

CURB INLET NOTES

GENERAL

1. ALL STORM SEWER STRUCTURES SHALL BE PRE-CAST OR POURED IN PLACE. IF PRE-CAST STRUCTURES ARE USED FOR PUBLICLY FINANCED, MAINTAINED OR ADMINISTERED CONSTRUCTION, THE TOPS SHALL BE POURED IN PLACE AND THE WALL STEEL SHALL BE LEFT EXPOSED TO A HEIGHT 2" BELOW THE FINISH TOP ELEVATION, OR AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
2. PRE-CAST SHOP DRAWINGS ARE TO BE APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) FOR PUBLICLY FINANCED OR ADMINISTERED PROJECTS.
3. DO NOT SCALE THESE DRAWINGS FOR DIMENSIONS OR CLEARANCES. ANY QUESTIONS REGARDING DIMENSIONS SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) PRIOR TO CONSTRUCTION.
4. ON-GRADE INLETS SHALL CONFORM TO THE STREET GRADE AND SUMP INLETS SHALL BE LEVEL.
5. THE FIRST DIMENSION LISTED IN THE CONSTRUCTION NOTES IS THE "L" DIMENSION. THE SECOND DIMENSION IS THE "W" DIMENSION. THE CONCRETE THICKNESS AND REINFORCEMENT SHOWN IS FOR BOXES WITH ("L"+"H") AND ("W"+"H") LESS THEN OR EQUAL TO 20. FOR BOXES WITH EITHER OF THESE CALCULATIONS GREATER THAN 20, A SPECIAL DESIGN IS REQUIRED.

CONCRETE

6. CONCRETE USED IN THIS WORK SHALL BE KDOT GRADE 4.0. SEE SPECIFICATION 32 16 19.
7. CONCRETE CONSTRUCTION SHALL MEET THE APPLICABLE REQUIREMENTS OF STANDARD SPECIFICATIONS FOR STATE ROAD AND BRIDGE CONSTRUCTION, KANSAS DEPARTMENT OF TRANSPORTATION, LATEST EDITION, EXCEPT AS MODIFIED IN THE ATCHISON MUNICIPAL CODE.
8. INLET FLOORS SHALL BE SHAPED WITH NON-REINFORCED CONCRETE INVERTS TO PROVIDE SMOOTH FLOW.
9. BEVEL ALL EXPOSED EDGES WITH 3/4" TRIANGULAR MOLDING.

REINFORCING STEEL

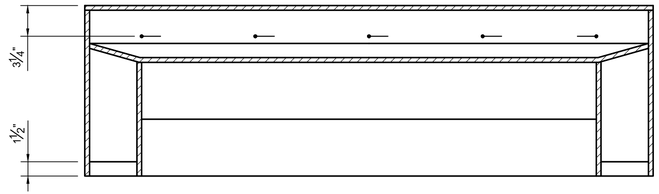
10. REINFORCING STEEL SHALL BE NEW BILLET, MINIMUM GRADE 40 AS PER ASTM A615M, AND SHALL BE BENT COLD.
11. ALL DIMENSIONS RELATIVE TO REINFORCING STEEL ARE TO CENTERLINE OF BARS. 2" CLEARANCE SHALL BE PROVIDED THROUGHOUT UNLESS NOTED OTHERWISE. TOLERANCE OF +/- 1/8" SHALL BE PERMITTED.
12. ALL LAP SPLICES NOT SHOWN SHALL BE A MINIMUM OF 40 BAR DIAMETERS IN LENGTH.
13. ALL REINFORCING STEEL SHALL BE SUPPORTED ON FABRICATED STEEL BAR SUPPORTS @ 3'-0" MAXIMUM SPACING.
14. ALL DOWELS SHALL BE ACCURATELY PLACED AND SECURELY TIED IN PLACE PRIOR TO PLACEMENT OF BOTTOM SLAB CONCRETE. STICKING OF DOWELS INTO FRESH OR PARTIALLY HARDENED CONCRETE WILL NOT BE ACCEPTABLE.

CONSTRUCTION

15. THE BOTTOM SLAB SHALL BE AT LEAST 24 HOURS OLD BEFORE PLACING SIDEWALL CONCRETE. ALL SIDEWALL FORMS SHALL REMAIN IN PLACE A MINIMUM OF 24 HOURS AFTER SIDEWALLS ARE POURED BEFORE REMOVAL, AND AFTER REMOVAL SHALL BE IMMEDIATELY TREATED WITH MEMBRANE CURING COMPOUND.
16. ALL CURB INLET TOPS ARE TO BE CONSTRUCTED AFTER FINAL CURB STRING LINE HAS BEEN APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) AND PRIOR TO CURB CONSTRUCTION, OR AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
17. PIPE CONNECTIONS TO PRE-CAST STRUCTURES SHALL HAVE A MINIMUM OF 6" OF CONCRETE AROUND THE ENTIRE PIPE WITHIN 2' OF THE STRUCTURE.
18. MATERIAL SELECTION AND COMPACTION REQUIREMENTS FOR BACKFILL AROUND STRUCTURES SHALL BE AS SPECIFIED IN THE MANUAL OF INFRASTRUCTURE STANDARDS, AS PROMULGATED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).

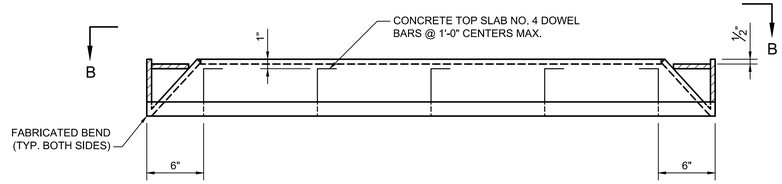
STEEL INLET FRAME NOTES

1. ALL WELDS SHALL BE PERFORMED IN ACCORDANCE WITH APPROPRIATE AWS SPECIFICATIONS AND PROCEDURES.
2. ALL STEEL SHALL BE 7 GAGE OR 3/16" THICK.
3. ALL WELDS ON EXPOSED SURFACES SHALL BE DRESSED SO AS TO PROVIDE A PLEASING FINISHED APPEARANCE.
4. THE ENTIRE FRAME SHALL BE HOT DIP ZINC COATED IN ACCORDANCE WITH ASTM A-123.



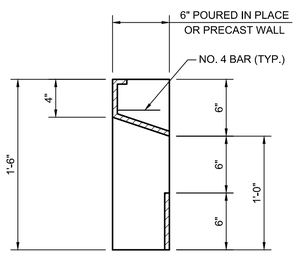
REAR VIEW (SECTION B-B)

NOT TO SCALE



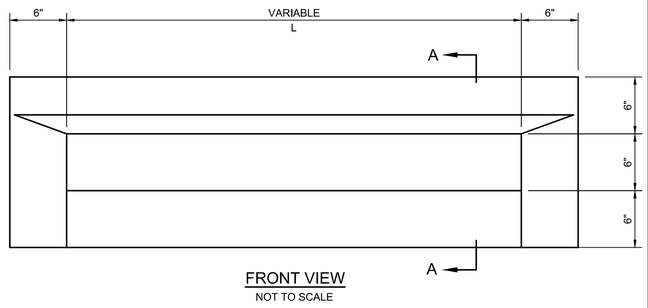
TOP VIEW

NOT TO SCALE



SIDE VIEW (SECTION A-A)

NOT TO SCALE



FRONT VIEW

NOT TO SCALE

CURB INLET FRAME (6" THROAT)

A1 NON-SETBACK CURB INLET DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID

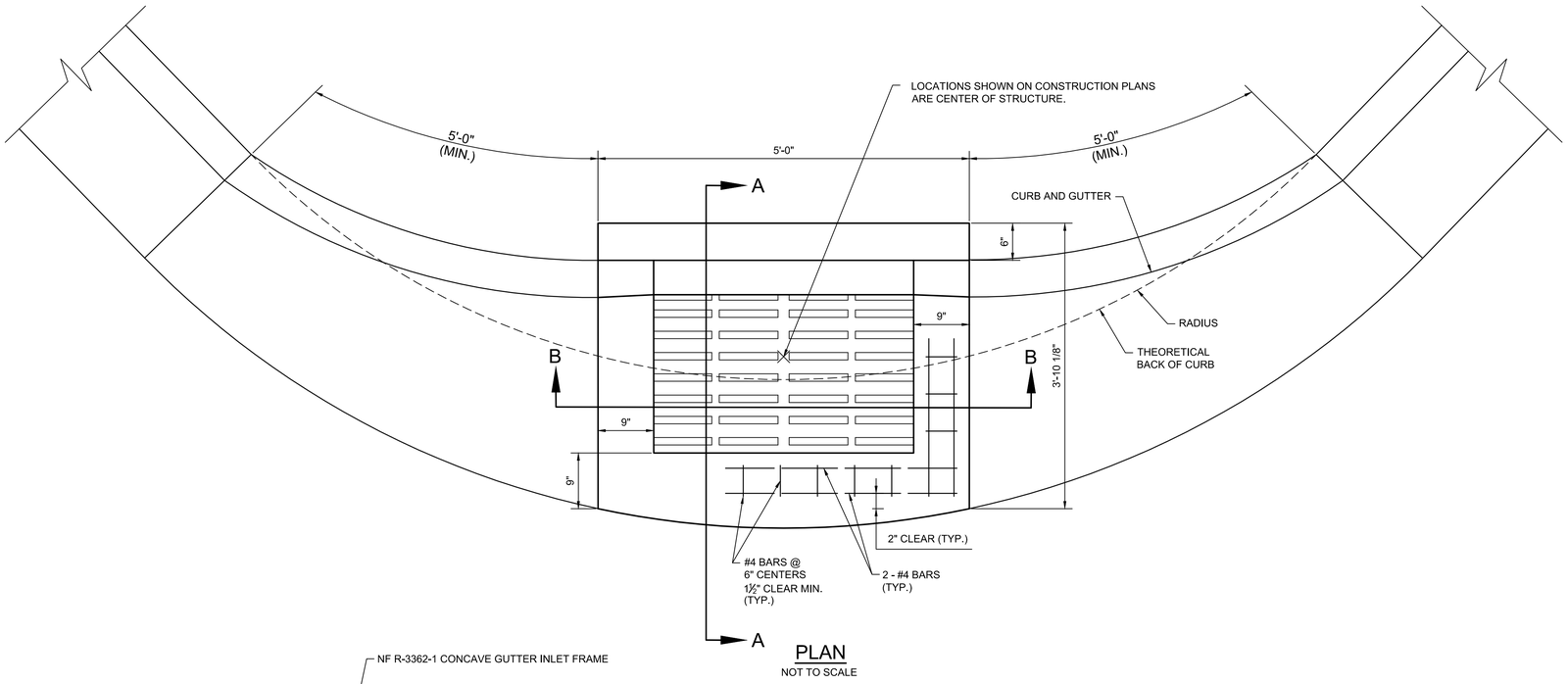
C-512

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C512-NON-SETBK.dwg
9/9/2026 4:24:40 PM

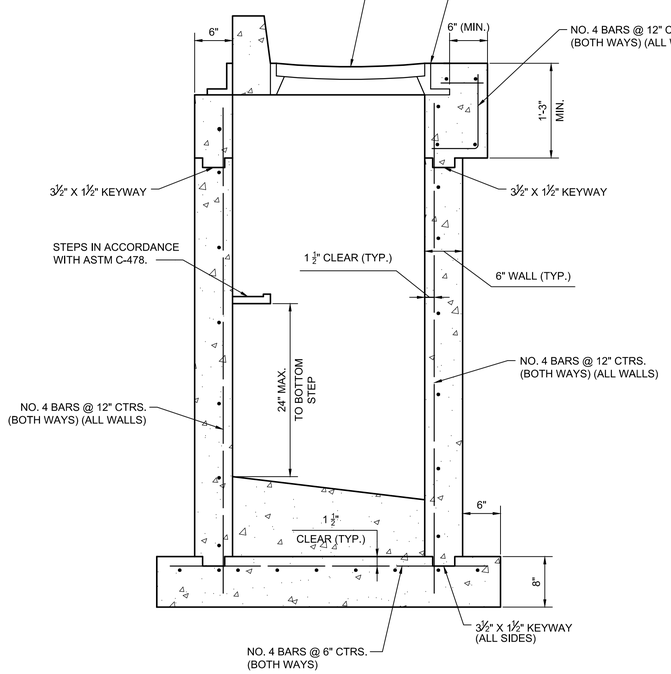
READY TO ADVERTISE

SHEET NOTES

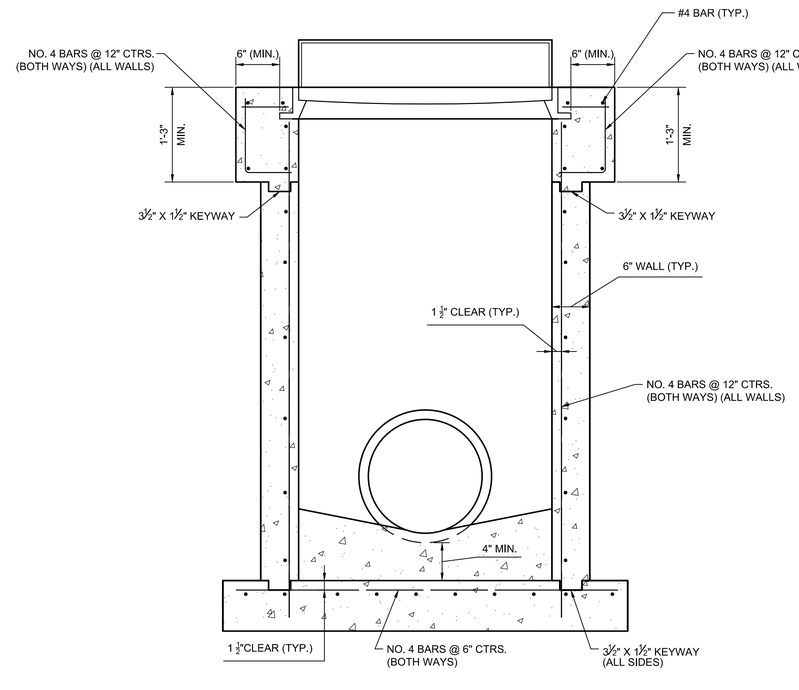
1. SEE SHEET C-510 FOR ADDITIONAL CURB INLET NOTES AND DETAILS.



PLAN
NOT TO SCALE



SECTION A-A
NOT TO SCALE



SECTION B-B
NOT TO SCALE

A1 SPECIAL GRATE INLET DETAILS
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

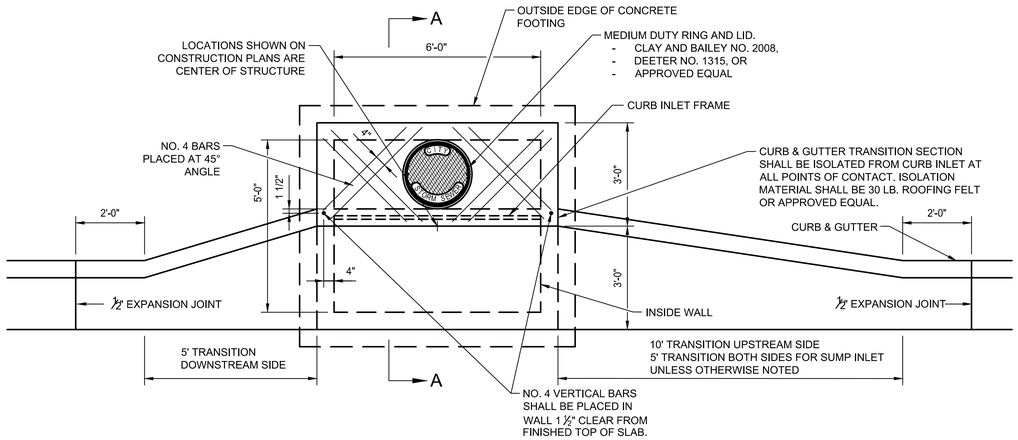
SHEET ID
C-513

T:\24-2309-011\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C513-SPCL INLET.dwg
9/9/2026 4:24:59 PM

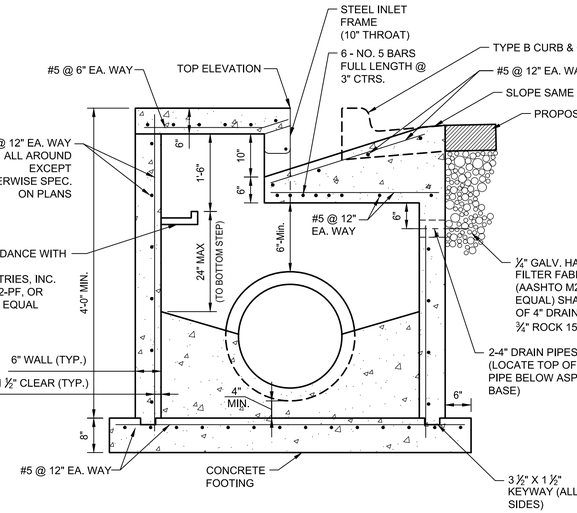
READY TO ADVERTISE

SHEET NOTES

1. SEE SHEET C-510 FOR ADDITIONAL CURB INLET NOTES AND DETAILS.



PLAN



SECTION A-A

6' X 5' SPECIAL CURB INLET DETAIL

Storm Sewer Structure

A1 SPECIAL CURB INLET DETAIL
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

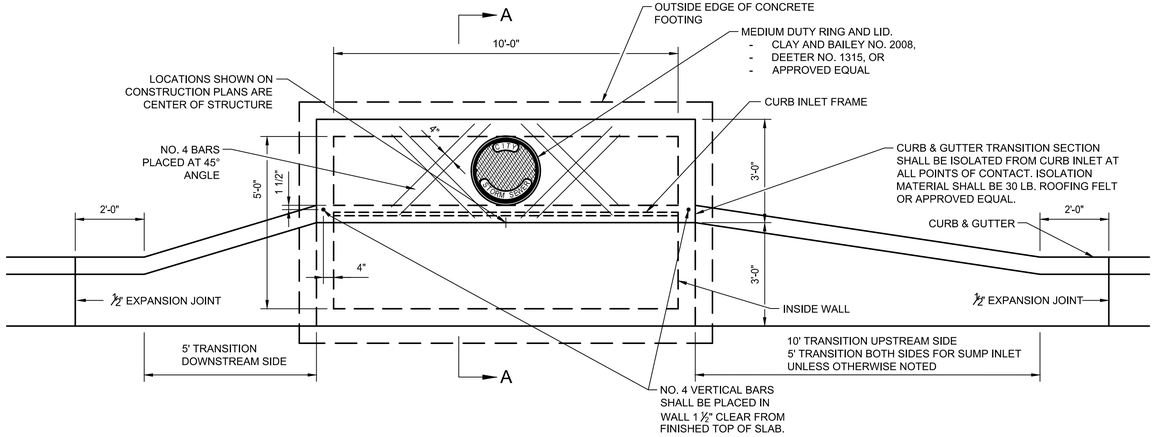
SHEET ID
C-514

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C514-SPCL CI.dwg
9/9/2026 4:25:20 PM

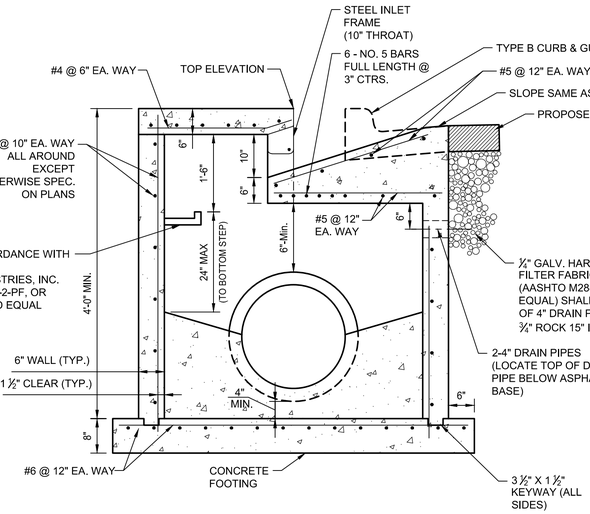
READY TO ADVERTISE

SHEET NOTES

1. SEE SHEET C-510 FOR ADDITIONAL CURB INLET NOTES AND DETAILS.



PLAN



SECTION A-A

10' X 5' SPECIAL CURB INLET DETAIL

Storm Sewer Structure

A1 SPECIAL CURB INLET DETAIL

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

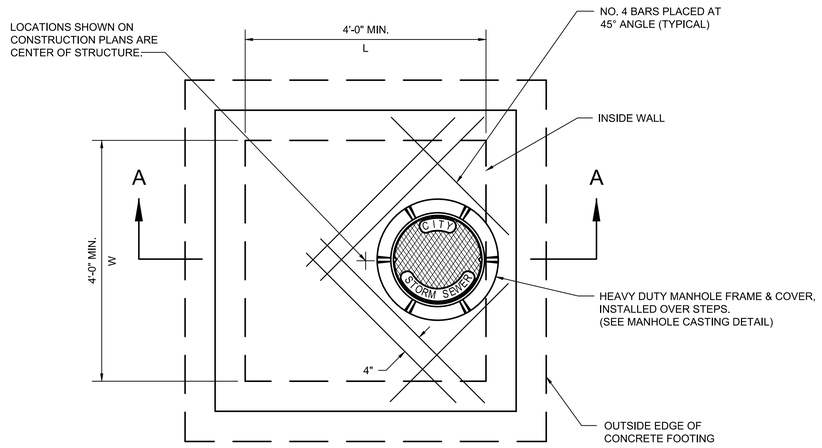
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-515

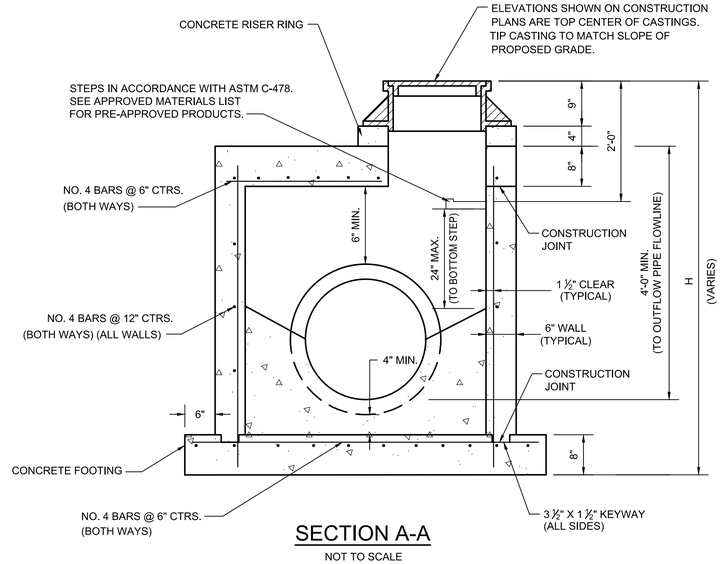
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C515-SPCL CI.dwg
9/9/2026 4:25:42 PM

READY TO ADVERTISE



PLAN

NOT TO SCALE



SECTION A-A

NOT TO SCALE

JUNCTION BOX NOTES

GENERAL

1. ALL STORM SEWER STRUCTURES SHALL BE PRE-CAST OR POURED IN PLACE. IF PRE-CAST STRUCTURES ARE USED, THE TOPS SHALL BE POURED IN PLACE AND THE WALL STEEL SHALL BE LEFT EXPOSED TO A HEIGHT 2" BELOW THE FINISH TOP ELEVATION, OR AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR).
2. PRE-CAST SHOP DRAWINGS ARE TO BE APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) FOR PUBLICLY FINANCED OR ADMINISTERED PROJECTS.
3. DO NOT SCALE THESE DRAWINGS FOR DIMENSIONS OR CLEARANCES. ANY QUESTIONS REGARDING DIMENSIONS SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) PRIOR TO CONSTRUCTION.
4. THE FIRST DIMENSION LISTED IN THE CONSTRUCTION NOTES IS THE "L" DIMENSION. THE SECOND DIMENSION IS THE "W" DIMENSION. THE CONCRETE THICKNESS AND REINFORCEMENT SHOWN IS FOR BOXES WITH ("L"+"H") AND ("W"+"H") LESS THAN OR EQUAL TO 20. FOR BOXES WITH EITHER OF THESE CALCULATIONS GREATER THAN 20, A SPECIAL DESIGN IS REQUIRED.

CONCRETE

5. CONCRETE USED IN THIS WORK SHALL BE KDOT GRADE 4.0. SEE SPECIFICATION 32 16 19.
6. INLET FLOORS SHALL BE SHAPED WITH NON-REINFORCED CONCRETE INVERTS TO PROVIDE SMOOTH FLOW.
7. BEVEL ALL EXPOSED EDGES WITH 3/4" TRIANGULAR MOLDING.

REINFORCING STEEL

8. REINFORCING STEEL SHALL BE NEW BILLET, MINIMUM GRADE 40 AS PER ASTM A615, AND SHALL BE BENT COLD.
9. ALL DIMENSIONS RELATIVE TO REINFORCING STEEL ARE TO CENTERLINE OF BARS. 2" CLEARANCE SHALL BE PROVIDED THROUGHOUT UNLESS NOTED OTHERWISE. TOLERANCE OF +/- 1/8" SHALL BE PERMITTED.
10. ALL LAP SPLICES NOT SHOWN SHALL BE A MINIMUM OF 40 BAR DIAMETERS IN LENGTH.
11. ALL REINFORCING STEEL SHALL BE SUPPORTED ON FABRICATED STEEL BAR SUPPORTS @ 3'-0" MAXIMUM SPACING.
12. ALL DOWELS SHALL BE ACCURATELY PLACED AND SECURELY TIED IN PLACE PRIOR TO PLACEMENT OF BOTTOM SLAB CONCRETE. STICKING OF DOWELS INTO FRESH OR PARTIALLY HARDENED CONCRETE WILL NOT BE ACCEPTABLE.

CONSTRUCTION

13. THE BOTTOM SLAB SHALL BE AT LEAST 24 HOURS OLD BEFORE PLACING SIDEWALL CONCRETE. ALL SIDEWALL FORMS SHALL REMAIN IN PLACE A MINIMUM OF 24 HOURS AFTER SIDEWALLS ARE POURED BEFORE REMOVAL, AND AFTER REMOVAL SHALL BE IMMEDIATELY TREATED WITH MEMBRANE CURING COMPOUND.
14. PIPE CONNECTIONS TO PRE-CAST STRUCTURES SHALL HAVE A MINIMUM OF 6" OF CONCRETE AROUND THE ENTIRE PIPE WITHIN 2' OF THE STRUCTURE.
15. TAMPED BACKFILL SHALL BE USED AROUND STRUCTURES, EXCEPT UNDER PAVED AREAS. TAMPED BACKFILL SHALL BE FINELY DIVIDED JOB EXCAVATED MATERIAL FREE FROM DEBRIS, ORGANIC MATERIAL AND STONES, COMPACTED TO 95% MAXIMUM DENSITY AS DETERMINED BY AASHTO STANDARD, METHOD T-99.
16. REMOVABLE FLOWABLE FILL IS REQUIRED UNDER ANY PORTION OF PAVED RIGHT OF WAY, INCLUDING THE AREA WITHIN EXISTING OR FUTURE PUBLIC STREET PAVEMENT AND UNDER EXISTING OR FUTURE CURB AND GUTTER, MEDIAN, ASPHALT PATH, CONCRETE SIDEWALK.

A1 JUNCTION BOX DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

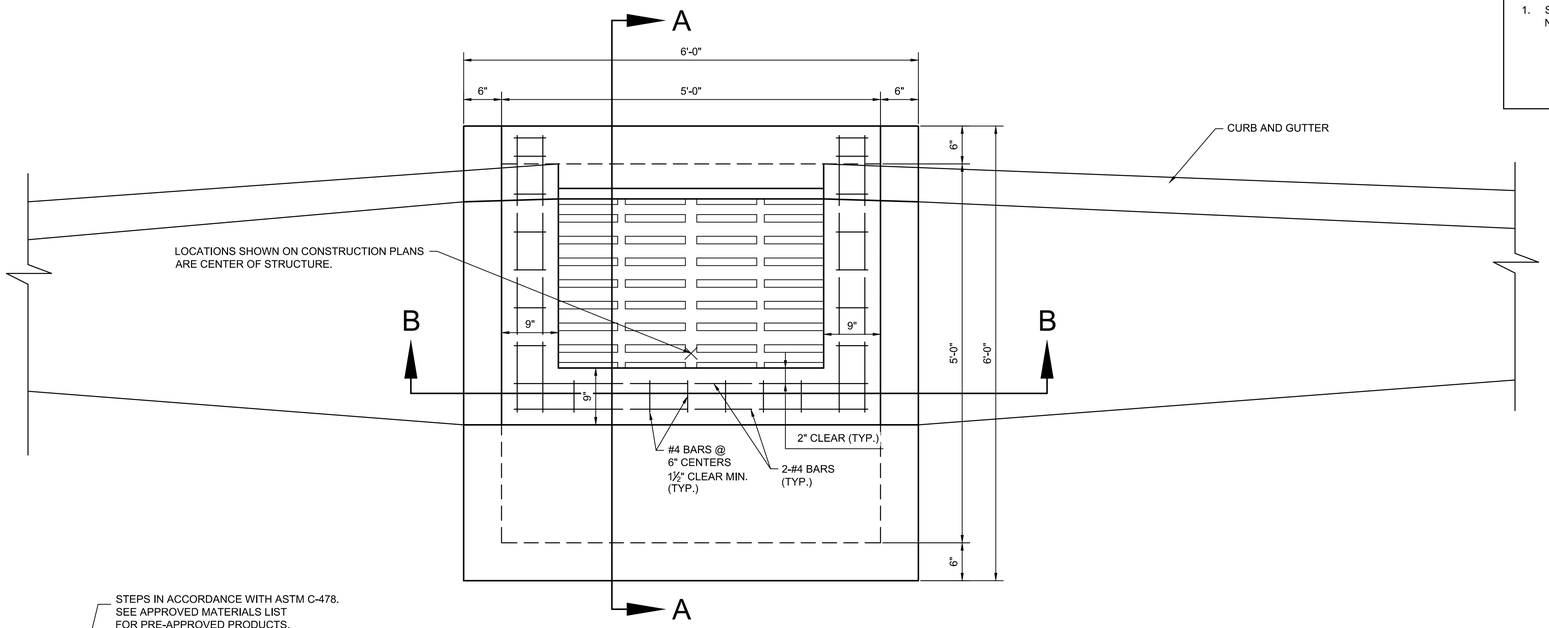
SHEET ID
C-516

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C516-JBOX.dwg
9/9/2026 4:26:00 PM

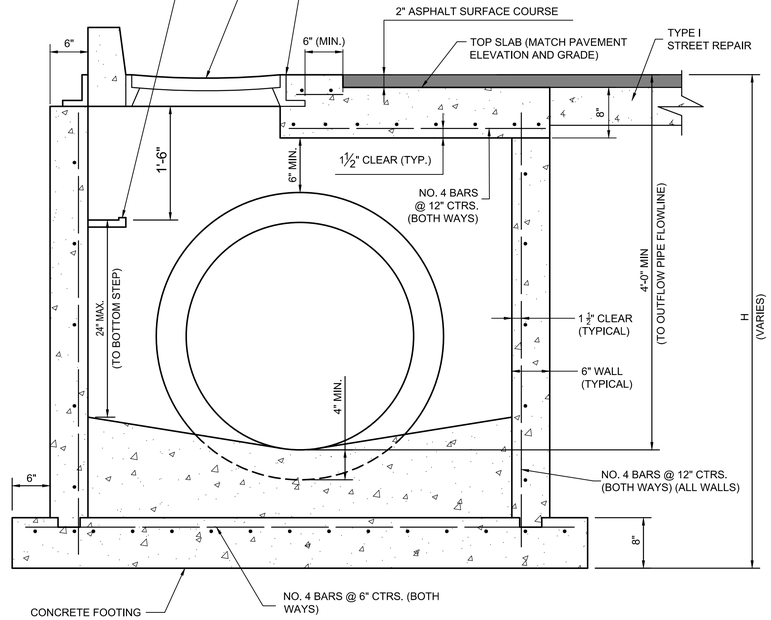
READY TO ADVERTISE

SHEET NOTES

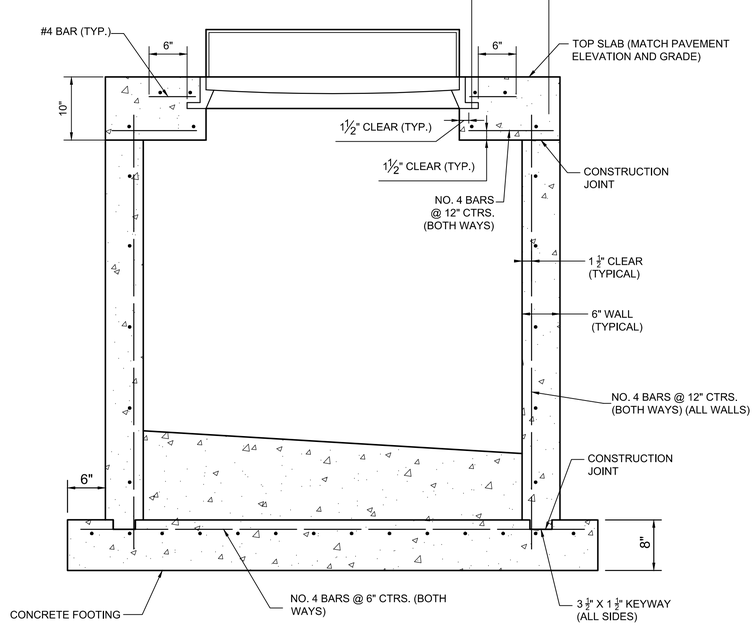
1. SEE SHEET C-510 FOR ADDITIONAL CURB INLET NOTES AND DETAILS.



PLAN
NOT TO SCALE



SECTION A-A
NOT TO SCALE



SECTION B-B
NOT TO SCALE

A1 CURB INLET WITH GRATE DETAILS
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

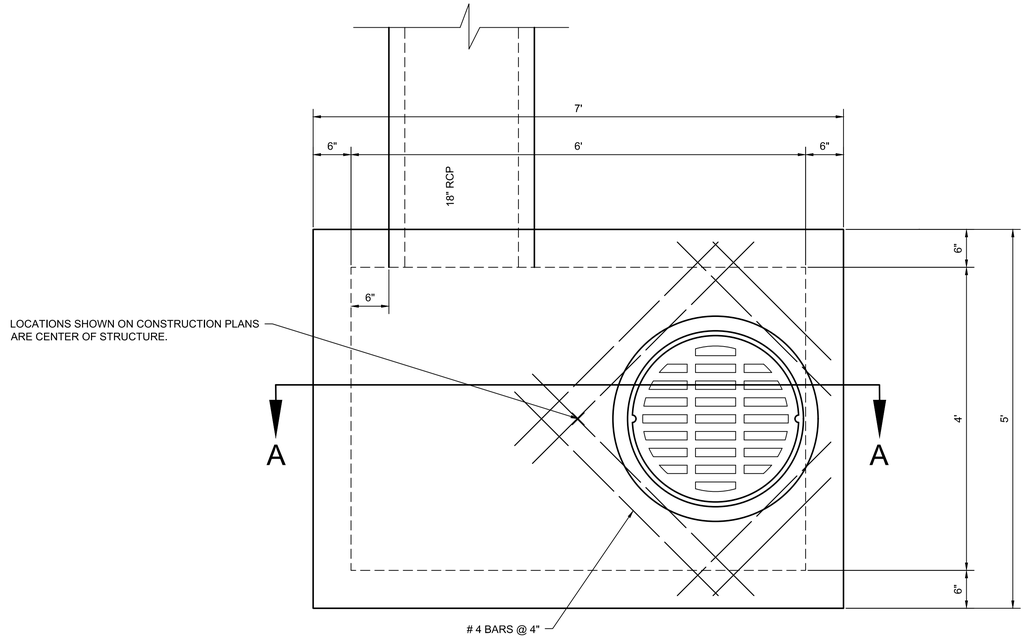
SHEET ID
C-517

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C517-CIGRATE.dwg
9/9/2026 4:26:20 PM

READY TO ADVERTISE

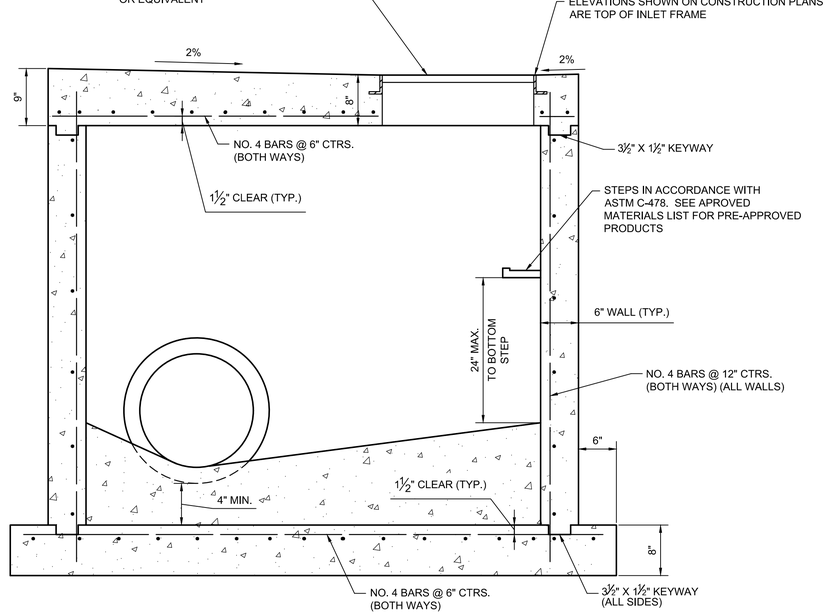
SHEET NOTES

1. SEE SHEET C-510 FOR ADDITIONAL CURB INLET NOTES AND DETAILS.



PLAN

NOT TO SCALE



SECTION A-A

NOT TO SCALE

A1 GRATE INLET DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

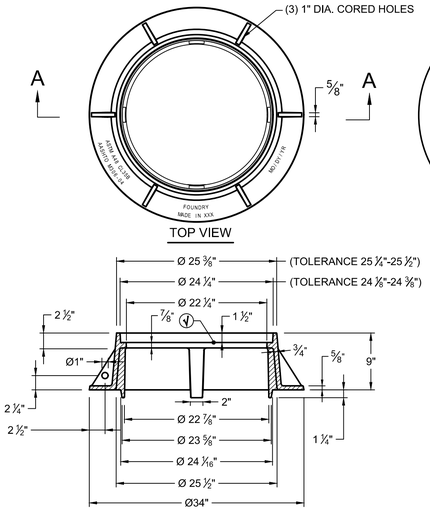
CIVIL DETAILS

SHEET ID

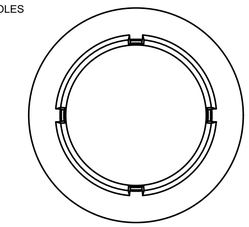
C-518

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C518-GINET.dwg
9/9/2026 4:26:41 PM

READY TO ADVERTISE

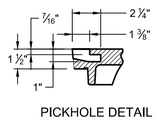


TOP VIEW

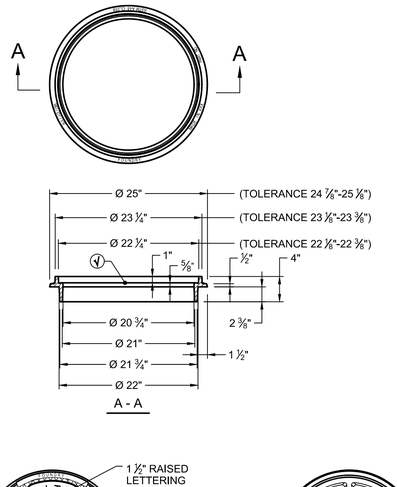


BOTTOM VIEW

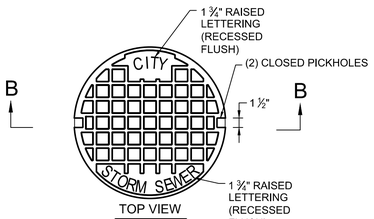
A - A



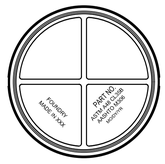
PICKHOLE DETAIL



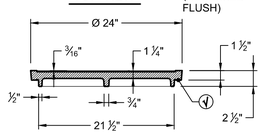
A - A



TOP VIEW



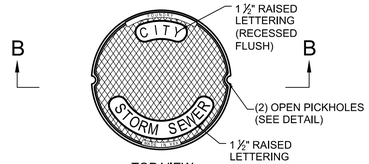
BOTTOM VIEW



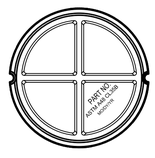
B - B

WEIGHT
COVER 161 LBS. (±8 LBS.)
FRAME 234 LBS. (±11.75 LBS.)

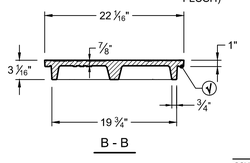
HEAVY DUTY FRAME & COVER



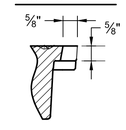
TOP VIEW



BOTTOM VIEW



B - B



PICKHOLE DETAIL

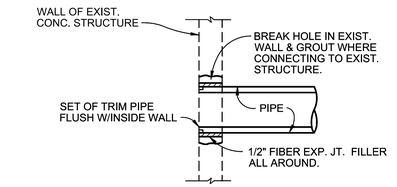
WEIGHT
COVER 134 LBS. (±6.75 LBS.)
FRAME 52 LBS. (±2.5 LBS.)

MEDIUM DUTY FRAME & COVER

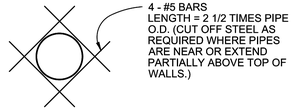
MANHOLE CASTING NOTES:

1. CASTING SHALL COMPLY WITH ASTM A-48 AND AASHTO M306-04.
2. AS-CAST DIMENSIONS MAY VARY ± 1/16 INCH PER FOOT.
3. EACH CASTING SHALL BE MARKED IN ACCORDANCE WITH AASHTO M 306-04, SEC.9.
4. RAISED SURFACES SHALL BE CAST AS NON-SKID FINISH.
5. THE SURFACES LABELED ON THESE DETAILS AS √ SHALL BE MACHINED.
6. FOR PRIVATE SYSTEMS OMIT LETTERING.
7. PAINT IS OPTIONAL UNLESS SPECIFIED.
8. YEAR AND/OR DATE SHOULD BE DATE OF MANUFACTURE.

A1 MANHOLE FRAME & COVER DETAILS
SCALE: NOT TO SCALE



H15 PIPE CONNECTION AT EXISTING CONCETE STRUCTURE
SCALE: NOT TO SCALE



E15 ADDITIONAL REINFORCING AT NEW CONCETE STRUCTURE
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
DATE:
ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

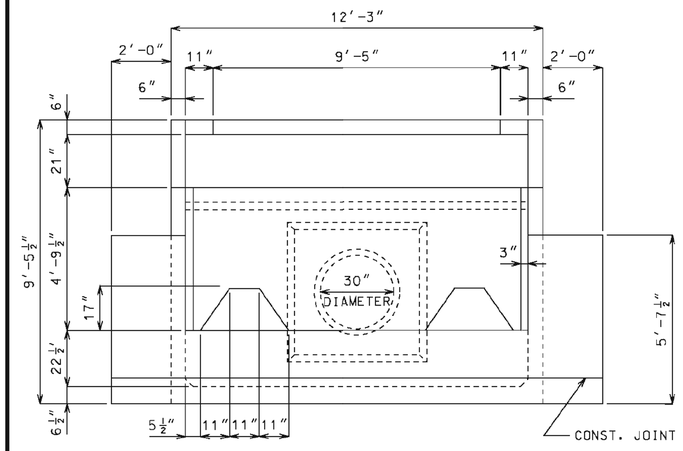
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

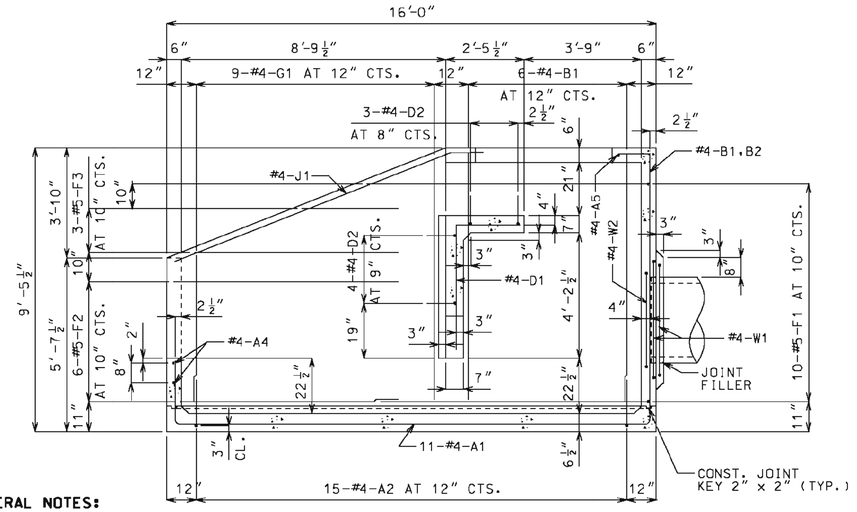
SHEET ID
C-519

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C519-MHC.dwg
9/9/2026 4:27:10 PM

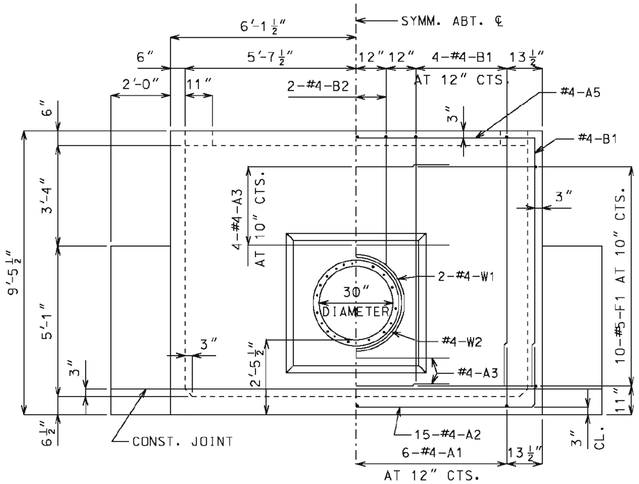
READY TO ADVERTISE



DOWNSTREAM ELEVATION

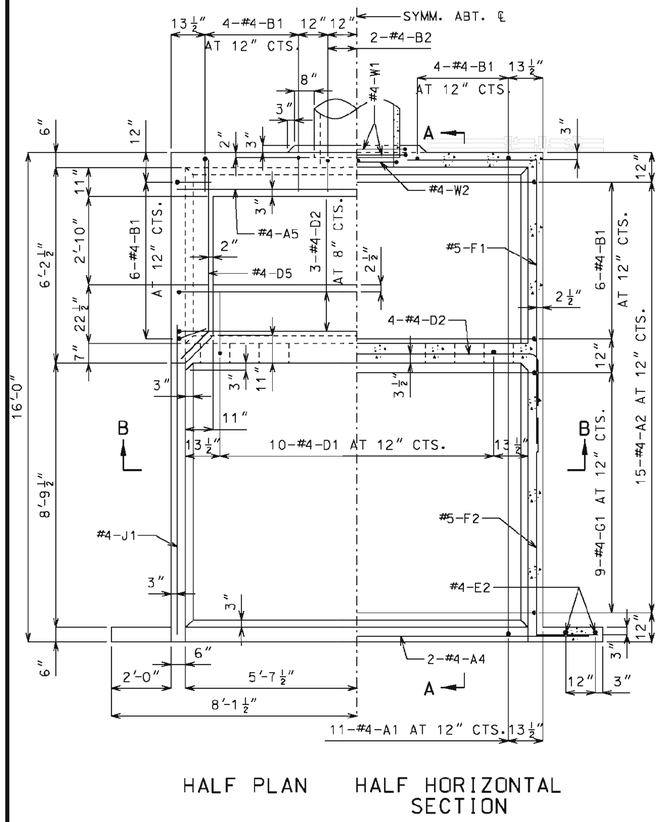


SECTION A - A



UPSTREAM ELEVATION

NOTE: BEND OR CUT A1 BARS IN FIELD TO CLEAR PIPE.



HALF PLAN HALF HORIZONTAL SECTION

GENERAL NOTES:

DESIGN UNIT STRESSES

CLASS B CONCRETE f'c = 3,000 psi
REINFORCING STEEL (GRADE 60) fy = 60,000 psi

REINFORCING STEEL

MINIMUM CLEARANCE TO REINFORCING STEEL SHALL BE 1½" UNLESS SHOWN OTHERWISE.

DIMENSIONS

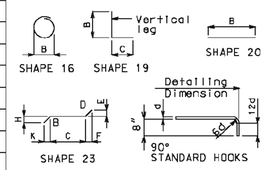
DRAWINGS ARE NOT TO SCALE. FOLLOW DIMENSIONS.

ESTIMATED QUANTITIES		
ITEM		TOTAL
CLASS B CONCRETE	CU. YD.	13.2
REINFORCING STEEL	LBS.	1,170

COMPLETE BILL OF REINFORCING STEEL

NO. REQ'D	MARK NO. SIZE	MARK	SHAPE NO.	VARIES (V)	NO. EACH	DIMENSIONS B	C	D	E	F	H	K	NOMINAL LENGTH	ACTUAL LENGTH	WEIGHT
						FT. IN.	FT. IN.	FT. IN.	FT. IN.	FT. IN.	FT. IN.	FT. IN.	FT. IN.	FT. IN.	LBS.
11	4	A1	11				2-0½	15-6	0-20½				19-3	19-1	140
15	4	A2	11				0-20½	11-10	0-20½				15-3	15-1	151
6	4	A3	20			5-4							5-4	5-4	21
2	4	A4	20			13-3							13-3	13-3	18
1	4	A5	20			11-9							11-9	11-9	8
20	4	B1	19			8-5½	0-13						9-7	9-5	126
3	4	B2	19			4-0	0-13						5-1	5-0	10
10	4	D1	19			4-4	2-0½						6-5	6-3	42
7	4	D2	11				0-12	11-10	0-12				13-10	13-8	64
1	4	D3	15			2-11	4-7	2-11	2-6	0-18	2-6	0-18	10-5	10-4	7
2	4	D4	14			0-12	0-7	3-0			2-6	0-20	4-7	4-5	6
2	4	D5	23			0-18⅜	5-7				0-13	0-13	7-1	7-1	9
2	4	E1	20			3-0							3-0	3-0	4
4	4	E2	20			5-3							5-3	5-3	14
20	5	F1	19			8-8⅜	4-2½						12-11	12-9	266
12	5	F2	19			2-1	8-1						10-2	10-1	126
6	5	F3	20	V	2	3-0⅞							3-1	3-1	
						7-0⅞	(INCREMENT = 24 INCHES)						7-1	7-1	32
18	4	G1	20	V	2	5-1⅛							5-1	5-1	
						8-5⅛	(INCREMENT = 5 INCHES)						8-5	8-5	81
2	4	J1	20			10-11							10-11	10-11	15
2	4	W1	16			3-6							11-10	11-10	16
1	4	W2	16			2-11							10-0	10-0	7

BENDING DIAGRAMS



ALL STANDARD HOOKS AND BENDS OTHER THAN 180 DEG. TO BE BENT WITH SAME PROCEDURE AS FOR 90 DEG. STANDARD HOOKS.

HOOKS AND BENDS SHALL BE IN ACCORDANCE WITH THE PROCEDURES AS SHOWN ON THIS SHEET.

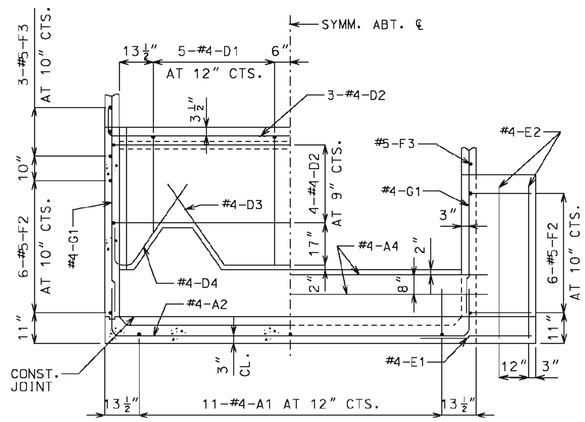
NOMINAL LENGTHS ARE BASED ON OUT TO OUT DIMENSIONS SHOWN IN BENDING DIAGRAMS AND ARE LISTED FOR FABRICATORS USE.

PAYWEIGHTS ARE BASED ON ACTUAL LENGTHS.

LENGTH = TOTAL LENGTHS ARE MEASURED ALONG CENTERLINE BAR TO THE NEAREST INCH.

V - BAR DIMENSIONS VARY IN EQUAL IN-CREMENTS BETWEEN DIMENSIONS SHOWN ON THIS LINE AND THE FOLLOWING LINE.

NO.EA. = NUMBER OF BARS OF EACH LENGTH.



HALF SECTION B-B

HALF ELEVATION OF LOWER BAFFLE AND WING

NOTE: BEND OR CUT D1 BARS IN FIELD TO CLEAR NOTCH IN BAFFLE WALL.

MISSOURI HIGHWAYS AND TRANSPORTATION COMMISSION
105 WEST CAPITOL
JEFFERSON CITY, MO 65102
1-888-ASK-MODOT (1-888-275-6636)

STATE OF MISSOURI
DENNIS W. HECKMAN
NUMBER PE-27141
PROFESSIONAL ENGINEER
THIS SHEET HAS BEEN SIGNED, SEALED AND DATED ELECTRONICALLY.

PIPE CULVERT HEADWALL

ENERGY DISSIPATOR (IMPACT TYPE) FOR 30" CONCRETE PIPE

DATE EFFECTIVE: 07/01/2001
DATE PREPARED: 9/3/2009

604.12E

SHEET NO. 1 OF 1

US Army Corps of Engineers®

IF A SEAL IS PRESENT ON THIS SHEET IT HAS BEEN ELECTRONICALLY SEALED AND DATED.

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

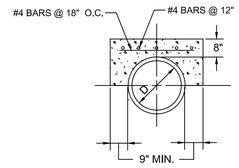
CIVIL DETAILS

SHEET ID C-520

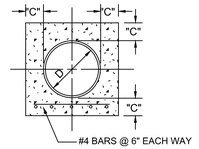
A1 PIPE CULVERT HEADWALL DETAILS
SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C520-HDWL.dwg
9/9/2026 4:27:38 PM

READY TO ADVERTISE

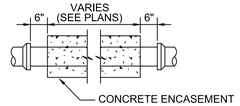


CLASS "A" EMBEDMENT
STANDARD REINFORCED CONCRETE
(CAP ENCASEMENT)



CLASS "A" EMBEDMENT
STANDARD REINFORCED CONCRETE
(ENCASEMENT)

TYPE	P	FACTOR
REINFORCED	0.40%	3.4
REINFORCED	1.00%	4.8
PLAIN		2.8



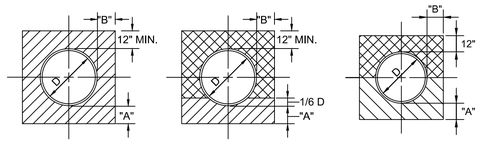
"C" = 6" FOR PIPE 18" & LESS
8" FOR PIPE 21" THRU 36"

NOTE: ENCASEMENT CONCRETE SHALL BE KDOT GRADE 4.0. SEE SPECIFICATION 32 16 19.

NOTE: CONTRACTOR SHALL PROVIDE DOCUMENTATION SUPPORTING THE SELECTED STRENGTH CLASS OF THE PIPE BASED ON EARTH LOADINGS AND THE CHOSEN PIPE BEDDING.

L7 REINFORCED CONCRETE ENCASEMENTS

SCALE: NOT TO SCALE



FLEXIBLE PIPE EMBEDMENT
(PIPE: PVC, DIP, FIBERGLASS, HDPE)

CLASS C
(RIGID PIPE)

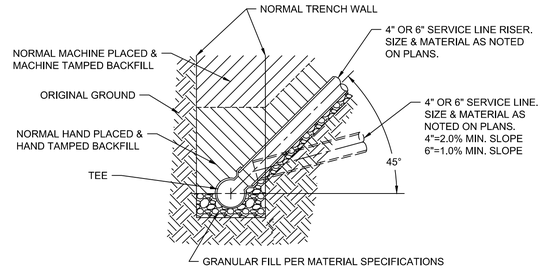
CLASS "B" EMBEDMENT
(PIPE: RCP OPTIONAL)

NOTE: IF THE CONTRACTOR USES BLASTING AS A MEANS OF EXCAVATION THE CONTRACTOR SHALL REMOVE ALL FRACTURED MATERIALS TO THE LIMITS OF WHERE MATERIAL IS UNDISTURBED BEFORE INSTALLING IMPERVIOUS DITCH CHECK.

TABLE OF BEDDING DEPTHS & SIDE CLEARANCES				
	FLEXIBLE		RIGID	
D	A	B	A	B
4"-27"	6"	9"	9"	16"
< 27"	6"	9"	9"	18"

HAND PLACED & HAND TAMPED BACKFILL
GRANDULAR FILL
CONCRETE

D NOMINAL PIPE SIZE
A FILL BELOW PIPE (SEE TABLE)
B SIDE CLEARANCES (SEE TABLE)
P AREA TRANSVERSE STEEL EXPRESSED AS A % OF AREA OF CONCRETE AT CROWN

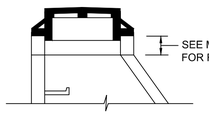


Note:

1. CONCRETE ANCHORS WILL BE REQUIRED WHERE THE SERVICE LINE IS INSTALLED AT A GRADE OF 30' OR GREATER. SEE SECTION 2536 OF THE SPECIFICATIONS AND DETAIL ON THIS SHEET.
2. NO SERVICE LINE SHALL BE INSTALLED WITH A GRADE GREATER THAN 45'.

K14 TEE ORIENTATION AND RISER DETAIL

SCALE: NOT TO SCALE



NOTES:

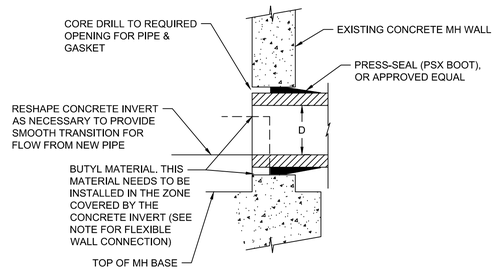
1. ALL MANHOLE RINGS SHALL BE SET IN A MINIMUM OF TWO (2) ROWS OF ¾ TO 1-INCH PRE-FORMED BUTYL JOINT SEALER.
2. MAXIMUM GRADE ADJUSTMENT ALLOWABLE IS 8". MINIMUM ALLOWABLE THICKNESS FOR PRECAST CONCRETE GRADE ADJUSTMENT RING IS 4".
3. RISER RINGS:
   A. MANHOLE IN PAVEMENT: THE THICKNESS OF THE RECYCLED RUBBER RISER RINGS SHALL NOT BE LESS THAN ONE (1) INCH NOR GREATER THAN FOUR (4) INCHES. IF THE REQUIRED THICKNESS OF RISER RINGS EXCEEDS 4 INCHES, A 4-INCH OR 6-INCH PRECAST CONCRETE RISER RING MAYBE INSTALLED BETWEEN THE RUBBER RISER RING REQUIRED TO ACHIEVE THE NECESSARY ADJUSTMENT UP TO A MAXIMUM OF 12 INCHES. THE MINIMUM NUMBER OF RISER RINGS REQUIRED TO ACHIEVE THE NECESSARY ADJUSTMENT SHALL BE USED. THE RUBBER RISER RINGS SHALL BE TAPERED TO MATCH THE SLOPE OF THE EXISTING OR PROPOSED PAVEMENT AT THE MANHOLE. THE JOINTS BETWEEN THE CONE, RUBBER RISER RINGS, AND COATING SHALL BE SEALED WITH THE MANUFACTURER-SUPPLIED SEALANT.
   B. BRICK AND MORTAR ADJUSTMENT WILL NOT BE ALLOWED.

D1 MANHOLE ADJUSTMENT

SCALE: NOT TO SCALE

G7 STANDARD EMBEDMENTS

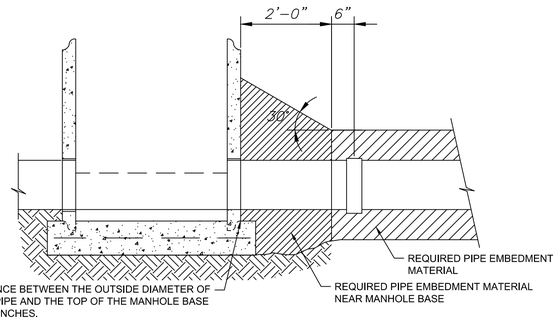
SCALE: NOT TO SCALE



NOTE: CONNECTIONS TO EXISTING BRICK OR BLOCK MANHOLES SHALL BE AS DIRECTED BY CONTRACTING OFFICER'S REPRESENTITIVE (COR)

D9 EXISTING WALL CONNECTION DETAIL

SCALE: NOT TO SCALE



NOTES:

1. FIRST JOINT OF RCP OR VCP PIPE SHALL BE EMBEDDED IN CONCRETE TO WITHIN 6" OF THE FIRST JOINT.
2. IF FLEXIBLE OR SEMI-FLEXIBLE PIPE IS USED, FLEXIBLE WALL CONNECTOR MUST BE USED.
3. IF A FLEXIBLE WALL CONNECTION IS USED IN CONJUNCTION WITH PVC OR DUCTILE IRON PIPE, A STANDARD FLEXIBLE EMBEDMENT SHALL BE USED.
4. FLEXIBLE WALL CONNECTIONS SHALL BE PRESS A-LOC X-CEL, Z-LOC, PRESS-SEAL (PSX BOOT-TYPE GASKET) OR APPROVED EQUAL.

C14 MANHOLE BASE SECTION

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D224F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

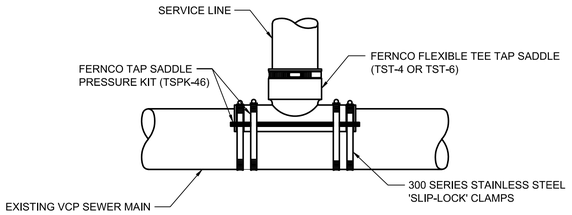
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

+SHEET TITLE 2

SHEET ID
C-521

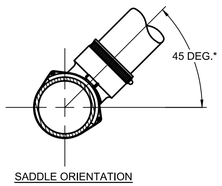
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C521-SAN-DET.dwg
9/9/2026 4:27:56 PM

READY TO ADVERTISE



NOTES:

1. SADDLE SHALL BE INSTALLED IN ACCORDANCE WITH SPECIFICATIONS FOR SANITARY SEWERS AND THE MANUFACTURER'S INSTALLATION INSTRUCTIONS. CONTRACTING OFFICER'S REPRESENTATIVE (COR) INSPECTION AND APPROVAL IS REQUIRED PRIOR TO BACKFILL.
2. ALL TAPS FOR SADDLES SHALL BE MACHINE TAPPED. NO 'CHIPPING' OR BREAK-IN' METHODS ALLOWED. REFER TO MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR CORE BIT OR HOLE SAW SIZING REQUIREMENTS.
3. IF INSTALLING AT AN EXISTING CONNECTION, SAW CUTTING AND REMOVING THE BRANCH OF THE EXISTING WYE/TEE CONNECTION IS REQUIRED. NO 'CHIPPING' OR 'BREAKING' METHODS ALLOWED.
4. IF INSTALLING ON A SEWER MAIN WITH CIPP LINING, THE EXISTING VCP SHALL BE CHIPPED OFF AND REMOVED AND THE SADDLE SHALL BE INSTALLED DIRECTLY ONTO THE CIPP LINER.

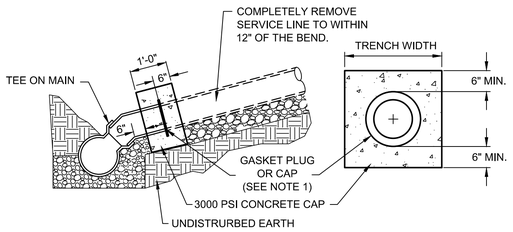


SADDLE ORIENTATION

* SADDLE SHALL BE INSTALLED AT 45 DEGREES AND NOT LESS THAN 30 DEGREES WITH PIPE SPRINGLINE.

G1 SADDLE CONNECTION TO VCP MAIN DETAIL

SCALE: NOT TO SCALE

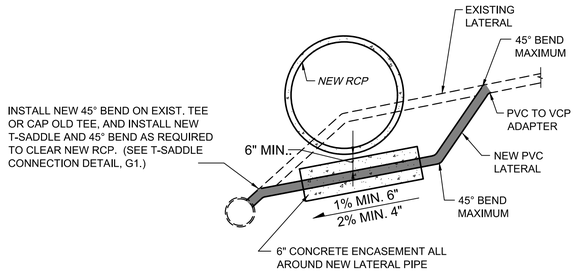


NOTES:

1. GASKET PLUG OR CAP SHALL MATCH PIPE MATERIAL. NATURAL RUBBER SEALS SHALL NOT BE USED. PLUGS ON VCP MAINS SHALL USE A LOW PRESSURE NYLON EXPANSION PLUG WITH A SYNTHETIC GASKET CONFORMING TO ASTM F477.
2. PROJECT REPRESENTATIVE INSPECTION OF THE GASKET PLUG OR CAP INSTALLATION IS REQUIRED PRIOR TO PLACING THE CONCRETE CAP.

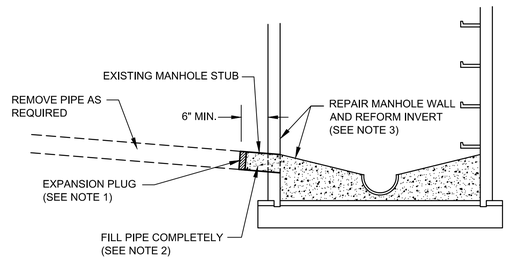
B1 TEE CAP-OFF DETAIL

SCALE: NOT TO SCALE



H8 SANITARY LATERAL RELOCATION

SCALE: NOT TO SCALE

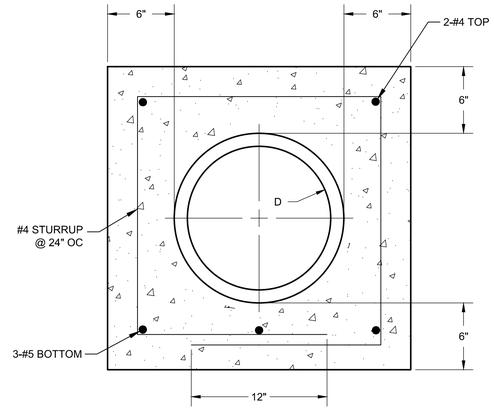


NOTES:

1. PLUG SERVICE LINE WITH A LOW PRESSURE NYLON EXPANSION PLUG WITH A SYNTHETIC GASKET CONFORMING TO ASTM F477. THE PLUG AND PLACEMENT SHALL BE INSPECTED PRIOR TO FILLING THE PIPE.
2. THE PIPE SHALL BE FILLED COMPLETELY WITH QUICK SETTING CORROSION RESISTANT CEMENTITIOUS MATERIAL MIXED AND APPLIED ACCORDING TO THE MANUFACTURER'S RECOMMENDATIONS AND SHALL HAVE THE FOLLOWING MINIMUM REQUIREMENTS: COMPRESSIVE STRENGTH: ASTM C579B, 1400 PSI, 6HRS; BOND: ASTM C321, 1,000 PSI, 24 HRS; APPLIED DENSITY: 120 LBS+/- 5 LBS PCF; SHRINKAGE: ASTM C157, 0% AT 90% R.H. PROJECT REPRESENTATIVE SHALL INSPECT THE EXCAVATION OR WITNESS THE PLACEMENT OF THE CEMENTITIOUS PRODUCT PRIOR TO CONCRETE INSTALLATION.
3. AFTER CLEANING, THE WALL OPENING SHALL BE REPAIRED AND THE INVERT SHALL BE REFORMED USING KODT GRADE 4.0 CONCRETE. CONTRACTING OFFICER'S REPRESENTATIVE (COR) INSPECTION OF THE COMPLETED WORK IS REQUIRED.
4. IF SERVICE CONNECTION IS ACTIVELY LEAKING AT THE MANHOLE CONNECTION, THE LEAK AND PIPE SHALL BE GROUTED. THE AREA WHERE THE GROUT WILL BE APPLIED SHALL BE BE POWER WASHED, PREPPED AND CLEANED PRIOR TO APPLICATION. SETTING CEMENTITIOUS PRODUCT, SPECIFICALLY FORMULATED FOR LEAK CONTROL, WHICH SHALL BE MIXED AND APPLIED ACCORDING TO MANUFACTURER'S RECOMMENDATIONS AND SHALL HAVE THE FOLLOWING MINIMUM REQUIREMENTS: COMPRESSIVE STRENGTH: ASTM C579B, 600 PSI, 1 HR., 1,000 PSI, 24 HRS. BOND: ASTM C321, 30 PSI,1 HR., 80 PSI, 24 HRS. CONTRACTING OFFICER'S REPRESENTATIVE (COR) INSPECTION OF THE GROUT PROCEDURE IS REQUIRED.

A8 MANHOLE CAP-OFF DETAIL

SCALE: NOT TO SCALE



15" PIPE

NOTE: ENCASEMENT CONCRETE SHALL BE KDOT GRADE 4.0

J14 CAGED REINFORCED CONCRETE ENCASEMENT (CRCE)

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

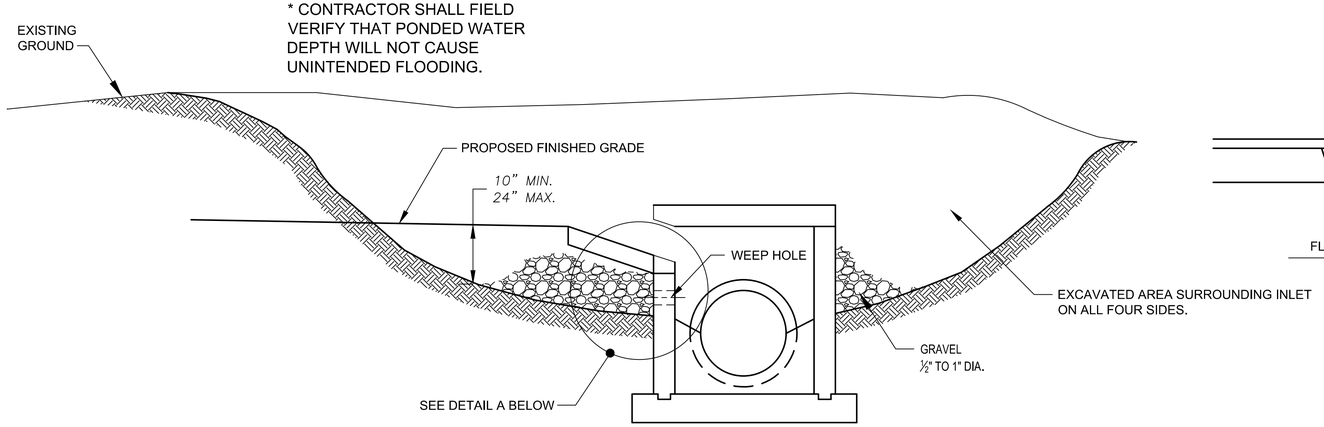
CIVIL DETAILS

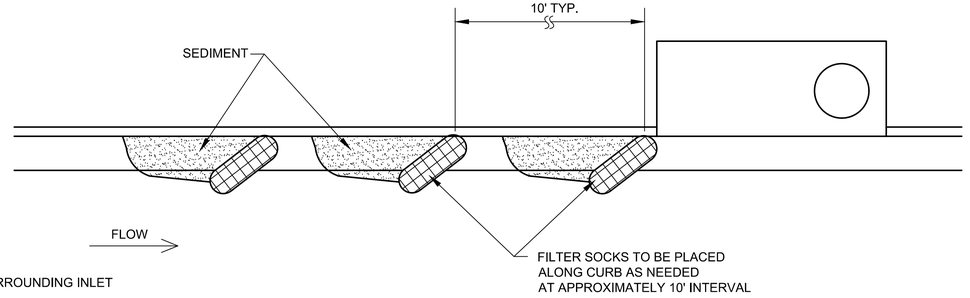
SHEET ID

C-522

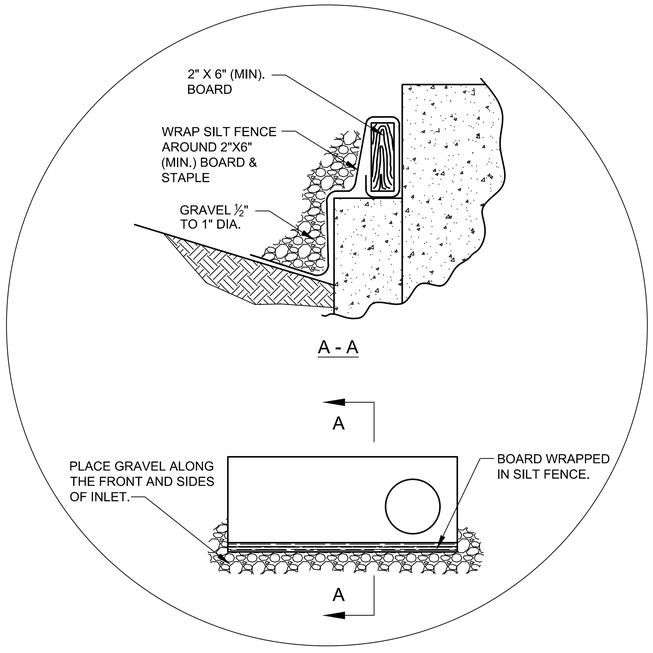
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C522-SAN-DET-2.dwg
9/9/2026 4:28:18 PM

READY TO ADVERTISE





ON GRADE CURB INLET PROTECTION



DETAIL A

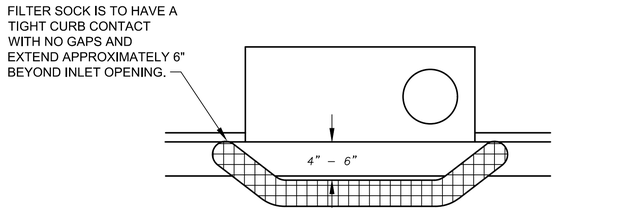
CONDITION A
OPEN BOX AND PRIOR TO POURING CURB AND INLET THROAT

NOTES:

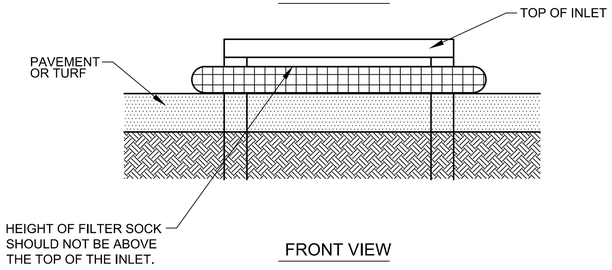
1. IMMEDIATELY FOLLOWING INLET CONSTRUCTION AND PRIOR TO CONSTRUCTION OF CURB AND INLET THROAT, PROTECT INLET OPENING BY INSTALLING 2" X 6" (MIN.) BOARD WRAPPED IN SILT FENCE. STRUCTURES SHALL HAVE EXCAVATED STORAGE AREA ON ALL FOUR SIDES TO ALLOW SETTLING OF SEDIMENT (CONDITION A).
2. WHEN INLET IS COMPLETED AND CURB POURED, FILTER SOCKS OR APPROVED EQUAL SHOULD BE USED (CONDITION B). STRAW WATTLES ARE NOT APPROVED FOR CURB INLET USE.
3. CONTRACTOR TO FIELD VERIFY PONDING WATER SHALL NOT CREATE A TRAFFIC HAZARD.

MAINTENANCE:

REMOVE DEPOSITED SEDIMENT FROM EXCAVATED STORAGE AREAS WHEN AVAILABLE STORAGE HAS BEEN REDUCED BY 20%. REMOVE DEPOSITED SEDIMENT FROM FILTER SOCKS OR SIMILAR WHEN ANY ACCUMULATION OF SEDIMENT IS VISIBLE. REPAIR OR REPLACE AS NECESSARY TO MAINTAIN FUNCTION AND INTEGRITY OF INSTALLATION.



TOP VIEW



FRONT VIEW

SUMP INLET SEDIMENT FILTER

CONDITION B
AFTER POURING CURB AND INLET THROAT

A1 CURB INLET PROTECTION DETAIL
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

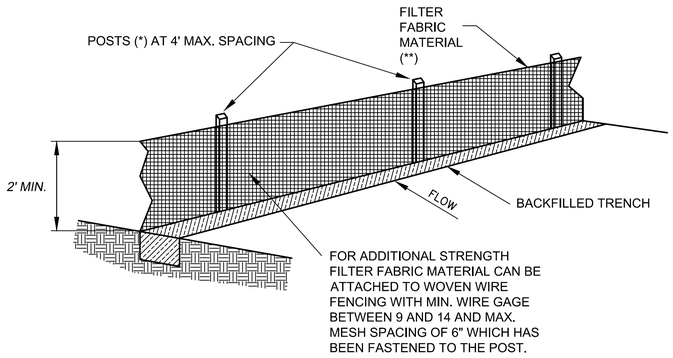
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

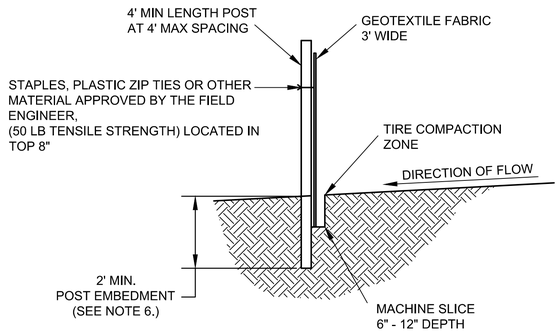
CIVIL DETAILS

SHEET ID
C-523

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C523-EC-CI.dwg
9/9/2026 4:28:37 PM

READY TO ADVERTISE





(*) POSTS

- MIN. LENGTH 4'
- HARDWOOD 1 3⁄16" X 1 3⁄16"
- NO.2 SOUTHERN PINE 2 5⁄8" X 2 5⁄8"
- STEEL 1.33 LB/FT

(**) - GEOTEXTILE FABRIC SHALL MEET THE REQUIREMENTS OF AASHTO M288

NOTES:

1. IN ORDER TO CONTAIN WATER, THE ENDS OF THE SILT FENCE MUST BE TURNED UPHILL (FIGURE A).
2. LONG PERIMETER RUNS OF SILT FENCE MUST BE LIMITED TO 100'. RUNS SHOULD BE BROKEN UP INTO SEVERAL SMALLER SEGMENTS TO MINIMIZE WATER CONCENTRATIONS (FIGURE A).
3. LONG SLOPES SHOULD BE BROKEN UP WITH INTERMEDIATE ROWS OF SILT FENCE TO SLOW RUNOFF VELOCITIES.
4. ATTACH FABRIC TO UPSTREAM SIDE OF POST.
5. INSTALL POSTS A MINIMUM OF 2' INTO THE GROUND.
6. TRENCHING WILL ONLY BE ALLOWED FOR SMALL OR DIFFICULT INSTALLATION WHERE SLICING MACHINE CANNOT BE REASONABLY USED.

MAINTENANCE:

REMOVE AND DISPOSE OF SEDIMENT DEPOSITS WHEN THE DEPOSIT APPROACHES ⅓ THE HEIGHT OF SILT FENCE. REPAIR AS NECESSARY TO MAINTAIN FUNCTION AND STRUCTURE.

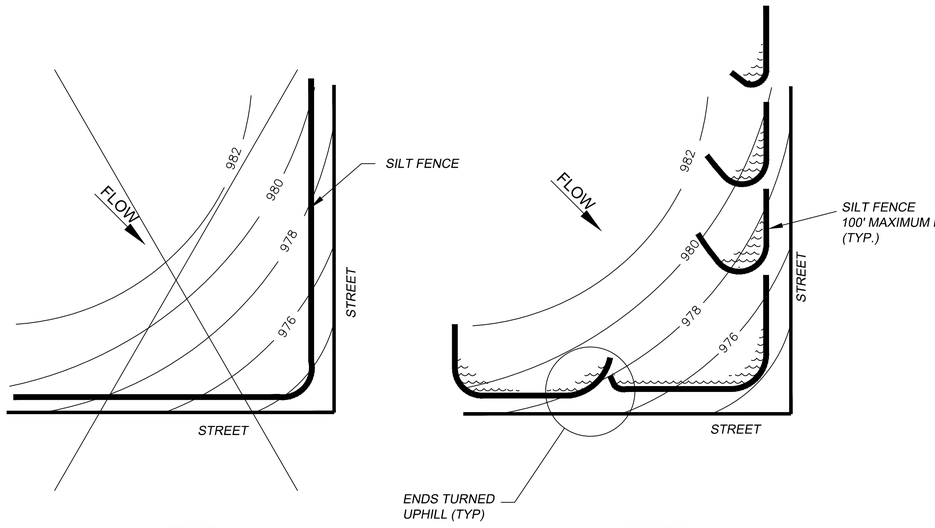
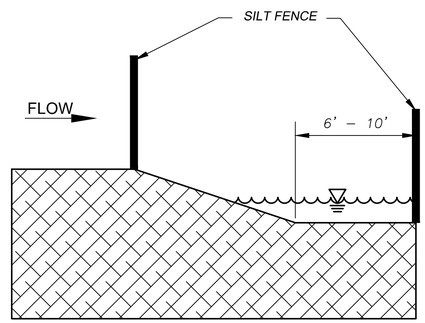


FIGURE A

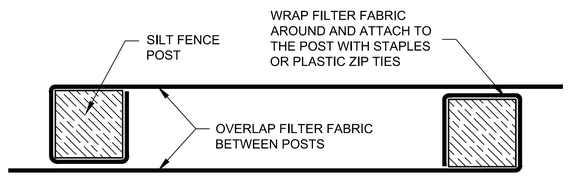
SILT FENCE LAYOUT

NOT TO SCALE

(2015 EDITION)



INSTALL SILT FENCE AT THE TOP OF THE SLOPE TO SLOW VELOCITY AND VOLUME OF WATER AND 6' TO 10' AWAY FROM THE TOW TO CREATE A SEDIMENT STORAGE AREA.



JOINING FENCE SECTIONS

NOT TO SCALE

A1 SILT FENCE DETAIL

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

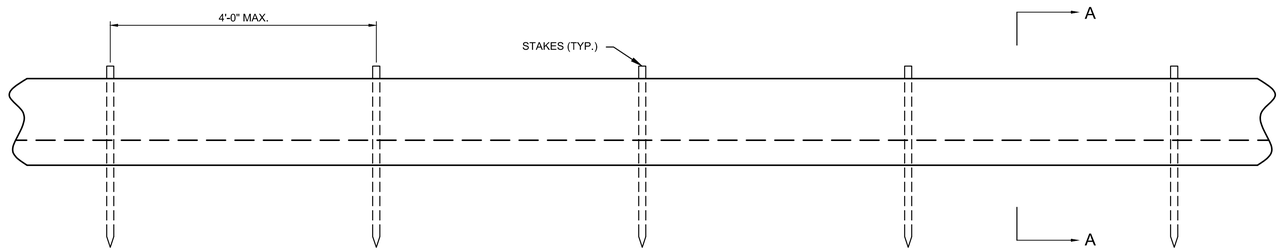
CIVIL DETAILS

SHEET ID

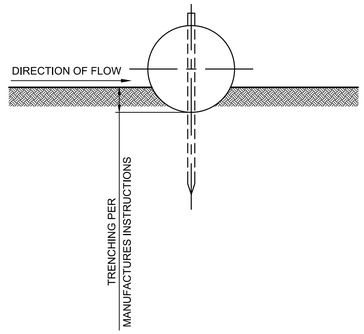
C-524

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C524-EC-FNC.dwg
9/9/2026 4:28:55 PM

READY TO ADVERTISE



TYPICAL ELEVATION



SECTION A-A

NOTES FOR WATTLES AND BIODEGRADABLE LOG SLOPE PROTECTION:

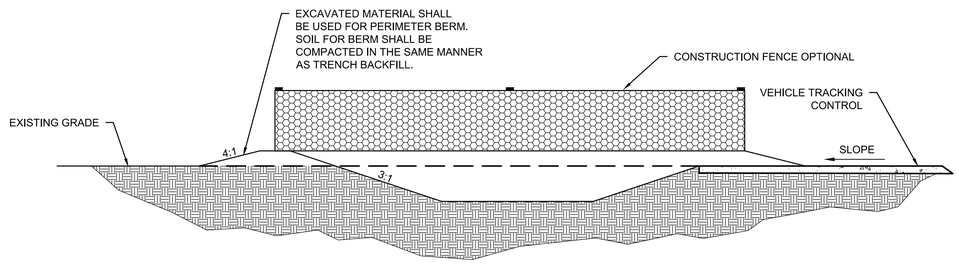
1. THE SLOPE BARRIERS SHALL BE PLACED ALONG CONTOUR LINES, WITH A SHORT SECTION TURNED UPGRADE AT EACH END OF THE BARRIER. THE MAXIMUM LENGTH OF THE SLOPE BARRIER SHALL NOT EXCEED 250 FEET, AND THE BARRIER ENDS NEED TO BE STAGGERED.

2. INSTALL WATTLES AND BIODEGRADABLE LOGS PER MANUFACTURER'S INSTRUCTIONS.

3. SPACING OF STAKES PER MANUFACTURER'S INSTRUCTIONS WITH 4' MAX. SPACING. LENGTH OF STAKES SHALL BE A MINIMUM OF 2 TIMES THE DIAMETER OF THE LOG WITH MINIMUM OF 24".

G1 WATTLE AND BIODEGRADABLE LOG

SCALE: NOT TO SCALE



CONCRETE WASHOUT

NOTES FOR CONCRETE WASHOUT:

1. CONCRETE WASHOUT AREAS SHALL BE INSTALLED PRIOR TO ANY CONCRETE PLACEMENT ON SITE.
2. CONCRETE WASHOUT AREA SHALL INCLUDE A FLAT SUBSURFACE PIT SIZED RELATIVE TO THE AMOUNT OF CONCRETE TO BE PLACED ON SITE. THE SLOPES LEADING OUT OF THE SUBSURFACE PIT SHALL BE 3:1. THE CHICLE TRACKING PAD SHALL BE SLOPED TOWARDS THE CONCRETE WASHOUT AREA.
3. VEHICLE TRACKING CONTROL IS REQUIRED AT THE ACCESS POINT TO ALL CONCRETE WASHOUT AREAS.
4. SIGNS SHALL BE PLACED AT THE CONSTRUCTION SITE ENTRANCE, WASHOUT AREA AND ELSEWHERE AS NECESSARY TO CLEARLY INDICATE THE LOCATION(S) OF THE CONCRETE WASHOUT AREA(S) TO OPERATORS OF CONCRETE TRUCK AND PUMP RIGS.
5. A ONE-PIECE IMPERVIOUS LINER MAY BE REQUIRED ALONG THE BOTTOM AND SIDES OF THE SUBSURFACE PIT IN SANDY OR GRAVELLY SOILS.

MAINTENANCE FOR CONCRETE WASHOUT:

1. CONCRETE WASHOUT MATERIALS SHALL BE REMOVED ONCE THE MATERIALS HAVE FILLED THE WASHOUT TO APPROXIMATELY 75% FULL.
2. CONCRETE WASHOUT AREAS SHALL BE ENLARGED AS NECESSARY TO MAINTAIN CAPACITY FOR WASTED CONCRETE.
3. CONCRETE WASHOUT WATER, WASTED PIECES OF CONCRETE AND ALL OTHER DEBRIS IN THE SUBSURFACE PIT SHALL BE TRANSPORTED FROM THE JOB SITE IN A WATER-TIGHT CONTAINER AND DISPOSED OF PROPERLY.
4. CONCRETE WASHOUT AREAS SHALL REMAIN IN PLACE UNTIL ALL CONCRETE FOR THE PROJECT IS PLACED.
5. WHEN CONCRETE WASHOUT AREAS ARE REMOVED, EXCAVATIONS SHALL BE FILLED WITH SUITABLE COMPACTED BACKFILL AND TOPSOIL, ANY DISTURBED AREAS ASSOCIATED WITH THE INSTALLATION, MAITENANCE, AND/OR REMOVAL OF THE CONCRETE WASHOUT AREAS SHALL BE STABILIZED.

A1 CONCRETE WASHOUT DETAILS

SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

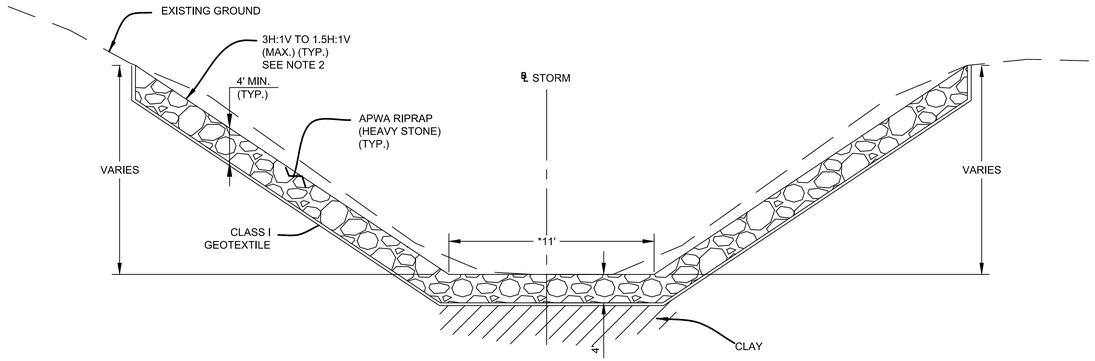
SHEET ID
C-525

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C525-EC-WASH.dwg
9/9/2026 4:29:14 PM

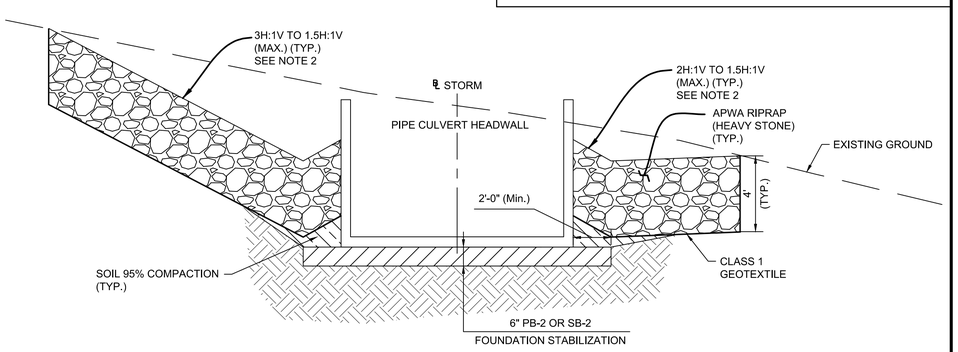
READY TO ADVERTISE

SHEET NOTES

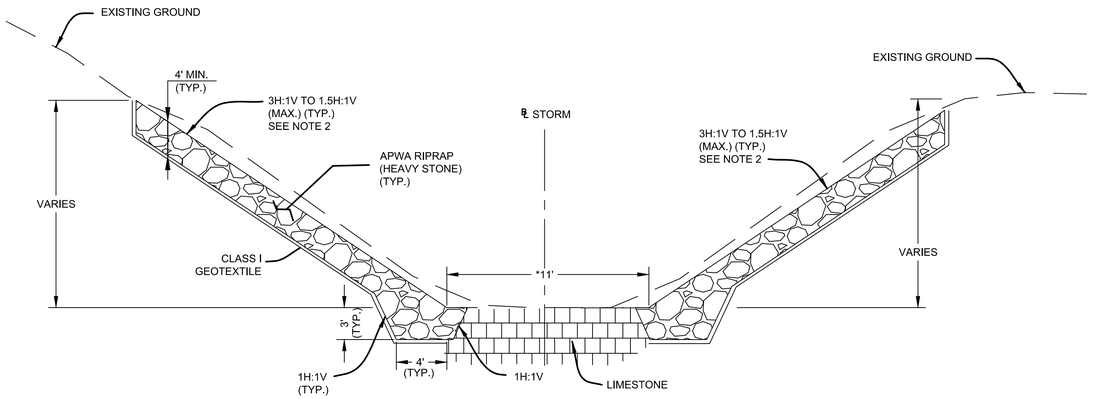
1. FOUNDATION STABILIZATION: PB-2 OR SB-2 USED FOR PIPE CULVERT HEADWALL FOUNDATION STABILIZATION SHALL NOT BE PAID FOR DIRECTLY BUT SHALL BE SUBSIDIARY TO PIPE CULVERT HEADWALL CONSTRUCTION.
2. SEE CIVIL GRADING PLAN CG102, CROSS SECTION SHEETS CU306 & CU307



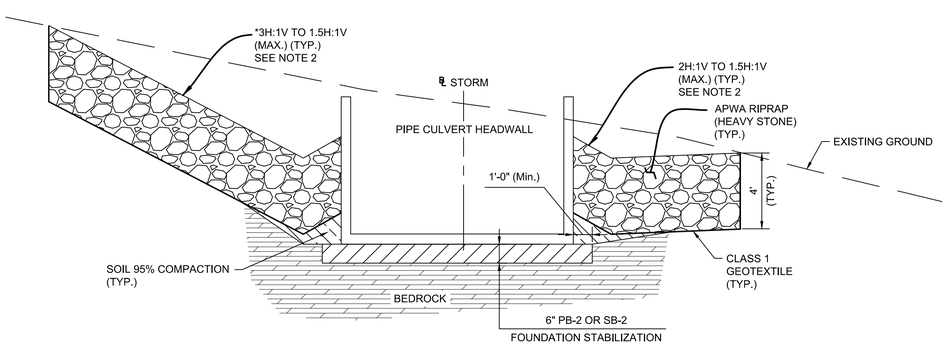
K1 SECTION - RIPRAP TYPICAL SECTION IN CLAY (LOOKING UPSTREAM)
SCALE: NOT TO SCALE



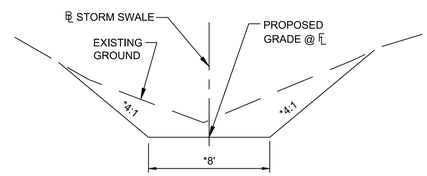
K13 FOUNDATION STABILIZATION ON SOIL (LOOKING UPSTREAM)
SCALE: NOT TO SCALE



D1 SECTION - RIPRAP TYPICAL SECTION IN LIMESTONE (LOOKING UPSTREAM)
SCALE: NOT TO SCALE



E13 FOUNDATION STABILIZATION ON ROCK (LOOKING UPSTREAM)
SCALE: NOT TO SCALE



B15 TYPICAL SWALE SECTION
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

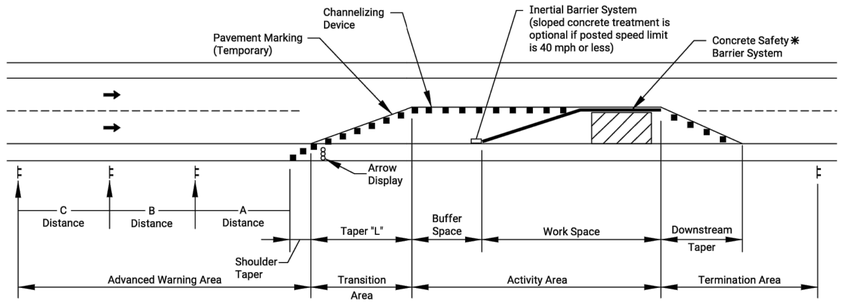
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-526

READY TO ADVERTISE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C526-GRADE-TYP.dwg
9/9/2026 4:29:34 PM



TYPICAL WORK ZONE COMPONENTS

Taper Length (L)

✳✳ Posted speed prior to work starting

Speed✳✳ (MPH)	Width of Offset (ft)												
	1	2	3	4	5	6	7	8	9	10	11	12	13
20	7	13	20	27	33	40	47	53	60	67	73	80	87
25	10	21	31	42	52	63	73	83	94	104	115	125	135
30	15	30	45	60	75	90	105	120	135	150	165	180	195
35	20	41	61	82	102	123	143	163	184	204	225	245	265
40	27	53	80	107	133	160	187	213	240	267	293	320	347
45	45	90	135	180	225	270	315	360	405	450	495	540	585
50	50	100	150	200	250	300	350	400	450	500	550	600	650
55	55	110	165	220	275	330	385	440	495	550	605	660	715
60	60	120	180	240	300	360	420	480	540	600	660	720	780
65	65	130	195	260	325	390	455	520	585	650	715	780	845
70	70	140	210	280	350	420	490	560	630	700	770	840	910
75	75	150	225	300	375	450	525	600	675	750	825	900	975

✳When concrete barrier system is used, portable channelizing devices are not needed along the tangent barrier section.

Taper Formulas:

L = WS for speeds of 45 MPH or more

L = WS²/60 for speeds of 40 MPH or less

Where: L = Minimum length of taper in feet
S = Numerical value of posted speed prior to work starting in MPH
W = Width in offset feet

Shifting Taper=1/2 L
Shoulder Taper=1/3 L

A distance of at least 2L should separate merging and shifting tapers

Advance Warning Sign Spacing

✳✳ Posted speed prior to work starting

SPEED (MPH)✳✳	A	B	C
URBAN (40MPH OR LOWER)	100	100	100
URBAN (45 MPH OR HIGHER)	350	350	350
RURAL (55 MPH OR LOWER)	500	500	500
RURAL (60 MPH OR HIGHER)	750	750	750
EXPRESSWAY/FREEWAY	1000	1500	2640

Minimum advance warning sign spacing (in feet)

The minimum spacing between signs shall be no less than 100', unless directed by the engineer.

The spacing between any signs may be increased beyond the minimum values in the table above as approved by the engineer in order to maximize visibility.

Channelizer Placement:

(1) The spacing between devices in transition area (taper) should not exceed a distance in feet equal to 1/2 the posted speed limit in mph prior to work starting.

(2) The spacing between devices in the advanced warning area and the activity area should not exceed a distance in feet equal to two times the posted speed limit in mph prior to work starting.

(3) Channelizing devices shall be placed for optimum visibility, normally at right angles to the traffic flow.

(4) Place directional indicator barricades in series to direct traffic onto the new path. The arrow sign should not be visible to opposing traffic.

(5) Alternating diagonal orange and white striping must slope downward in the direction traffic is expected to pass.

Buffer Spacing

✳✳ Posted speed prior to work starting

SPEED (MPH)✳✳	*20*	*25*	*30*	*35*	*40*	*45*	*50*	*55*	*60*	*65*	*70*	*75*
LENGTH (ft)	*115*	*155*	*200*	*250*	*305*	*360*	*425*	*495*	*570*	*645*	*730*	*820*

Neither work activity nor storage of equipment, vehicles, or material should occur in the buffer space. When a protection vehicle is placed in advance of the work space, only the space upstream of the vehicle constitutes the buffer space.

If temporary concrete safety barrier system is used to separate approaching traffic from the work space, the barrier system shall be considered part of the activity area. A full lane width should be available throughout the length of the buffer space. See typical work zone components above.

1) Design Speed: Those items delegated to temporary traffic control should be designed and installed using the posted/legal speed of the roadway prior to work starting.

2) Minimum Lane Width: Lane widths shall be a minimum of 11', measured between centerlines of pavement markings (centerline, edge line and lane line) or as directed by the engineer. A lane width less than 10' should be closed according to the currently adopted MUTCD. Lane widths that fall between 11' and 10' should include restricted roadway signing (W5-1, W5-2, W5-4)

3) TTC plans should include bicyclists and pedestrians, including those with disabilities, and should be provided with accesses and passage through the TTC
zone.

3a) Adequate warning, delineation, and channelization should be provided to assist in guiding road users in advance of and through the TTC zone with information in a usable format for pedestrians with vision disabilities. Provisions for effective continuity of accessible circulation paths for pedestrians should be incorporated into the TTC plan.

3b) When existing pedestrian facilities including transit are disrupted, closed, or relocated in a TTC zone, the temporary facilities shall be detectable and include accessibility features consistent with the features present in the existing pedestrian facility. A pedestrian channelizing device that is detectable by a person with a vision disability traveling with the aid of a long cane shall be placed across the full width of the closed pedestrian facility. Pedestrian and bicycles are not allowed on Interstate.

3c) SIDEWALK CLOSED CROSS HERE signs shall include audible information devices to provide adequate communication to pedestrians with vision disabilities.

4) When the driving surface open to traffic is milled, or is a temporary surface made of loose material, or when directed by the engineer, a W8-15 (Grooved Pavement) or W8-7 (Loose Gravel) sign shall be used on mainline approaches. This sign should be placed a "C" distance after the W20-1 (Road Work Ahead) sign. A W8-15p motorcycle plaque shall be used to supplement the W8-15 or W8-7 signs. All signs shall be displayed as long as the condition is present.

5) Alternative temporary rumble strip options may be available. Please contact the Temporary Traffic Control Unit for more information at 785-296-3618

6) When detour routes impact railroad grade crossings, verify that there is no potential to cause queuing, if so, install "DO NOT STOP ON TRACKS" (R8-8).

7) When Temporary Traffic Control plans impact a railroad grade crossing, a temporary traffic signal will need to be approved by KDOT Traffic engineering and the affected railroad company.

8) Temporary traffic control signs shall not restrict the motorist visibility to railroad devices.

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL GENERAL NOTES

TE700

FHWA APPROVAL		12-05-25	APP'D.	Nick Rogers
DESIGNED B.A.H.	DETAILED R.W.B.	QUANTITIES	TRACED	
DESIGN CK.	DETAIL CK.	QUAN.CK.	TRACE CK.	

KDOT Graphics Certified 12-17-2025 Sh. No. 0

KDOT Graphics Certified

Plotted by : KDOT#CADD.Support_ks.gov 19-DEC-2025 01:24
File : TE700.dgn

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

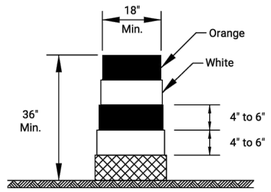
SHEET ID
C-527

A1 TRAFFIC CONTROL GENERAL NOTES
SCALE: NOT TO SCALE

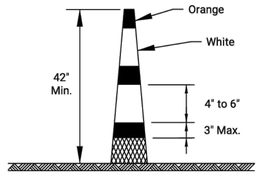
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C527-TC-NOTES.dwg
9/16/2026 1:37:52 PM

READY TO ADVERTISE

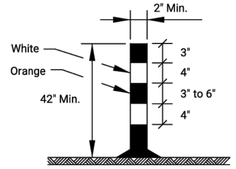
STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0



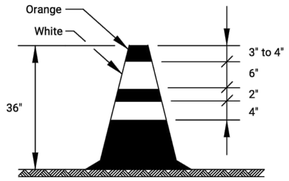
DRUM



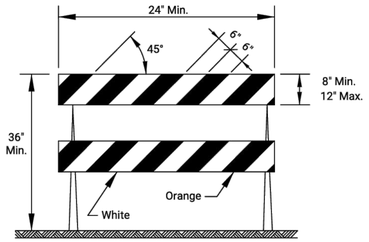
CONICAL DELINEATOR



TUBULAR MARKER

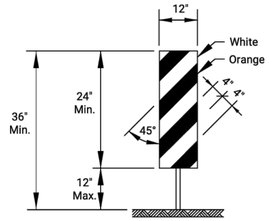


TRAFFIC CONE



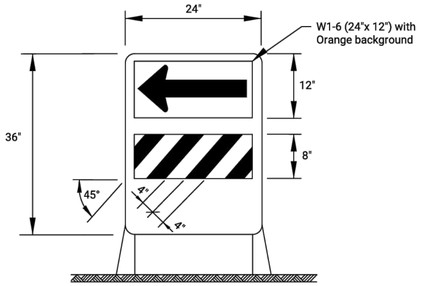
TYPE 2 BARRICADE

For rails less than 36" long, 4" wide stripes may be used. All stripes shall slope downward to the traffic side for channelization.



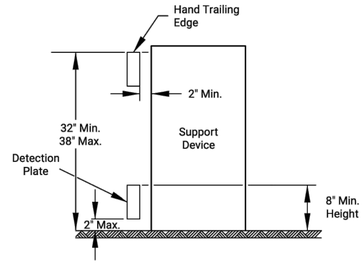
VERTICAL PANEL

The stripes shall slope downward to the traffic side for channelization.



DIRECTION INDICATOR BARRICADE

The stripes shall slope downward in the direction traffic is to pass. The direction indicator barricade shall be used in series to direct the motorist into the intended lane of travel.



PEDESTRIAN CHANNELIZER

1. Support device shall not project beyond the detection plate into the pathway.
2. Hand trailing edges and detection plates are optional for continuous walls.
3. Interconnect pedestrian channelizers to prevent displacement and to provide continuous guidance through or around work.
4. Alternate pathways shall be firm, stable, and slip resistant.
5. Treat height differentials > 1/2" in the surfaces of alternate paths with a firm, stable, and slip resistant temporary ramp having a slope of 12:1 or flatter and having a width equal to the alternate path.
6. Use alternating orange/white on interconnected devices.

Item	Location	Cross-overs	Shoofly Diversions	Tangents	Tapers	Ramps	Head to Head	Object Identifier	Lead-in Devices	Gores
Portable										
	Drums	Yes	Yes	Yes	Yes	Yes	(1)	Yes	Yes	Yes
	Conical Delineators	Yes	Yes	Yes	Yes	Yes	(1)	Yes	Yes	Yes
	Vertical Panels	Yes	Yes	Yes	Yes	Yes	(1)	Yes	Yes	Yes
	Direction Indicator Barricade	No	No	No	Yes	No	No	No	No	No
	Type 2 Barricade	Yes	Yes	Yes	Yes	No	No	Yes	No	No
	Traffic Cones	No	No	(3)	(3)	(3)	No	(3)	(3)	(3)
Fixed										
	Tubular Markers	(2)	(2)	(2)	No	(2)	Yes	No	Yes	Yes
	Vertical Panels	(2)	(2)	(2)	(2)	(2)	(2)	Yes	(2)	Yes

(1) Not allowed on centerline delineation along freeways or expressways.
(2) May be used upon the approval of the engineer.
(3) Daytime operations only.

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL CHANNELIZING DEVICES

TE702

FHWA APPROVAL		12-05-25	APP'D.	Nick Rogers	
DESIGNED L.E.R.	DETAILED R.W.B.		QUANTITIES	TRACED	
DESIGN CK.	DETAIL CK.		QUAN.CK.	TRACE CK.	

KDOT Graphics Certified 12-08-2025 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D0219D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-528

Plotted by: KDOT#CADD.Support_ks.gov 19-DEC-2025 01:01
File : TE702.dgn

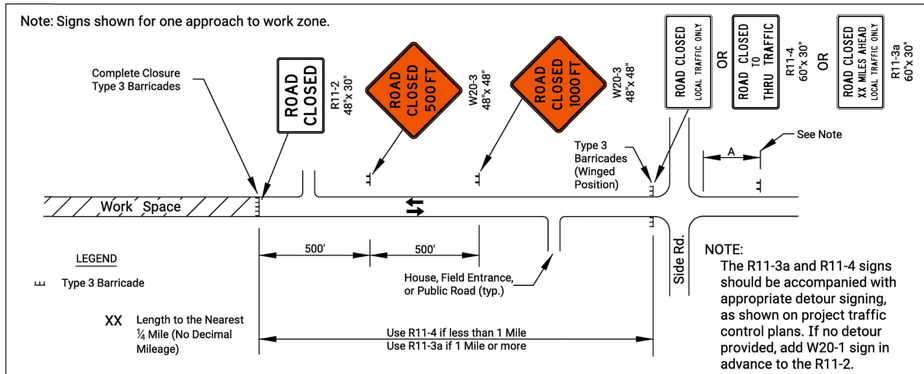
A1 TRAFFIC CONTROL CHANNELIZING DEVICES
SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C528-TC-DEVICE.dwg
9/16/2026 1:37:59 PM

READY TO ADVERTISE

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

Note: Signs shown for one approach to work zone.



NOTE:
The R11-3a and R11-4 signs should be accompanied with appropriate detour signing, as shown on project traffic control plans. If no detour provided, add W20-1 sign in advance to the R11-2.

FIGURE 1: TYPICAL SIGNING FOR ROAD CLOSURE (MAINLINE OR SIDE ROAD)

Note: Sign shown for one approach to intersection (work zone).

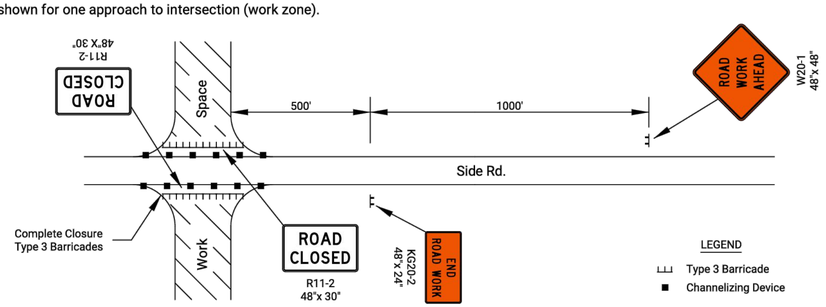


FIGURE 2: TYPICAL SIGNING FOR SIDE ROAD OPEN

Note: Signs shown for one approach to work zone.

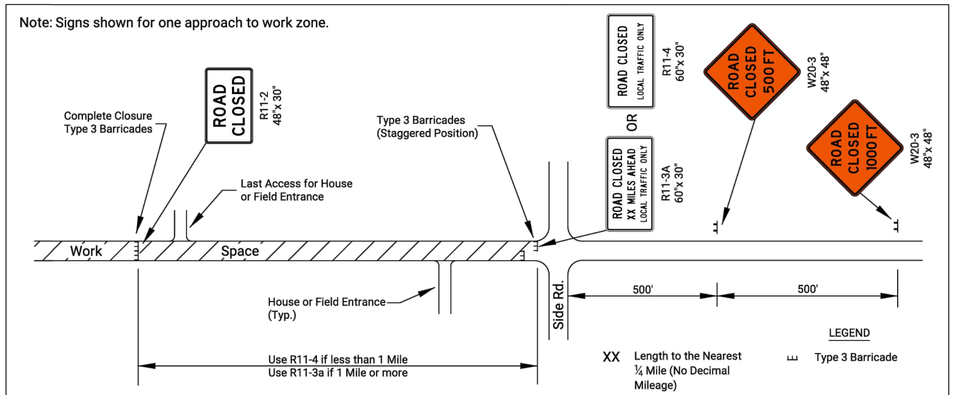


FIGURE 3: TYPICAL SIGNING FOR ROAD CLOSURE - LOCAL TRAFFIC ACCESS

ROAD CLOSED GENERAL NOTES

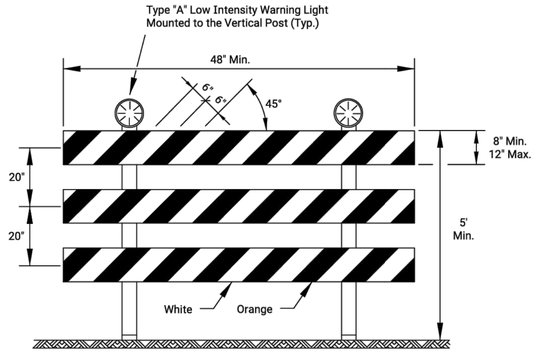
As shown in Figure 1, at the point where thru traffic must detour and local traffic can proceed to the location where the roadway is completely closed, the R11-3a (ROAD CLOSED # MILES AHEAD LOCAL TRAFFIC ONLY) or R11-4 (ROAD CLOSED LOCAL TRAFFIC ONLY or ROAD CLOSED TO THRU TRAFFIC) sign shall be used with Type 3 barricades (winged position), placed on the shoulders of roadway.

As shown in Figure 3, when local traffic must be allowed access into the work zone, Type 3 barricades shall be longitudinally staggered to maintain the appearance of a closed roadway. A second line of end-to-end Type 3 barricades shall be placed just beyond the last access point in the work zone, to completely close the roadway.

The R11-4 (ROAD CLOSED TO THRU TRAFFIC or ROAD CLOSED LOCAL TRAFFIC ONLY) sign shall be used when the distance to the point of complete closure of the roadway is less than 1 mile.

The R11-3a (ROAD CLOSED # MILES AHEAD LOCAL TRAFFIC ONLY) sign shall be used when the distance to the point of complete closure of the roadway is 1 mile or greater.

The words "BRIDGE OUT" (or BRIDGE CLOSED) may be substituted for the words "ROAD CLOSED" on the R11-3a or R11-4 sign where applicable.



TYPE 3 BARRICADE WITH LIGHTS

Approved signs mounted on Type 3 barricades should not cover more than 50% of the top two rails or 33% of the total area of the three rails.

When barricades are placed end-to-end or staggered, a Type "A" low intensity warning light shall be mounted to the vertical post near each outside corner of the end barricades.

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL CLOSURES

TE704

FHWA APPROVAL		12-05-25	APP'D.	Nick Rogers	
DESIGNED	B.A.H.	DETAILED	R.W.B.	QUANTITIES	TRACED
DESIGN CK.		DETAIL CK.		QUAN.CK.	TRACE CK.

KDOT Graphics Certified 12-08-2025 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SEE ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.: TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-529

A1 TRAFFIC CONTROL CLOSURES
SCALE: NOT TO SCALE

Plotted by : KDOT#CADD.Support_ks.gov 19-DEC-2025 01:03
File : TE704.dgn

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C529-TC-CLS.dwg
9/16/2026 1:38:06 PM

READY TO ADVERTISE

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

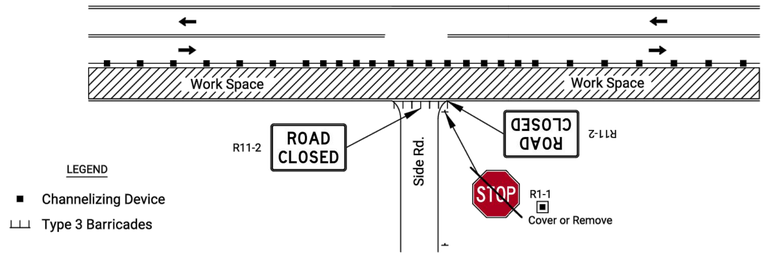


FIGURE 1: SIDE ROAD OR ENTRANCE CLOSED THROUGH WORK AREA

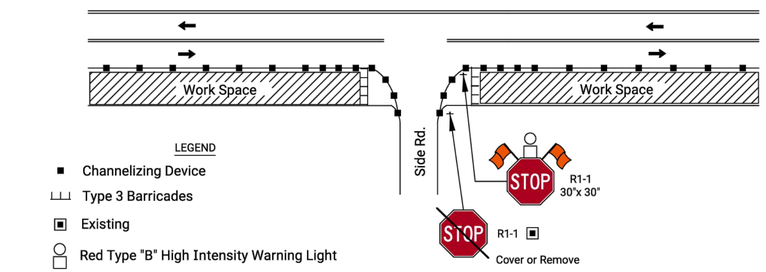


FIGURE 2: SIDE ROAD OR ENTRANCE OPEN THROUGH WORK AREA

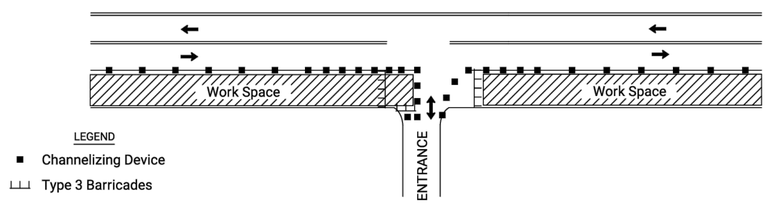


FIGURE 3: LOW VOLUME ENTRANCE CONTRUCTED HALF AT A TIME

Note: Consider large vehicles making right turns into and out of entrance
and use figure 4 as needed

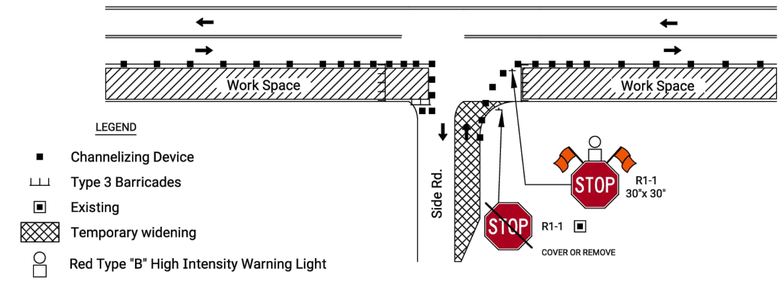


FIGURE 4: SIDE ROAD OR ENTRANCE CONSTRUCTED HALF AT A TIME:
TWO WAY TRAFFIC REQUIRED

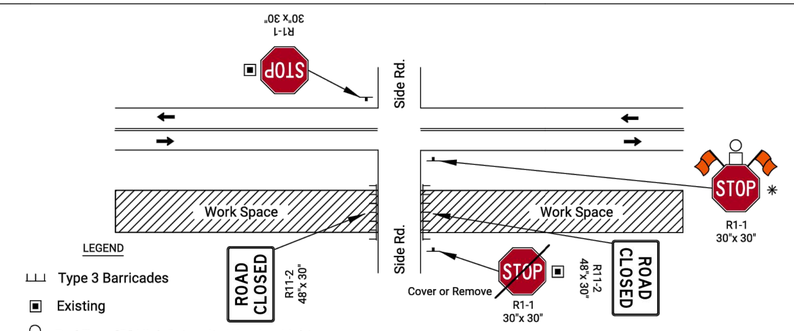


FIGURE 5: SIDE ROAD OPEN THROUGH WORK AREA ON DIVIDED ROADWAY

Note: Yield signs shall be treated as
Stop signs shown in the figures above

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL
ACCESS THROUGH THE WORK AREA

TE705

FHWA APPROVAL			12-05-25	APP'D.	Nick Rogers
DESIGNED	R.W.B.	DETAILED	R.W.B.	QUANTITIES	TRACED
DESIGN CK.		DETAIL CK.		QUAN CK.	TRACE CK.

KDOT Graphics Certified 12-08-2025 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-530

Plotted by : KDOT#CADD.Support_ks.gov 19-DEC-2025 01:04
File : TE705.dgn

A1 TRAFFIC CONTROL ACCESS THROUGH THE WORK AREA
SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C530-TC-WRKZN.dwg
9/16/2026 1:38:11 PM

READY TO ADVERTISE

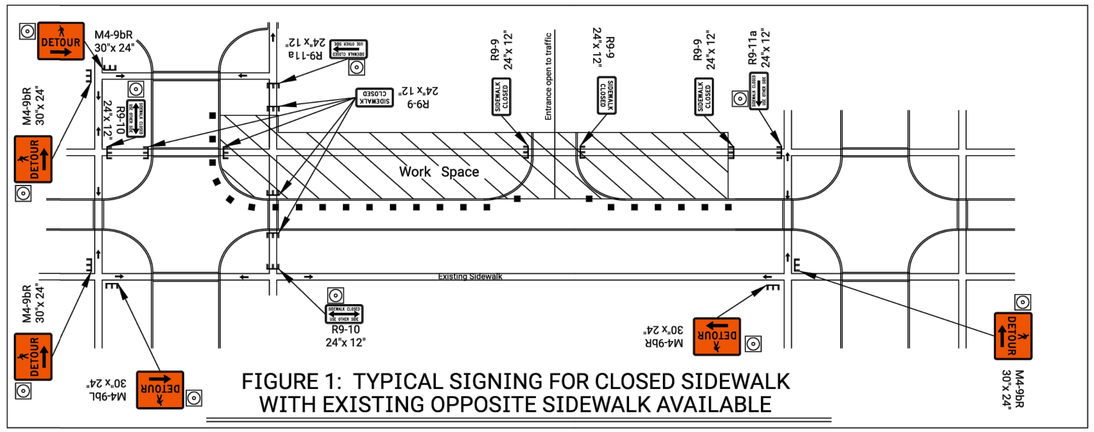


FIGURE 1: TYPICAL SIGNING FOR CLOSED SIDEWALK WITH EXISTING OPPOSITE SIDEWALK AVAILABLE

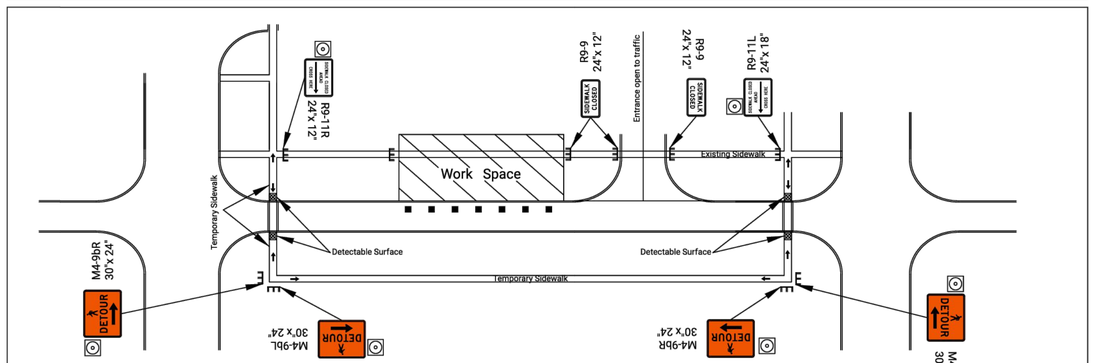


FIGURE 2: TYPICAL SIGNING FOR CLOSED SIDEWALK WITH TEMPORARY SIDEWALK CROSSINGS

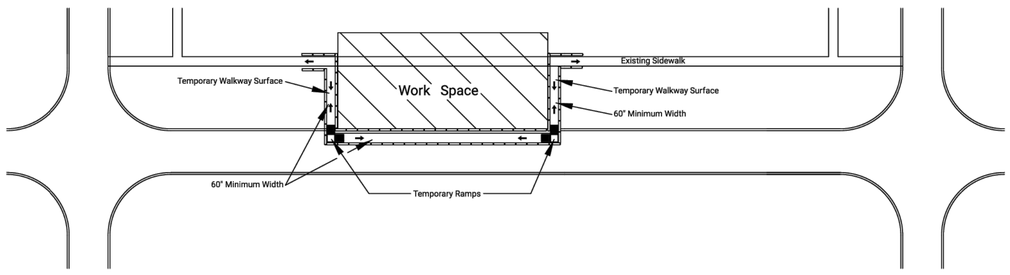


FIGURE 3: TYPICAL SIGNING FOR CLOSED SIDEWALK WITH FOUR LANE OR TWO LANE WITH PARKING

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

NOTE:

When existing pedestrian facilities are disrupted, closed, or relocated in a TTC zone, the temporary facilities shall be detectable and include accessibility features consistent with the currently adopted MUTCD.

Temporary walkway surface shall provide adequate support to cover rough, soft or uneven ground or hazard as approved by the Engineer.

When used, temporary ramps shall provide a 12:1 (8.33%) or flatter slope, with a slip-resistant surface. The ramp landing area shall provide a 48" x 48" minimum area with a 2% or flatter cross-slope.

If TTC for the pedestrian facility encroaches on the roadway, use the applicable traffic control layout sheet and KDOT Standards.

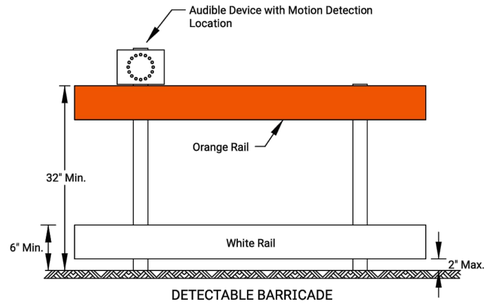
All SIDEWALK CLOSED CROSS HERE, and PEDESTRIAN DETOUR signing shall include audible information devices to provide adequate communication to pedestrians with vision disabilities.

Audible Device (with motion detection) shall be installed per manufacturer and approved by the Engineer.

ROAD WORK AHEAD and END ROAD WORK signs shall be a minimum of 100' from any existing signs.

LEGEND

- Pedestrian Barricade
- Audible Device (with motion detection)
- Pedestrian Channelizer
- Pedestrian Ramp
- Detectable Surface



DETECTABLE BARRICADE

1. Support device shall not project beyond the detection plate into the pathway.
2. Barricades shall be used to close the entire width of the pathway.
3. Do not use warning lights on pedestrian barricades.
4. Do not use warning lights on audible devices.
5. SIDEWALK CLOSED CROSS HERE, SIDEWALK CLOSED AHEAD CROSS HERE, and PEDESTRIAN DETOUR signs along with audible devices shall be installed on Detectable Barricades.
6. A pedestrian channelizing device that is detectable by a person with a vision disability traveling with the aid of a long cane shall be placed across the full width of the closed sidewalk.

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL
SIDEWALK CLOSURES

TE706

FHWA APPROVAL		12-05-25 APP'D.	Nick Rogers
DESIGNED R.W.B.	DETAILED R.W.B.	QUANTITIES	TRACED
DESIGN CK.	DETAIL CK.	QUAN.CK.	TRACE CK.

KDOT Graphics Certified 01-22-2026 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROWL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
SIZE:	ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-531

A1 TRAFFIC CONTROL ACCESS THROUGH THE WORK AREA
SCALE: NOT TO SCALE

Plotted by : Robert.Bartron@ks.gov 22-JAN-2026 21:51
File : te706.dgn

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C531-TC-SWDTR.dwg
9/16/2026 1:38:18 PM

READY TO ADVERTISE

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

SIGN LAYOUT INFORMATION

KG20-2

Std. Size Expwy/Freeway
6" C
48"x 24"

KG20-5

Std. Size Expwy/Freeway
6" C
48"x 24"

KM4-2OP

Std. Size	Expwy/Freeway
3" C	6" C
24"x 6"	48"x 12"

KW21-25

Std. Size Expwy/Freeway
7" C
48"x 48"

KW21-26

Std. Size Expwy/Freeway
6" C
48"x 48"

KR2-6

Std. Size Expwy/Freeway
8" D
48"x 36"

KR2-11

Std. Size Expwy/Freeway
6" C
24"x 48"

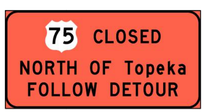
KR2-11 sign shall be used whenever KR2-6 sign is used in a Work Zone.

KE5-25P

Std. Size Expwy/Freeway
12" D
96" x 30"

SP-01
(Special Sign)

Std. Size
6" C
18" Shield
Expwy/Freeway
10" D
36" Shield

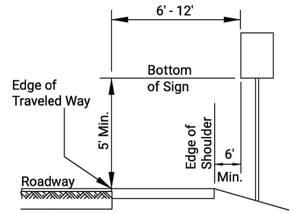


SP-02
(Special Sign)

Std. Size
Uppercase: 6" C
Lowercase: 4.5" C
18" Shield
Expwy/Freeway
Uppercase: 10" D
Lowercase: 8" D
36" Shield

All city names and street names on special signs and destination signs must have upper and lower case letters.

Routes need to be shown using the correct Shield on all Special signs.

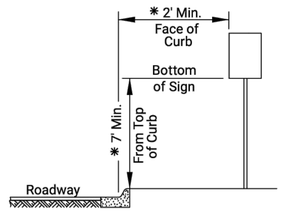


RURAL

1) Ground-mounted signs shall be mounted at a minimum height of 5' measured from the bottom of sign to the near edge of the pavement.

2) Large signs having an area exceeding 50 square feet installed on multiple breakaway posts shall be mounted a minimum of 7' above the ground.

3) The height of the secondary sign mounted below another sign may be 4' measured from the bottom of the sign to the near edge of the pavement. Signs shall not overlap each other.



URBAN

1) Signs shall be mounted at a minimum height of 7' measured from the bottom of sign to the near edge of the pavement.

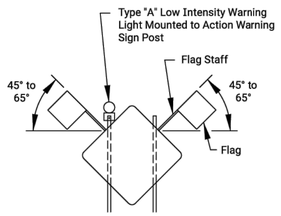
2) Neither portable nor permanent sign supports shall be located on sidewalks or areas designated for pedestrian or bicycle traffic.

3) Signs mounted lower than 7' should not project more than 4" into pedestrian facilities.

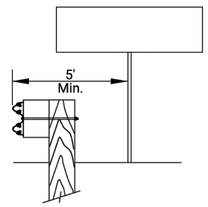
4) The height of the secondary sign mounted below another sign may be 6' measured from the bottom of sign to the near edge of the pavement. Signs shall not overlap each other.

5) Large signs having an area exceeding 50 square feet installed on multiple breakaway posts shall be mounted a minimum of 7' above the ground.

* 6) Pedestrian detour signing shall be a minimum of 2' measured from the top of the pedestrian pathway to the bottom of the sign and shall not protrude into the walkway nor shall it project beyond the back of curb.

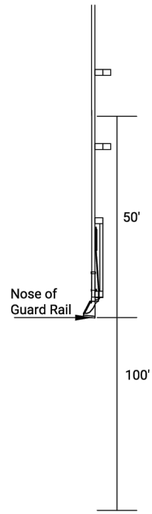


In the case of hitting rock when driving posts
1. Shift the sign location. Do not violate minimum sign spacing.
2. With the engineer's approval, use acceptable alternative sign stands.



1) The edge of the sign is not to extend beyond the back side of the guard rail. The sign post shall not be located within 5' of the guard rail.

2) The mounting height for signs behind the guard rail shall be the same as the mounting height for shoulder mounted signs.



NO SIGN PLACEMENT AREA

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL SIGN INFORMATION

TE710

FHWA APPROVAL			12-05-25	APP'D.		Nick Rogers	
DESIGNED	R.W.B.	DETAILED	R.W.B.	QUANTITIES		TRACED	
DESIGN OK		DETAIL OK		QUAN.OK		TRACE OK	

KDOT Graphics Certified 01-22-2026 Sh. No. 0

KDOT Graphics Certified

Plotted by : Robert.Bartron@ks.gov 22-JAN-2026 21:51
File : TE710.dgn

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

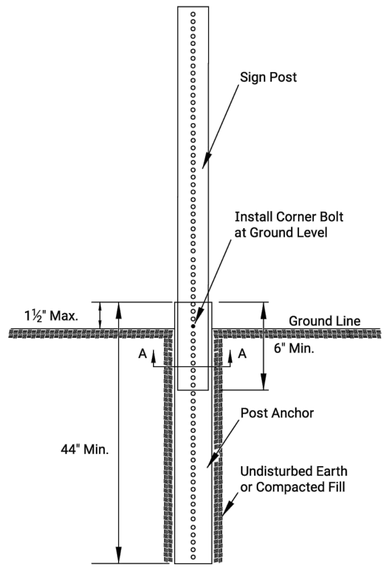
SHEET ID
C-532

A1 TRAFFIC CONTROL SIGN INFORMATION
SCALE: NOT TO SCALE

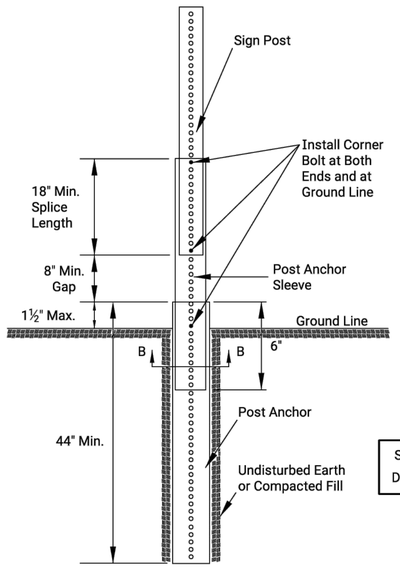
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C532-TC-SGNINF.dwg
9/16/2026 1:38:24 PM

READY TO ADVERTISE

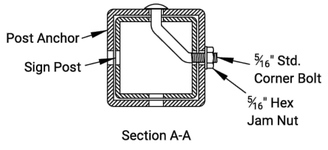
PERFORATED SQUARE STEEL TUBE (P.S.S.T.) POST SETUP



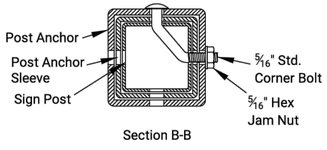
P.S.S.T. Detail



Telescoping P.S.S.T. Detail



Section A-A

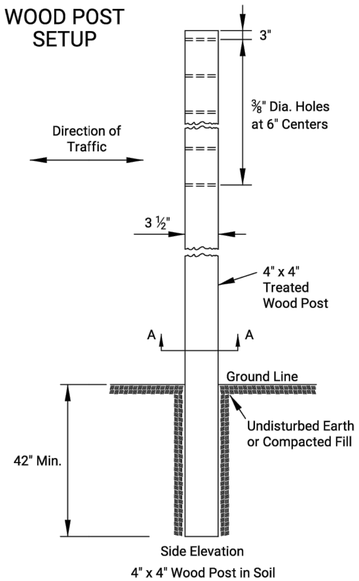


Section B-B

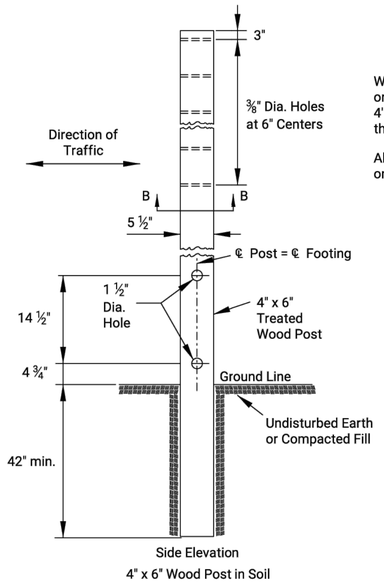
Details for 2", 2 ¼", or 2 ½" sign posts

Place bolts in the same corner along each sign post.

WOOD POST SETUP



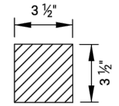
Side Elevation
4" x 4" Wood Post in Soil



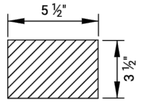
Side Elevation
4" x 6" Wood Post in Soil

When the sign width is equal to or greater than 9', three or more wood posts should be used with a minimum of 4' between the centerline of each post. All signs less than 9' in width shall use a maximum of two wood posts.

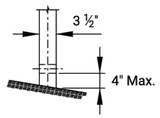
All Special signs larger than 16 sq. ft. shall be installed on wood posts.



Section A-A



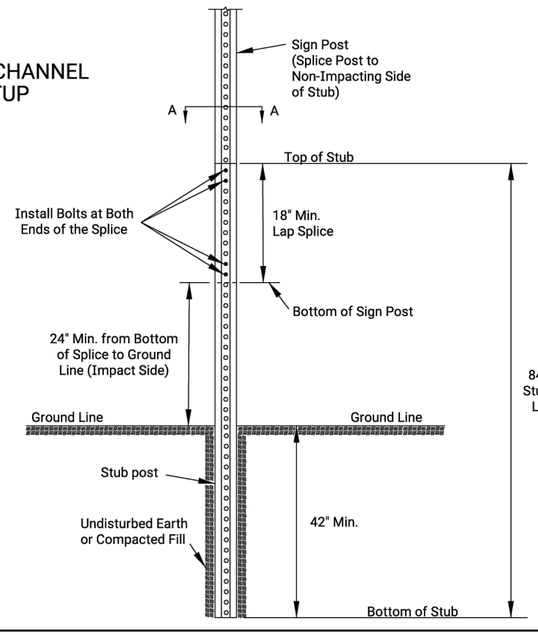
Section B-B



Front Elevation

See TE710 for Additional Details and Requirements

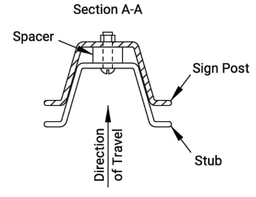
3 LB/F U-CHANNEL SETUP



Notes:

Place two bolts at both ends of the splice through the holes nearest the ends of the splice.

Use manufacturer recommended spacers over the bolts between the spliced pieces of U-Channel.



NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL SIGN POSTS

TE712

FHWA APPROVAL		12-05-25	APP'D.	Nick Rogers
DESIGNED	B.A.H.	DETAILED R.W.B.	QUANTITIES	TRACED
DESIGN CK.		DETAIL CK.	QUAN.CK.	TRACE CK.

KDOT Graphics Certified 12-17-2025 Sh. No. 0

KDOT Graphics Certified

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

Plotted by : KDOT#CADD.Support_ks.gov 19-DEC-2025 01:05
File : TE712.dgn

A1 TRAFFIC CONTROL SIGN POSTS
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

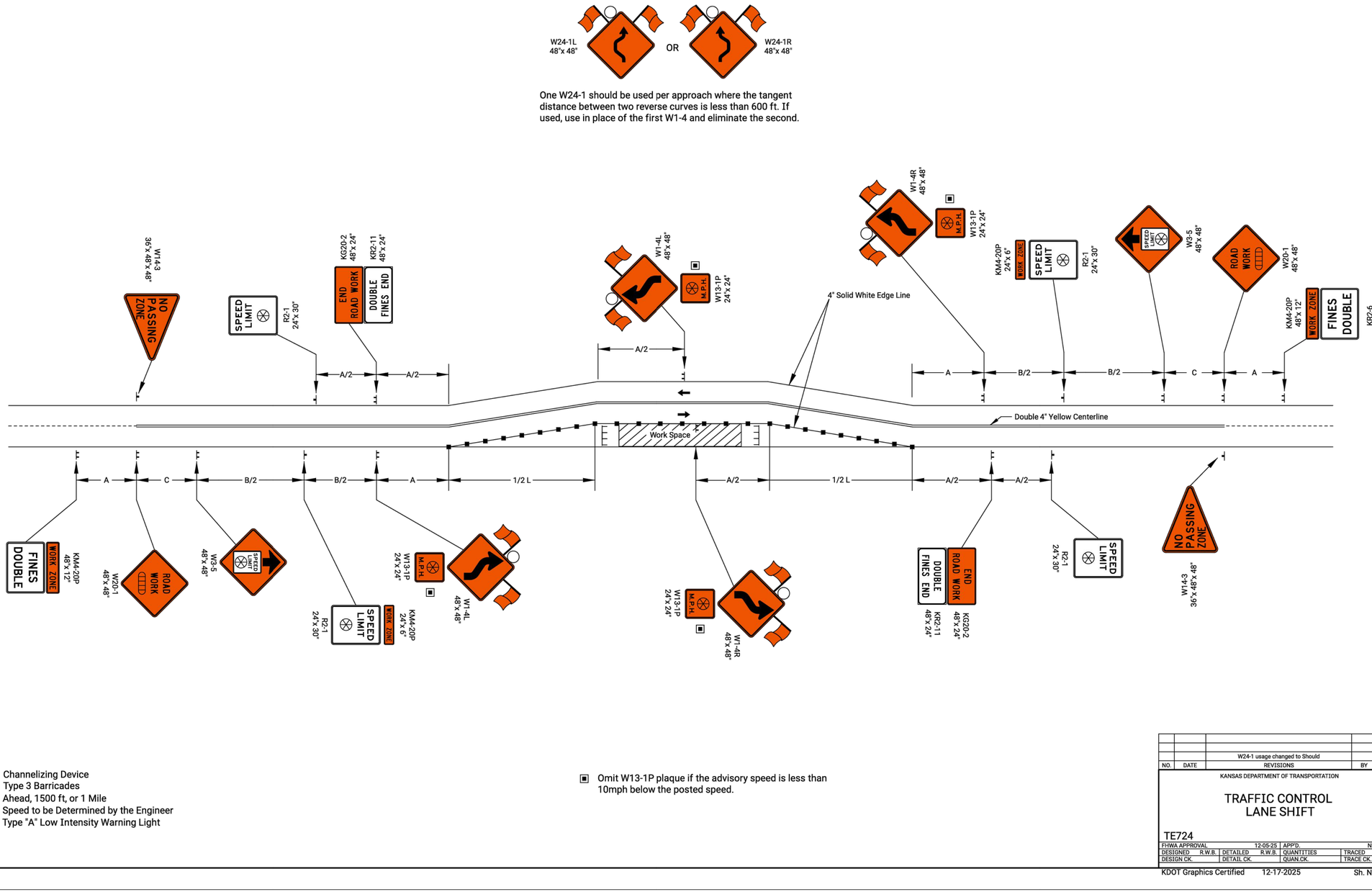
CIVIL DETAILS

SHEET ID
C-533

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C533-TC-SGNPST.dwg
9/16/2026 1:38:31 PM

READY TO ADVERTISE

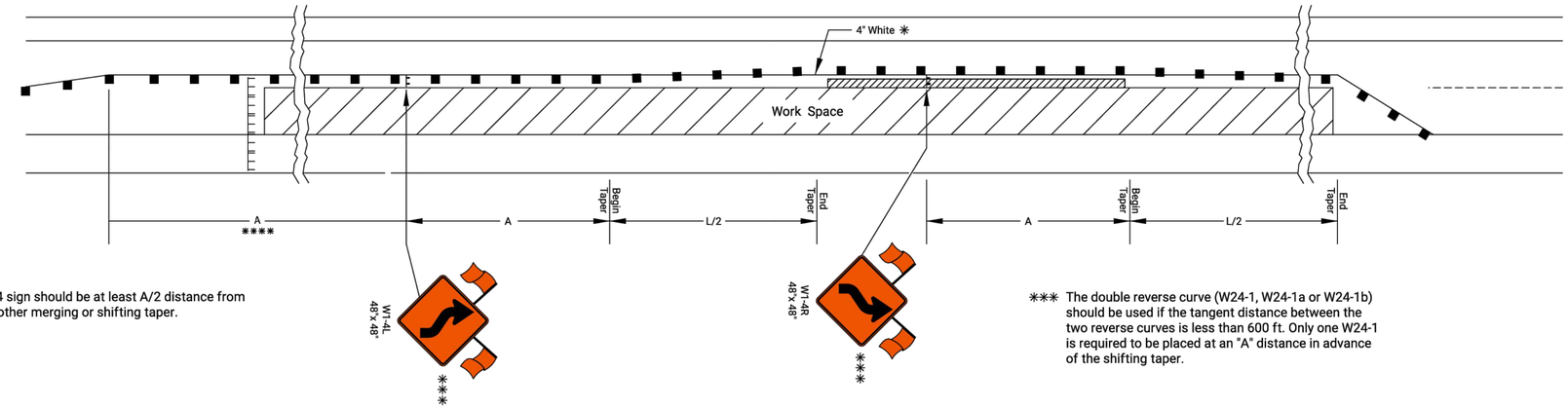
STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0


W24-1L
48"x 48"
OR
W24-1R
48"x 48"
One W24-1 should be used per approach where the tangent distance between two reverse curves is less than 600 ft. If used, use in place of the first W1-4 and eliminate the second.
W1-4R
48"x 48"
W13-1P
24"x 24"
KM4-20P
24"x 6"
WORK ZONE
SPEED LIMIT
R2-1
24"x 30"
W3-5
48"x 48"
ROAD WORK
W20-1
48"x 48"
KM4-20P
48"x 12"
WORK ZONE
FINES DOUBLE
KR2-6
48"x 36"
W1-4L
48"x 48"
W13-1P
24"x 24"
KG20-2
48"x 24"
END ROAD WORK
KR2-11
48"x 24"
DOUBLE FINES END
W1-4R
48"x 48"
W13-1P
24"x 24"
W1-4L
48"x 48"
W13-1P
24"x 24"
KM4-20P
24"x 6"
WORK ZONE
R2-1
24"x 30"
SPEED LIMIT
W3-5
48"x 48"
SPEED LIMIT
W20-1
48"x 48"
ROAD WORK
KM4-20P
48"x 12"
WORK ZONE
KR2-6
48"x 36"
FINES DOUBLE
W1-4L
48"x 48"
W13-1P
24"x 24"
KG20-2
48"x 24"
END ROAD WORK
KR2-11
48"x 24"
DOUBLE FINES END
R2-1
24"x 30"
SPEED LIMIT

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

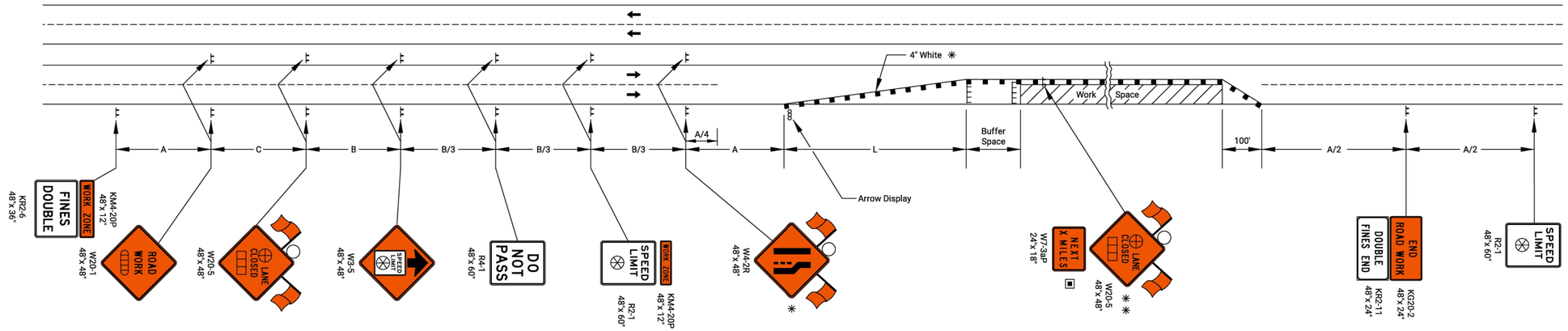
SHIFTING TAPER DETAIL

Add signs and devices as shown for work inside a closed lane that extends near to (or into) the open traffic lane.



✳✳✳✳ W1-4 sign should be at least A/2 distance from any other merging or shifting taper.

✳✳✳ The double reverse curve (W24-1, W24-1a or W24-1b) should be used if the tangent distance between the two reverse curves is less than 600 ft. Only one W24-1 is required to be placed at an "A" distance in advance of the shifting taper.



- Type 3 Barricades
- X Length to the Nearest Whole Mile
- ■ Channelizing Device
- Ahead, 1500 ft, or 1 mile
- Ahead, 1000 ft, 1500 ft, or ½ mile
- ⊕ Right or Left
- ⊗ Speed to be determined by the Engineer
- ○ Type "A" Low Intensity Warning Light

✳ For left lane closures use W4-2L and yellow edge line along channelizing devices.

✳✳ The W20-5 (⊕Lane Closed) and W7-3aP (Next X Miles) signs should be placed at 2 mile increments on a project of 4 miles or longer.

▣ Mileage to be Determined by the Engineer.

Left-side signs shall be omitted for a four-lane undivided highway.

At least one Type 3 barricade should be placed at least every 500' in a closed lane.

Shoulder Taper is required if paved shoulder is 6' or wider.

NO.	DATE	REVISIONS	BY	APP'D

KANSAS DEPARTMENT OF TRANSPORTATION

TRAFFIC CONTROL
LANE CLOSURE ON MULTILANE HWY

TE744

FHWA APPROVAL		12-05-25	APP'D.	Nick Rogers	
DESIGNED B.A.H.	DETAILED R.W.B.		QUANTITIES		TRACED
DESIGN OK	DETAIL OK		QUAN. OK		TRACE OK

KDOT Graphics Certified 12-17-2025 Sh. No. 0

KDOT Graphics Certified

Plotted by : KDOT#CADD.Support_ks.gov 19-DEC-2025 01:19
File : TE744.dgn

A1 TRAFFIC CONTROL SIGN INFORMATION
SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C535-TC-MLHWY.dwg
9/16/2026 1:38:44 PM

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROWL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
SIZE:	ANSI D
ISSUE DATE:	23 FEBRUARY 2026
SOLICITATION NO.:	
TASK ORDER # W912D24F1019	
CONTRACT NO.:	W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

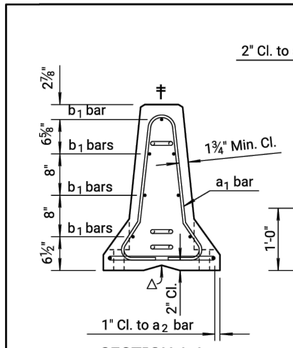
ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

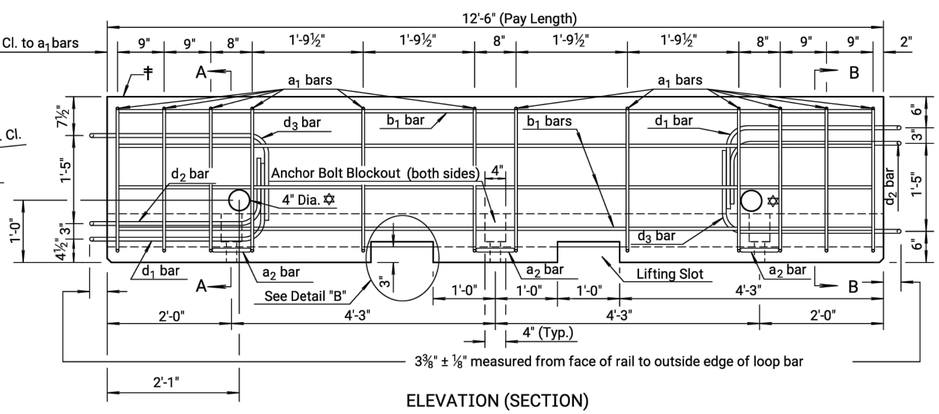
SHEET ID
C-535

READY TO ADVERTISE

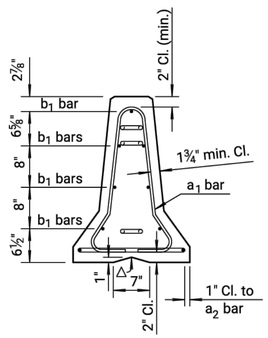
STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0



SECTION A-A
TYPICAL SECTION
(Stirrup Placement)

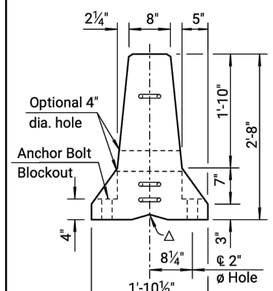


ELEVATION (SECTION)

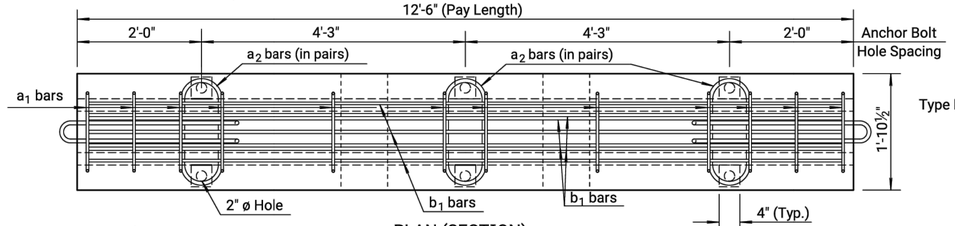


SECTION B-B
TYPICAL SECTION
(Stirrup Placement)

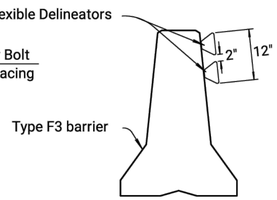
✝ Marked End
✡ 4" diameter - 11 gauge steel round mechanical tubing sleeve. These holes are optional.
△ V Notch is optional



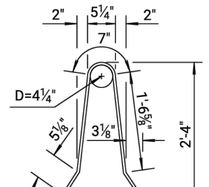
END VIEW

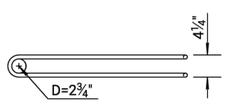


PLAN (SECTION)

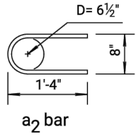


DELINEATOR DETAILS

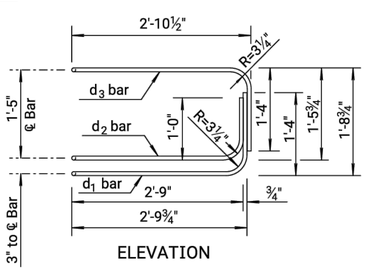




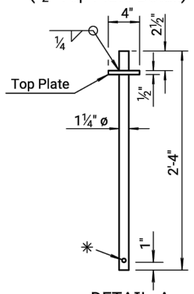
PLAN
LOOP BAR ASSEMBLY
(Marked end shown, invert for other end)
(Material as stated in General Notes)



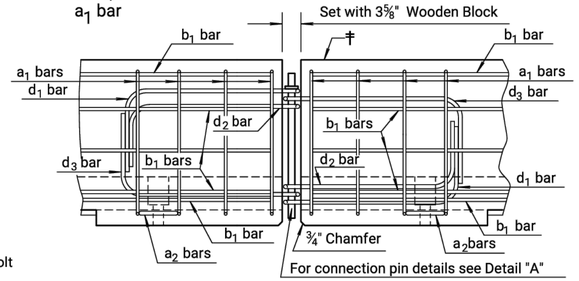
a2 bar



ELEVATION



DETAIL A
CONNECTION PIN
(A36 Steel) 10.9 lbs. each
✳ ⅝" ø hole for retainer bolt. The retainer bolt & nut are installed at Contractor's option.
Note: Retainer bolt & nut required with Tie Down Strap.

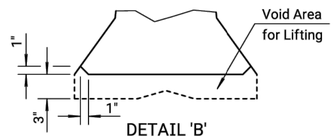


DETAILS OF BARRIER CONNECTION

Per 12'-6" Barrier Section

REINFORCING A615 Gr. 60					
Bar	Bar Size	Shape	No. of Bars	Length Ft.	Weight Lbs.
a_1	#4		12	6'-0"	48.1
a_2	#6		6	2'-11"	26.3
b_1	#5		7	12'-2"	88.8
LOOP ASSEMBLY					
d_1	#6		2	8'-5"	25.3
d_2	#6		2	7'-7"	22.8
d_3	#6		2	8'-6"	25.5

Concrete Quantity = 1.3 C.Y.
(Dimensions are out to out of bars unless otherwise noted.)



DETAIL 'B'
LIFTING SLOT DETAIL
(1" Chamfer to prevent spalling)

NOTE: At no time shall the barriers be lifted, moved, etc. by use of the loop bars: d_1, d_2 or d_3.

GENERAL NOTES:

MATERIAL: Use ASTM A615, Grade 60 reinforcing bars, except for the loop bars (d_1, d_2 and d_3). The loop bars (d_1, d_2 and d_3) shall be ¾" smooth steel bars with a minimum yield of 60 ksi, a tensile strength of not less than 1.25 times the yield strength but a minimum of 80 ksi, a minimum 14% elongation in 8 inches, and passing a 180 degree bend test using a 3.5 D pin bend diameter. The loops shall be installed within ⅛" of the plan dimensions.
Use air-entrained concrete with f'c = 5,000 p.s.i.

SECTION: The section furnished must generally comply with dimensions shown. Requests for minor variations in section geometry and attachments may be submitted to the Engineer for approval.

LIFTING SLOTS: Lifting slots shall be constructed where specified on the plans to facilitate the drainage of water after installation on the roadway.

TEMPORARY CONCRETE SAFETY BARRIER: Furnishing and placing of all materials when required and all labor and equipment required to position the temporary barrier shall be included in the Contract unit price bid for "Concrete Safety Barrier (Type F3)(Temporary)". Any relocation of the barrier required for the project shall be paid in accordance with the Special Provisions under the bid item "Concrete Safety Barrier (Type F3) (Temporary-Relocate)". Unless otherwise noted on the Plans, the Temporary Concrete Safety Barrier shall become the property of the Contractor and shall be removed from the site upon acceptance of the completed project.
Approximate weight of one unit equals 2.7 tons.

PLACEMENT: Barrier shall be placed on a paved surface. All loose dirt and sand shall be removed from the roadway surface just prior to placement of the barrier.
After the barrier is placed and the connection pin is inserted, tension or pull the barrier such that the installation is taut and the connection pin cannot freely move vertically. If the connection pin or loop bar assembly are damaged during the tensioning process, it is the responsibility of the Contractor to repair the damaged area or replace the temporary barrier section.

MARKING: The left end (✝) of each barrier shall be permanently marked by stamping or forming into the barrier the following information:
- Type F3
- Manufacturer code (as specified by KDOT Bureau of Const. & Maint.)
- Date manufactured (month and year)

DELINEATION: Delineators shall be spaced on 50' centers, except through curves where they shall be spaced on 25' centers. See Standard Drawing RD610 for additional details.
The delineation shall be mounted on the side of the Temporary Concrete Safety Barrier with two delineators at each location. Each delineator shall have a minimum height-to-width ratio of 1.75, and a minimum reflective surface area of 7 sq. in.. The delineators shall be affixed to the Temporary Concrete Safety Barrier as recommended by the manufacturer.
Delineators shall be attached to bridge rail or other structures in construction zones when roadway is narrowed and traffic is adjacent to the structure. The method and location of placement shall be similar to permanent barrier delineation.
When traffic flow is in one direction, the delineators shall be yellow when used on the left, white when used on the right. When traffic flow is in both directions delineators shall be placed back-to-back, and shall correspond to the color of the edge line.
The work and materials required for the installation of delineators as mentioned shall be subsidiary to the bid item "Concrete Safety Barrier (Type F3) (Temporary).
If necessary, include Standard Drawing RD622A for Taper Section, Standard drawing RD622B for anchor and tie down details, Standard Drawing RD622C for Bridges with thermal expansion of 1½" or greater and Standard Drawing RD622D for Barrier Layouts.
The Contractor shall be responsible for maintaining a clear area, shown as dimension "A" on Standard Drawing RD622B. The clear area is located behind the Temporary Concrete Safety Barrier and shall be kept free of any equipment, material stockpiles or other obstacles. For non-anchored roadway applications, dimension "A" shall be a minimum of 2'-0".

NO.	DATE	REVISIONS	BY	APP'D
07	09-11-17	Revised Markers	A.L.R.	S.W.K.
06	07-17-17	Revised General Note	A.L.R.	S.W.K.
05	08-27-15	Added Note, Pay Length	K.E.K.	S.W.K.

KANSAS DEPARTMENT OF TRANSPORTATION

TEMPORARY CONCRETE SAFETY BARRIER TYPE F3

RD622

FHWA APPROVAL	03-05-18	APP'D.	Scott W. King
DESIGNED	DETAILED	QUANTITIES	TRACED
DESIGN CK.	DETAIL CK.	QUAN.CK.	TRACE CK.

KDOT Graphics Certified 11-13-2024 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D21D19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-536

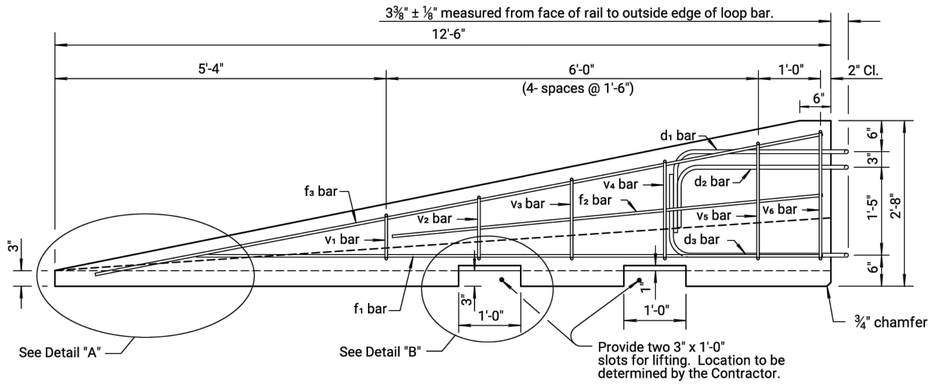
Plotted by : Amy.Rockers@ks.gov 13-NOV-2024 19:49
File : rd622.dgn

A1 TRAFFIC CONTROL: TEMPORARY CONCRETE SAFETY BARRIER(TCSB) (TYPE F3)
SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C536-TC-TCSB.dwg
9/9/2026 4:33:11 PM

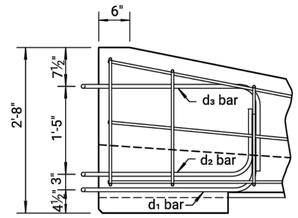
READY TO ADVERTISE

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS		0	0	



ELEVATION (SECTION)

(For connection to left end of Barrier)



ELEVATION (SECTION)

(For connection to right end of Barrier)

GENERAL NOTES:

MATERIAL: Use ASTM A615, Grade 60 reinforcing bars, except for the loop bars (d_1, d_2, and d_3). The loop bars (d_1, d_2, and d_3) shall be 3#4" smooth steel bars with a minimum yield of 60 ksi, a tensile strength of not less than 1.25 times the yield strength but a minimum of 80 ksi, a minimum 14% elongation in 8 inches, and passing a 180 degree bend test using a 3.5" D pin bend diameter. The loops shall be installed within 1#8" of the plan dimensions.
Use air-entrained concrete with f'c = 5,000 p.s.i.

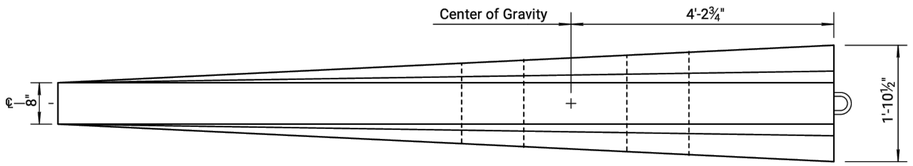
SECTION: The section furnished must generally comply with dimensions shown. Requests for minor variations in section geometry and attachments may be submitted to the Engineer for approval.

LIFTING SLOTS: Lifting slots shall be constructed where specified on the plans to facilitate the drainage of water after installation on the roadway.

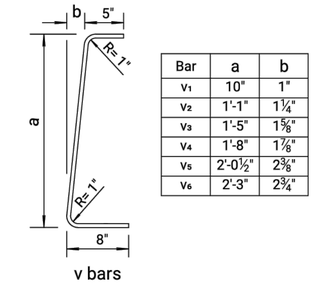
TEMPORARY CONCRETE SAFETY BARRIER: One section of Taper Barrier shall be bid as one section of Type F3 Barrier. Type F3 barrier taper sections shall be used only for low speed (40 mph or less) applications or where a barrier terminates beyond the roadway clear zone. Where a barrier terminates within the clear zone of a high speed roadway, an appropriate impact attenuator shall be installed on the approach end. Furnishing and placing of all materials when required and all labor and equipment required to position the temporary barrier shall be included in the Contract unit price bid for "Concrete Safety Barrier (Type F3)(Temporary)". Any relocation of the barrier required for the project shall be paid in accordance with the Special Provisions under the bid item "Concrete Safety Barrier (Type F3)(Temporary-Relocate)". Unless otherwise noted on the Plans, the Temporary Concrete Safety Barrier shall become the property of the Contractor and shall be removed from the site upon acceptance of the completed project. Approximate weight of one unit equals 1.3 tons.

SURFACE PREPARATION: Barrier shall be placed on a paved surface. All loose dirt and sand shall be removed from the roadway surface just prior to placement of the barrier.

MARKING: Each barrier shall be permanently marked by stamping or forming into the barrier the follow -ing information:
- Type F3
- Manufacturer code (as specified by KDOT Bureau of Const. & Maint.)
- Date manufactured (month and year)



PLAN

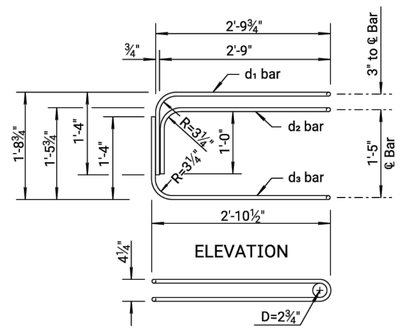


DETAIL "A" BENT BAR DETAIL

v bars

2 at each size required for stirrup assembly

Bar	a	b
v_1	10"	1"
v_2	1'-1"	1¼"
v_3	1'-5"	1⅝"
v_4	1'-8"	1⅞"
v_5	2'-0½"	2⅜"
v_6	2'-3"	2¾"

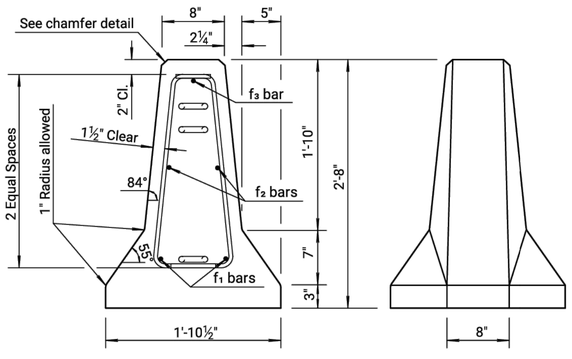


PLAN

LOOP BAR ASSEMBLY

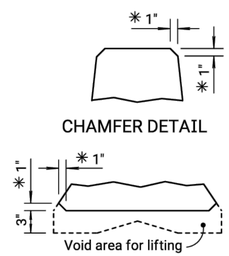
(Left Barrier Connection shown, invert for other end)

Note: At no time shall the barriers be lifted, moved, etc. by use of the loop bars: d_1, d_2, or d_3.



END SECTION

END VIEW



DETAIL "B" LIFTING SLOT DETAIL

✱ 1" Chamfer to prevent spalling

Per 12'-6" Barrier Taper Section

REINFORCING A615 Gr. 60					
Bar	Bar Size	Shape	No. of Bars	Length ft.	Weight lbs.
v_1	#4	⌊	2	1'-11"	2.6
v_2	#4	⌊	2	2'-2"	2.9
v_3	#4	⌊	2	2'-6"	3.3
v_4	#4	⌊	2	2'-9"	3.7
v_5	#4	⌊	2	3'-2"	4.2
v_6	#4	⌊	2	3'-4"	4.5
f_1	#4	—	2	12'-0"	16.0
f_2	#4	—	2	7'-6"	10.0
f_3	#5	—	1	11'-9"	12.3
LOOP ASSEMBLY					
d_1	#6	⌐	1	8'-5"	12.6
d_2	#6	⌐	1	7'-7"	11.4
d_3	#6	⌐	1	8'-6"	12.8

Concrete Quantity = 0.6 C.Y.

NO.	DATE	REVISIONS	BY	APP'D
01	01-10-07	Revised layout & notes	S.W.K.	J.O.B.

KANSAS DEPARTMENT OF TRANSPORTATION

TEMPORARY CONCRETE SAFETY BARRIER TAPER SECTION TYPE F3

RD622A

FHWA APPROVAL	01-19-07	APP'D.	James O. Brewer
DESIGNED	DETAILED	QUANTITIES	TRACED B.N.B.
DESIGN CK.	DETAIL CK.	QUAN.CK.	TRACE CK. S.W.K.

KDOT Graphics Certified 07-18-2022 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
DATE: 23 FEBRUARY 2026
ANSI D

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID

C-537

Plotted by: Kylee.Sims@ks.gov 22-JUL-2022 14:32
File: rd622a.dgn

A1 TRAFFIC CONTROL: TEMPORARY CONCRETE SAFETY BARRIER-TAPER SECTION

SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C537-TC-TCSB.dwg
9/9/2026 4:33:33 PM

READY TO ADVERTISE

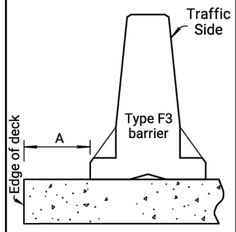
Note to Designer: For use on Haunched slab bridges, the Road Designer shall coordinate with the Bridge Designer for "corridor in the reinforcing steel layout to accommodate barrier anchoring".
Road Designer shall coordinate barrier layout with Bridge Designer to accomodate for expansion during construction.

GENERAL NOTES:
INSTALLATION: Holes into the pavement to anchor the concrete safety barrier may be drilled after positioning barrier. When anchoring with 3 bolts on traffic side, install barrier with through anchor bolt where possible, use grouted anchor bolts where through bolt can't be used. Do not drill into or otherwise damage support beams, girders, or expansion joints. All work & materials required for the installation of the anchors are subsidiary to the bid item "Concrete Safety Barrier".
UTILITIES & STRUCTURES (Stakes): Verify buried utilities & structures within stake depth. If conflicts between stake & buried elements exist, up to 2 stakes maximum in a single barrier may be omitted if adjacent barriers have 3 stakes each.

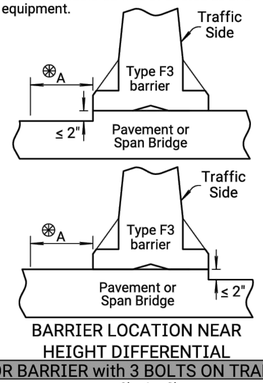
ANCHORAGE: Use galvanized grouted anchor bolts, through anchor bolts, nuts & washers that meet standard specifications. Install 3 anchor bolts or asphalt pins per barrier on the traffic side except on transition barrier as shown.
BARRIER REMOVAL: Completely remove all anchor systems. Remove grouted or drop-in anchor system by drilling the anchor with a core barrel 2x the diameter of the insert. Core to a depth equal to the installed depth & remove the core, prepare the hole by removing dust & debris. Fill hole with material that meets KDOT Pre-qualified "Non-shrink grouts for grouting anchor bolts & reinforcing into previously poured concrete". Follow the manufacturer's procedures for mixing, hole preparation & curing. To fill through bolt anchor or screw-in anchor system, remove & completely fill the hole using instructions for drop-in anchors except no coring is required.
For removed or relocated barrier on flexible pavement, fill stake holes completely with hot or cold asphalt patch material. Work & materials required to remove & patch anchor holes are subsidiary to the bid item "Concrete Safety Barrier".
TEMPORARY BARRIERS: Temporary Barriers shown in the details of this drawing are not allowed for permanent installations.
See KDOT Standard Drawing "Temporary Concrete Safety Barrier Type F3 Transition Layouts" for transition details between anchored & free-standing barriers. See KDOT Standard Drawing "Temporary Concrete Safety Barrier Type F3" for details & quantities not shown on this sheet.
SIGNING: For sign spacing, traffic control device details & reference notes, see Index of Sheets.

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS		0	0	

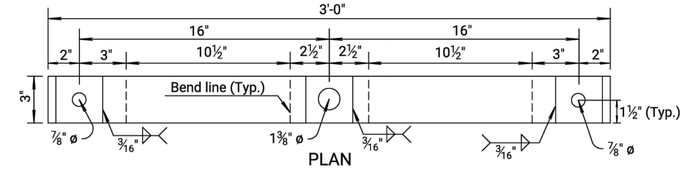
NO ANCHORAGE REQUIRED unless shown on plans		
"A" DISTANCE	A ≥ 4'	A ≥ 2'
LOCATION	Span Bridge	⊛ Box Culverts Roadways - Flexible or Rigid Pvmnt.



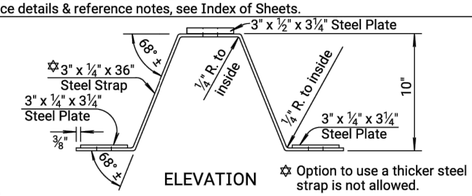
⊛Applies on span bridges when the action creates a height differential of ≤ 2". Measured from the toe of the barrier, the "A" distance should be free of obstacles and equipment.



BARRIER LOCATION NEAR HEIGHT DIFFERENTIAL



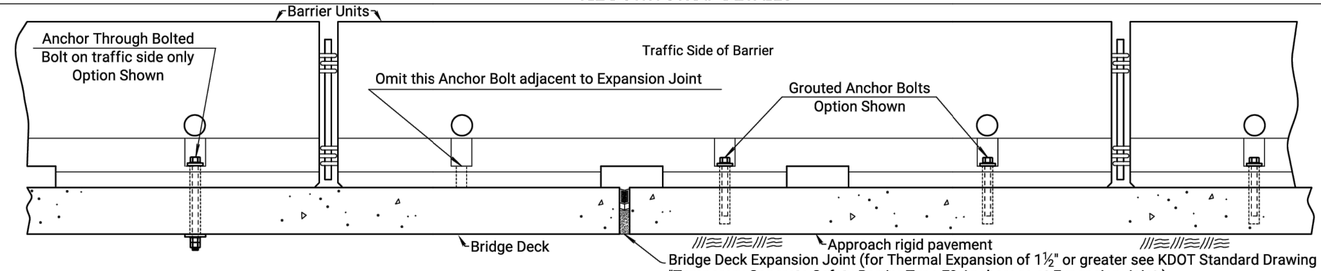
PLAN



ELEVATION

✡ Option to use a thicker steel strap is not allowed.

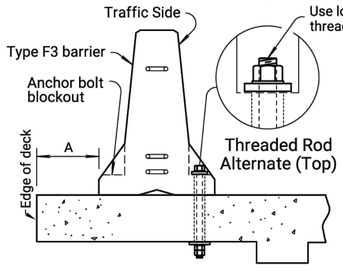
TIE-DOWN STRAP DETAILS

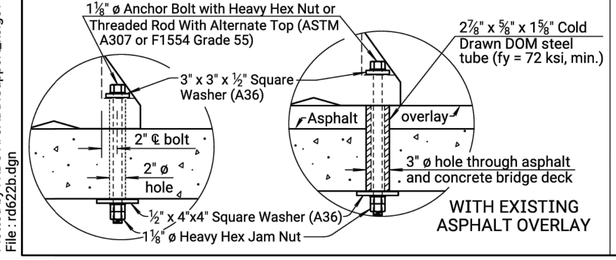


ELEVATION - TREATMENT AT BRIDGE DECK EXPANSION JOINT SCHEMATIC (Expansion < 1½")

ANCHORAGE	ANCHOR BARRIER with 3 BOLTS ON TRAFFIC SIDE	ANCHOR BARRIER with 3 BOLTS ON TRAFFIC SIDE		ANCHOR BARRIER with TIE-DOWN STRAP		ANCHOR BARRIER with STAKES
"A" DISTANCE	0' ≤ A < 2'	0' ≤ A < 2'		△ 2' ≤ A < 4'	6" ≤ A < 2'	6" ≤ A < 2'
LOCATION	Span Bridge	Span Bridge	Box Culverts Roadways - RIGID Pavement	Span Bridge	Box Culverts Roadways - RIGID Pavement	Box Culverts (ceiling below stake depth) Roadways - FLEXIBLE Pavement

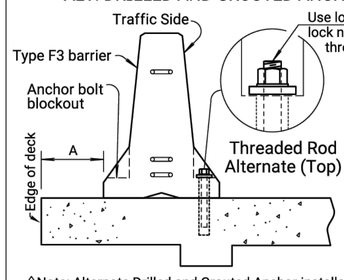
THROUGH BOLT (Preferred)



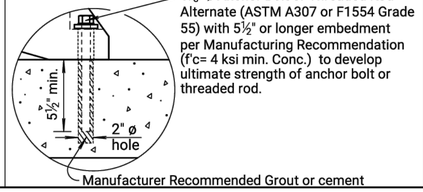


WITH EXISTING ASPHALT OVERLAY

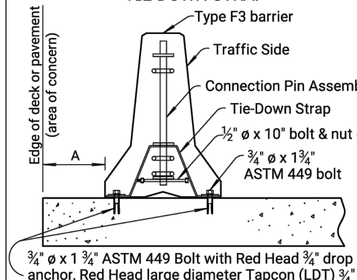
✡ALT. DRILLED AND GROUTED ANCHOR



✡Note: Alternate Drilled and Grouted Anchor installation avoids damage to the support beams, girders or expansion joint. The State Bridge Office shall approve the use of the Alternate Drilled and Grouted Anchor installation for bridge applications.



TIE-DOWN STRAP

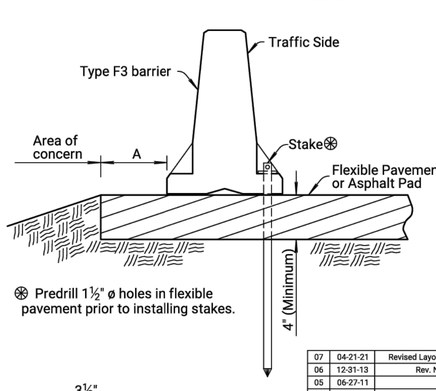


¾" ø x 1 ¾" ASTM 449 Bolt with Red Head ¾" drop in anchor, Red Head large diameter Tapcon (LDT) ¾" ø x 4 ½" long, or Simpson Titen HD ¾" ø x 5" long

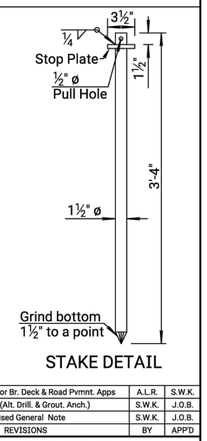
△"A" distance may be reduced to 6" if traffic does not travel under the bridge.

With prior approval from the State Bridge Office, anchoring the barrier with 3 bolts (Through Bolt OR Drilled & Grouted Anchor oprtions) on the traffic side may be used in lieu of tie-down straps.

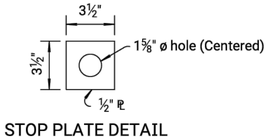
STAKED DOWN



⊛ Predrill 1½" ø holes in flexible pavement prior to installing stakes.



STAKE DETAIL



STOP PLATE DETAIL

NO.	DATE	REVISIONS	BY	APP'D
07	04-21-21	Revised Layouts for Br. Deck & Road Pvmnt. Apps	A.L.R.	S.W.K.
06	12-31-13	Rev. Note (Alt. Drill. & Grout. Anch.)	S.W.K.	J.O.B.
05	06-27-11	Revised General Note	S.W.K.	J.O.B.

KANSAS DEPARTMENT OF TRANSPORTATION

TEMPORARY CONCRETE SAFETY BARRIER TYPE F3 ANCHORAGE

RD622B

FHWA APPROVAL	04-21-21	APP'D.	Scott W. King
DESIGNED	DETAILED	QUANTITIES	TRACED
DESIGN CK.	DETAIL CK.	QUAN.CK.	TRACE CK.

KDOT Graphics Certified 05-13-2022 Sh. No. 0

KDOT Graphics Certified

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROWL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
ISSUE DATE:	23 FEBRUARY 2026
SOLICITATION NO.:	
TASK ORDER # W912D24F1019	
CONTRACT NO.:	W912DQ19D1012
ANSI D	

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-538

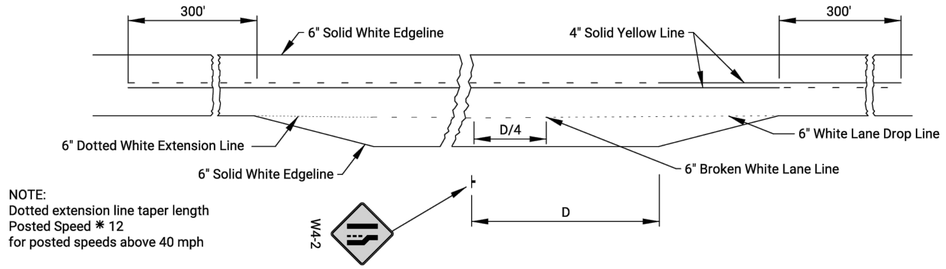
A1 TRAFFIC CONTROL: TEMPORARY CONCRETE SAFETY BARRIER(TCSB) (TYPE F3)-ANCHORAGE
SCALE: NOT TO SCALE

Plotted by : KDOT#CADD.Support_ks.gov 18-JUN-2022 00:09
File : rd622b.dgn

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C538-TC-ANCHOR.dwg
9/9/2026 4:33:58 PM

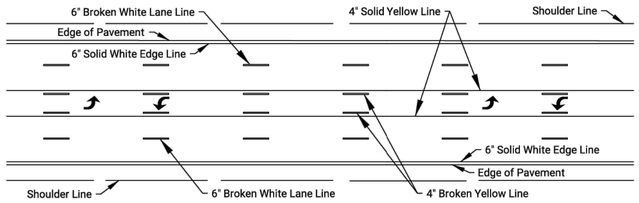
READY TO ADVERTISE

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

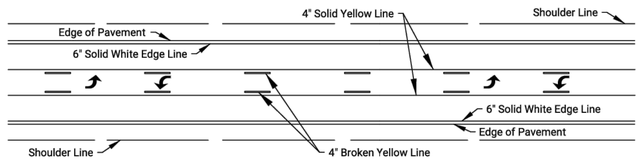


NOTE:
Dotted extension line taper length
Posted Speed ✳ 12
for posted speeds above 40 mph

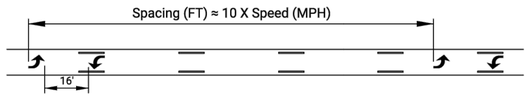
TYPICAL MARKING FOR AUXILIARY PASSING LANE



TWO-WAY LEFT TURN DETAIL FOR FIVE LANE ROADWAY

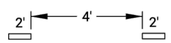


TWO-WAY LEFT TURN DETAIL FOR THREE LANE ROADWAY

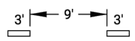


TWO-WAY LEFT TURN ARROW SPACING DETAIL

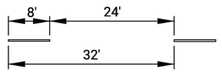
NOTE:
If arrows are used space the arrows as shown in
the spacing detail.



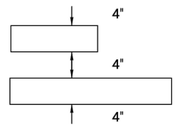
TYPICAL SPACING FOR DOTTED EXTENSION LINES, UNLESS OTHERWISE NOTED ON PLANS.



TYPICAL SPACING FOR LANE DROP, UNLESS OTHERWISE NOTED ON PLANS.



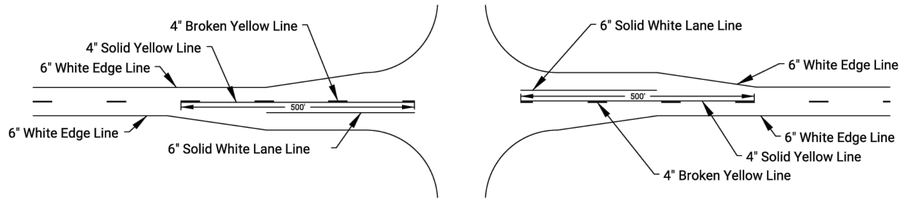
TYPICAL SPACING FOR BROKEN LINES, UNLESS OTHERWISE NOTED ON PLANS



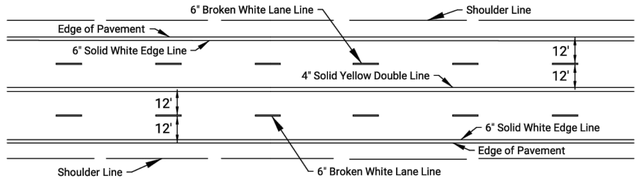
TYPICAL SPACING FOR NO PASSING LINES, UNLESS OTHERWISE NOTED ON PLANS

NOTE:
All pavement markings shall be broken at
cross roads.

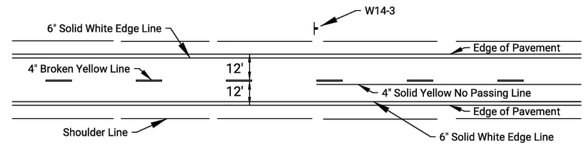
For highway junctions the no passing
zone will extend 1000' from intersection.



TYPICAL ROAD JUNCTION MARKINGS WITH BYPASS LANES



TYPICAL MARKINGS FOR FOUR LANE ROADWAY



TYPICAL TWO LANE MARKINGS

NOTE:
Longitudinal pavement marking lines shall be offset
a minimum of 2" from longitudinal pavement joints.

On non I, US, and K routes, 4" edge lines may be installed.
6" edge lines are not required on non I, US, and K routes.

NO.	DATE	REVISIONS	BY	APP'D
03	05-25-12	Added Dotted Extension and Lane Drop Lines	B.A.H.	B.D.G.
02	09-20-05	Removed Aux. Passing Lane Dotted Ext. Line	J.F.F.	B.D.G.
01	07-26-05	New FHWA Approval Date	J.F.F.	B.D.G.

KANSAS DEPARTMENT OF TRANSPORTATION

TYPICAL PAVEMENT MARKING DETAILS FOR UNDIVIDED ROADWAYS

TE308

FHWA APPROVAL		05-25-12	APP'D.	Brian D. Gower
DESIGNED	J.F.F.	DETAILED	J.F.F.	QUANTITIES	TRACED
DESIGN CK.	B.D.G.	DETAIL CK.	B.D.G.	QUAN.CK.	TRACE CK.

KDOT Graphics Certified 07-19-2022 Sh. No. 0

KDOT Graphics Certified

Plotted by: KDOT#CADD.Support_ks.gov 12-SEP-2023 00:04
File : te308.dgn

A1 PAVEMENT MARKING: DETAILS FOR UNDIVIDED ROADWAYS
SCALE: NOT TO SCALE

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C539-PM-UNDVD.dwg
9/17/2026 1:28:15 PM

READY TO ADVERTISE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY:	M. CROVL
DRAWN BY:	T. ROUSH
CHECKED BY:	J. DAVIS
SUBMITTED BY:	S. BRISCOE
SIZE:	ANSI D
ISSUE DATE:	23 FEBRUARY 2026
SOLICITATION NO.:	
TASK ORDER # W912D2024F1019	
CONTRACT NO.:	W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPORATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

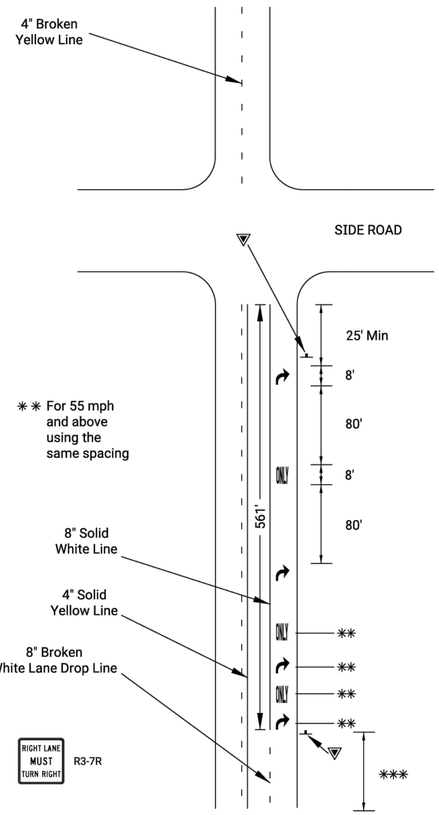
ATCHISON, KANSAS
COMBINED SEWER RESTORATION PROJECT
PZ 363708

CIVIL DETAILS

SHEET ID
C-539

STATE	PROJECT NO.	YEAR	SHEET NO.	TOTAL SHEETS
KANSAS	0	0	0	0

TYPICAL SIGNING AND MARKING FOR RIGHT LANE MUST TURN RIGHT



RAILROAD CROSSING MARKING

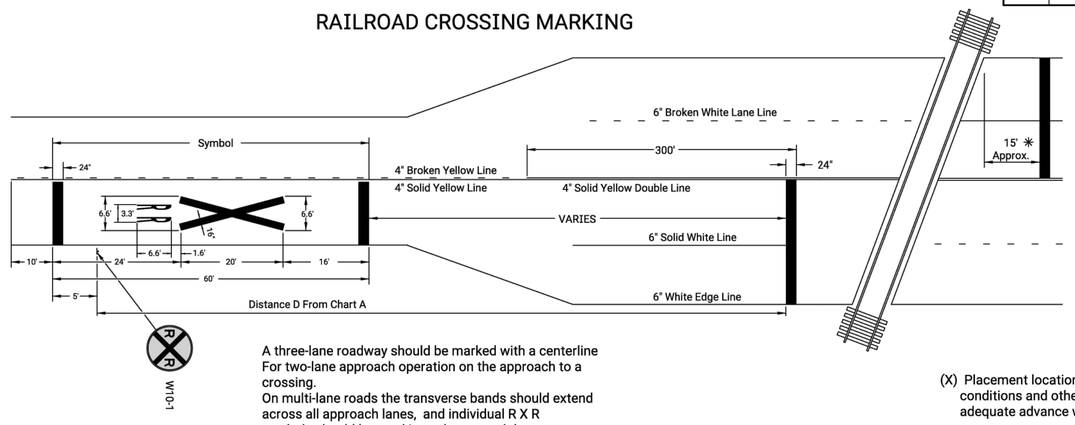


CHART "A"

SPEED MPH	DISTANCE D (feet)
75	850
70	750
65	650
60	550
55	450
50	375
45	300
40	225
35	150
30	(X)
25	(X)
20	(X)

All distances are minimum.

A three-lane roadway should be marked with a centerline
For two-lane approach operation on the approach to a crossing.
On multi-lane roads the transverse bands should extend across all approach lanes, and individual R X R symbols should be used in each approach lane.
Refer to standard alphabet for highway signs and markings for R X R symbols details.

✳Stop line 8' from near edge of gate or cantilever, if present.

NOTE:
On non I, US, and K routes, 4" edge lines may be installed.
6" edge lines are not required on non I, US, and K routes.

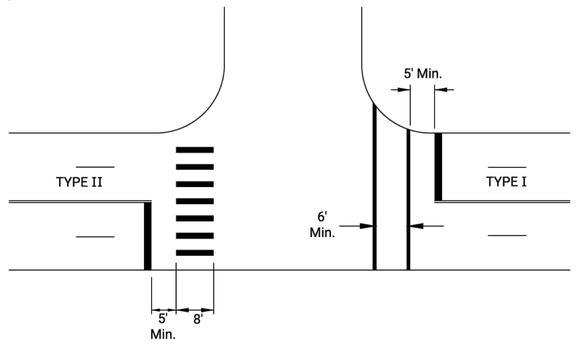
(X) Placement location is dependent on site conditions and other signing to provide adequate advance warning to the driver

TYPICAL CROSSWALKS

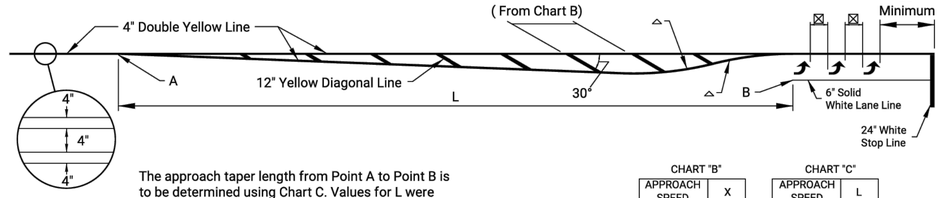
TYPE I: Crosswalk lines shall be 12" solid white lines. They shall be spaced a minimum of 6' apart from inside edge to inside edge.

TYPE II: These lines should be solid white 24" wide placed parallel to the direction of traffic flow. The line placement is determined by lane line, center line, and wheel path in such a manner as to minimize traffic wear. The crosswalk width should be not less than 8'. The transverse crosswalk lines may be added.

When required, stop lines shall be installed a minimum of 5' from crosswalks.



TYPICAL APPROACH TAPER DETAIL



The approach taper length from Point A to Point B is to be determined using Chart C. Values for L were calculated using the equations below and increased to the next higher 5 mph increment.

- SPEEDS < 45 MPH $L = \frac{W \ast S^2}{60}$
- SPEEDS = 45 MPH $L = W \ast S$

☒ If arrows are used and unless otherwise specified the space between lines should be at least four times the height of the characters for low speed roads but not more than ten times the height of the characters, under any conditions.

△ For speeds less than or equal to 40 mph, R=150'. for speeds greater than or equal to 45 mph, R=300'.

CHART "B"

APPROACH SPEED	X
20 MPH	20'
25 MPH	25'
30 MPH	30'
35 MPH	35'
40 MPH	40'
45 MPH	45'
50 MPH	50'
55 MPH	55'
60 MPH	60'
65 MPH	65'
70 MPH	70'

CHART "C"

APPROACH SPEED	L
20 MPH	80'
25 MPH	125'
30 MPH	180'
35 MPH	245'
40 MPH	320'
45 MPH	540'
50 MPH	600'
55 MPH	660'
60 MPH	720'
65 MPH	780'
70 MPH	840'

NO.	DATE	REVISIONS	BY	APP'D
03	05-25-12	Updated Chart B and Lane Drop Lines	B.A.H.	B.D.G.
02	10-20-06	RR Xing Symbol Changed from 18" to 16"	T.L.H.	B.D.G.
01	09-20-05	Added 4" Solid Yellow Double Line to RRxing	J.F.F.	B.D.G.

KANSAS DEPARTMENT OF TRANSPORTATION

TYPICAL MISCELLANEOUS PAVEMENT MARKING DETAIL SHEET

TE309

FHWA APPROVAL		07-26-05	APP'D.	Brian D. Gower	
DESIGNED	J.F.F.	DETAILED	J.F.F.	QUANTITIES	TRACED
DESIGN CK.	B.D.G.	DETAIL CK.	B.D.G.	QUAN.CK.	TRACE CK.

KDOT Graphics Certified 07-05-2022 Sh. No. 0

KDOT Graphics Certified

Plotted by: KDOT#CADD.Support_ks.gov 12-SEP-2023 00:05
File : te309.dgn

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912DQ24F1018
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPRATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
PZ 553708

CIVIL DETAILS

SHEET ID
C-540

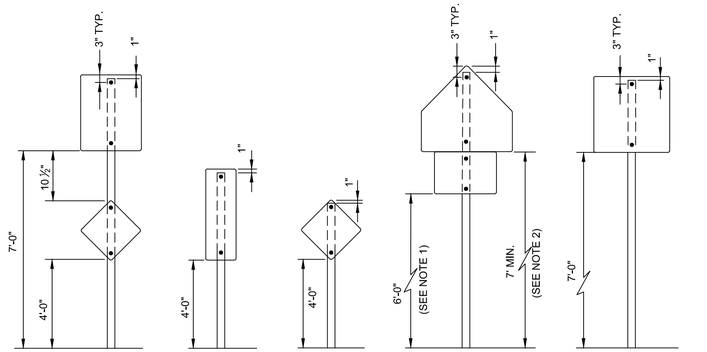
A1 PAVEMENT MARKING: TYPICAL MISCELLANEOUS PAVEMENT MARKING DETAILS
SCALE: NOT TO SCALE

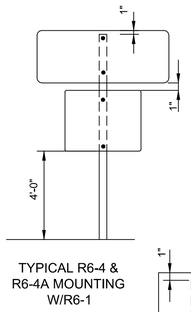
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C540-PM-MISC.dwg
9/17/2026 1:28:22 PM

READY TO ADVERTISE

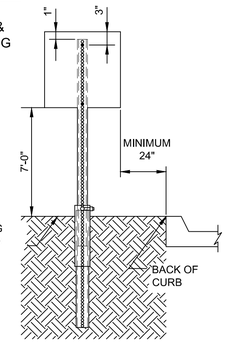
PERMANENT SIGNING GENERAL NOTES

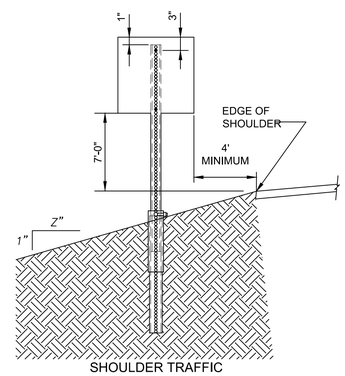
1. ALL PERMANENT SIGNING SHALL BE PROVIDED AND INSTALLED BY THE CONTRACTOR AS INDICATED IN THE PLANS AND SPECIFICATIONS. ALL LETTER, NUMBER AND SYMBOL SIZES, SPACING AND SIGN COLORS SHALL CONFORM TO THE CURRENT MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES (MUTCD).
2. ALL SCHOOL SIGNING SHALL HAVE A FLUORESCENT YELLOW-GREEN BACKGROUND WITH A BLACK LEGEND AND BORDER. ALL OTHER WARNING SIGNS SHALL HAVE A STANDARD YELLOW BACKGROUND.
3. THE LOCATIONS OF EXISTING UTILITIES, IF SHOWN, ARE APPROXIMATE ONLY AND HAVE NOT BEEN INDEPENDENTLY VERIFIED. THE CONTRACTOR SHALL BE RESPONSIBLE FOR CONTACTING ALL UTILITY COMPANIES FOR LOCATIONS OF ALL UNDERGROUND LINES PRIOR TO EXCAVATION AND BE FULLY RESPONSIBLE FOR ANY AND ALL DAMAGES, WHICH MIGHT OCCUR AS A RESULT OF THE CONTRACTORS FAILURE TO EXACTLY LOCATE AND PRESERVE ANY AND ALL UNDERGROUND UTILITIES.
4. THE CONTRACTOR SHALL CALL 811 TO OBTAIN LOCATES FOR STREETLIGHTING, TRAFFIC SIGNAL, AND FIBER OPTIC CONDUITS/CABLES.
5. ALL NEW SIGNS SHALL BE LOCATED WITHIN PUBLIC RIGHT-OF-WAY.
6. ALL RETRO-REFLECTIVE MICRO-ENCAPSULATED PRISMATIC SHEETING SHALL BE ASTM TYPE XI. ALL SIGNS SHALL BE MADE FROM A COMBINATION RETRO-REFLECTIVE SHEETING BACKGROUND WITH ELECTRONIC CUTTABLE FILM APPLIED TO THE SURFACE. (UNLESS OTHERWISE INDICATED IN THE PLANS OR SPECIFICATIONS). PROCESS COLOR INKS OR SILK SCREENING SIGNS ARE NOT ALLOWED.
7. ALL SIGN BLANK MATERIAL SHALL BE MADE OF 0.08" ALUMINUM EXCEPT ALL OVERHEAD STREET NAME SIGNS SHALL BE MADE OF 0.125" ALUMINUM.
8. EXISTING STREET NAME SIGNS IN THE WAY OF CONSTRUCTION:
   THE STREET NAME SIGNS SHALL BE RELOCATED OUT OF THE WAY OF CONSTRUCTION BUT IN A CONSPICUOUS LOCATION FOR THE DRIVING PUBLIC AND EMERGENCY PROVIDERS. THE STREET NAME SIGNS SHALL BE REINSTALLED IN THEIR PROPER LOCATION AS SOON AS POSSIBLE UNLESS OTHERWISE INDICATED. THE REMOVAL AND RE-INSTALLATION OF EXISTING SIGNS WILL BE CONSIDERED SUBSIDIARY TO OTHER BID ITEMS.
9. SIGNS SHOWN TO BE INSTALLED ON THE SIDE OF POLES SHALL BE MOUNTED BY STAINLESS STEEL MOUNTING BANDS AS DETAILED IN THE PLANS. ALL R10 SERIES SIGNS INSTALLED ON A TRAFFIC SIGNAL MAST ARM SHALL BE MOUNTED WITH APPROVED MOUNTING BRACKETS AS SPECIFIED.
10. ALL POST MOUNTED SIGNS SHALL BE MOUNTED ON BREAK-AWAY SIGN POSTS ACCORDING TO THE STANDARD DETAILS.
11. ALL SIGNS AND POSTS SHOWN IN THE PLANS SHALL BE NEW UNLESS OTHERWISE INDICATED IN THE PLANS OR BY THE TRAFFIC SERVICE INSPECTOR.
12. ALL EXISTING REGULATORY SIGNS, WARNING SIGNS, AND STREET NAME SIGNS SHALL BE USED IN PLACE DURING CONSTRUCTION AND PROTECTED FROM DAMAGE UNLESS OTHERWISE INDICATED IN THE PLANS. IF THE CONTRACTOR DAMAGES ANY EXISTING SIGN OR POSTS DURING CONSTRUCTION, HE SHALL BE REQUIRED TO REINSTALL NEW SIGNS AND POSTS OF THE SAME TYPE TO REPLACE THE DAMAGED EQUIPMENT.
13. ANY EXISTING PERMANENT SIGNS SHOWN TO BE REMOVED BY THE CONTRACTOR FOR CONSTRUCTION PURPOSES OTHER THAN STOP SIGNS, YIELD SIGNS OR STREET NAME SIGNS SHALL BE SALVAGED. SEE INSTRUCTIONS FOR DISASSEMBLY AND RETURN OF TRAFFIC SIGN EQUIPMENT.
14. ALL STOP, YIELD, WARNING SIGNS, AND STREET NAME SIGNS SHALL BE MAINTAINED IN A CONSPICUOUS LOCATION FOR THE DRIVING PUBLIC. ALL STOP AND YIELD SIGNS REMOVED FOR CONSTRUCTION PURPOSES CAN BE TEMPORARILY ERECTED (NO LESS THAN 7 FEET VERTICAL FROM GRADE) UNTIL THEY CAN BE PERMANENTLY RE-INSTALLED. ANY TEMPORARY STOP OR YIELD SIGN INSTALLATION TO BE LEFT IN PLACE OVERNIGHT WILL REQUIRE PRIOR APPROVAL FROM THE TRAFFIC SERVICES INSPECTOR.
15. ALL EXISTING SIGNS, OTHER THAN STOP, YIELD, WARNING SIGNS, OR STREET NAME SIGNS, SHOWN TO BE USED IN PLACE SHALL BE PROTECTED FROM DAMAGE BY THE CONTRACTOR. THE CONTRACTOR MAY TEMPORARILY REMOVE THE SIGN AND POST TO PREVENT DAMAGE AT THE APPROVAL OF THE INSPECTOR. STORAGE OF THE SIGNS & POSTS IS THE RESPONSIBILITY OF THE CONTRACTOR.
16. ALL SIGN POSTS INSTALLED IN CONCRETE, ASPHALT OR BRICK PAVER ISLANDS OR MEDIANS SHALL HAVE A 6" PVC SLEEVE OR 6" CORE DRILLED HOLE COMPLETELY THROUGH THE FULL DEPTH OF PAVEMENT TO THE TOP OF SUBGRADE. SEE MEDIAN NOSE DETAILS
17. MINOR ADJUSTMENTS IN THE LOCATION OF SIGN POSTS SHOULD BE MADE IN THE FIELD DURING CONSTRUCTION IN ORDER TO MAINTAIN 4'-0" CLEARANCE FROM THE CENTERLINE OF ANY FIRE HYDRANT TO THE FACE OF THE SIGN POST.
18. IN THE EVENT OF UTILITY CONFLICTS WITH SIGN POST LOCATIONS AND THE SIGN POST CANNOT BE RELOCATED, THE CONTRACTOR SHALL BE REQUIRED TO HAND DIG THE SIGN POST AND BACKFILL WITH CONCRETE TO PROVIDE A STURDY POST INSTALLATION. THE COST SHALL BE SUBSIDIARY TO OTHER SIGN BID ITEMS.
19. THE CONTRACTOR SHALL FLAG SIGN LOCATIONS FOR INSTALLATION AND SHALL NOT INSTALL SIGNS UNTIL FINAL LOCATION HAS BEEN APPROVED BY THE ENGINEER OR INSPECTOR.
20. SLIP BASES MAY BE REQUIRED FOR ANY SIGN INSTALLATION REQUIRING MORE THAN ONE POST.



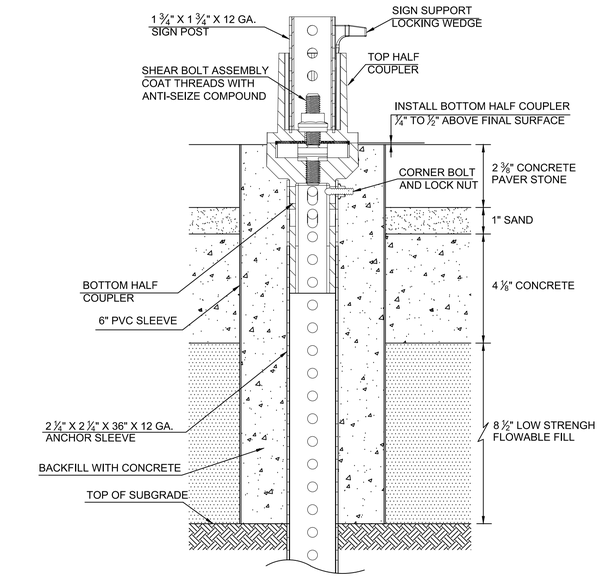


NOTES:
1. THE HEIGHT TO THE BOTTOM OF A SUPPLEMENTAL PLAQUE MOUNTED BELOW ANOTHER SIGN WHEN IT IS LOCATED IN A PEDESTRIAN WALKWAY, OR EXTENDS MORE THAN 4" INTO A PEDESTRIAN WALKWAY, SHALL BE A MINIMUM OF 80" IN COMPLIANCE WITH THE AMERICANS WITH DISABILITY ACT (ADA).
2. IF THE SUPPLEMENTAL PLAQUE IS LESS THAN 12" HIGH, THE HEIGHT TO THE BOTTOM OF THE PRIMARY SIGN SHOULD BE A MINIMUM OF 7'-0".
3. ALL VERTICAL INSTALLATION DIMENSIONS ARE FROM THE BOTTOM OF THE SIGN TO THE TOP OF CURB OR EDGE TRAVEL WAY AS APPLICABLE.
4. Z = SIDE SLOPE, E.G. 4 FOR 4:1 SIDE SLOPE



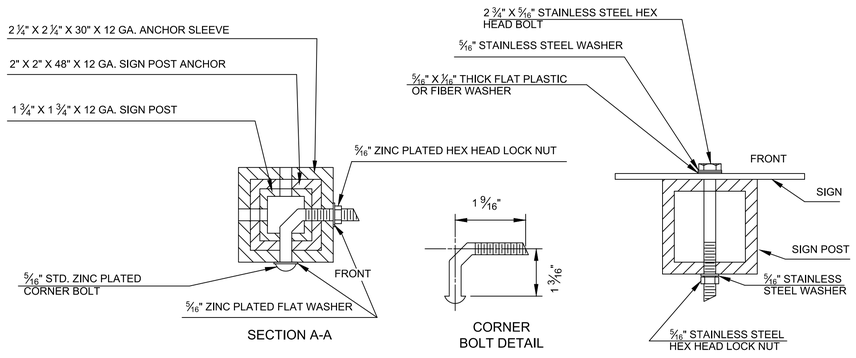


SIGN MOUNTING DETAILS



Break-Away Sign Post Anchor Detail
(Paver Stone Mount)

Break-Away Sign Post Anchor Detail
(Concrete Mount)

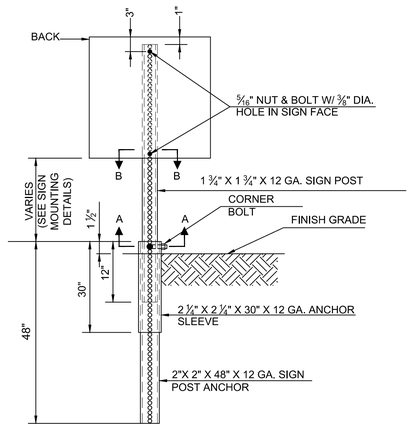


NOTES:
THE DETAIL DRAWING SHOWS THE BOLT ENTERING THE BACK.

SIGN POST INSTALLATION SEQUENCE

1. SIGN POST ANCHOR DRIVEN PARTIALLY INTO SUBGRADE USING A DRIVE CAP WITH SLEDGE OR POWER EQUIPMENT.
2. ANCHOR SLEEVE SLIPPED OVER ANCHOR AND DRIVEN INTO SUBGRADE TOGETHER WITH THE SIGN POST ANCHOR.
3. INSERT SIGN POST INTO THE POST ANCHOR AND BOLT IN PLACE.

NOTE:
FOR IN-PAVEMENT INSTALLATION, COMPLETE STEPS 1 & 2 PRIOR TO THE PLACEMENT OF THE PAVEMENT.
THE FIRST HOLE ABOVE THE FINISHED GRADE LEVEL IN ALL THREE UNITS MUST BE IN LINE FOR INSERTION OF THE CORNER BOLT.
ALL CORNER BOLTS AND NUTS FOR FASTENING THE SIGNS AND SIGN POST ASSEMBLY AND ALL WASHERS SHALL COMPLY WITH APPROPRIATE SECTIONS OF THE STANDARD SPECIFICATIONS (LATEST EDITION) AND SHALL BE A SUBSIDIARY ITEM.



BREAK-AWAY SIGN POST DETAIL

A1 SIGNING: GENERAL NOTES
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912D21D19D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

CIVIL DETAILS

SHEET ID
C-541

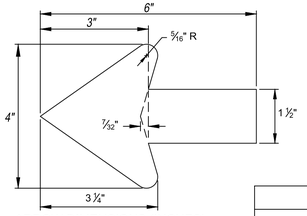
T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C541-SGN-NOTE.dwg
9/9/2026 4:34:57 PM

READY TO ADVERTISE

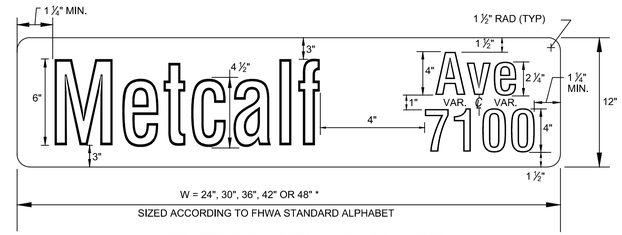
TABLE 1 (ALPHA STREETS)	
STANDARD ABBREVIATION LIST	
AVENUE	AVE
BOULEVARD	BLVD
CIRCLE	CIR
COURT	CT
CREEK	CRK
DRIVE	DR
HIGHWAY	HWY
LANE	LN
PARKWAY	PKWY
PLACE	PL
PLAZA	PLZ
ROAD	RD
STREET	ST
TERRACE	TER
TRAIL	TR
WAY	WAY

TABLE 2 (NUMBERED STREETS)	
STANDARD ABBREVIATION LIST	
FIRST	ST
SECOND	ND
THIRD	RD
FOURTH TO NINTH	TH

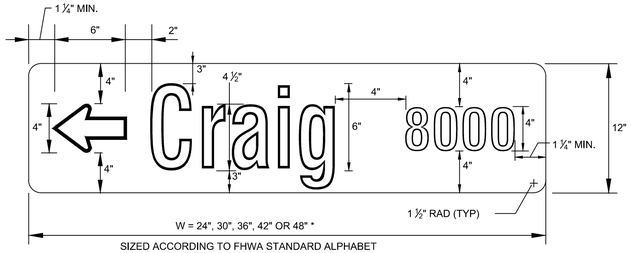
*NOTE: SERIES B 2000 LETTERS SHALL BE USED IN LIEU OF HIGHWAY SERIES C TO AVOID EXCEEDING A 48" SIGN BLANK.



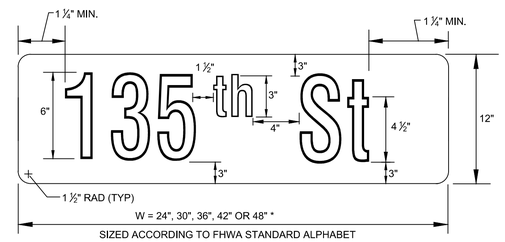
ARROW DIMENSIONS (INCHES)



NAMED STREET SIGN DETAIL W/ BLOCK NUMBER



NAMED STREET SIGN DETAIL / DIRECTIONAL



NUMBERED STREET SIGN DETAIL

SIGN MATERIAL NOTES:

SIGN BLANK: 0.080 GAUGE, ALODIZED 6061-T6 OR 5052-H38

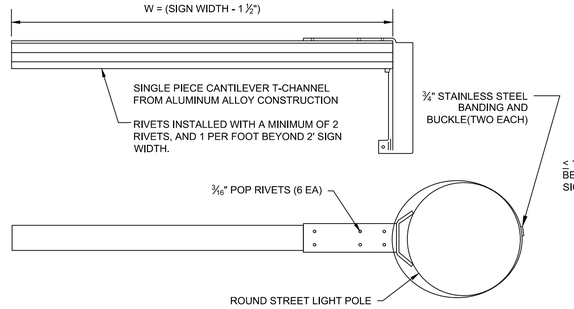
LEGEND: WHITE (NO BORDER)

BACKGROUND: GREEN (FEDERAL COLOR STDS 595A, COLOR NO.14109

TEXT SERIES: SERIES C 2000 AS INDICATED IN THE EXAMPLES

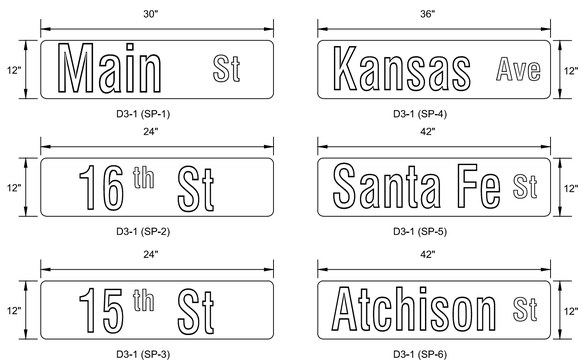
SHEETING: MICRO-ENCAPSULATED, RETRO-REFLECTIVE PRISMATIC SHEETING (TYPE XI)

PROCESS: ELECTRO-CUTABLE FILM

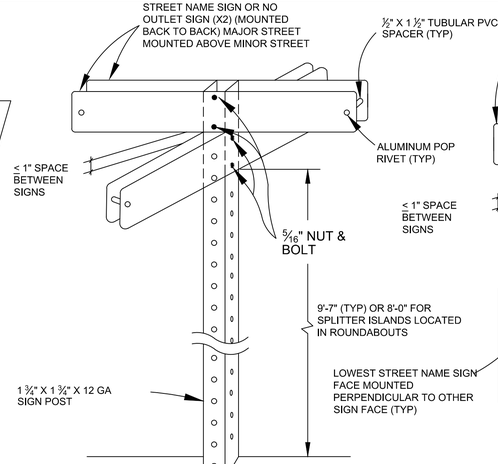


STREET NAME SIGN WING BRACKET DETAIL FOR ROUND TUBULAR SUPPORTS

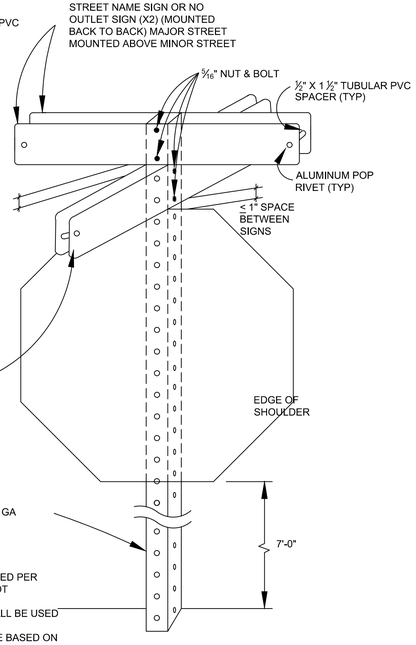
STREET NAME SIGN QUANTITY TABLE				
SIGN DESIGNATION	SIZE (W X H)	AREA (S.F.)	NUMBER	QUANTITY (S.F.)
D3-1 (SP-1)	X 12"	_	-	-
D3-1 (SP-2)	X 12"			
D3-1 (SP-3)	X 12"			
D3-1 (SP-4)	X 12"			
D3-1 (SP-5)	X 12"			
D3-1 (SP-6)	X 12"			
TOTAL				



PROJECT SIGN DETAILS

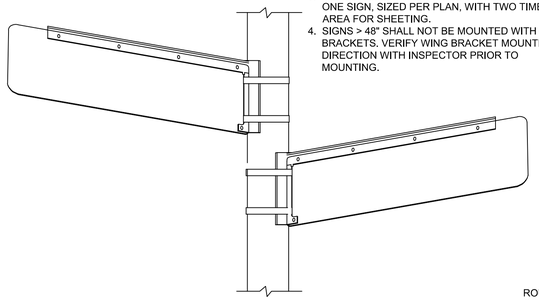


STREET NAME SIGN ASSEMBLY (MOUNTED SEPARATELY)

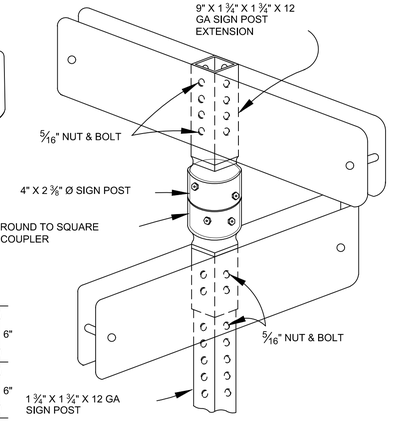


SIGN ASSEMBLY (MOUNTED WITH OTHER SIGN)

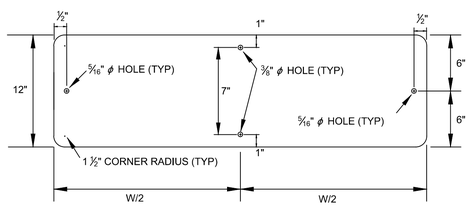
NOTE:
1. ONE WING BRACKET SHALL BE INSTALLED PER EACH SIGN. CROSS BRACKETS ARE NOT PERMITTED.
2. ONE DOUBLED SIDED SIGN BLADE SHALL BE USED WITH WING BRACKETS.
3. MEASUREMENT AND PAYMENT WILL BE BASED ON ONE SIGN, SIZED PER PLAN, WITH TWO TIMES THE AREA FOR SHEETING.
4. SIGNS > 48" SHALL NOT BE MOUNTED WITH WING BRACKETS. VERIFY WING BRACKET MOUNTING DIRECTION WITH INSPECTOR PRIOR TO MOUNTING.



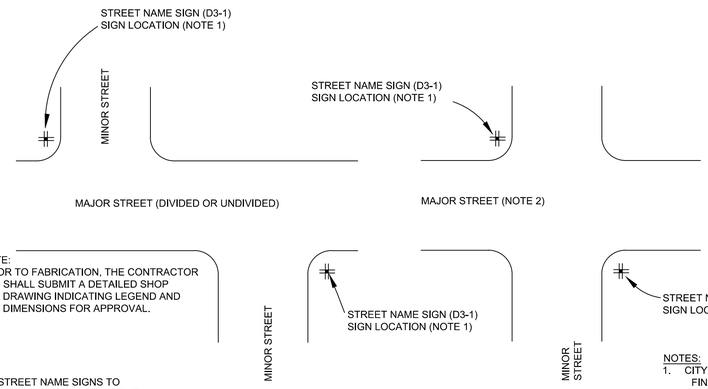
TUBULAR SUPPORT MOUNTING DETAIL



STREET NAME SIGN BRACKET SQUARE TUBULAR SUPPORTS (SKEWED INTERSECTIONS ONLY)



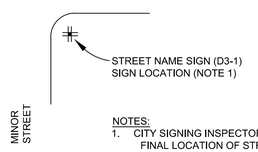
SIGN MOUNTING HOLE DETAIL (FOR MOUNTING ON SQUARE TUBULAR SUPPORTS)



NOTE:
PRIOR TO FABRICATION, THE CONTRACTOR SHALL SUBMIT A DETAILED SHOP DRAWING INDICATING LEGEND AND DIMENSIONS FOR APPROVAL.

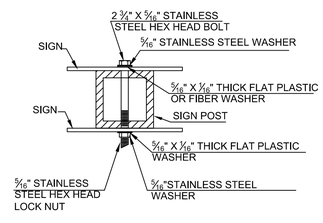
GENERAL NOTE:
WHEN INSTALLING STREET NAME SIGNS TO BRIDGES FOR THE TRAIL SYSTEM, INSTALL THE NEW STREET NAME SIGN TO BRIDGE FACIA WITH 0.25" X 1.25" STAINLESS STEEL TAPCON SCREWS WITH 5/16" STAINLESS STEEL WASHERS AND 5/16" X 1/8" THICK FLAT PLASTIC OR FIBER WASHER.

STREET NAME SIGN LOCATION DETAIL ("T" INTERSECTION)

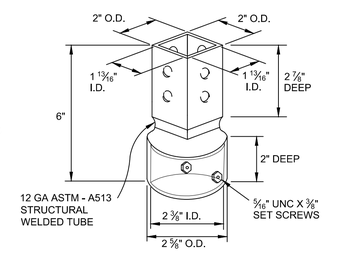


STREET NAME SIGN LOCATION DETAIL (4-LEG INTERSECTION)

NOTES:
1. CITY SIGNING INSPECTOR SHALL APPROVE FINAL LOCATION OF STREET NAME SIGN.
2. IF MAJOR STREET IS LESS THAN 36' BACK TO BACK, ONLY ONE STREET NAME SIGN ASSEMBLY WILL BE INSTALLED IN LOCATION APPROVED BY THE CITY SIGNING INSPECTOR.



DOUBLE SIGN FACE MOUNTING DETAIL



ROUND TO SQUARE COUPLER DETAIL

A1 SIGNING: GROUND MOUNT STREET NAME SIGN DETAILS
SCALE: NOT TO SCALE

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.:
W912D219D1012

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

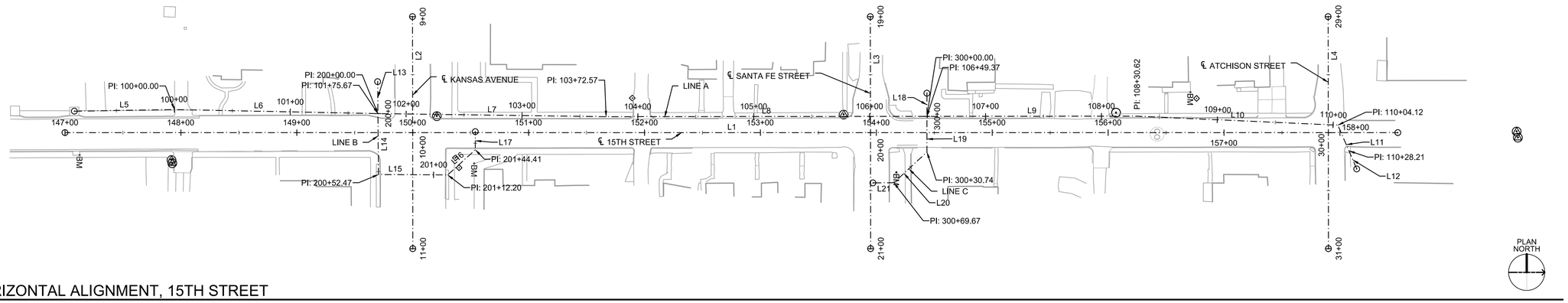
CIVIL DETAILS

SHEET ID
C-542

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\KDOT DETAILS\Traffic\53708_ATCH_CSO_C542-SGN-GNDMT.dwg
9/9/2026 4:35:14 PM

READY TO ADVERTISE

HORIZONTAL ALIGNMENT - 15TH STREET								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L1	147+00.00	1150.00	N1° 09' 39"W	460352.853	2119149.845	158+50.00	461502.617	2119126.546



F1 HORIZONTAL ALIGNMENT, 15TH STREET
SCALE:1"=50'

0 50' 100'

HORIZONTAL ALIGNMENT - SANTA FE STREET								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L3	19+00.00	200.00	N88° 45' 44"E	461045.550	2119035.788	21+00.00	461049.871	2119235.741

HORIZONTAL ALIGNMENT - KANSAS AVENUE								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L2	9+00.00	200.00	N88° 45' 44"E	460650.631	2119043.791	11+00.00	460654.952	2119243.744

HORIZONTAL ALIGNMENT - ATCHISON STREET								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L4	29+00.00	200.00	N88° 45' 44"E	461440.470	2119027.785	31+00.00	461444.790	2119227.738

A1 HORIZONTAL ALIGNMENTS: KANSAS AVENUE, SANTA FE AND ATCHISON STREET
SCALE:1"=50'

HORIZONTAL ALIGNMENT - STORM LINE A (15TH STREET)								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L5	99+13.71	86.29	N2° 06' 25"W	460360.566	2119131.193	100+00.00	460446.802	2119128.021
L6	100+00.00	175.67	N0° 01' 03"W	460446.802	2119128.021	101+75.67	460622.474	2119127.967
L7	101+75.67	196.89	N0° 23' 40"W	460622.474	2119127.967	103+72.57	460819.364	2119126.611
L8	103+72.57	276.80	N1° 12' 47"W	460819.364	2119126.611	106+49.37	461096.103	2119120.751
L9	106+49.37	181.25	N1° 12' 49"W	461096.103	2119120.751	108+30.62	461277.316	2119116.912
L10	108+30.62	173.50	N1° 23' 44"E	461277.316	2119116.912	110+04.12	461450.763	2119121.138
L11	110+04.12	24.09	N65° 32' 23"E	461450.763	2119121.138	110+28.21	461460.737	2119143.064
L12	110+28.21	17.01	N65° 32' 23"E	461460.737	2119143.064	110+45.22	461467.782	2119158.552

HORIZONTAL ALIGNMENT - STORM LINE B (15TH STREET)								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L13	199+72.50	27.50	N88° 23' 59"E	460621.706	2119100.480	200+00.00	460622.474	2119127.967
L14	200+00.00	52.47	N88° 18' 27"E	460622.474	2119127.967	200+52.47	460624.024	2119180.418
L15	200+52.47	59.73	N0° 47' 36"W	460624.024	2119180.418	201+12.20	460683.745	2119179.591
L16	201+12.20	32.21	N43° 27' 29"W	460683.745	2119179.591	201+44.41	460707.127	2119157.435
L17	201+44.41	15.59	S88° 45' 22"W	460707.127	2119157.435	201+60.00	460706.788	2119141.849

HORIZONTAL ALIGNMENT - STORM LINE C (15TH STREET)								
NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L18	299+80.00	20.00	N88° 49' 44"E	461095.695	2119100.758	300+00.00	461096.103	2119120.751
L19	300+00.00	30.74	N88° 45' 04"E	461096.103	2119120.751	300+30.74	461096.773	2119151.482
L20	300+30.74	38.93	S43° 54' 43"E	461096.773	2119151.482	300+69.67	461068.724	2119178.485
L21	300+69.67	18.01	S1° 32' 49"E	461068.724	2119178.485	300+87.68	461050.721	2119178.971

A1 HORIZONTAL ALIGNMENTS: STORM LINE A, B, AND C
SCALE:1"=50'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

15TH STREET
HORIZONTAL ALIGNMENT

SHEET ID
C-601

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C601-C602_ALIGN.dwg
9/9/2026 4:36:07 PM

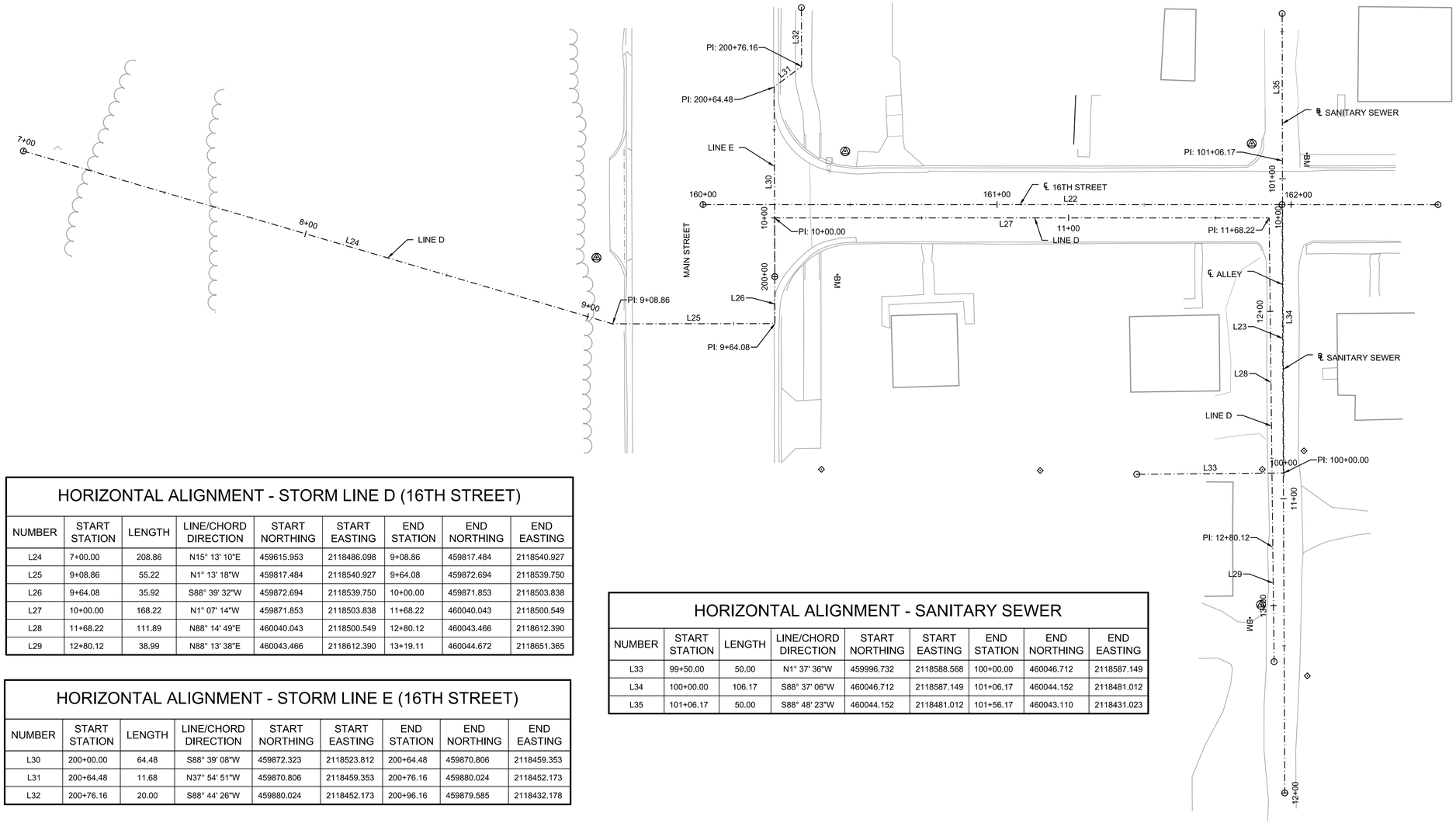
READY TO ADVERTISE

HORIZONTAL ALIGNMENT - 16TH STREET

NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L22	160+00.00	250.00	N1° 07' 01"W	459847.444	2118499.790	162+50.00	460097.397	2118494.917

HORIZONTAL ALIGNMENT - ALLEY

NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L23	10+00.00	200.00	N88° 36' 45"E	460044.342	2118495.947	12+00.00	460049.185	2118695.889



HORIZONTAL ALIGNMENT - STORM LINE D (16TH STREET)

NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L24	7+00.00	208.86	N15° 13' 10"E	459615.953	2118486.098	9+08.86	459817.484	2118540.927
L25	9+08.86	55.22	N1° 13' 18"W	459817.484	2118540.927	9+64.08	459872.694	2118539.750
L26	9+64.08	35.92	S88° 39' 32"W	459872.694	2118539.750	10+00.00	459871.853	2118503.838
L27	10+00.00	168.22	N1° 07' 14"W	459871.853	2118503.838	11+68.22	460040.043	2118500.549
L28	11+68.22	111.89	N88° 14' 49"E	460040.043	2118500.549	12+80.12	460043.466	2118612.390
L29	12+80.12	38.99	N88° 13' 38"E	460043.466	2118612.390	13+19.11	460044.672	2118651.365

HORIZONTAL ALIGNMENT - STORM LINE E (16TH STREET)

NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L30	200+00.00	64.48	S88° 39' 08"W	459872.323	2118523.812	200+64.48	459870.806	2118459.353
L31	200+64.48	11.68	N37° 54' 51"W	459870.806	2118459.353	200+76.16	459880.024	2118452.173
L32	200+76.16	20.00	S88° 44' 26"W	459880.024	2118452.173	200+96.16	459879.585	2118432.178

HORIZONTAL ALIGNMENT - SANITARY SEWER

NUMBER	START STATION	LENGTH	LINE/CHORD DIRECTION	START NORTHING	START EASTING	END STATION	END NORTHING	END EASTING
L33	99+50.00	50.00	N1° 37' 36"W	459996.732	2118588.568	100+00.00	460046.712	2118587.149
L34	100+00.00	106.17	S88° 37' 06"W	460046.712	2118587.149	101+06.17	460044.152	2118481.012
L35	101+06.17	50.00	S88° 48' 23"W	460044.152	2118481.012	101+56.17	460043.110	2118431.023

PLAN NORTH

0 20' 40'

A1 HORIZONTAL ALIGNMENTS: 16TH STREET, ALLEY, STORM LINE D AND STORM LINE E

SCALE:1"=20'

US Army Corps of Engineers®

MARK	DESCRIPTION	DATE

DESIGNED BY: M. CROWL
DRAWN BY: T. ROUSH
CHECKED BY: J. DAVIS
SUBMITTED BY: S. BRISCOE
SIZE: ANSI D
ISSUE DATE: 23 FEBRUARY 2026
SOLICITATION NO.:
TASK ORDER # W912D24F1019
CONTRACT NO.: W912DQ19D1012

U.S. ARMY CORPS OF ENGINEERS
KANSAS CITY DISTRICT
KANSAS CITY, MO

AFFINIS CORP
8900 INDIAN CREEK PARKWAY
CORPATE WOODS BUILDING 6
OVERLAND PARK, KS 66210

ATCHISON, KANSAS
COMBINED SEWER SEPARATION PROJECT
P2 503708

16TH STREET
HORIZONTAL ALIGNMENT

SHEET ID
C-602

T:\24-2309-01\ProjectData\CAD_BIM_Sheets\53708_ATCH_CSO_C601-C602_ALIGN.dwg
9/9/2026 4:36:20 PM

READY TO ADVERTISE